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| Danish                 | 103, 1     |                      |        | asutosh kumar Biswal   | 5     |
| Manu Thakur            | 99, 1      |                      |        | Anu007                 | 5     |
| Rajesh Pradhan         | 96, 6      |                      |        | sanjay                 | 5     |
| Prashant               | 96, 7      |                      |        | Satbir                 | 5     |
| Singh                  |            |                      |        | Singh                  |       |
| neha pawar             | 94, 2      |                      |        | Akhil Nadh PC          | 5     |
| Mithlesh               | 94, 2      |                      |        | Ramayya                | 5     |
| Upadhyay               |            |                      |        | shekhar                | 5     |
| kvkumar                | 93, 13     |                      |        | chauhan                |       |
| gate_asp               | 90, 1      |                      |        | Deepak Poonia          | 4     |
| Arpit Dhuriya          | 82, 2      |                      |        | Dhananjay Kumar Sharma | 4     |
| Shobhit                | 80, 1      |                      |        | Shubham Sharma         | 4     |
| Prasanna Ranganathan   | 78, 1      |                      |        | Gaurav Sharma          | 4     |
| Rahul Lalitkumar Jain  | 77, 43     |                      |        |                        |       |
| Shashank Kumar         | 74, 1      |                      |        |                        |       |
| Prateek                | 73, 1      |                      |        |                        |       |
| Arora                  |            |                      |        |                        |       |
| Himanshu Agarwal       | 72, 1      |                      |        |                        |       |
| IgnitorSandeep         | 68, 1      |                      |        |                        |       |
| Sambit Kumar           | 68, 7      |                      |        |                        |       |
| Soumya Jain            | 67, 6      |                      |        |                        |       |
| abhishek kumar         | 67, 13     |                      |        |                        |       |
| Kalpna Bhargav         | 64, 1      |                      |        |                        |       |
| Ravi                   | 64, 5      |                      |        |                        |       |
| Singh                  |            |                      |        |                        |       |
| Vicky Bajoria          | 62, 1      |                      |        |                        |       |
| sonveer                | 61, 11     |                      |        |                        |       |
| tomar                  |            |                      |        |                        |       |
| Ashish Patel           | 59, 1      |                      |        |                        |       |
| rameshababu            | 59, 7      |                      |        |                        |       |
| Umang Raman            | 58, 6      |                      |        |                        |       |
| anshu                  | 56, 2      |                      |        |                        |       |

|                         |        |
|-------------------------|--------|
| Happy Mittal            | 55, 1  |
| Sourav Roy              | 55, 1  |
| "Shrowd"                | 55, 4  |
| Prateek Dwivedi         | 54, 3  |
| vamsi2376               | 53, 5  |
| paradox                 | 53, 5  |
| gshivam63               | 45, 3  |
| Keith Kr                | 44, 1  |
| Shubham Sharma          | 44, 23 |
| Praveen Saini           | 43, 3  |
| Krishanveer Gangwar     | 42, 4  |
| pawan sahu              | 42, 6  |
| dd                      | 42, 7  |
| Tuhin Dutta             | 41, 8  |
| sumit kumar singh dixit | 39, 1  |
| ranjan621               | 39, 1  |
| habedo007               | 39, 7  |
| Ankesh Gautam           | 38, 1  |
| Catalan 1920            | 38, 2  |
| sanjeev_zero            | 37, 1  |
| Achintya Desai          | 36, 2  |
| Pranay Datta            | 35, 2  |
| ANKUR MAHIWAL           | 34, 1  |
| Mojo Jojo               | 34, 2  |
| Gate_15_isHere          | 33, 1  |
| REGGIE S                | 33, 4  |
| ponagraj                | 31, 1  |
| Jithin Jayan            | 31, 2  |
| gatecse                 | 29, 1  |



## 1.0.1 ISRO-DEC2017-20



Match the following and choose the correct answer for the order  $A, B, C, D$

|                                   |                          |
|-----------------------------------|--------------------------|
| A. Strassen matrix multiplication | p. Decrease and Conquer  |
| B. Insertion sort                 | q. Dynamic Programming   |
| C. Guassian Elimination           | r. Divide and Conquer    |
| D. Floyd shortest path algorithm  | s. Transform and Conquer |

A. a-r, b-s, c-p, d-q

B. a-r, b-p, c-s, d-q

C. a-q, b-s, c-p, d-r

D. a-s, b-p, c-q, d-r

isrodec2017 algorithms

Answer key

## 1.0.2 ISRO-DEC2017-44



A sorting technique is called stable if

A. If it takes  $O(n \log n)$  time.

B. It uses divide and conquer technique.

C. Relative order of occurrence of non-distinct elements is maintained.

D. It takes  $O(n)$  space.

isrodec2017

Answer key

## 1.1

## Algorithm Design Technique (5)

## 1.1.1 Algorithm Design Technique: GATE CSE 1994 | Question: 1.19, ISRO2016-31



Algorithm design technique used in quicksort algorithm is?

A. Dynamic programming

B. Backtracking

C. Divide and conquer

D. Greedy method

gate1994 algorithms algorithm-design-technique quick-sort easy isro2016

Answer key

## 1.1.2 Algorithm Design Technique: GATE CSE 1998 | Question: 1.21, ISRO2008-16



Which one of the following algorithm design techniques is used in finding all pairs of shortest distances in a graph?

A. Dynamic programming

B. Backtracking

C. Greedy

D. Divide and Conquer

gate1998 algorithms algorithm-design-technique easy isro2008

Answer key

## 1.1.3 Algorithm Design Technique: ISRO CSE 2007 | Question: 55



Selection sort algorithm design technique is an example of

A. Greedy method

B. Divide-and-conquer

## C. Dynamic Programming

isro2007 algorithms algorithm-design-technique easy

Answer key 

## D. Backtracking

### 1.1.4 Algorithm Design Technique: ISRO CSE 2011 | Question: 66



Which of the following algorithm design technique is used in merge sort?

- A. Greedy method
- B. Backtracking
- C. Dynamic programming
- D. Divide and Conquer

isro2011 algorithms algorithm-design-technique merge-sort

Answer key 

### 1.1.5 Algorithm Design Technique: ISRO CSE 2018 | Question: 16



The following paradigm can be used to find the solution of the problem in minimum time:

Given a set of non-negative integer and a value  $K$ , determine if there is a subset of the given set with sum equal to  $K$ :

- a. Divide and Conquer
- b. Dynamic Programming
- c. Greedy Algorithm
- d. Branch and Bound

isro2018 algorithm-design-technique algorithms

Answer key 

## 1.2 Asymptotic Notation (1)

### 1.2.1 Asymptotic Notation: ISRO-DEC2017-18



Consider the recurrence equation

$$T(n) = \begin{cases} 2T(n-1), & \text{if } n > 0 \\ 1, & \text{otherwise} \end{cases}$$

Then  $T(n)$  is (in *big O* order)

- A.  $O(n)$
- B.  $O(2^n)$
- C.  $O(1)$
- D.  $O(\log n)$

isrodec2017 recurrence-relation asymptotic-notation

Answer key 

## 1.3 Binary Search (2)

### 1.3.1 Binary Search: ISRO CSE 2007 | Question: 59



The time taken by binary search algorithm to search a key in a sorted array of  $n$  elements is

- A.  $O(\log_2 n)$
- B.  $O(n)$
- C.  $O(n \log_2 n)$
- D.  $O(n^2)$

isro2007 algorithms binary-search time-complexity

Answer key 

### 1.3.2 Binary Search: ISRO CSE 2014 | Question: 28



Suppose there are 11 items in sorted order in an array. How many searches are required on the average, if binary search is employed and all searches are successful in finding the item?

- A. 3.00
- B. 3.46
- C. 2.81
- D. 3.33

Answer key 

## 1.4

## Decision Table (1)

## 1.4.1 Decision Table: ISRO CSE 2007 | Question: 69



A rule in a limited entry decision table is a

- A. row of the table consisting of condition entries
- B. row of the table consisting of action entries
- C. column of the table consisting of condition entries and the corresponding action entries
- D. columns of the table consisting of conditions of the stub

isro2007 algorithms decision-table

Answer key 

## 1.5

## Dijkstras Algorithm (2)

## 1.5.1 Dijkstras Algorithm: ISRO CSE 2007 | Question: 80



Dijkstra's algorithm is used to

- A. Create LSAs
- B. Flood an internet with information
- C. Calculate the routing tables
- D. Create a link state database

isro2007 algorithms graph-algorithms shortest-path dijkstras-algorithm

Answer key 

## 1.5.2 Dijkstras Algorithm: ISRO CSE 2023 | Question: 20



Match the following:

|                    |                                        |
|--------------------|----------------------------------------|
| (A) Floyd Warshall | (i) shortest path between two vertices |
| (B) Dijkstra       | (ii) single source shortest path       |
| (C) Kruskal's      | (iii) Minimum spanning tree            |
| (D) Bellman ford   | (iv) solving all pair shortest path    |

- A. (A)-(iii), (B)-(ii), (C)-(iv), (D)-(i)
- B. (A)-(iv), (B)-(i), (C)-(iii), (D)-(ii)
- C. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
- D. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

isro-cse-2023 minimum-spanning-tree dijkstras-algorithm

## 1.6

## Graph Algorithms (3)

## 1.6.1 Graph Algorithms: ISRO CSE 2017 | Question: 17



Which of the following data structure is useful in traversing a given graph by breadth first search?

- A. Stack
- B. Queue
- C. List
- D. None of the above

isro2017 data-structures graph-algorithms breadth-first-search easy

Answer key 



### 1.6.2 Graph Algorithms: ISRO CSE 2017 | Question: 76



Which of the following algorithms solves the all pair shortest path problem?

- A. Prim's algorithm
- B. Dijkstra's algorithm
- C. Bellman ford algorithm
- D. Floyd warshalls algorithm

isro2017 algorithms graph-algorithms

Answer key

### 1.6.3 Graph Algorithms: ISRO CSE 2018 | Question: 33



Which of the following is application of Breath First Search on the graph?

- A. Finding diameter of the graph
- B. Finding bipartite graph
- C. Both (a) and (b)
- D. None of the above

isro2018 graph-algorithms breadth-first-search algorithms

Answer key

## 1.7 Hashing (3)

### 1.7.1 Hashing: GATE CSE 1989 | Question: 1-vii, ISRO2015-14



A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols  $S_1$  to  $S_7$  initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

|   |    |
|---|----|
| 0 | S7 |
| 1 | S1 |
| 2 |    |
| 3 | S4 |
| 4 | S2 |
| 5 |    |
| 6 | S5 |
| 7 |    |
| 8 | S6 |
| 9 | S3 |

- A. 4
- B. 5
- C. 6
- D. 3

hashing isro2015 gate1989 algorithms normal

Answer key

### 1.7.2 Hashing: ISRO CSE 2020 | Question: 24



In linear hashing, if blocking factor  $bfr$ , loading factor  $i$  and file buckets  $N$  are known, the number of records will be

- A.  $cr = i + bfr + N$
- B.  $r = i - bfr - N$
- C.  $r = i + bfr - N$
- D.  $r = i * bfr * N$

isro-2020 algorithms hashing normal

Answer key

### 1.7.3 Hashing: ISRO-DEC2017-50



The characters of the string K R P C S N Y T J M are inserted into a hash table of the size of size 10 using a hash function

$$h(x) = (ord(x) - ord(A) + 1) \bmod 10$$

If linear probing is used to resolve collisions, then the following insertion causes the collision

A. *Y*B. *C*C. *M*D. *P*

isrodec2017 hashing easy

Answer key 

## 1.8

## Huffman Code (1)

## 1.8.1 Huffman Code: ISRO CSE 2020 | Question: 34



Huffman tree is constructed for the following data :  $\{A, B, C, D, E\}$  with frequency  $\{0.17, 0.11, 0.24, 0.33 \text{ and } 0.15\}$  respectively. 100 00 01101 is decoded as

A. *BACE*B. *CADE*C. *BAD*D. *CADD*

isro-2020 algorithms huffman-code normal

Answer key 

## 1.9

## Identify Function (6)

## 1.9.1 Identify Function: ISRO CSE 2007 | Question: 23



Study the following program

```
//precondition: x>=0
public void demo(int x)
{
    System.out.print(x % 10);
    if (x % 10 != 0)
    {
        demo(x/10);
    }
    System.out.print(x%10);
}
```

Which of the following is printed as a result of the call demo (1234)?

A. 1441

B. 3443

C. 12344321

D. 43211234

isro2007 algorithms identify-function

Answer key 

## 1.9.2 Identify Function: ISRO CSE 2008 | Question: 69



What is the value of  $F(4)$  using the following procedure:

```
function F(K : integer)
integer;
begin
if (k<3) then F:=k else F:=F(k-1)*F(k-2)+F(k-3)
end;
```

A. 5

B. 6

C. 7

D. 8

isro2008 algorithms recursion identify-function

Answer key 

## 1.9.3 Identify Function: ISRO CSE 2011 | Question: 40



Consider the following pseudocode

```
x:=1;
i:=1;
while ( x <= 500)
begin
x:=2^x;
```

```
i:=i+1;
end
```

What is the value of  $i$  at the end of the pseudocode?

- A. 4                      B. 5                      C. 6                      D. 7

isro2011   algorithms   identify-function

Answer key 

#### 1.9.4 Identify Function: ISRO CSE 2018 | Question: 12



An array  $A$  consists of  $n$  integers in locations  $A[0], A[1], \dots, A[n-1]$ . It is required to shift the elements of the array cyclically to the left by  $k$  places, where  $1 \leq k \leq (n-1)$ . An incomplete algorithm for doing this in linear time, without using another array is given below. Complete the algorithm by filling in the blanks.

```
min=n; i=0;
while( ) {
    temp= A[i]; j=i;
    while( ) {
        A[j]= ;
        j=(j+k) mod n;
        if(j<min) then
            min = j;
        }
    A[(n+i-k) mod n]= ;
    i= ;
}
```

- a.  $i > \text{min}; \quad j! = (n+1) \bmod n; \quad A[j+k] \quad \text{temp}; \quad i+1;$   
 b.  $i < \text{min}; \quad j! = (n+i) \bmod n; \quad A[j+k] \quad \text{temp}; \quad i+1;$   
 c.  $i > \text{min}; \quad j! = (n+i+k) \bmod n; \quad A[j+k] \quad \text{temp}; \quad i+1;$   
 d.  $i < \text{min}; \quad j! = (n+i-k) \bmod n; \quad A[(j+k) \bmod n] \quad \text{temp}; \quad i+1;$

isro2018   algorithms   identify-function

Answer key 

#### 1.9.5 Identify Function: ISRO CSE 2018 | Question: 7



Assume  $A$  and  $B$  are non-zero positive integers. The following code segment:

```
while (A!=B) {
    if*(A> B)
        A -= B;
    else
        B -= A;
}
cout<<A; // printing the value of A
```

- a. Computes the *LCM* of two numbers                      b. Divides the larger number by the smaller number  
 c. Computes the *GCD* of two numbers                      d. Finds the smaller of two numbers

isro2018   algorithms   identify-function

Answer key 

#### 1.9.6 Identify Function: ISRO CSE 2018 | Question: 72



Consider the following C code segment

```
int f(int x)
{
    if(x<1) return 1;
    else return (if(x-1)+g(x));
}
int g(int x)
```

```
{
    if(x<2) return 2;
    else return (if(x-1)+g(x/2));
}
```

Of the following, which best describes the growth of  $f(x)$  as a function of  $x$ ?

- a. Linear                      b. Exponential                      c. Quadratic                      d. Cubic

isro2018 algorithms identify-function time-complexity

Answer key 

## 1.10

### Linear Probing (3)

#### 1.10.1 Linear Probing: ISRO CSE 2014 | Question: 64



Consider a 13 element hash table for which  $f(\text{key}) = \text{key} \bmod 13$  is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order?

- A. 0                      B. 1                      C. 11                      D. 12

isro2014 hashing linear-probing

Answer key 

#### 1.10.2 Linear Probing: ISRO CSE 2016 | Question: 29



A Hash Function  $f$  defined as  $f(\text{key}) = \text{key} \bmod 7$ . With linear probing while inserting the keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0, in which location key 11 will be stored (Count table index 0 as 0<sup>th</sup> location)?

- A. 3                      B. 4                      C. 5                      D. 6

hashing isro2016 linear-probing

Answer key 

#### 1.10.3 Linear Probing: ISRO CSE 2018 | Question: 36



A hash table with 10 buckets with one slot per bucket is depicted here. The symbols,  $S_1$  to  $S_7$  are initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

|   |       |
|---|-------|
| 0 | $S_7$ |
| 1 | $S_1$ |
| 2 |       |
| 3 | $S_4$ |
| 4 | $S_2$ |
| 5 |       |
| 6 | $S_5$ |
| 7 |       |
| 8 | $S_6$ |
| 9 | $S_3$ |

- A. 4                      B. 5                      C. 6                      D. 3

isro2018 algorithms hashing linear-probing

Answer key 

1.11

## Machine Learning (1)

## 1.11.1 Machine Learning: ISRO CSE 2023 | Question: 16



Which of the following clustering technique is used by K- Means Algorithm:

- A. Hierarchical Technique
- B. Partitional technique
- C. Divisive
- D. Agglomerative

isro-cse-2023 non-gatecse machine-learning

Answer key

1.12

## Master Theorem (1)

## 1.12.1 Master Theorem: ISRO CSE 2020 | Question: 21



The master theorem

- A. assumes the subproblems are unequal sizes
- B. can be used if the subproblems are of equal size
- C. cannot be used for divide and conquer algorithms
- D. cannot be used for asymptotic complexity analysis

isro-2020 algorithms master-theorem easy

Answer key

1.13

## Matrix Chain Ordering (1)

## 1.13.1 Matrix Chain Ordering: ISRO CSE 2020 | Question: 79



Consider product of three matrices  $M_1, M_2$  and  $M_3$  having  $w$  rows and  $x$  columns,  $x$  rows and  $y$  columns, and  $y$  rows and  $z$  columns. Under what condition will it take less time to compute the product as  $(M_1 M_2) M_3$  than to compute  $M_1 (M_2 M_3)$ ?

- A. Always take the same time
- B.  $(1/x + 1/z) < (1/w + 1/y)$
- C.  $x > y$
- D.  $(w + x) > (y + z)$

isro-2020 algorithms matrix-chain-ordering normal

Answer key

1.14

## Merge Sort (3)

## 1.14.1 Merge Sort: GATE CSE 1999 | Question: 1.14, ISRO2015-42



If one uses straight two-way merge sort algorithm to sort the following elements in ascending order:

20, 47, 15, 8, 9, 4, 40, 30, 12, 17

then the order of these elements after second pass of the algorithm is:

- A. 8, 9, 15, 20, 47, 4, 12, 17, 30, 40
- B. 8, 15, 20, 47, 4, 9, 30, 40, 12, 17
- C. 15, 20, 47, 4, 8, 9, 12, 30, 40, 17
- D. 4, 8, 9, 15, 20, 47, 12, 17, 30, 40

gate1999 algorithms merge-sort normal isro2015 sorting

Answer key

### 1.14.2 Merge Sort: ISRO CSE 2007 | Question: 58



The average case and worst case complexities for Merge sort algorithm are

- A.  $O(n^2), O(n^2)$
- B.  $O(n^2), O(n \log_2 n)$
- C.  $O(n \log_2 n), O(n^2)$
- D.  $O(n \log_2 n), O(n \log_2 n)$

isro2007 algorithms sorting merge-sort

Answer key

### 1.14.3 Merge Sort: ISRO CSE 2018 | Question: 35



Given two sorted list of size  $m$  and  $n$  respectively. The number of comparisons needed the worst case by the merge sort algorithm will be:

- A.  $m \times n$
- B. maximum of  $m$  and  $n$
- C. minimum of  $m$  and  $n$
- D.  $m + n - 1$

isro2018 algorithms merge-sort

Answer key

## 1.15 Minimum Spanning Tree (1)

### 1.15.1 Minimum Spanning Tree: ISRO2015-41



The number of spanning trees for a complete graph with seven vertices is

- A.  $2^5$
- B.  $7^5$
- C.  $3^5$
- D.  $2^{2 \times 5}$

isro2015 algorithms minimum-spanning-tree

Answer key

## 1.16 Non Gatecse (1)

### 1.16.1 Non Gatecse: ISRO-2017



Let  $S$  be an NP-complete problem.  $Q$  and  $R$  are other two problems not known to be NP.  $Q$  is polynomial time reducible to  $S$  and  $S$  is polynomial time reducible to  $R$ . Which of the following statements is true ?

- A)  $R$  is NP-complete
- B)  $R$  is NP-hard
- C)  $Q$  is NP-complete
- D)  $Q$  is NP-hard

isro2017 non-gatecse

Answer key

## 1.17 P NP NPC NPH (2)

### 1.17.1 P NP NPC NPH: GATE CSE 2004 | Question: 30, ISRO2017-10



The problem 3-SAT and 2-SAT are

- A. both in P
- B. both NP complete
- C. NP-complete and in P respectively
- D. undecidable and NP complete respectively

gatecse-2004 algorithms p-np-npc-nph easy isro2017 out-of-gatecse-syllabus

Answer key

### 1.17.2 P NP NPC NPH: GATE CSE 2006 | Question: 16, ISRO-DEC2017-27



Let S be an NP-complete problem and Q and R be two other problems not known to be in NP. Q is polynomial time reducible to S and S is polynomial-time reducible to R. Which one of the following statements is true?

- A. R is NP-complete
- B. R is NP-hard
- C. Q is NP-complete
- D. Q is NP-hard

gatecse-2006 algorithms p-np-npc-nph normal isrodec2017 out-of-gatecse-syllabus

Answer key

### 1.18 Quick Sort (1)

#### 1.18.1 Quick Sort: ISRO2015-12



A machine needs a minimum of 100 sec to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately

- A. 50.2 sec
- B. 6.7 sec
- C. 72.7 sec
- D. 11.2 sec

isro2015 algorithms quick-sort

Answer key

### 1.19 Recurrence Relation (5)

#### 1.19.1 Recurrence Relation: GATE CSE 1994 | Question: 1.7, ISRO2017-14



The recurrence relation that arises in relation with the complexity of binary search is:

- A.  $T(n) = 2T\left(\frac{n}{2}\right) + k$ , k is a constant
- B.  $T(n) = T\left(\frac{n}{2}\right) + k$ , k is a constant
- C.  $T(n) = T\left(\frac{n}{2}\right) + \log n$
- D.  $T(n) = T\left(\frac{n}{2}\right) + n$

gate1994 algorithms recurrence-relation easy isro2017

Answer key

#### 1.19.2 Recurrence Relation: GATE CSE 2004 | Question: 83, ISRO2015-40



The time complexity of the following C function is (assume  $n > 0$ )

```
int recursive (int n) {
    if(n == 1)
        return (1);
    else
        return (recursive (n-1) + recursive (n-1));
}
```

- A.  $O(n)$
- B.  $O(n \log n)$
- C.  $O(n^2)$
- D.  $O(2^n)$

gatecse-2004 algorithms recurrence-relation time-complexity normal isro2015

Answer key

#### 1.19.3 Recurrence Relation: GATE CSE 2006 | Question: 51, ISRO2016-34



Consider the following recurrence:

$$T(n) = 2T(\sqrt{n}) + 1, T(1) = 1$$

Which one of the following is true?

- A.  $T(n) = \Theta(\log \log n)$
- B.  $T(n) = \Theta(\log n)$
- C.  $T(n) = \Theta(\sqrt{n})$
- D.  $T(n) = \Theta(n)$

Answer key **1.19.4 Recurrence Relation: ISRO CSE 2011 | Question: 56**

Let  $T(n)$  be defined by  $T(1) = 10$  and  $T(n+1) = 2n + T(n)$  for all integers  $n \geq 1$ . Which of the following represents the order of growth of  $T(n)$  as a function of  $n$ ?

- A.  $O(n)$                       B.  $O(n \log n)$                       C.  $O(n^2)$                       D.  $O(n^3)$

isro2011 algorithms time-complexity recurrence-relation

Answer key **1.19.5 Recurrence Relation: ISRO CSE 2018 | Question: 37**

The running time of an algorithm is given by:

$$T(n) = T(n-1) + T(n-2) - T(n-3), \text{ if } n > 3$$

$$= n, \text{ otherwise}$$

Then what should be the relation between  $T(1), T(2), T(3)$ , so that the order of the algorithm is constant?

- a.  $T(1) = T(2) = T(3)$                       b.  $T(1) + T(3) = 2T(2)$   
c.  $T(1) - T(3) = T(2)$                       d.  $T(1) + T(2) = T(3)$

isro2018 algorithms recurrence-relation time-complexity

Answer key **1.20****Relations (1)****1.20.1 Relations: ISRO CSE 2017 | Question: 80**

The time complexity of computing the transitive closure of a binary relation on a set of  $n$  elements is known to be

- a.  $O(n \log n)$   
b.  $O(n^{3/2})$   
c.  $O(n^3)$   
d.  $O(n)$

isro2017 relations algorithms time-complexity

Answer key **1.21****Searching (2)****1.21.1 Searching: GATE CSE 1996 | Question: 2.13, ISRO2016-28**

The average number of key comparisons required for a successful search for sequential search on  $n$  items is

- A.  $\frac{n}{2}$                       B.  $\frac{n-1}{2}$                       C.  $\frac{n+1}{2}$                       D. None of the above

gate1996 algorithms easy isro2016 searching

Answer key 



### 1.21.2 Searching: ISRO CSE 2011 | Question: 70



Number of comparisons required for an unsuccessful search of an element in a sequential search organized, fixed length, symbol table of length  $L$  is

- A.  $L$                       B.  $L/2$                       C.  $(L+1)/2$                       D.  $2L$

isro2011   algorithms   searching

Answer key

### 1.22                      Sorting (11)

#### 1.22.1 Sorting: GATE CSE 2006 | Question: 14, ISRO2011-14



Which one of the following in place sorting algorithms needs the minimum number of swaps?

- A. Quick sort                      B. Insertion sort                      C. Selection sort                      D. Heap sort

gatecse-2006   algorithms   sorting   easy   isro2011

Answer key

#### 1.22.2 Sorting: ISRO CSE 2008 | Question: 77



How many comparisons are needed to sort an array of length 5 if a straight selection sort is used and array is already in the opposite order?

- A. 1                      B. 10                      C. 15                      D. 20

isro2008   algorithms   sorting   selection-sort

Answer key

#### 1.22.3 Sorting: ISRO CSE 2014 | Question: 62



Consider the following sorting algorithms.

- I. Quicksort
- II. Heapsort
- III. Mergesort

Which of them perform in least time in the worst case?

- A. I and II only                      B. II and III only                      C. III only                      D. I, II and III

isro2014   algorithms   sorting

Answer key

#### 1.22.4 Sorting: ISRO CSE 2017 | Question: 15



Which one of the following in-place sorting algorithms needs the minimum number of swaps?

- A. Insertion Sort                      B. Quick Sort                      C. Heap Sort                      D. Selection Sort

isro2017   algorithms   sorting

Answer key

#### 1.22.5 Sorting: ISRO CSE 2017 | Question: 49



The number of swappings needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order using bubble sort is

A. 11

B. 12

C. 13

D. 10

isro2017 algorithms sorting

Answer key **1.22.6 Sorting: ISRO CSE 2018 | Question: 66**

Of the following sorting algorithms, which has a running time that is least dependent on the initial ordering of the input?

- a. Mege Sort                      b. Insertion Sort                      c. Selection Sort                      d. Quick Sort

isro2018 algorithms sorting

Answer key **1.22.7 Sorting: ISRO CSE 2020 | Question: 33**

If an array  $A$  contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array  $A$  after third pass of insertion sort?

- A. 67, 12, 10, 5, 4, 7, 23                      B. 4, 7, 10, 23, 67, 12, 5  
C. 4, 5, 7, 67, 10, 12, 23                      D. 10, 7, 4, 67, 23, 12, 5

isro-2020 algorithms sorting normal

Answer key **1.22.8 Sorting: ISRO CSE 2020 | Question: 65**

Of the following sort algorithms, which has execution time that is least dependant on initial ordering of the input?

- A. Insertion sort                      B. Quick sort                      C. Merge sort                      D. Selection sort

isro-2020 algorithms sorting normal

Answer key **1.22.9 Sorting: ISRO CSE 2023 | Question: 6**

Worst case time complexity of for  $n$  elements

- A.  $O(n \log n)$                       B.  $O(\log n)$   
C.  $O(n^2)$                       D.  $O(n)$

isro-cse-2023 algorithms heap-sort sorting

Answer key **1.22.10 Sorting: ISRO-2013-12**

Which of the following sorting algorithms has the minimum running time complexity in the best and average case?

- A. Insertion sort, Quick sort                      B. Quick sort, Quick sort  
C. Quick sort, Insertion sort                      D. Insertion sort, Insertion sort

isro2013 sorting

Answer key **1.22.11 Sorting: ISRO-DEC2017-48**

Quick-sort is run on 2 inputs shown below to sort in ascending order :

A.  $1, 2, 3, \dots, n$

B.  $n, n-1, n-2, \dots, 1$

Let  $C_1$  and  $C_2$  be the number of comparisons made for **A** and **B** respectively. Then,

A.  $C_1 > C_2$

B.  $C_1 = C_2$

C.  $C_1 < C_2$

D. Cannot say anything for arbitrary  $n$

isrodec2017 sorting

Answer key 

1.23

Symbol Table (1)

1.23.1 Symbol Table: ISRO CSE 2016 | Question: 39



Access time of the symbolic table will be logarithmic if it is implemented by

A. Linear list

B. Search tree

C. Hash table

D. Self organization list

isro2016 algorithms symbol-table

Answer key 

1.24

Time Complexity (3)

1.24.1 Time Complexity: GATE CSE 2007 | Question: 15, ISRO2016-26



Consider the following segment of C-code:

```
int j, n;  
j = 1;  
while (j <= n)  
    j = j * 2;
```

The number of comparisons made in the execution of the loop for any  $n > 0$  is:

A.  $\lceil \log_2 n \rceil + 1$

B.  $n$

C.  $\lceil \log_2 n \rceil$

D.  $\lfloor \log_2 n \rfloor + 1$

gatecse-2007 algorithms time-complexity normal isro2016

Answer key 

1.24.2 Time Complexity: ISRO CSE 2014 | Question: 67



What is the time complexity for the following C module? Assume that  $n > 0$ .

```
int module(int n)  
{  
    if (n == 1)  
        return 1;  
    else  
        return (n + module(n-1));  
}
```

A.  $O(n)$

B.  $O(\log n)$

C.  $O(n^2)$

D.  $O(n!)$

isro2014 algorithms time-complexity

Answer key 

1.24.3 Time Complexity: ISRO CSE 2020 | Question: 36



What is the complexity of the following code?

```
sum=0;  
for(i=1; i<=n; i*=2)  
    for(j=1; j<=n; j++)  
        sum++;
```

Which of the following is not a valid string?

A.  $O(n^2)$

B.  $O(n \log n)$

C.  $O(n)$

D.  $O(n \log n \log n)$

isro-2020 algorithms time-complexity normal

Answer key 

## Answer Keys

|        |   |        |     |         |     |         |   |        |     |
|--------|---|--------|-----|---------|-----|---------|---|--------|-----|
| 1.0.1  | B | 1.0.2  | C   | 1.1.1   | C   | 1.1.2   | A | 1.1.3  | X   |
| 1.1.4  | D | 1.1.5  | B   | 1.2.1   | B   | 1.3.1   | A | 1.3.2  | A   |
| 1.4.1  | C | 1.5.1  | C   | 1.5.2   | Q-Q | 1.6.1   | B | 1.6.2  | D   |
| 1.6.3  | C | 1.7.1  | B   | 1.7.2   | D   | 1.7.3   | X | 1.8.1  | A   |
| 1.9.1  | D | 1.9.2  | A   | 1.9.3   | B   | 1.9.4   | D | 1.9.5  | C   |
| 1.9.6  | B | 1.10.1 | B   | 1.10.2  | C   | 1.10.3  | B | 1.11.1 | Q-Q |
| 1.12.1 | B | 1.13.1 | B   | 1.14.1  | B   | 1.14.2  | D | 1.14.3 | D   |
| 1.15.1 | B | 1.16.1 | Q-Q | 1.17.1  | C   | 1.17.2  | B | 1.18.1 | B   |
| 1.19.1 | B | 1.19.2 | D   | 1.19.3  | B   | 1.19.4  | C | 1.19.5 | X   |
| 1.20.1 | C | 1.21.1 | C   | 1.21.2  | A   | 1.22.1  | C | 1.22.2 | B   |
| 1.22.3 | B | 1.22.4 | D   | 1.22.5  | D   | 1.22.6  | A | 1.22.7 | B   |
| 1.22.8 | C | 1.22.9 | Q-Q | 1.22.10 | A   | 1.22.11 | B | 1.23.1 | B   |
| 1.24.1 | D | 1.24.2 | A   | 1.24.3  | X   |         |   |        |     |

## 2.1.1 Neural Network: ISRO CSE 2011 | Question: 2



Which of the following is an unsupervised neural network?

- A. RBS
- B. Hopfield
- C. Back propagation
- D. Kohonen

isro2011   neural-network   non-gatecse

Answer key

## 2.1.2 Neural Network: ISRO CSE 2018 | Question: 75



In neural network, the network capacity is defined as:

- a. The traffic (tarry capacity) of the network
- b. The total number of nodes in the network
- c. The number of patterns that can be stored and recalled in a network
- d. None of the above

isro2018   non-gatecse   neural-network

Answer key

## 2.2.1 Non Gatecse: ISRO CSE 2023 | Question: 79



A hybrid Bayesian network contains

- A. Only continuous variables
- B. Only discrete variables
- C. Both discrete and discontinuous variables
- D. Both continuous and discrete variables

isro-cse-2023   non-gatecse

Answer key

## Answer Keys

|       |   |       |   |       |     |
|-------|---|-------|---|-------|-----|
| 2.1.1 | D | 2.1.2 | C | 2.2.1 | Q-Q |
|-------|---|-------|---|-------|-----|



### 3.0.1 ISRO 2015 Regsisters [EE]



Resistor of microprocessor which keeps track of the execution of program and which contain the memory address of next instruction to be executed is called

- (a) Index resistor (b) Program counter (c) Memory address resistor (d) Instruction resistor

isro-ee

Answer key

### 3.0.2 ISRO-DEC2017-38



A CPU has a  $32\text{ KB}$  direct mapped cache with  $128 - \text{byte}$  block size. Suppose A is a 2-d array of size  $512 \times 512$  with elements that occupy  $8\text{ bytes}$  each. Consider the code segment

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        x += A[i][j];
    }
}
```

Assuming that array is stored in order  $A[0][0], A[0][1], A[0][2] \dots$ , the number of cache misses is

- A. 16384                      B. 512                      C. 2048                      D. 1024

isrodec2017

Answer key

### 3.0.3 ISRO-DEC2017-42



In designing a computer's cache system, the cache block (or cache line) size es an important parameter. Which one of the following statements is correct in this context?

- A. Smaller block size incurs lower cache miss penalty.  
 B. Smaller block size implies better spatial locality.  
 C. Smaller block size implies smaller cache tag.  
 D. Smaller block size implies lower cache hit time.

isrodec2017

Answer key

## 3.1

### 8085 Microprocessor (8)

#### 3.1.1 8085 Microprocessor: ISRO CSE 2008 | Question: 39



Consider the following Assembly language program

```
MVIA 30 H
ACI 30 H
XRA A
POP H
```

After the execution of the above program, the contents of the accumulator will be

- A. 30 H                      B. 60 H                      C. 00 H                      D. contents of stack

Answer key **3.1.2 8085 Microprocessor: ISRO CSE 2008 | Question: 45**

The TRAP is one of the interrupts available in INTEL 8085. Which one of the following statements is true of TRAP ?

- A. it is level triggered  
 B. it is negative edge triggered  
 C. it is +ve edge triggered  
 D. it is both +ve and -ve edges triggered

isro2008 co-and-architecture 8085-microprocessor non-gatecse

Answer key **3.1.3 8085 Microprocessor: ISRO CSE 2017 | Question: 20**

Which interrupt in 8085 Microprocessor is unmaskable?

- a. RST 5.5                      b. RST 7.5                      c. TRAP                      d. Both (a) and (b)

isro2017 8085-microprocessor non-gatecse

Answer key **3.1.4 8085 Microprocessor: ISRO ECE 2012 | 8085 processor**

The complement Accumulator (CMA) instruction in 8085 processor on execution affects

- a. Zero flag                      b. Sign flag                      c. Overflow flag                      d. none of the above

isro2012-ece isro-ece co-and-architecture 8085-microprocessor

Answer key **3.1.5 8085 Microprocessor: ISRO ECE 2012 | Computer Architecture**

The register in 8085 that is used to keep track of the memory address of the next opcode to be run in the program is the:

- a. Stack Pointer                      b. Program Counter  
 c. Accumulator                      d. Non of the above

isro-ece co-and-architecture isro2012-ece 8085-microprocessor

Answer key **3.1.6 8085 Microprocessor: ISRO-2013-35**

How many number of times the instruction sequence below will loop before coming out of the loop?

MOV AL, 00H

A1: INC AL

JNZ A1

- A. 1                      B. 255  
 C. 256                      D. Will not come out of the loop.

isro2013 8085-microprocessor non-gatecse

Answer key **3.1.7 8085 Microprocessor: ISRO-2013-36**

In 8085 microprocessor, the ISR for handling trap interrupt is at which location?

A. 3CH

B. 34H

C. 74H

D. 24H

isro2013 8085-microprocessor non-gatecse

Answer key 

### 3.1.8 8085 Microprocessor: ISRO2015-61



The contents of the flag register after execution of the following program by 8085 microprocessor will be

**Program**

SUB A

MVI B,(01)<sub>H</sub>

DCR B

HLT

A. (54)<sub>H</sub>

B. (00)<sub>H</sub>

C. (01)<sub>H</sub>

D. (45)<sub>H</sub>

8085-microprocessor non-gatecse isro2015

Answer key 

## 3.2

### Addressing Modes (11)

#### 3.2.1 Addressing Modes: GATE CSE 1996 | Question: 1.16, ISRO2016-42



Relative mode of addressing is most relevant to writing:

A. Co – routines

B. Position – independent code

C. Shareable code

D. Interrupt Handlers

gate1996 co-and-architecture addressing-modes easy isro2016

Answer key 

#### 3.2.2 Addressing Modes: GATE CSE 2008 | Question: 33, ISRO2009-80



Which of the following is/are true of the auto-increment addressing mode?

I. It is useful in creating self-relocating code

II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation

III. The amount of increment depends on the size of the data item accessed

A. I only

B. II only

C. III only

D. II and III only

gatecse-2008 addressing-modes co-and-architecture normal isro2009

Answer key 

#### 3.2.3 Addressing Modes: GATE IT 2006 | Question: 39, ISRO2009-42



Which of the following statements about relative addressing mode is FALSE?

A. It enables reduced instruction size

B. It allows indexing of array element with same instruction

C. It enables easy relocation of data

D. It enables faster address calculation than absolute addressing

gateit-2006 co-and-architecture addressing-modes normal isro2009

Answer key 



### 3.2.4 Addressing Modes: ISRO CSE 2009 | Question: 21, UGCNET-Dec2012-II: 12



In which addressing mode, the effective address of the operand is generated by adding a constant value to the content of a register?

- A. Absolute mode
- B. Indirect mode
- C. Immediate mode
- D. Index mode

isro2009 co-and-architecture ugcnetcse-dec2012-paper2 addressing-modes

Answer key

### 3.2.5 Addressing Modes: ISRO CSE 2011 | Question: 5



MOV [BX], AL type of data addressing is called?

- A. register
- B. immediate
- C. register indirect
- D. register relative

isro2011 co-and-architecture addressing-modes

Answer key

### 3.2.6 Addressing Modes: ISRO CSE 2017 | Question: 19



The most appropriate matching for the following pairs :

- |                              |                |
|------------------------------|----------------|
| X. Indirect Addressing       | i. Loop        |
| Y. Immediate Addressing      | ii. Pointers   |
| Z. Auto Decrement Addressing | iii. Constants |

- A. X-iii, Y-ii, Z-i
- B. X-ii, Y-iii, Z-i
- C. X-iii, Y-i, Z-ii
- D. X-ii, Y-i, Z-iii

isro2017 co-and-architecture addressing-modes match-the-following easy

Answer key

### 3.2.7 Addressing Modes: ISRO CSE 2020 | Question: 1



The immediate addressing mode can be used for

1. Loading internal registers with initial values
2. Perform arithmetic or logical operation on data contained in instructions

Which of the following is true?

- A. Only 1
- B. Only 2
- C. Both 1 and 2
- D. Immediate mode refers to data in cache

isro-2020 co-and-architecture normal addressing-modes

Answer key

### 3.2.8 Addressing Modes: ISRO CSE 2020 | Question: 15



A stack organized computer is characterised by instructions with

- A. indirect addressing
- B. direct addressing
- C. zero addressing
- D. index addressing

isro-2020 co-and-architecture addressing-modes normal

Answer key

### 3.2.9 Addressing Modes: ISRO CSE 2020 | Question: 3



Which of the following affects the processing power assuming they do not influence each other

1. Data bus capability
2. Address scheme
3. Clock speed

A. 3 only                      B. 1 and 3 only                      C. 2 and 3 only                      D. 1,2 and 3

isro-2020   co-and-architecture   addressing-modes   normal

Answer key 

### 3.2.10 Addressing Modes: ISRO CSE 2020 | Question: 8



Consider a 32-bit processor which supports 70 instructions. Each instruction is 32 bit long and has 4 fields namely opcode, two register identifiers and an immediate operand of unsigned integer type. Maximum value of the immediate operand that can be supported by the processor is 8191. How many registers the processor has?

A. 32                      B. 64                      C. 128                      D. 16

isro-2020   co-and-architecture   addressing-modes   normal

Answer key 

### 3.2.11 Addressing Modes: ISRO-DEC2017-43



Consider an instruction of the type LW R1, 20(R2) which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by adding a constant 20 and contents of R2. Which one best reflects the source operand?

- |                                 |                        |
|---------------------------------|------------------------|
| A. Immediate addressing         | B. Register addressing |
| C. Register Indirect addressing | D. Indexed addressing  |

isrodec2017   addressing-modes

Answer key 

## 3.3 Analog Signal (1)

### 3.3.1 Analog Signal: ISRO ECE 2012 | Computer architecture



Minimum number of bits required to represent maximum value of an analog signal with accuracy of 0.1% is

- a. 8 bits                      b. 9 bits                      c. 10 bits                      d. 12 bits

co-and-architecture   isro-ece   isro2012-ece   analog-signal

## 3.4 CO and Architecture (27)

### 3.4.1 CO and Architecture: ISRO 2006- ECE



If the memory chip size is 256 x 1 bits, then the number of chips required to make up 1K bytes of memory is

- a) 32
- b) 24
- c) 12
- d) 8

isro-ece   co-and-architecture

Answer key 

### 3.4.2 CO and Architecture: ISRO 2006-ECE WAIT states



**WAIT** states are used to

- a) Make the processor wait during a DMA operation
- b) Make the processor wait during a power interrupt processing
- c) Make the processor wait during a power shutdown
- d) Interface slow peripherals to the processor

isro-ece co-and-architecture

Answer key 

### 3.4.3 CO and Architecture: ISRO 2007- ECE Program Counter



When a program is being executed in an 8085 microprocessor, its Program Counter contains

- a) The number of instructions in the current program that have already been executed
- b) The total number of instructions in the program being executed
- c) The memory address of the instruction that is being currently executed
- d) The memory address of the instruction that is to be executed next

isro-ece co-and-architecture

Answer key 

### 3.4.4 CO and Architecture: ISRO 2007- ECE Time division MUX



In time division multiplexing

- a) Time is doubled between bits of a byte
- b) Time slicing at CPU level takes place
- c) Total time available in the channel is divided between several users and each users is allotted a time slice.
- d) None of the above

isro-ece co-and-architecture

Answer key 

### 3.4.5 CO and Architecture: ISRO 2007-ECE Memory



A memory system of size 16K bytes is required to be designed using memory chips, which have 12 address lines and 4 data lines each. The number of such chips required to design the memory system is

- a) 2
- b) 4
- c) 8
- d) 16

isro-ece co-and-architecture

Answer key 

### 3.4.6 CO and Architecture: ISRO 2008- ECE Computer Architecture



The performance gain that can be obtained by improving some portion of a computer can be calculated using

- (a) Moore's law
- (b) Dijkstra's algorithm
- (c) Amdahl's law
- (d) Murphy's law

isro-ece co-and-architecture

Answer key 

### 3.4.7 CO and Architecture: ISRO 2008- ECE Computer architecture



----- machines tend to make use of internal resources of the processor, a rich set of registers and a pipelined organization.

- (a) CISC
- (b) Parallel processor
- (c) RISC
- (d) Array processor

isro-ece co-and-architecture

Answer key 

### 3.4.8 CO and Architecture: ISRO 2008-ECE Cache Memory



The advantage of write (copy) back data cache organization over write through organization is

- (a) Main memory consistency
- (b) Write allocate on write miss
- (c) Less memory bandwidth
- (d) Higher capacity requirement

isro-ece co-and-architecture

Answer key 

### 3.4.9 CO and Architecture: ISRO 2008-ECE Computer Architecture



Micro programming refers to

- (a) Emulation
- (b) Programming at micro level
- (c) The use of storage to implement the control unit
- (d) Array processing

isro-ece co-and-architecture

Answer key 

### 3.4.10 CO and Architecture: ISRO 2008-ECE Computer architecture



Digital signal processors use a computer architecture derived from

- (a) Harvard Architecture
- (b) Von-Neumann Architecture
- (c) Cray Architecture
- (d) None of the above

isro-ece co-and-architecture

Answer key

### 3.4.11 CO and Architecture: ISRO 2008-ECE Digital Logic



Which type of memory has fast erase and write times

- (a) EPROM
- (b) EEPROM
- (c) Flash memory
- (d) None of these

isro-ece co-and-architecture

Answer key

### 3.4.12 CO and Architecture: ISRO 2008-ECE Interrupt request in daisy chain



In a daisy chained connection to the CPU, the peripheral whose interrupt request has the highest priority is the one

- (a) With the largest vector address
- (b) With highest speed of operation
- (c) Electrically nearest to the CPU
- (d) Electrically farthest from the CPU

isro-ece co-and-architecture

Answer key

### 3.4.13 CO and Architecture: ISRO 2008-ECE array processor



An array processor is a machine

- (a) SIMD
- (b) MIMD
- (c) SISD
- (d) MISD

isro-ece co-and-architecture

Answer key 

#### 3.4.14 CO and Architecture: ISRO 2009- ECE Computer Architecture



Consider the following program for 8085

```
XRA A
LXI B, 0007H
LOOP : DCX B
      JNZ LOOP
```

The loop will be executed

- a) 8 times
- b) once
- c) 7 times
- d) infinite times

co-and-architecture isro-ece

Answer key 

#### 3.4.15 CO and Architecture: ISRO 2009- ECE Computer Architecture



Interrupt latency is the time elapsed between:

- a) Occurrence of an interrupt and its detection by the CPU
- b) Assertion of an interrupt and the start of the associated ISR
- c) Assertion of an interrupt and the completion of the associated ISR
- d) Start and completion of associated ISR

isro-ece co-and-architecture

Answer key 

#### 3.4.16 CO and Architecture: ISRO 2009- ECE Computer architecture



An 8-bit microcontroller has an external RAM with the memory map from 8000H to 9FFFH.  
The number of bytes this RAM can store is

- a) 8193
- b) 8192
- c) 8191
- d) 8000

isro-ece co-and-architecture

Answer key 

#### 3.4.17 CO and Architecture: ISRO 2009-ECE : Computer Architecture



A micro controller differs from a microprocessor in terms of

- a) I/O interfaces and instruction decoding
- b) Memory configuration and I/O interfaces

- c) Data bus width and clock speed
- d) Memory configuration and instruction decoding

isro-ece co-and-architecture

Answer key 

### 3.4.18 CO and Architecture: ISRO 2010-ECE Cache memory



A microprocessor has a cache memory with access time of 2 ns and a main memory with access time of 10 ns. If the cache miss ratio is 0.6, what is the average memory access time?

- a) 6.8 ns
- b) 6 ns
- c) 5.2 ns
- d) 12 ns

isro-ece co-and-architecture

Answer key 

### 3.4.19 CO and Architecture: ISRO 2011- ECE Computer Architecture



Which of the following statements with reference to a **generic microprocessor** is correct?

- a. Instruction cycle time period is exactly equal to machine cycle time period
- b. Instruction cycle time period is shorter than machine cycle time period
- c. Machine cycle time period is shorter than instruction cycle time period
- d. Instruction cycle time period is exactly half of machine cycle time period

isro-ece isro2011-ece co-and-architecture

Answer key 

### 3.4.20 CO and Architecture: ISRO 2011-ECE Computer Architecture



A microprocessor with 12 bit address bus will be able to access \_\_\_\_\_ kilobytes of memory

- a) 0.4
- b) 2
- c) 10
- d) 4

isro-ece isro2011-ece co-and-architecture

Answer key 

### 3.4.21 CO and Architecture: ISRO 2011-ECE Computer Architecture



In a 8085 microprocessor system with memory mapped I/O

- a. I/O devices have 8 bit address

- b. I/O devices are accessed using IN and OUT instructions.
- c. There can be maximum 256 input and 256 output devices
- d. Arithmetic and logic operations can be directly performed with I/O data

isro-ece isro2011-ece co-and-architecture

#### 3.4.22 CO and Architecture: ISRO 2011-ECE Computer Architecture



When a microprocessor interfaces with a peripheral or memory device, the normal timing of the microprocessor may need to be altered by introducing\_\_\_\_\_

- a. Latching
- b. Wait states
- c. Tristate logics
- d. None of the above

isro-ece isro2011-ece co-and-architecture

Answer key

#### 3.4.23 CO and Architecture: ISRO 2011-ECE Computer Architecture



Which of the following statement regarding a constant is not true

- a. Constant defined in a package can be referenced by any entity or architecture for which package is used.
- b. The value of constant can be changed with in the design description
- c. Constant defined in an architecture is visible only to that architecture
- d. Constant defined in a process declarative region is not visible outside that process

isro-ece isro2011-ece co-and-architecture

#### 3.4.24 CO and Architecture: ISRO 2015- Main memory [EE]



Techniques that automatically move program and data blocks into physical main memory when they are required for execution are called

- (a) Main memory techniques
- (b) Cache memory techniques
- (c) Virtual memory techniques
- (d) Associate memory techniques

isro-ee co-and-architecture

Answer key

#### 3.4.25 CO and Architecture: ISRO CSE 2009 | Question: 34



The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields. a micro operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?



A. 10,3,1024

B. 8,5,256

C. 5,8,2048

D. 10,3,512

isro2009 co-and-architecture control-unit

Answer key 

### 3.4.26 CO and Architecture: ISRO CSE 2020 | Question: 2



Statements associated with registers of a CPU are given. Identify the false statement.

- A. The program counter holds the memory address of the instruction in execution
- B. Only opcode is transferred to the control unit
- C. An instruction in the instruction register consists of the opcode and the operand
- D. The value of the program counter is incremented by 1 once its value has been read to the memory address register

isro-2020 co-and-architecture control-unit normal

Answer key 

### 3.4.27 CO and Architecture: ISRO ECE 2012 | Computer Architecture



"Cycle Stealing" in microprocessor parlance refers to

- a. Special type of DMA access by an external device
- b. RESET operation at power ON by microprocessor
- c. Interrupt Acknowledge cycle
- d. none of above.

co-and-architecture isro2012-ece isro-ece

## 3.5 Cache Memory (12)

### 3.5.1 Cache Memory: GATE CSE 2001 | Question: 1.7, ISRO2008-18



More than one word are put in one cache block to:

- A. exploit the temporal locality of reference in a program
- B. exploit the spatial locality of reference in a program
- C. reduce the miss penalty
- D. none of the above

gatecse-2001 co-and-architecture easy cache-memory isro2008

Answer key 

### 3.5.2 Cache Memory: GATE IT 2004 | Question: 12, ISRO2016-77



Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

- A. 13.0
- B. 12.8
- C. 12.6
- D. 12.4

gateit-2004 co-and-architecture cache-memory normal isro2016

Answer key 

### 3.5.3 Cache Memory: ISRO CSE 2007 | Question: 14



The principal of the locality of reference justifies the use of

- A. virtual memory
- B. interrupts
- C. main memory
- D. cache memory

isro2007 co-and-architecture cache-memory

Answer key 

### 3.5.4 Cache Memory: ISRO CSE 2007 | Question: 46



Consider a small 2-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently (LRU) scheme. The number of cache misses for the following sequence of block addresses is 8, 12, 0, 12, 8

- A. 2                      B. 3                      C. 4                      D. 5

isro2007 co-and-architecture cache-memory

Answer key

### 3.5.5 Cache Memory: ISRO CSE 2011 | Question: 16



Consider a direct mapped cache with 64 blocks and a block size of 16 bytes. To what block number does the byte address 1206 map to

- A. does not map              B. 6                      C. 11                      D. 54

isro2011 co-and-architecture cache-memory

Answer key

### 3.5.6 Cache Memory: ISRO CSE 2011 | Question: 42



The search concept used in associative memory is

- A. Parallel search                      B. Sequential search  
C. Binary search                      D. Selection search

isro2011 co-and-architecture cache-memory

Answer key

### 3.5.7 Cache Memory: ISRO CSE 2017 | Question: 21



A cache memory needs an access time of 30 ns and main memory 150 ns, what is average access time of CPU (assume hit ratio = 80%)?

- A. 60 ns                      B. 30 ns                      C. 150 ns                      D. 70 ns

isro2017 co-and-architecture cache-memory

Answer key

### 3.5.8 Cache Memory: ISRO CSE 2018 | Question: 73



For a multi-processor architecture, in which protocol a write transaction is forwarded to only those processors that are known to possess a copy of newly altered cache line?

- a. Snoopy bus protocol                      b. Cache coherency protocol  
c. Directory based protocol                      d. None of the above

isro2018 cache-memory

Answer key

### 3.5.9 Cache Memory: ISRO CSE 2020 | Question: 43



Which of the following is an efficient method of cache updating?

- A. Snoopy writes                      B. Write through  
C. Write within                      D. Buffered write

isro-2020 co-and-architecture cache-memory normal

Answer key

### 3.5.10 Cache Memory: ISRO CSE 2020 | Question: 47



How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?

- A. 2 Mbits                      B. 1.7 Mbits                      C. 2.5 Mbits                      D. 1.5 Mbits

isro-2020   co-and-architecture   cache-memory   direct-mapping   normal

Answer key

### 3.5.11 Cache Memory: ISRO-2013-16



How much speed do we gain by using the cache, when cache is used 80% of the time? Assume cache is faster than main memory.

- A. 5.27                      B. 2.00                      C. 4.16                      D. 6.09

isro2013   co-and-architecture   cache-memory

Answer key

### 3.5.12 Cache Memory: ISRO-DEC2017-37



A two-way set associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 – bits. The physical address space is 4 GB. The number of bits in the TAG, SET fields are

- A. 20,7                      B. 19,8                      C. 20,8                      D. 21,9

isrodec2017   cache-memory

Answer key

## 3.6

## Cpu (2)

### 3.6.1 Cpu: ISRO CSE 2011 | Question: 41



If a microcomputer operates at 5 MHz with an 8-bit bus and a newer version operates at 20 MHz with a 32-bit bus, the maximum speed-up possible approximately will be

- A. 2                      B. 4                      C. 8                      D. 16

isro2011   co-and-architecture   cpu

Answer key

### 3.6.2 Cpu: ISRO CSE 2014 | Question: 10



The number of logical CPUs in a computer having two physical quad-core chips with hyper threading enabled is \_\_\_\_\_

- A. 1                      B. 2                      C. 8                      D. 16

co-and-architecture   isro2014   cpu   non-gatecse

Answer key

## 3.7

## DMA (1)

### 3.7.1 DMA: ISRO-2013-14



A processor is fetching instructions at the rate of 1 MIPS. A DMA module is used to transfer characters to RAM from a device transmitting at 9600 bps. How much time will the processor be slowed down due to DMA activity?

- A. 9.6ms                      B. 4.8ms                      C. 2.4ms                      D. 1.2ms

isro2013 dma

Answer key 

### 3.8 Disk (1)

#### 3.8.1 Disk: ISRO CSE 2020 | Question: 46



A magnetic disk has 100 cylinders, each with 10 tracks of 10 sectors. If each sector contains 128 bytes, what is the maximum capacity of the disk in kilobytes?

- A. 1,280,000                      B. 1280                      C. 1250                      D. 128,000

isro-2020 co-and-architecture disk normal

Answer key 

### 3.9 Floating Point Representation (2)

#### 3.9.1 Floating Point Representation: ISRO CSE 2007 | Question: 53



In the Big-Endian system, the computer stores

- A. MSB of data in the lowest memory address of data unit  
 B. LSB of data in the lowest memory address of data unit  
 C. MSB of data in the highest memory address of data unit  
 D. LSB of data in the highest memory address of data unit

isro2007 co-and-architecture floating-point-representation little-endian-big-endian

Answer key 

#### 3.9.2 Floating Point Representation: ISRO CSE 2020 | Question: 5



An array of 2 two byte integers is stored in big endian machine in byte addresses as shown below. What will be its storage pattern in little endian machine ?

| Address        | Data |
|----------------|------|
| $0 \times 104$ | 78   |
| $0 \times 103$ | 56   |
| $0 \times 102$ | 34   |
| $0 \times 101$ | 12   |

A.

| Address        | Data |
|----------------|------|
| $0 \times 104$ | 12   |
| $0 \times 103$ | 56   |
| $0 \times 102$ | 34   |
| $0 \times 101$ | 78   |

B.

| Address        | Data |
|----------------|------|
| $0 \times 104$ | 12   |
| $0 \times 103$ | 34   |
| $0 \times 102$ | 56   |
| $0 \times 101$ | 78   |

C.

| Address        | Data |
|----------------|------|
| $0 \times 104$ | 56   |
| $0 \times 103$ | 78   |
| $0 \times 102$ | 12   |
| $0 \times 101$ | 34   |

D.

| Address        | Data |
|----------------|------|
| $0 \times 104$ | 56   |
| $0 \times 103$ | 12   |
| $0 \times 102$ | 78   |
| $0 \times 101$ | 34   |

isro-2020 co-and-architecture floating-point-representation normal little-endian-big-endian

Answer key 

### 3.10

### IO Handling (6)

#### 3.10.1 IO Handling: GATE CSE 2008 | Question: 64, ISRO2009-13



Which of the following statements about synchronous and asynchronous I/O is NOT true?

- A. An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O
- B. In both synchronous and asynchronous I/O, an ISR (Interrupt Service Routine) is invoked after completion of the I/O
- C. A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O
- D. In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

gatecse-2008 operating-system io-handling normal isro2009

Answer key 

#### 3.10.2 IO Handling: ISRO CSE 2008 | Question: 36



An interrupt in which the external device supplies its address as well as the interrupt requests is known as

- A. vectored interrupt
- B. maskable interrupt

C. non maskable interrupt

D. designated interrupt

isro2008 co-and-architecture io-handling interrupts

Answer key 

### 3.10.3 IO Handling: ISRO CSE 2009 | Question: 78



On receiving an interrupt from an I/O device, the CPU

- A. Halts for a predetermined time
- B. Branches off to the interrupt service routine after completion of the current instruction
- C. Branches off to the interrupt service routine immediately
- D. Hands over control of address bus and data bus to the interrupting device

isro2009 co-and-architecture io-handling

Answer key 

### 3.10.4 IO Handling: ISRO CSE 2011 | Question: 39



Two control signals in microprocessor which are related to Direct Memory Access (DMA) are

- A. INTR & INTA
- B. RD & WR
- C. S0 & S1
- D. HOLD & HLDA

isro2011 co-and-architecture io-handling dma

Answer key 

### 3.10.5 IO Handling: ISRO CSE 2011 | Question: 58



In DMA transfer scheme, the transfer scheme other than burst mode is

- A. cycle technique
- B. stealing technique
- C. cycle stealing technique
- D. cycle bypass technique

isro2011 co-and-architecture io-handling dma

Answer key 

### 3.10.6 IO Handling: ISRO CSE 2018 | Question: 65



Of the following, which best characterizes computers that use memory-mapped I/O?

- a. The computer provides special instructions for manipulating I/O ports
- b. I/O ports are placed at addresses on the bus and are accessed just like other memory locations
- c. To perform I/O operations, it is sufficient to place the data in an address register and call channel to perform the operation
- d. I/O can be performed only when memory management hardware is turned on

isro2018 co-and-architecture io-handling

Answer key 

## 3.11

### Instruction Format (5)

#### 3.11.1 Instruction Format: ISRO CSE 2009 | Question: 79



Compared to CISC processors, RISC processors contain

- A. More register and smaller instruction set
- B. larger instruction set
- C. less registers and smaller instruction set
- D. more transistor elements

isro2009 co-and-architecture instruction-format

Answer key 

### 3.11.2 Instruction Format: ISRO CSE 2016 | Question: 24



In which class of Flynn's taxonomy, Von Neumann architecture belongs to?

- A. SISD                      B. SIMD                      C. MIMD                      D. MISD

co-and-architecture   isro2016   instruction-format

Answer key

### 3.11.3 Instruction Format: ISRO CSE 2018 | Question: 31



A byte addressable computer has a memory capacity of  $2^m KB$  ( $k$  bytes) and can perform  $2^n$  operations. An instruction involving 3 operands and one operator needs maximum of:

- a.  $3m$  bits                      b.  $3m + n$  bits                      c.  $m + n$  bits                      d. none of the above

isro2018   co-and-architecture   instruction-format

Answer key

### 3.11.4 Instruction Format: ISRO CSE 2018 | Question: 6



A data driven machine is one that executes an instruction if the needed data is available. The physical ordering of the code listing does not dictate the course of execution. Consider the following pseudo-code:

- A. Multiply  $E$  by 0.5 to get  $F$                       B. Add  $A$  and  $B$  to get  $E$   
C. Add  $B$  with 0.5 to get  $D$                       D. Add  $E$  and  $F$  to get  $G$   
E. Add  $A$  with 10.5 to get  $C$                       Assume  $A, B, C$  are already assigned values and the  
desired output is  $G$ . Which of the following sequence of execution is valid?

- a. B, C, D, A, E                      b. C, B, E, A, D  
c. A, B, C, D, E                      d. E, D, C, B, A

isro2018   co-and-architecture   instruction-format

Answer key

### 3.11.5 Instruction Format: ISRO CSE 2020 | Question: 49



One instruction tries to write an operand before it is written by previous instruction. This may lead to a dependency called

- A. True dependency                      B. Anti-dependency  
C. Output dependency                      D. Control Hazard

isro-2020   co-and-architecture   instruction-format   normal

Answer key

## 3.12 Interrupts (3)

### 3.12.1 Interrupts: ISRO CSE 2008 | Question: 37



The ability to temporarily halt the CPU and use this time to send information on buses is called

- A. direct memory access                      B. vectoring the interrupt  
C. polling                      D. cycle stealing

isro2008   co-and-architecture   interrupts

Answer key

### 3.12.2 Interrupts: ISRO CSE 2009 | Question: 22



A certain microprocessor requires 4.5 microseconds to respond to an interrupt. Assuming that the three interrupts  $I_1$ ,  $I_2$  and  $I_3$  require the following execution time after the interrupt is recognized:

- i.  $I_1$  requires 25 microseconds
- ii.  $I_2$  requires 35 microseconds
- iii.  $I_3$  requires 20 microseconds

$I_1$  has the highest priority and  $I_3$  has the lowest. What is the possible range of time for  $I_3$  to be executed assuming that it may or may not occur simultaneously with other interrupts?

- A. 24.5 microseconds to 39.5 microseconds
- B. 24.5 microseconds to 93.5 microseconds
- C. 4.5 microseconds to 24.5 microseconds
- D. 29.5 microseconds to 93.5 microseconds

isro2009 co-and-architecture interrupts

Answer key 

### 3.12.3 Interrupts: ISRO CSE 2023 | Question: 32



Which of the following is false about interrupts?

- A. Interrupts can be triggered by a hardware or by a software
- B. Hardware interrupts may be triggered by sending a signal to CPU through a system bus
- C. Software interrupts may be triggered by executing system calls
- D. Trap is a hardware generated interrupt

isro-cse-2023 co-and-architecture interrupts

Answer key 

### 3.13 Machine Instruction (3)

#### 3.13.1 Machine Instruction: GATE CSE 2006 | Question: 09, ISRO2009-35



A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

- A. 400
- B. 500
- C. 600
- D. 700

gatecse-2006 co-and-architecture machine-instruction easy isro2009

Answer key 

#### 3.13.2 Machine Instruction: ISRO CSE 2008 | Question: 42



Which of the following architecture is/are not suitable for realising SIMD?

- A. Vector processor
- B. Array processor
- C. Von Neumann
- D. All of the above

isro2008 co-and-architecture machine-instruction

Answer key 

#### 3.13.3 Machine Instruction: ISRO CSE 2020 | Question: 16



A computer which issues instructions in order, has only 2 registers and 3 opcodes ADD, SUB and MOV. Consider 2 different implementations of the following basic block :

| Case 1          | Case 2          |
|-----------------|-----------------|
| $t1 = a + b;$   | $t2 = c + d;$   |
| $t2 = c + d;$   | $t3 = e - t2;$  |
| $t3 = e - t2;$  | $t1 = a + b;$   |
| $t4 = t1 - t2;$ | $t4 = t1 - t2;$ |



Assume that all operands are initially in memory. Final value of computation also has to reside in memory. Which one is better in terms of memory accesses and by how many MOV instructions?

- A. Case 2,2                      B. Case 2,3                      C. Case 1,2                      D. Case 1,3

isro-2020   co-and-architecture   machine-instruction   normal

Answer key 

### 3.14

#### Memory Interfacing (9)

##### 3.14.1 Memory Interfacing: ISRO CSE 2007 | Question: 37



In comparison with static RAM memory, the dynamic Ram memory has

- A. lower bit density and higher power consumption  
B. higher bit density and higher power consumption  
C. lower bit density and lower power consumption  
D. higher bit density and lower power consumption

isro2007   co-and-architecture   memory-interfacing

Answer key 

##### 3.14.2 Memory Interfacing: ISRO CSE 2009 | Question: 23



The process of organizing the memory into two banks to allow 8-and 16-bit data operation is called

- A. Bank switching                      B. Indexed mapping  
C. Two-way memory interleaving                      D. Memory segmentation

isro2009   co-and-architecture   memory-interfacing

Answer key 

##### 3.14.3 Memory Interfacing: ISRO CSE 2011 | Question: 54



Number of chips ( $128 \times 8$  RAM) needed to provide a memory capacity of 2048 bytes

- A. 2                      B. 4                      C. 8                      D. 16

isro2011   co-and-architecture   memory-interfacing

Answer key 

##### 3.14.4 Memory Interfacing: ISRO CSE 2014 | Question: 1



Consider a 33 MHz cpu based system. What is the number of wait states required if it is interfaced with a 60 ns memory? Assume a maximum of 10 ns delay for additional circuitry like buffering and decoding.

- A. 0                      B. 1  
C. 2                      D. 3

co-and-architecture   isro2014   memory-interfacing

Answer key 

##### 3.14.5 Memory Interfacing: ISRO CSE 2014 | Question: 17



If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a  $4 \times 6$  array, where each chip is  $8\text{ K} \times 4$  bits?

- A. 13                      B. 15                      C. 16                      D. 17

co-and-architecture   memory-interfacing   isro2014

Answer key 

### 3.14.6 Memory Interfacing: ISRO CSE 2014 | Question: 40



Assume that 16-bit CPU is trying to access a double word starting at an odd address. How many memory operations are required to access the data?

- A. 1                      B. 2                      C. 3                      D. 4

isro2014   co-and-architecture   memory-interfacing

Answer key 

### 3.14.7 Memory Interfacing: ISRO CSE 2017 | Question: 18



How many  $128 \times 8$  bit RAMs are required to design  $32 \text{ K} \times 32$  bit RAM?

- A. 512                      B. 1024                      C. 128                      D. 32

isro2017   co-and-architecture   memory-interfacing

Answer key 

### 3.14.8 Memory Interfacing: ISRO ECE 2013 | Computer architecture



A memory system of size 16 K bytes is required to be designed using memory chips which have 12 address lines and 4 data lines each. The number of such chips required to design the system is:

- a. 2                      b. 4                      c. 8                      d. 16

isro-ece   isro2013-ece   co-and-architecture   memory-interfacing

### 3.14.9 Memory Interfacing: ISRO2015-77



In  $X = (M + N \times O) / (P \times Q)$ , how many one-address instructions are required to evaluate it?

- A. 4                      B. 6                      C. 8                      D. 10

memory-interfacing   co-and-architecture   machine-instruction   isro2015

Answer key 

## 3.15 Memory Management (1)

### 3.15.1 Memory Management: ISRO2015-62



The minimum time delay between the initiation of two independent memory operations is called

- A. Access time                      B. Cycle time                      C. Rotational time                      D. Latency time

isro2015   co-and-architecture   memory-management

Answer key 

## 3.16 Microprocessors (3)

### 3.16.1 Microprocessors: GATE CSE 1998 | Question: 2.10, ISRO2008-17



The address space of 8086 CPU is

- A. one Megabyte                      B. 256 Kilobytes  
C. 1 K Megabytes                      D. 64 Kilobytes

gate1998   co-and-architecture   microprocessors   out-of-syllabus-now   isro2008

Answer key 

### 3.16.2 Microprocessors: ISRO CSE 2011 | Question: 37



Find the memory address of the next instruction executed by the microprocessor (8086), when operated in real mode for CS=1000 and IP=E000

- A. 10E00                      B. 1E000                      C. F000                      D. 1000E

isro2011 co-and-architecture non-gatecse 8086 microprocessors

Answer key

### 3.16.3 Microprocessors: ISRO ECE 2012 | Computer Architecture



Principle of "locality" is used in context of

- a. Addressing lowest Memory address by microprocessor
- b. Addressing Highest Memory address by microprocessor
- c. Accessing Cache memory locations.
- d. None of above.

co-and-architecture isro2012-ece microprocessors

Answer key

## 3.17 Microprogramming (1)

### 3.17.1 Microprogramming: ISRO CSE 2018 | Question: 34



Micro program is:

- a. the name of a source program in micro computers
- b. set of microinstructions that defines the individual operations in response to a machine-language instruction
- c. a primitive form of macros used in assembly language programming
- d. a very small segment of machine code

isro2018 co-and-architecture microprogramming

Answer key

## 3.18 Non Gatecse (1)

### 3.18.1 Non Gatecse: ISRO-2013-34



In 8086, the jump condition for the instruction *JNBE* is?

- A. CF = 0 or ZF = 0                      B. ZF = 0 and SF = 1                      C. CF = 0 and ZF = 0                      D. CF = 0

isro2013 8086 non-gatecse

Answer key

## 3.19 Parallel Programming (1)

### 3.19.1 Parallel Programming: ISRO CSE 2018 | Question: 71



A particular parallel program computation requires 100 sec when executed on a single processor, if 40% of this computation is inherently sequential (i.e. will not benefit from additional processors), then theoretically best possible elapsed times of this program running with 2 and 4 processors, respectively, are:

- a. 20 sec and 10 sec                      b. 30 sec and 15 sec                      c. 50 sec and 25 sec                      d. 70 sec and 55 sec

isro2018 co-and-architecture parallel-programming

## 3.20

## Pipelining (12)

## 3.20.1 Pipelining: GATE CSE 2002 | Question: 2.6, ISRO2008-19



The performance of a pipelined processor suffers if:

- A. the pipeline stages have different delays
- B. consecutive instructions are dependent on each other
- C. the pipeline stages share hardware resources
- D. All of the above

gatecse-2002 co-and-architecture pipelining easy isro2008

Answer key 

## 3.20.2 Pipelining: GATE CSE 2003 | Question: 10, ISRO-DEC2017-41



For a pipelined CPU with a single ALU, consider the following situations

- I. The  $j + 1^{st}$  instruction uses the result of the  $j^{th}$  instruction as an operand
- II. The execution of a conditional jump instruction
- III. The  $j^{th}$  and  $j + 1^{st}$  instructions require the ALU at the same time.

Which of the above can cause a hazard

- A. I and II only
- B. II and III only
- C. III only
- D. All the three

gatecse-2003 co-and-architecture pipelining normal isrodec2017

Answer key 

## 3.20.3 Pipelining: GATE CSE 2007 | Question: 37, ISRO2009-37



Consider a pipelined processor with the following four stages:

- IF: Instruction Fetch
- ID: Instruction Decode and Operand Fetch
- EX: Execute
- WB: Write Back

The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

|            |            |                         |
|------------|------------|-------------------------|
| <b>ADD</b> | R2, R1, R0 | $R2 \leftarrow R1 + R0$ |
| <b>MUL</b> | R4, R3, R2 | $R4 \leftarrow R3 * R2$ |
| <b>SUB</b> | R6, R5, R4 | $R6 \leftarrow R5 - R4$ |

- A. 7
- B. 8
- C. 10
- D. 14

gatecse-2007 co-and-architecture pipelining normal isro2009

Answer key 

## 3.20.4 Pipelining: GATE CSE 2012 | Question: 20, ISRO2016-23



Register renaming is done in pipelined processors:

- A. as an alternative to register allocation at compile time
- B. for efficient access to function parameters and local variables
- C. to handle certain kinds of hazards
- D. as part of address translation

gatecse-2012 co-and-architecture pipelining easy isro2016

Answer key 

### 3.20.5 Pipelining: GATE IT 2007 | Question: 6, ISRO2011-25



A processor takes 12 cycles to complete an instruction  $I$ . The corresponding pipelined processor uses 6 stages with the execution times of 3, 2, 5, 4, 6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

- A. 1.83
- B. 2
- C. 3
- D. 6

gateit-2007 co-and-architecture pipelining normal isro2011

Answer key 

### 3.20.6 Pipelining: ISRO 2008-ECE Pipeline



Assuming ideal conditions, the speed up obtained from a balanced  $N$  stage pipeline is

- (a)  $2N$
- (b)  $N^2$
- (c)  $N$
- (d)  $N!$

isro-ece co-and-architecture pipelining

Answer key 

### 3.20.7 Pipelining: ISRO CSE 2016 | Question: 19



Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is

- A. 3.2
- B. 3.0
- C. 2.2
- D. 2.0

co-and-architecture pipelining isro2016

Answer key 

### 3.20.8 Pipelining: ISRO CSE 2020 | Question: 6



A non-pipelined CPU has 12 general purpose registers ( $R_0, R_1, R_2, \dots, R_{12}$ ). Following operations are supported

- ADD  $R_a, R_b, R_r$  Add  $R_a$  to  $R_b$  and store the result in  $R_r$
- MUL  $R_a, R_b, R_r$  Multiply  $R_a$  to  $R_b$  and store the result in  $R_r$

MUL operation takes two clock cycles, ADD takes one clock cycle.

Calculate minimum number of clock cycles required to compute the value of the expression  $XY + XYZ + YZ$ . The variable  $X, Y, Z$  are initially available in registers  $R_0, R_1$  and  $R_2$  and contents of these registers must not be modified.

- A. 5
- B. 6
- C. 7
- D. 8

Answer key **3.20.9 Pipelining: ISRO CSE 2020 | Question: 7**

Consider a 5-segment pipeline with a clock cycle time 20 ns in each sub operation. Find out the approximate speed-up ratio between pipelined and non-pipelined system to execute 100 instructions. (if an average, every five cycles, a bubble due to data hazard has to be introduced in the pipeline)

- A. 5                                      B. 4.03                                      C. 4.81                                      D. 4.17

isro-2020 co-and-architecture pipelining normal

Answer key **3.20.10 Pipelining: ISRO ECE 2015 | Pipelining**

Pipelining technique is used in microprocessor to improve which of the following parameter?

- a. Power dissipation                                      b. Interrupt latency  
c. Die size                                                      d. Maximum clock frequency

isro2015-ece isro-ece co-and-architecture pipelining

Answer key **3.20.11 Pipelining: ISRO-2013-15**

A pipeline  $P$  operating at 400 MHz has a speedup factor of 6 and operating at 70% efficiency. How many stages are there in the pipeline?

- A. 5                                      B. 6                                      C. 8                                      D. 9

isro2013 co-and-architecture pipelining

Answer key **3.20.12 Pipelining: ISRO2010-ECE Pipeline Hazard**

Consider the following assembly code for a hypothetical RISC processor with a 4-stage pipeline (Instruction Fetch, Decode/Register Read, Execute and Write).

```
add r1,r2,r3      // r1 = r2+r3
sub r4,r1,r3      // r4 = r1 - r3
mul r5,r2,r3      // r5 = r2*r3
```

Identify the possible pipeline hazard and the affected instruction.

- a. Read after write hazard during mul                                      b. Read after write hazard during sub  
c. Read after write hazard during add                                      d. Write after write hazard during mul

isro-ece isro2011-ece co-and-architecture pipelining

Answer key **3.21****Registers (2)****3.21.1 Registers: ISRO CSE 2007 | Question: 08**

A read bit can be read

- A. and written by CPU                                      B. and written by peripheral  
C. by peripheral and written by CPU                                      D. by CPU and written by the peripheral

isro2007 co-and-architecture registers

Answer key 

### 3.21.2 Registers: ISRO CSE 2008 | Question: 55



The Memory Address Register

- A. is a hardware memory device which denotes the location of the current instruction being executed.
- B. is a group of electrical ckt, that performs the intent of instructions fetched from memory
- C. contains the address of the memory location that is to be read from or stored into
- D. contains a copy of the designated memory location specified by the MAR after a "read" or the new contents of the memory prior to a "write"

isro2008 co-and-architecture registers

Answer key

## 3.22

### Runtime Environment (1)

#### 3.22.1 Runtime Environment: GATE CSE 2008 | Question: 37, ISRO2009-38



The use of multiple register windows with overlap causes a reduction in the number of memory accesses for:

- I. Function locals and parameters
- II. Register saves and restores
- III. Instruction fetches

- A. I only
- B. II only
- C. III only
- D. I, II and III

gatecse-2008 co-and-architecture normal isro2009 runtime-environment

Answer key

## Answer Keys

|        |     |        |      |        |     |        |     |        |     |
|--------|-----|--------|------|--------|-----|--------|-----|--------|-----|
| 3.0.1  | Q-Q | 3.0.2  | A    | 3.0.3  | A   | 3.1.1  | C   | 3.1.2  | D   |
| 3.1.3  | C   | 3.1.4  | Q-Q  | 3.1.5  | Q-Q | 3.1.6  | C   | 3.1.7  | D   |
| 3.1.8  | A   | 3.2.1  | B    | 3.2.2  | C   | 3.2.3  | D   | 3.2.4  | D   |
| 3.2.5  | C   | 3.2.6  | B    | 3.2.7  | C   | 3.2.8  | C   | 3.2.9  | D   |
| 3.2.10 | B   | 3.2.11 | D    | 3.3.1  | Q-Q | 3.4.1  | Q-Q | 3.4.2  | Q-Q |
| 3.4.3  | Q-Q | 3.4.4  | Q-Q  | 3.4.5  | Q-Q | 3.4.6  | Q-Q | 3.4.7  | Q-Q |
| 3.4.8  | Q-Q | 3.4.9  | Q-Q  | 3.4.10 | Q-Q | 3.4.11 | Q-Q | 3.4.12 | Q-Q |
| 3.4.13 | Q-Q | 3.4.14 | Q-Q  | 3.4.15 | Q-Q | 3.4.16 | Q-Q | 3.4.17 | Q-Q |
| 3.4.18 | Q-Q | 3.4.19 | Q-Q  | 3.4.20 | Q-Q | 3.4.21 | Q-Q | 3.4.22 | Q-Q |
| 3.4.23 | Q-Q | 3.4.24 | Q-Q  | 3.4.25 | A   | 3.4.26 | A   | 3.4.27 | Q-Q |
| 3.5.1  | B   | 3.5.2  | C    | 3.5.3  | D   | 3.5.4  | C   | 3.5.5  | C   |
| 3.5.6  | A   | 3.5.7  | A    | 3.5.8  | C   | 3.5.9  | A   | 3.5.10 | D   |
| 3.5.11 | C   | 3.5.12 | B    | 3.6.1  | B   | 3.6.2  | D   | 3.7.1  | D   |
| 3.8.1  | C   | 3.9.1  | A, D | 3.9.2  | C   | 3.10.1 | B   | 3.10.2 | A   |
| 3.10.3 | B   | 3.10.4 | D    | 3.10.5 | C   | 3.10.6 | B   | 3.11.1 | A   |
| 3.11.2 | A   | 3.11.3 | D    | 3.11.4 | B   | 3.11.5 | C   | 3.12.1 | D   |

|         |     |
|---------|-----|
| 3.12.2  | B   |
| 3.14.1  | B   |
| 3.14.6  | C   |
| 3.16.1  | A   |
| 3.19.1  | D   |
| 3.20.5  | B   |
| 3.20.10 | Q-Q |
| 3.22.1  | A   |

|         |     |
|---------|-----|
| 3.12.3  | Q-Q |
| 3.14.2  | C   |
| 3.14.7  | B   |
| 3.16.2  | Q-Q |
| 3.20.1  | D   |
| 3.20.6  | Q-Q |
| 3.20.11 | D   |

|         |     |
|---------|-----|
| 3.13.1  | C   |
| 3.14.3  | D   |
| 3.14.8  | Q-Q |
| 3.16.3  | Q-Q |
| 3.20.2  | D   |
| 3.20.7  | A   |
| 3.20.12 | Q-Q |

|        |   |
|--------|---|
| 3.13.2 | C |
| 3.14.4 | D |
| 3.14.9 | C |
| 3.17.1 | B |
| 3.20.3 | B |
| 3.20.8 | B |
| 3.21.1 | D |

|        |   |
|--------|---|
| 3.13.3 | A |
| 3.14.5 | D |
| 3.15.1 | B |
| 3.18.1 | C |
| 3.20.4 | C |
| 3.20.9 | B |
| 3.21.2 | C |



**4.0.1 ISRO-DEC2017-66**

Which languages necessarily need heap allocation in the runtime environment?

- A. Those that support recursion
- B. Those that use dynamic scoping
- C. Those that use global variables
- D. Those that allow dynamic data structures

isrodec2017

Answer key

**4.0.2 ISRO CSE 2020 | Question: 60**

A given grammar is called ambiguous if

- A. two or more productions have the same non-terminal on the left hand side
- B. a derivation tree has more than one associated sentence
- C. there is a sentence with more than one derivation tree corresponding to it
- D. brackets are not present in the grammar

isro-2020 compiler-design ambiguous-grammar easy

Answer key

**4.1****Assembler (2)****4.1.1 Assembler: ISRO CSE 2016 | Question: 43**

A simple two-pass assembler does which of the following in the first pass:

- A. Checks to see if the instructions are legal in the current assembly mode
- B. It allocates space for the literals.
- C. It builds the symbol table for the symbols and their values.
- D. All of these

compiler-design assembler isro2016

Answer key

**4.1.2 Assembler: ISRO CSE 2020 | Question: 14**

In a two-pass assembler, resolution of subroutine calls and inclusion of labels in the symbol table is done during

- A. second pass
- B. first pass and second pass respectively
- C. second pass and first pass respectively
- D. first pass

isro-2020 compiler-design assembler easy

Answer key

**4.2****Code Optimization (9)****4.2.1 Code Optimization: ISRO CSE 2008 | Question: 68**

Which of the following class of statement usually produces no executable code when compiled?

- A. declaration
- B. assignment statements

C. input and output statements

D. structural statements

isro2008 compiler-design code-optimization

Answer key 

#### 4.2.2 Code Optimization: ISRO CSE 2009 | Question: 43



Substitution of values for names (whose values are constants) is done in

A. Local optimization

B. Loop optimization

C. Constant folding

D. Strength reduction

isro2009 compiler-design code-optimization

Answer key 

#### 4.2.3 Code Optimization: ISRO CSE 2011 | Question: 13



Which of the following statements about peephole optimization is False?

A. It is applied to a small part of the code

B. It can be used to optimize intermediate code

C. To get the best out of this, it has to be applied repeatedly

D. It can be applied to the portion of the code that is not contiguous

isro2011 compiler-design code-optimization

Answer key 

#### 4.2.4 Code Optimization: ISRO CSE 2011 | Question: 3



In compiler terminology reduction in strength means

A. Replacing run time computation by compile time computation

B. Removing loop invariant computation

C. Removing common subexpressions

D. replacing a costly operation by a relatively cheaper one

isro2011 compiler-design code-optimization

Answer key 

#### 4.2.5 Code Optimization: ISRO CSE 2016 | Question: 44



Peephole optimization is form of

A. Loop optimization

B. Local optimization

C. Constant folding

D. Data flow analysis

compiler-design code-optimization isro2016

Answer key 

#### 4.2.6 Code Optimization: ISRO CSE 2018 | Question: 29



DU-chains(Definition-Use) in compiler design:

a. consist of a definition of a variable and all its uses, reachable from that definition

b. are created using a form static code analysis

c. are prerequisite for many compiler optimization including constant propagation and common sub-expression elimination

d. All of the above

Answer key **4.2.7 Code Optimization: ISRO CSE 2018 | Question: 30**

Which of the following comment about peep-hole optimization is true?

- a. It is applied to small part of the code and applied repeatedly
- b. It can be used to optimize intermediate code
- c. It can be applied to a portion of the code that is not contiguous
- d. It is applied in symbol table to optimize the memory requirements.

Answer key **4.2.8 Code Optimization: ISRO CSE 2020 | Question: 42**

Which of the following is a type of a out-of-order execution, with the reordering done by a compiler

- |                       |                          |
|-----------------------|--------------------------|
| A. loop unrolling     | B. dead code elimination |
| C. strength reduction | D. software pipelining   |

Answer key **4.2.9 Code Optimization: ISRO-DEC2017-67**

Consider the code segment

```
int i, j, x, y, m, n;
n = 20;
for (i=0; i<n; i++)
{
    for (j=0; j<n; j++)
    {
        if (i%2)
        {
            x += ((4*j)+5*i);
            y += (7+4*j)
        }
    }
}
m=x+y;
```

Which one of the following is false?

- A. The code contains loop invariant computation
- B. There is scope of common sub-expression elimination in this code
- C. There is scope of strength reduction in this code
- D. There is scope of dead code elimination in this code

Answer key **4.3 Compilation Phases (1)****4.3.1 Compilation Phases: ISRO CSE 2020 | Question: 17**

Which one indicates a technique of building cross compilers?

- |               |                   |                  |            |
|---------------|-------------------|------------------|------------|
| A. Beta cross | B. Canadian cross | C. Mexican cross | D. X-cross |
|---------------|-------------------|------------------|------------|

Answer key **4.4 Compiler tokenization (1)****4.4.1 Compiler tokenization: ISRO CSE 2020 | Question: 13**

The number of tokens in the following C code segment is

```
switch(inputvalue)
{
    case 1 : b =c*d; break;
    default : b =b++; break;
}
```

- A. 27                      B. 29                      C. 26                      D. 24

Answer key **4.5 Context Free Grammar (1)****4.5.1 Context Free Grammar: ISRO CSE 2018 | Question: 64**

A particular BNF definition for a "word is given by the following rules.

```
<word>      :: = <letter> I <letter> <charpair> I <letter> <intpair>
<charpair>  :: = <letter> <letter> I <charpair> <letter> <letter>
<intpair>   :: = <integer> <integer> I <intpair> <integer> <integer>
<letter>    :: = a I b I c I .....I Y I Z
<integer>   :: = 0 I 1 I 2 I .....I 9
```

Which of the following lexical entries can be derived from `< word >`?

- I. pick  
II. picks  
III. c44

- A. I, II and III                      B. I and II only                      C. I and III only                      D. II and III only

Answer key **4.6 Expression Evaluation (1)****4.6.1 Expression Evaluation: ISRO CSE 2009 | Question: 32**

The expression  $1 * 2 \wedge 3 * 4 \wedge 5 * 6$  will be evaluated as

- A.  $32^{30}$   
B.  $162^{30}$   
C. 49152  
D. 173458

In original question 5 is not given.

Answer key **4.7 Grammar (5)**

#### 4.7.1 Grammar: GATE CSE 2006 | Question: 32, ISRO2016-35



Consider the following statements about the context free grammar

$$G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$$

- I.  $G$  is ambiguous
- II.  $G$  produces all strings with equal number of  $a$ 's and  $b$ 's
- III.  $G$  can be accepted by a deterministic PDA.

Which combination below expresses all the true statements about  $G$ ?

- A. I only                      B. I and III only                      C. II and III only                      D. I, II and III

gatecse-2006   compiler-design   grammar   normal   isro2016

Answer key

#### 4.7.2 Grammar: ISRO CSE 2014 | Question: 52



Consider the following grammar.

- $S \rightarrow AB$
- $A \rightarrow a$
- $A \rightarrow BaB$
- $B \rightarrow bbA$

Which of the following statements is FALSE?

- A. The length of every string produced by this grammar is even  
B. No string produced by this grammar has three consecutive  $a$ 's  
C. The length of substring produced by  $B$  is always odd  
D. No string produced by this grammar has four consecutive  $b$ 's

isro2014   compiler-design   grammar   parsing

Answer key

#### 4.7.3 Grammar: ISRO CSE 2020 | Question: 35



Given the grammar

- $s \rightarrow T^*S \mid T$
- $T \rightarrow U + T \mid U$
- $U \rightarrow a \mid b$

Which of the following statements is wrong?

- A. Grammar is not ambiguous                      B. Priority of  $+$  over  $*$  is ensured  
C. Right to left evaluation of  $*$  and  $+$  happens                      D. None of these

isro-2020   compiler-design   grammar   easy

Answer key

#### 4.7.4 Grammar: ISRO CSE 2020 | Question: 75



A grammar is defined as

- $A \rightarrow BC$
- $B \rightarrow x \mid Bx$
- $C \rightarrow B \mid D$

- $D \rightarrow y \mid Ey$
- $E \rightarrow z$

The non terminal alphabet of the grammar is

- $\{A, B, C, D, E\}$
- $\{B, C, D, E\}$
- $\{A, B, C, D, E, x, y, z\}$
- $\{x, y, z\}$

isro-2020 compiler-design grammar parsing easy

Answer key 

#### 4.7.5 Grammar: ISRO-2013-6



Which of the following productions eliminate left recursion in the productions given below:

$S \rightarrow Aa \mid b$

$A \rightarrow Ac \mid Sd \mid \epsilon$

- $S \rightarrow Aa \mid b, A \rightarrow bdA', A' \rightarrow A'c \mid A'ba \mid A \mid \epsilon$
- $S \rightarrow Aa \mid b, A \rightarrow A' \mid bdA', A' \rightarrow cA' \mid adA' \mid \epsilon$
- $S \rightarrow Aa \mid b, A \rightarrow A'c \mid A'd, A' \rightarrow bdA' \mid cA \mid \epsilon$
- $S \rightarrow Aa \mid b, A \rightarrow cA' \mid adA' \mid bdA', A' \rightarrow A \mid \epsilon$

isro2013 parsing grammar

Answer key 

### 4.8

#### Intermediate Code (2)

##### 4.8.1 Intermediate Code: ISRO CSE 2008 | Question: 38



Relative to the program translated by a compiler, the same program when interpreted runs

- Faster
- Slower
- At the same speed
- May be faster or slower

isro2008 compiler-design intermediate-code

Answer key 

##### 4.8.2 Intermediate Code: ISRO CSE 2018 | Question: 28



Incremental-Compiler is a compiler:

- which is written in a language that is different from the source language
- compiles the whole source code to generate object code afresh
- compiles only those portion of source code that have been modified
- that runs on one machine but produces object code for another machine

isro2018 compiler-design intermediate-code

Answer key 

### 4.9

#### Lexical Analysis (2)

##### 4.9.1 Lexical Analysis: GATE CSE 2000 | Question: 1.18, ISRO2015-25



The number of tokens in the following C statement is

```
printf("i=%d, &i=%x", i, &i);
```

A. 3

B. 26

C. 10

D. 21

gatecse-2000 compiler-design lexical-analysis easy isro2015

Answer key **4.9.2 Lexical Analysis: ISRO 2017-79**

The output of a lexical analyzer is

- A. A parse tree  
C. Machine code

- B. Intermediate code  
D. A stream of tokens

isro2017 compiler-design lexical-analysis

Answer key **4.10****Operator Grammar (1)****4.10.1 Operator Grammar: ISRO2015-26**Which grammar rules violate the requirement of the operator grammar?  $A, B, C$  are variables and  $a, b, c$  are terminals

- i.  $A \rightarrow BC$   
ii.  $A \rightarrow CcBb$   
iii.  $A \rightarrow BaC$   
iv.  $A \rightarrow \epsilon$

- a. (i) only                      b. (i) and (ii)                      c. (i) and (iii)                      d. (i) and (iv)

isro2015 compiler-design operator-grammar parsing

Answer key **4.11****Operator Precedence (1)****4.11.1 Operator Precedence: GATE CSE 2000 | Question: 2.21, ISRO2015-24**

Given the following expression grammar:

$$E \rightarrow E * F \mid F + E \mid F$$

$$F \rightarrow F - F \mid id$$

Which of the following is true?

- A.  $*$  has higher precedence than  $+$                       B.  $-$  has higher precedence than  $*$   
C.  $+$  and  $-$  have same precedence                      D.  $+$  has higher precedence than  $*$

gatecse-2000 operator-precedence normal compiler-design isro2015

Answer key **4.12****Parameter Passing (2)****4.12.1 Parameter Passing: GATE CSE 2004 | Question: 2, ISRO2017-54**

Consider the following function

```
void swap(int a, int b)
{
    int temp;
```

```

temp = a;
a = b;
b = temp;
}

```

In order to exchange the values of two variables  $x$  and  $y$ .

- A. call  $swap(x, y)$
- B. call  $swap(\&x, \&y)$
- C.  $swap(x, y)$  cannot be used as it does not return any value
- D.  $swap(x, y)$  cannot be used as the parameters are passed by value

gatecse-2004 compiler-design programming-in-c parameter-passing easy isro2017 runtime-environment

Answer key 

#### 4.12.2 Parameter Passing: ISRO2015-80



Consider the following code fragment

```

void foo(int x, int y)
{
    x+=y;
    y+=x;
}
main()
{
    int x=5.5;
    foo(x, x);
}

```

What is the final value of  $x$  in both call by value and call by reference, respectively?

- A. 5 and 16
- B. 5 and 12
- C. 5 and 20
- D. 12 and 20

isro2015 compiler-design runtime-environment parameter-passing

Answer key 

#### 4.13 Parsing (8)

##### 4.13.1 Parsing: ISRO CSE 2011 | Question: 43



Which variable does not drive a terminal string in grammar?

- $S \rightarrow AB$
- $A \rightarrow a$
- $B \rightarrow b$
- $B \rightarrow C$

- A. A
- B. B
- C. C
- D. S

isro2011 compiler-design parsing

Answer key 

##### 4.13.2 Parsing: ISRO CSE 2014 | Question: 5



What is the number of steps required to derive the string  $((()())())$  for the following grammar?

- $S \rightarrow SS$
- $S \rightarrow (S)$
- $S \rightarrow \epsilon$



- A. 10
- C. 15

- B. 12
- D. 16

isro2014 compiler-design parsing

Answer key 

#### 4.13.3 Parsing: ISRO CSE 2016 | Question: 40



Recursive descent parsing is an example of

- A. Top-down parsers
- C. Predictive parsers

- B. Bottom-up parsers
- D. None of the above

isro2016 compiler-design parsing

Answer key 

#### 4.13.4 Parsing: ISRO CSE 2016 | Question: 41



A top-down parser generates

- A. Rightmost Derivation
- C. Leftmost derivation

- B. Rightmost derivation in reverse
- D. Leftmost derivation in reverse

isro2016 compiler-design parsing

Answer key 

#### 4.13.5 Parsing: ISRO-2013-5



Shift reduce parsing belongs to a class of

- A. Bottom up parsing.
- C. Recursive parsing.

- B. Top down parsing.
- D. Predictive parsing.

isro2013 parsing

Answer key 

#### 4.13.6 Parsing: ISRO2015-27



Which one of the following is a top-down parser?

- A. Recursive descent parser
- C.  $SLR(k)$  parser

- B. Shift left associative parser
- D.  $LR(k)$  parser

isro2015 compiler-design parsing

Answer key 

#### 4.13.7 Parsing: ISRO2015-28



Yacc stands for

- A. yet accept compiler constructs
- C. yet another compiler constructs

- B. yet accept compiler compiler
- D. yet another compiler compiler

isro2015 compiler-design parsing

Answer key 

#### 4.13.8 Parsing: ISRO2015-29



Which statement is true?

- A. LALR parser is more powerful and costly as compare to other parsers
- B. All CFG's are LP and not all grammars are uniquely defined
- C. Every SLR grammar is unambiguous but not every unambiguous grammar is SLR

D. LR(K) is the most general back tracking shift reduce parsing method

isro2015 compiler-design parsing

Answer key 

## 4.14

### Runtime Environment (1)

#### 4.14.1 Runtime Environment: GATE CSE 1998 | Question: 1.25, ISRO2008-41



In a resident – OS computer, which of the following systems must reside in the main memory under all situations?

- A. Assembler      B. Linker      C. Loader      D. Compiler

gate1998 compiler-design runtime-environment normal isro2008

Answer key 

## 4.15

### Symbol Table (1)

#### 4.15.1 Symbol Table: ISRO CSE 2011 | Question: 78



A symbol table of length 152 is processing 25 entries at any instant. What is occupation density?

- A. 0.164      B. 127      C. 8.06      D. 6.08

isro2011 compiler-design symbol-table

Answer key 

## 4.16

### Variable Binding (1)

#### 4.16.1 Variable Binding: ISRO-2013-7



Consider the following psuedocode:

```
x: integer := 1
y: integer := 2
procedure add
  x := x + y
procedure second (P: Procedure)
  x: integer := 2
  p()
procedure first
  y: integer := 3
  second (add)
first ()
write_integer(x)
```

What does it print if the language uses dynamic scoping with deep binding?

- A. 2      B. 3      C. 4      D. 5

isro2013 runtime-environment variable-binding

Answer key 

## Answer Keys

|       |   |       |   |       |   |       |   |       |   |
|-------|---|-------|---|-------|---|-------|---|-------|---|
| 4.0.1 | D | 4.0.2 | C | 4.1.1 | D | 4.1.2 | C | 4.2.1 | A |
| 4.2.2 | C | 4.2.3 | X | 4.2.4 | D | 4.2.5 | B | 4.2.6 | D |
| 4.2.7 | X | 4.2.8 | D | 4.2.9 | D | 4.3.1 | B | 4.4.1 | C |

|        |   |
|--------|---|
| 4.5.1  | D |
| 4.7.4  | A |
| 4.9.2  | D |
| 4.13.1 | C |
| 4.13.6 | A |
| 4.16.1 | C |

|        |   |
|--------|---|
| 4.6.1  | C |
| 4.7.5  | B |
| 4.10.1 | D |
| 4.13.2 | A |
| 4.13.7 | D |

|        |   |
|--------|---|
| 4.7.1  | B |
| 4.8.1  | B |
| 4.11.1 | B |
| 4.13.3 | A |
| 4.13.8 | C |

|        |   |
|--------|---|
| 4.7.2  | D |
| 4.8.2  | C |
| 4.12.1 | D |
| 4.13.4 | C |
| 4.14.1 | C |

|        |   |
|--------|---|
| 4.7.3  | D |
| 4.9.1  | C |
| 4.12.2 | C |
| 4.13.5 | A |
| 4.15.1 | A |

**5.0.1 ISRO CSE 2014 | Question: 58**

Assume the following information.

- Original timestamp value = 46
- Receive timestamp value = 59
- Transmit timestamp value = 60
- Timestamp at arrival of packet = 69

Which of the following statements is correct?

- A. Receive clock should go back by 3 milliseconds
- B. Transmit and Receive clocks are synchronized
- C. Transmit clock should go back by 3 milliseconds
- D. Receive clock should go ahead by 1 milliseconds

isro2014 computer-networks icmp

Answer key

**5.0.2 ISRO-2013-19**

What is the maximum number of characters (7 bits + parity) that can be transmitted in a second on a 19.2 Kbps line. This asynchronous transmission requires 1 start bit and 1 stop bit.

- A. 192
- B. 240
- C. 1920
- D. 1966

isro2013 serial-communication

Answer key

**5.0.3 ISRO-2013-44**

Which algorithm is used to shape the bursty traffic into a fixed rate traffic by averaging the data rate?

- A. Solid bucket algorithm
- B. Spanning tree algorithm
- C. Hocken helm algorithm
- D. Leaky bucket algorithm

isro2013 computer-networks

Answer key

**5.0.4 ISRO CSE 2009 | Question: 5**

What is the primary purpose of a VLAN?

- A. Demonstrating the proper layout for a network
- B. Simulating a network
- C. To create a virtual private network
- D. Segmenting a network inside a switch or device

isro2009 computer-networks

Answer key

**5.0.5 ISRO CSE 2007 | Question: 19**

Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through a synchronous serial line at 2400 baud rate, and with two stop bits is

- A. 109
- B. 216
- C. 218
- D. 219

isro2007 serial-communication

Answer key 

### 5.0.6 ISRO2015-51



How many characters per sec (7 bits + 1 parity) can be transmitted over a 2400 bps line if the transfer is synchronous (1 start and 1 stop bit)?

- A. 300                      B. 240                      C. 250                      D. 275

isro2015   computer-networks   serial-communication

Answer key 

### 5.0.7 ISRO ECE 2016 | Networks



A typical optical fiber has

- a. high refractive index core & low refractive index cladding
- b. Low refractive index core & high refractive index cladding
- c. Uniform refractive index core surrounded by variable refractive index cladding
- d. None of the above

isro2016-ece   computer-networks

Answer key 

### 5.0.8 ISRO ECE 2016 | Networks



An FM signal at 10.7 MHz IF needs to be digitized for demodulation in a digital domain. If the bandwidth of this signal is 200 kHz, the maximum usable sampling frequency is

- a. 200 kHz                      b. 600 kHz                      c. 400 kHz                      d. 800 kHz

isro2016-ece   computer-networks

Answer key 

### 5.0.9 ISRO ECE 2016 | Networks



Hamming codes are used for

- a. Burst error correction
- b. Signal error correction
- c. Both (a) and (b)
- d. None of the above

isro2016-ece   computer-networks

Answer key 

### 5.0.10 ISRO ECE 2015 | Networks



Which of the following modulation scheme requires minimum power for transmission?

- a. QPSK                      b. 16-QAM                      c. 8-PSK                      d. 64-QAM

isro2015-ece   computer-networks

### 5.0.11 ISRO ECE 2015 | Networks



A signal transmitted from an Earth station bounces back from a geostationary satellite, with an altitude of 35864 km from the Earth surface. The range from the Earth station, is 38200 km. The round trip time is about: (excluding processing delays and assuming  $c = 3 \times 10^8$  m/s)

- a. 255 ms                      b. 127 ms                      c. 239 ms                      d. 120 ms

computer-networks   isro2015-ece

Answer key 

### 5.0.12 ISRO ECE 2015 | Computer networks



A communication channel is having a bandwidth of 3000 Hz. The transmitted power is such that the received Signal-to-Noise ratio is 1023. The maximum data rate that can be transmitted error-free through the channel is:

- a. 3 Kbps                      b. 3 Mbps                      c. 30 Kbps                      d. 300 Kbps

isro2015-ece   computer-networks   isro-ece

Answer key 

### 5.0.13 ISRO ECE 2014 | Computer Network



Match the following:

|   |                       |   |                                          |
|---|-----------------------|---|------------------------------------------|
| A | Gaussian distribution | 1 | Calls on a telephone channel             |
| B | Rayleigh distribution | 2 | Random number                            |
| C | Poisson distribution  | 3 | Thermal noise                            |
| D | Uniform distribution  | 4 | Fading channel in wireless communication |

A. A-3, B-1, C-4, D-2

B. A-3, B-4, C-2, D-1

C. A-1, B-4, C-3, D-2

D. A-3, B-4, C-1, D-2

isro-ece   isro2014-ece   computer-networks

Answer key 

### 5.0.14 ISRO ECE 2014 | Computer Networks



In a fiber-optic cable, which phenomenon occurs for signal to propagate along the inner core:

- A. Modulation                      B. Reflection                      C. Refraction                      D. All of the above

computer-networks   isro2014-ece   isro-ece

Answer key 

### 5.0.15 ISRO ECE 2014 | : Network



In satellite communication, scrambling is mainly used for:

- a. Clock recovery                      b. Encryption  
c. limiting power spectral density                      d. bandwidth efficiency

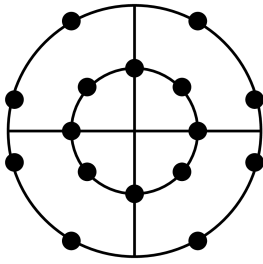
isro-ece   isro2014-ece   computer-networks

Answer key 

### 5.0.16 ISRO ECE 2012 | Networks



Following constellation diagram represents:



- a. 16PSK
- b. 16QAM
- c. 16 APSK
- d. All the above

isro-ece isro2012-ece computer-networks

### 5.0.17 ISRO 2011-ECE Networks



The channel capacity under the Gaussian noise environment for a discrete memory less channel with a bandwidth of 4 MHz and SNR 31 is

- a. 20 Mbps
- b. 4 Mbps
- c. 8 Kbps
- d. 4 Kbps

computer-networks isro2011-ece isro-ece

### 5.0.18 ISRO 2011-ECE Computer Networks



In satellite communication, frequency modulation is used because satellite channel has

- a. High Modulation index
- b. Small bandwidth and negligible noise
- c. Large bandwidth and severe noise
- d. Maximum Bandwidth and Minimum noise

computer-networks isro-ece isro2011-ece

### 5.0.19 ISRO 2011-ECE Computer Networks



The critical angle of an optical fiber is given by \_\_\_\_\_ where  $n_1$  is refractive index of medium 1 and  $n_2$  is the refractive index of medium 2

- a)  $\sin^{-1}(n_2/n_1)$
- b)  $\sin^{-1}(n_1/n_2)$
- c)  $\sin^{-1}(n_2 \cdot n_1)$
- d)  $\sin^{-1} n_2$

isro-ece isro2011-ece computer-networks

### 5.0.20 ISRO 2011-ECE Computer networks



The bandwidth of a 'N' bit binary coded PCM signal for modulating a signal having bandwidth of  $f$  Hz is

- a.  $f/N$  Hz.
- b.  $f$
- c.  $Nf$
- d.  $N$

isro-ece isro2011-ece computer-networks

Answer key 

#### 5.0.21 ISRO 2011-ECE Computer networks

The modulation normally used with digital data is



- a) AM
- b) FM
- c) SSB
- d) QPSK

isro-ece isro2011-ece computer-networks

#### 5.0.22 ISRO 2011-ECE Computer Networks

Go-stationary satellites are placed in equatorial orbits at the height approximately



- a. 1000 km
- b. 15000 km
- c. 25000 km
- d. 36000 km

isro-ece isro2011-ece computer-networks

Answer key 

#### 5.0.23 ISRO 2011-ECE Computer networks

In phase modulated signal, the frequency deviation is proportional to



- a. Frequency only
- b. Amplitude only
- c. Both (a) and (b)
- d. none of the above

isro-ece isro2011-ece computer-networks

Answer key 

#### 5.0.24 ISRO 2011-ECE Computer networks

For a fast communication which of the following requirements have to be met





- a. Large bandwidth
- b. High S/N ratio
- c. High channel capacity
- d. None of the above

isro-ece isro2011-ece computer-networks

#### 5.0.25 ISRO 2011- ECE Computer networks



\_\_\_\_\_ is used to describe the light gathering or light collecting ability of an optical fiber

- a. Critical angle
- b. Cut-off wavelength
- c. Numerical Aperture
- d. Acceptance angle

isro-ece isro2011-ece computer-networks

Answer key

#### 5.0.26 ISRO 2009-ECE Ethernet



The ethernet protocol uses

- a) CSMA/CA
- b) SCPC
- c) CSMA/CD
- d) Slotted ALOHA

isro-ece computer-networks

Answer key

#### 5.0.27 ISRO 2009- ECE Code efficiency



A source produces 4 symbols with probabilities  $1/2, 1/4, 1/8, 1/8$ . For this source, a practical coding scheme has an average code- word length of 2 bits/symbol. The efficiency of the code is

- a) 1
- b)  $7/8$
- c)  $1/2$
- d)  $1/4$

isro-ece

Answer key

### 5.0.28 ISRO 2008- ECE Switch and Hub



Difference between a switch and Hub

- (a) Both are same
- (b) Switch avoids flooding
- (c) Hub avoids broadcast
- (d) None of the above

isro-ece computer-networks

Answer key

### 5.0.29 ISRO 2008- ECE 100 Base T



100 Base T refers to

- (a) Fibre connectivity
- (b) BNC connectors
- (c) SONET
- (d) Twisted pair

isro-ece computer-networks

Answer key

### 5.0.30 ISRO 2008-ECE RS 232 interface



**RS-232** interface

- (a) Uses only positive voltage
- (b) Cannot transmit signals over long distance .
- (c) Uses only negative voltage
- (d) A logic high uses positive voltage

isro-ece computer-networks

### 5.0.31 ISRO 2015 UDP/TCP [EE]



The transport layer protocol used for real time multimedia, file transfer, DNS and e-mail respectively are

- (a) TCP, UDP, UDP and TCP
- (b) UDP, TCP, UDP and TCP
- (c) UDP, TCP, TCP and UDP
- (d) TCP, UDP, TCP and UDP

isro-ee computer-networks

Answer key

### 5.0.32 ISRO 2015- Optical fiber [EE]



The **cladding** which surrounds the optic fiber line is used

- (a) To protect the fiber
- (b) To reduce optical interference
- (c) Because it help to guide light in the core
- (d) To ensure that refractive index remains constant

isro-ee computer-networks

Answer key

### 5.0.33 ISRO-DEC2017-31



In the IPv4 addressing format, the number of networks allowed under Class C addresses is

- A.  $2^{20}$                       B.  $2^{24}$                       C.  $2^{14}$                       D.  $2^{21}$

isrodec2017

Answer key

### 5.0.34 ISRO-DEC2017-30



Which of the following are used to generate a message digest by the network security protocols?

- (P) SHA-256
- (Q) AES
- (R) DES
- (S) MD5

- A. P and S only                      B. P and Q only                      C. R and S only                      D. P and R only

isrodec2017

Answer key

### 5.0.35 ISRO-DEC2017-33



Assume that Source  $S$  and Destination  $D$  are connected through an intermediate router  $R$ .

How many times a packet has to visit the network layer and data link layer during a transmission from  $S$  to  $D$ ?

- A. Network layer—4 times, Data link layer—4 times
- B. Network layer—4 times, Data link layer—6 times
- C. Network layer—2 times, Data link layer—4 times
- D. Network layer—3 times, Data link layer—4 times

isrodec2017

Answer key

### 5.0.36 ISRO-DEC2017-35



Consider the set of activities related to e-mail

- A: Send an e-mail from a mail client to mail server.

Download e-mail headers from mailbox and  
**B:** retrieve mails from server to cache

**C:** Checking e-mail through a web browser

The application level protocol used for each activity in the same sequence is

- A. SMTP,HTTPS,IMAP  
B. SMTP,POP,IMAP  
C. SMTP,IMAP,HTTPS  
D. SMTP,IMAP,POP

isrodec2017

Answer key 

#### 5.0.37 ISRO-DEC2017-36



Station  $A$  uses  $32 - \text{byte}$  packets to transmit messages to Station  $B$  using a sliding window protocol. The round trip time delay between  $A$  and  $B$  is  $40\text{ms}$  and the bottleneck bandwidth on the path  $A$  and  $B$  is  $64\text{kbps}$ . What is the optimal window size that  $A$  should use?

- A. 5  
B. 10  
C. 40  
D. 80

isrodec2017

Answer key 

#### 5.0.38 ISRO-DEC2017-34



Generally, TCP is reliable and UDP is not reliable. DNS which has to be reliable uses UDP because

- A. UDP is slower.  
B. DNS servers has to keep connections.  
C. DNS requests are generally very small and fit well within UDP segments.  
D. None of these.

isrodec2017

Answer key 

#### 5.0.39 ISRO CSE 2023 | Question: 59



Which of the following is the cause of Ping of death issue related to ICMP packets?

- A. Buffer overflow  
B. Divide by ZERO  
C. Missing input sanitisation  
D. Privilege escalation

isro-cse-2023 computer-networks icmp

Answer key 

#### 5.0.40 ISRO CSE 2023 | Question: 17



Match the following context of information Security which are closely associated

Matching

|       |                   |     |                                  |
|-------|-------------------|-----|----------------------------------|
| (i)   | Ingress Filtering | (P) | Data Leakage Prevention          |
| (ii)  | Egress Filtering  | (Q) | Hiding identity of systems       |
| (iii) | NAT               | (R) | Keep Track of TCP/IP connections |
| (iv)  | Stateful firewall | (S) | Malicious traffic prevention     |

- A. (i) - (S), (ii) - (P), (iii) - (Q), (iv) - (R)
- B. (i) - (P), (ii) - (R), (iii) - (S), (iv) - (Q)
- C. (i) - (S), (ii) - (R), (iii) - (P), (iv) - (Q)
- D. (i) - (R), (ii) - (Q), (iii) - (S), (iv) - (P)

isro-cse-2023 computer-networks match-the-following

## 5.1 Application Layer Protocols (1)

### 5.1.1 Application Layer Protocols: GATE CSE 2008 | Question: 14, ISRO2016-74



What is the maximum size of data that the application layer can pass on to the TCP layer below?

- A. Any size
- B.  $2^{16}$  bytes - size of TCP header
- C.  $2^{16}$  bytes
- D. 1500 bytes

gatecse-2008 easy computer-networks application-layer-protocols isro2016

Answer key

## 5.2 CRC Polynomial (4)

### 5.2.1 CRC Polynomial: GATE CSE 2007 | Question: 68, ISRO2016-73



The message 11001001 is to be transmitted using the CRC polynomial  $x^3 + 1$  to protect it from errors. The message that should be transmitted is:

- A. 11001001000
- B. 11001001011
- C. 11001010
- D. 110010010011

gatecse-2007 computer-networks error-detection crc-polynomial normal isro2016

Answer key

### 5.2.2 CRC Polynomial: ISRO CSE 2018 | Question: 11



\_\_\_\_\_ can detect burst error of length less than or equal to degree of the polynomial and detects burst errors that affect odd number of bits.

- A. Hamming Code
- B. CRC
- C. VRC
- D. None of the above

isro2018 computer-networks crc-polynomial

Answer key

### 5.2.3 CRC Polynomial: ISRO-2013-40



If the frame to be transmitted is 1101011011 and the CRC polynomial to be used for generating checksum is  $x^4 + x + 1$ , then what is the transmitted frame?

- A. 11010110111011
- B. 11010110111101
- C. 11010110111110
- D. 11010110111001

isro2013 computer-networks crc-polynomial

Answer key

#### 5.2.4 CRC Polynomial: ISRO2015-52



In CRC if the data unit is 100111001 and the divisor is 1011 then what is dividend at the receiver?

- A. 100111001101      B. 100111001011      C. 100111001      D. 100111001110

isro2015   computer-networks   crc-polynomial

Answer key

#### 5.3

#### CSMA CD (2)

##### 5.3.1 CSMA CD: ISRO CSE 2008 | Question: 6



Which of the following transmission media is not readily suitable to CSMA operation?

- A. Radio      B. Optical fibers      C. Coaxial cable      D. Twisted pair

isro2008   computer-networks   csma-cd

Answer key

##### 5.3.2 CSMA CD: ISRO CSE 2016 | Question: 69



In Ethernet CSMA/CD, the special bit sequence transmitted by media access management to handle collision is called

- A. Preamble      B. Postamble      C. Jam      D. None of these

computer-networks   csma-cd   isro2016

Answer key

#### 5.4

#### Communication (7)

##### 5.4.1 Communication: ISRO CSE 2007 | Question: 20



If the bandwidth of a signal is 5 kHz and the lowest frequency is 52 kHz, what is the highest frequency

- A. 5 kHz      B. 10 kHz      C. 47 kHz      D. 57 kHz

isro2007   communication

Answer key

##### 5.4.2 Communication: ISRO CSE 2007 | Question: 22



Phase transition for each bit are used in

- A. Amplitude modulation      B. Carrier modulation  
C. Manchester encoding      D. NRZ encoding

isro2007   computer-networks   communication

Answer key

##### 5.4.3 Communication: ISRO CSE 2007 | Question: 24, ISRO CSE 2016 | Question: 67



Bit stuffing refers to

- A. inserting a 0 in user stream to differentiate it with a flag  
B. inserting a 0 in flag stream to avoid ambiguity  
C. appending a nipple to the flag sequence  
D. appending a nipple to the use data stream

Answer key **5.4.4 Communication: ISRO CSE 2008 | Question: 9**

What is the bandwidth of the signal that ranges from 40 kHz to 4 MHz

- A. 36 MHz                      B. 360 kHz                      C. 3.96 MHz                      D. 396 kHz

isro2008 communication non-gatecse

Answer key **5.4.5 Communication: ISRO CSE 2011 | Question: 1**

The encoding technique used to transmit the signal in gigabit ethernet technology over fiber optic medium is

- A. Differential Manchester encoding                      B. Non return to zero  
C. 4B/5B encoding                      D. 8B/10B encoding

computer-networks isro2011 communication

Answer key **5.4.6 Communication: ISRO CSE 2014 | Question: 45**

Consider a 50 kbps satellite channel with a 500 milliseconds round trip propagation delay. If the sender wants to transmit 1000 bit frames, how much time will it take for the receiver to receive the frame?

- A. 250 milliseconds                      B. 20 milliseconds  
C. 520 milliseconds                      D. 270 milliseconds

isro2014 computer-networks communication

Answer key **5.4.7 Communication: ISRO2015-49**

What frequency range is used for microwave communications, satellite and radar?

- A. Low frequency: 30 kHz to 300 kHz  
B. Medium frequency: 300 kHz to 3 MHz  
C. Super high frequency: 3000 MHz to 30000 MHz  
D. Extremely high frequency: 30000 kHz

isro2015 computer-networks communication

Answer key **5.5 Cryptography (7)****5.5.1 Cryptography: ISRO CSE 2009 | Question: 7**

Advanced Encryption Standard (AES) is based on

- A. Asymmetric key algorithm                      B. Symmetric key algorithm  
C. Public key algorithm                      D. Key exchange

isro2009 computer-networks cryptography network-security

Answer key **5.5.2 Cryptography: ISRO CSE 2017 | Question: 35**

MD5 is a widely used hash function for producing hash value of

A. 64 bits

B. 128 bits

C. 512 bits

D. 1024 bits

isro2017 computer-networks network-security cryptography

Answer key 

### 5.5.3 Cryptography: ISRO CSE 2018 | Question: 78



Which one of the following algorithm is not used in asymmetric key cryptography?

a. RSA Algorithm

b. Gillie-Hellman Algorithm

c. Electronic Code Book Algorithm

d. None of the above

isro2018 computer-networks network-security cryptography

Answer key 

### 5.5.4 Cryptography: ISRO CSE 2020 | Question: 44



In a columnar transportation cipher, the plain text is “the tomato is a plant in the night shade family”, keyword is “TOMATO”. The cipher text is

A. "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"

B. "TINESAX / EOAHTFX / MAIIAIX / HTLTHEY / TAPNGDL / OSTNHMX"

C. "TINESAX / EOAHTFX / HTLTHEY / MAIIAIX / OSTNHMX / TAPNGDL"

D. "EOAHTFX / TINESAX / HTLTHEY / MAIIAIX / TAPNGDL / OSTNHMX"

isro-2020 computer-networks cryptography normal

Answer key 

### 5.5.5 Cryptography: ISRO CSE 2020 | Question: 45



Avalanche effect in cryptography refers

A. Large changes in cipher text when the keyword is changed minimally

B. Large changes in cipher text when the plain text is changed

C. Large Impact of keyword change to length of the cipher text

D. None of the above

isro-2020 computer-networks cryptography normal

Answer key 

### 5.5.6 Cryptography: ISRO-2013-21



What will be the cipher text produced by the following cipher function for the plain text ISRO with key  $k = 7$ . [ Consider ' $A' = 0, B' = 1, \dots, Z' = 25$ ]

$$C_k(M) = (kM + 13) \bmod 26$$

A. RJCH

B. QIBG

C. GQPM

D. XPIN

isro2013 cryptography

Answer key 

### 5.5.7 Cryptography: ISRO-2013-46



Which of the following encryption algorithms is based on the Feistel structure?

A. Advanced Encryption Standard

B. RSA public key cryptographic algorithm

C. Data Encryption standard

D. RC4

isro2013 cryptography



## 5.6

## Distance Vector Routing (1)

## 5.6.1 Distance Vector Routing: ISRO CSE 2017 | Question: 34



Match with the suitable one :

**List-1**

- (A) Multicast group membership
- (B) Interior gateway protocol
- (C) Exterior gateway protocol
- (D) RIP

**List-2**

- 1. Distance Vector routing
- 2. IGMP
- 3. OSPF
- 4. BGP

- A. A-2, B-3, C-4, D-1
- C. A-3, B-4, C-1, D-2

- B. A-2, B-4, C-3, D-1
- D. A-3, B-1, C-4, D-2

isro2017 computer-networks distance-vector-routing

## 5.7

## Dns (2)

## 5.7.1 Dns: ISRO CSE 2016 | Question: 80



When a DNS server accepts and uses incorrect information from a host that has no authority giving that information, then it is called

- A. DNS lookup
- C. DNS spoofing
- B. DNS hijacking
- D. None of the mentioned

isro2016 computer-networks dns

## 5.7.2 Dns: ISRO2015-44



The DNS maps the IP address to

- A. A binary address as strings
- C. A hierarchy of domain names
- B. A n alphanumeric address
- D. A hexadecimal address

isro2015 computer-networks dns

## 5.8

## Encoding (2)

## 5.8.1 Encoding: ISRO CSE 2007 | Question: 13



By using an eight bit optical encoder the degree of resolution that can be obtained is (approximately)

- A.  $1.8^\circ$
- B.  $3.4^\circ$
- C.  $2.8^\circ$
- D.  $1.4^\circ$

isro2007 communication encoding

## 5.8.2 Encoding: ISRO CSE 2020 | Question: 52



To send same bit sequence, NRZ encoding require

- A. Same clock frequency as Manchester encoding
- B. Half the clock frequency as Manchester encoding
- C. Twice the clock frequency as Manchester encoding

D. A clock frequency which depend on number of zeroes and ones in the bit sequence

isro-2020 computer-networks encoding normal

Answer key 

## 5.9

### Error Correction (2)

#### 5.9.1 Error Correction: ISRO CSE 2011 | Question: 48



The hamming distance between the octets of 0xAA and 0x55 is

- A. 7                      B. 5                      C. 8                      D. 6

isro2011 computer-networks error-correction hamming-code

Answer key 

#### 5.9.2 Error Correction: ISRO CSE 2023 | Question: 3



Which of the following is true?

- A. Both Parity and Cyclic Redundancy Check are error correcting codes  
B. Both Parity and Low Density Parity Check Code are error correcting codes  
C. Both Reed-Solomon code and Low Density Parity Check Code are error correcting codes  
D. Both Cyclic Redundancy Check and Low Density Parity Check Code are error correcting codes

isro-cse-2023 error-correction computer-networks

Answer key 

## 5.10

### Error Detection (3)

#### 5.10.1 Error Detection: ISRO CSE 2008 | Question: 79



Repeated execution of simple computation may cause compounding of

- A. round-off errors                      B. syntax errors  
C. run-time errors                      D. logic errors

isro2008 computer-networks error-detection

Answer key 

#### 5.10.2 Error Detection: ISRO CSE 2011 | Question: 50



Data is transmitted continuously at 2.048 Mbps rate for 10 hours and received 512 bits errors. What is the bit error rate?

- A. 6.9 e-9                      B. 6.9 e-6                      C. 69 e-9                      D. 4 e-9

isro2011 computer-networks error-detection

Answer key 

#### 5.10.3 Error Detection: ISRO-2013-18



How many check bits are required for 16 bit data word to detect 2 bit errors and single bit correction using hamming code?

- A. 5                      B. 6                      C. 7                      D. 8

isro2013 error-detection

Answer key 

## 5.11

## Ethernet (6)

### 5.11.1 Ethernet: ISRO CSE 2007 | Question: 21



An Ethernet hub

- A. functions as a repeater
- B. connects to a digital PBX
- C. connects to a token-ring network
- D. functions as a gateway

isro2007 computer-networks ethernet

Answer key 

### 5.11.2 Ethernet: ISRO CSE 2008 | Question: 5



In Ethernet, the source address field in the MAC frame is the \_\_\_\_\_ address.

- A. original sender's physical
- B. previous station's physical
- C. next destination's physical
- D. original sender's service port

isro2008 computer-networks ethernet

Answer key 

### 5.11.3 Ethernet: ISRO CSE 2014 | Question: 61



A mechanism or technology used in Ethernet by which two connected devices choose common transmission parameters such as speed, duplex mode and flow control is called

- A. Autosense
- B. Synchronization
- C. Ping
- D. Auto negotiation

isro2014 computer-networks ethernet

Answer key 

### 5.11.4 Ethernet: ISRO CSE 2017 | Question: 33



An Ethernet frame that is less than the IEEE 802.3 minimum length of 64 octets is called

- A. Short frame
- B. Small frame
- C. Mini frame
- D. Runt frame

isro2017 computer-networks ethernet

Answer key 

### 5.11.5 Ethernet: ISRO-2013-38



In the Ethernet, which field is actually added at the physical layer and is not part of the frame.

- A. Preamble
- B. CRC
- C. Address
- D. Location

isro2013 computer-networks ethernet

Answer key 

### 5.11.6 Ethernet: ISRO-2013-39



Ethernet layer- 2 switch is a network element type which gives.

- A. Different collision domain and same broadcast domain.
- B. Different collision domain and different broadcast domain.
- C. Same collision domain and same broadcast domain.
- D. Same collision domain and different broadcast domain.

Answer key 

## 5.12

## Firewall (1)

## 5.12.1 Firewall: ISRO CSE 2018 | Question: 77



What is one advantage of setting up a DMZ (Demilitarized Zone) with two firewalls?

- A. You can control where traffic goes in the three networks
- B. You can do statefull packet filtering
- C. You can do load balancing
- D. Improve network performance

isro2018 computer-networks network-security firewall

Answer key 

## 5.13

## Huffman Code (1)

## 5.13.1 Huffman Code: ISRO 2010- ECE Huffman code length



A zero memory source emits six messages with probabilities 0.3, 0.25, 0.15, 0.12, 0.1 and 0.08. If binary Huffman coding is used, what will be the average code length?

- a) 2.45 bits
- b) 3.45 bits
- c) 2.54 bits
- d) 3.54 bits

isro-ece computer-networks huffman-code

Answer key 

## 5.14

## IP Addressing (2)

## 5.14.1 IP Addressing: ISRO CSE 2008 | Question: 32



The network 198.78.41.0 is a

- |                    |                    |
|--------------------|--------------------|
| A. Class A network | B. Class B network |
| C. Class C network | D. Class D network |

isro2008 computer-networks ip-addressing

Answer key 

## 5.14.2 IP Addressing: ISRO CSE 2014 | Question: 6



The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called

- |                                   |                                      |
|-----------------------------------|--------------------------------------|
| A. Port address translation (PAT) | B. Network address translation (NAT) |
| C. Address mapping                | D. Port mapping                      |

isro2014 computer-networks ip-addressing

Answer key 

## 5.15

## IP Packet (3)

### 5.15.1 IP Packet: ISRO CSE 2009 | Question: 74



Use of IPSEC in tunnel mode results in

- A. IP packet with same header
- B. IP packet with new header
- C. IP packet without header
- D. No changes in IP packet

isro2009 computer-networks ip-packet

Answer key

### 5.15.2 IP Packet: ISRO CSE 2014 | Question: 31



A IP packet has arrived in which the fragmentation offset value is 100, the value of HLEN is 5 and the value of total length field is 200. What is the number of the last byte?

- A. 194
- B. 394
- C. 979
- D. 1179

computer-networks ip-packet isro2014

Answer key

### 5.15.3 IP Packet: ISRO CSE 2014 | Question: 55



An IP packet has arrived with the first 8 bits as 01000010. Which of the following is correct?

- A. The number of hops this packet can travel is 2
- B. The total number of bytes in header is 16 bytes
- C. The upper layer protocol is ICMP
- D. The receiver rejects the packet

isro2014 computer-networks ip-packet

Answer key

## 5.16 Inter Process Communication (1)

### 5.16.1 Inter Process Communication: ISRO CSE 2020 | Question: 58



Remote Procedure Calls are used for

- A. communication between two processes remotely different from each other on the same system
- B. communication between two processes on the same system
- C. communication between two processes on the separate systems
- D. none of the above

isro-2020 computer-networks inter-process-communication normal

Answer key

## 5.17 Ipv6 (1)

### 5.17.1 Ipv6: ISRO-2013-42



**IPv6** does not support which of the following addressing modes?

- A. Unicast addressing
- B. Multicast addressing
- C. Broadcast addressing
- D. Anycast addressing

isro2013 computer-networks ipv6

Answer key

## 5.18 LAN Technologies (1)

### 5.18.1 LAN Technologies: ISRO CSE 2008 | Question: 4



On a LAN, where are IP datagrams transported?

- A. In the LAN header
- B. In the application field
- C. In the information field of the LAN frame
- D. After the TCP header

isro2008 computer-networks lan-technologies

Answer key

### 5.19

#### Link State Routing (1)

### 5.19.1 Link State Routing: ISRO CSE 2007 | Question: 26



If there are five routers and six networks in intranet using link state routing, how many routing tables are there?

- A. 1
- B. 5
- C. 6
- D. 11

isro2007 computer-networks routing link-state-routing

Answer key

### 5.20

#### MAC Protocol (4)

### 5.20.1 MAC Protocol: ISRO CSE 2009 | Question: 4



Which of the following is a MAC address?

- A. 192.166.200.50
- B. 00056A:01A01A5CCA7FF60
- C. 568, Airport Road
- D. 01:A5:BB:A7:FF:60

isro2009 computer-networks mac-protocol

Answer key

### 5.20.2 MAC Protocol: ISRO CSE 2011 | Question: 20



The IEEE standard for WiMax technology is

- A. IEEE 802.16
- B. IEEE 802.36
- C. IEEE 812.16
- D. IEEE 806.16

isro2011 non-gatecse computer-networks mac-protocol

Answer key

### 5.20.3 MAC Protocol: ISRO CSE 2014 | Question: 71



Which of the following is not a valid multicast MAC address?

- A. 01:00:5E:00:00:00
- B. 01:00:5E:00:00:FF
- C. 01:00:5E:00:FF:FF
- D. 01:00:5E:FF:FF:FF

computer-networks mac-protocol isro2014

Answer key

### 5.20.4 MAC Protocol: ISRO CSE 2017 | Question: 32



Which media access control protocol is used by IEEE 802.11 wireless LAN?

- A. CDMA
- B. CSMA/CA
- C. ALOHA
- D. None of the above

isro2017 computer-networks mac-protocol

Answer key

## 5.21

### Network Layer (2)

#### 5.21.1 Network Layer: ISRO CSE 2009 | Question: 2



In networking, UTP stands for

- A. Unshielded T-connector port
- B. Unshielded twisted pair
- C. Unshielded terminating pair
- D. Unshielded transmission process

isro2009 computer-networks network-layer

Answer key

#### 5.21.2 Network Layer: ISRO CSE 2017 | Question: 28



In networking terminology UTP means

- A. Uniquitous teflon port
- B. Uniformly terminating port
- C. Unshielded twisted pair
- D. Unshielded T-connector port

isro2017 computer-networks network-layer

Answer key

## 5.22

### Network Layering (3)

#### 5.22.1 Network Layering: ISRO CSE 2007 | Question: 75



When a host on network A sends a message to a host on network B, which address does the router look at?

- A. Port
- B. IP
- C. Physical
- D. Subnet mask

isro2007 computer-networks network-layering

Answer key

#### 5.22.2 Network Layering: ISRO CSE 2011 | Question: 32



In which layer of network architecture, the secured socket layer (SSL) is used?

- A. physical layer
- B. session layer
- C. application layer
- D. presentation layer

isro2011 computer-networks network-security network-layering

Answer key

#### 5.22.3 Network Layering: ISRO2015-58



Which layers of the OSI reference model are host-to-host layers?

- A. Transport, session, presentation, application
- B. Session, presentation, application
- C. Datalink, transport, presentation, application
- D. Physical, datalink, network, transport

isro2015 computer-networks network-layering

Answer key

## 5.23

### Network Protocols (9)

#### 5.23.1 Network Protocols: ISRO CSE 2007 | Question: 63



Which of these is not a feature of WAP 2.0

- A. Push and Pull Model
- B. Interface to a storage device

C. Multimedia messaging

D. Hashing

isro2007 network-protocols

Answer key 

### 5.23.2 Network Protocols: ISRO CSE 2008 | Question: 10



Which Project 802 standard provides for a collision-free protocol?

A. 802.2

B. 802.3

C. 802.5

D. 802.6

isro2008 computer-networks network-protocols

Answer key 

### 5.23.3 Network Protocols: ISRO CSE 2009 | Question: 3



The address resolution protocol (ARP) is used for

- A. Finding the IP address from the DNS
- B. Finding the IP address of the default gateway
- C. Finding the IP address that corresponds to a MAC address
- D. Finding the MAC address that corresponds to an IP address

isro2009 computer-networks network-protocols

Answer key 

### 5.23.4 Network Protocols: ISRO CSE 2011 | Question: 12



The network protocol which is used to get MAC address of a node by providing IP address is

A. SMTP

B. ARP

C. RIP

D. BOOTP

isro2011 computer-networks network-protocols

Answer key 

### 5.23.5 Network Protocols: ISRO CSE 2011 | Question: 69



Lightweight Directory Access protocol is used for

- A. Routing the packets
- B. Authentication
- C. obtaining IP address
- D. domain name resolving

isro2011 computer-networks network-protocols

Answer key 

### 5.23.6 Network Protocols: ISRO CSE 2016 | Question: 70



Which network protocol allows hosts to dynamically get a unique IP number on each bootup

A. DHCP

B. BOOTP

C. RARP

D. ARP

isro2016 computer-networks network-protocols

Answer key 

### 5.23.7 Network Protocols: ISRO CSE 2017 | Question: 31



Which of the following protocol is used for transferring electronic mail messages from one machine to another?

A. TELNET

B. FTP

C. SNMP

D. SMTP



Answer key **5.23.8 Network Protocols: ISRO-2013-47**

The protocol data unit for the transport layer in the internet stack is

- A. Segment                      B. Message                      C. Datagram                      D. Frame

isro2013 network-protocols

Answer key **5.23.9 Network Protocols: ISRO2015-56**

Which statement is false?

- A. PING is a TCP/IP application that sends datagrams once every second in the hope of an echo response from the machine being PINGED  
 B. If the machine is connected and running a TCP/IP protocol stack, it should respond to the PING datagram with a datagram of its own  
 C. If PING encounters an error condition, an ICMP message is not returned  
 D. PING display the time of the return response in milliseconds or one of several error message

isro2015 computer-networks network-protocols

Answer key **5.24****Network Security (13)****5.24.1 Network Security: ISRO CSE 2007 | Question: 68**

SSL is not responsible for

- A. Mutual authentication of client & server                      B. Secret communication  
 C. Data Integrity protection                      D. Error detection and correction

isro2007 computer-networks network-security

Answer key **5.24.2 Network Security: ISRO CSE 2007 | Question: 70**

The standard for certificates used on internet is

- A. X.25                      B. X.301                      C. X.409                      D. X.509

isro2007 computer-networks network-security

Answer key **5.24.3 Network Security: ISRO CSE 2007 | Question: 71**

Hashed message is signed by a sender using

- A. his public key                      B. his private key  
 C. receiver's public key                      D. receiver's private key

isro2007 computer-networks network-security

Answer key **5.24.4 Network Security: ISRO CSE 2008 | Question: 59**

A public key encryption system

- A. allows anyone to decode the transmissions
- B. allows only the correct sender to decode the data
- C. allows only the correct receiver to decode the data
- D. does not encode the data before transmitting it

isro2008 computer-networks network-security

Answer key 

#### 5.24.5 Network Security: ISRO CSE 2009 | Question: 6



SHA-1 is a

- A. encryption algorithm
- B. decryption algorithm
- C. key exchange algorithm
- D. message digest function

isro2009 computer-networks network-security

Answer key 

#### 5.24.6 Network Security: ISRO CSE 2011 | Question: 18



An example of poly-alphabetic substitution is

- A. P-box
- B. S-box
- C. Caesar cipher
- D. Vigenere cipher

isro2011 computer-networks network-security

Answer key 

#### 5.24.7 Network Security: ISRO CSE 2014 | Question: 24



In a system an RSA algorithm with  $p = 5$  and  $q = 11$ , is implemented for data security. What is the value of the decryption key if the value of the encryption key is 27?

- A. 3
- B. 7
- C. 27
- D. 40

isro2014 computer-networks network-security

Answer key 

#### 5.24.8 Network Security: ISRO CSE 2017 | Question: 36



Which protocol suite designed by IETF to provide security for a packet at the Internet layer?

- A. IPSec
- B. NetSec
- C. PacketSec
- D. SSL

isro2017 computer-networks network-security

Answer key 

#### 5.24.9 Network Security: ISRO CSE 2017 | Question: 37



Pretty Good Privacy (PGP) is used in:

- A. Browser security
- B. FTP security
- C. Email security
- D. None of the above

isro2017 computer-networks network-security

Answer key 

#### 5.24.10 Network Security: ISRO CSE 2018 | Question: 46



In cryptography, the following uses transposition ciphers and the keyword is LAYER. Encrypt the following message. (Spaces are omitted during encryption)

## WELCOME TO NETWORK SECURITY!

- a. WMEKREETSILTWETCOOCYONRU!
- b. EETSICOO CYWMEKRONRU!LTWET
- c. LTWETONRU!WMEKRCOOCYEETSI
- d. ONRU!COOCYLTWETEETSIWMEKR

isro2018 computer-networks network-security

Answer key 

### 5.24.11 Network Security: ISRO CSE 2018 | Question: 74



Avalanche effect in cryptography

- a. Is desirable property of cryptographic algorithm
- b. Is undesirable property of cryptographic algorithm
- c. Has no effect on encryption algorithm
- d. None of the above

isro2018 computer-networks network-security

Answer key 

### 5.24.12 Network Security: ISRO-2013-45



A packet filtering firewall can

- A. Deny certain users from accessing a service
- B. Block worms and viruses from entering the network
- C. Disallow some files from being accessed through FTP
- D. Block some hosts from accessing the network

isro2013 computer-networks network-security

Answer key 

### 5.24.13 Network Security: ISRO-DEC2017-29



Using public key cryptography,  $X$  adds a digital signature  $\sigma$  to a message  $M$ , encrypts  $\langle M, \sigma \rangle$  and sends it to  $Y$ , where it is decrypted. Which one of the following sequence of keys is used for operations ?

- A. Encryption:  $X$ 's private key followed by  $Y$ 's private key; Decryption:  $X$ 's public key followed by  $Y$ 's public key.
- B. Encryption:  $X$ 's private key followed by  $Y$ 's public key; Decryption:  $X$ 's public key followed by  $Y$ 's private key.
- C. Encryption:  $X$ 's private key followed by  $Y$ 's public key; Decryption:  $Y$ 's private key followed by  $X$ 's public key.
- D. Encryption:  $X$ 's public key followed by  $Y$ 's private key; Decryption:  $Y$ 's public key followed by  $X$ 's private key.

isrodec2017 network-security

Answer key 

## 5.25

### Network topologies (2)

### 5.25.1 Network topologies: ISRO CSE 2017 | Question: 27



Physical topology of FDDI is?

- A. Bus
- B. Ring
- C. Star
- D. None of the above

isro2017 computer-networks network-topologies

Answer key 

### 5.25.2 Network topologies: ISRO CSE 2017 | Question: 30



If there are  $n$  devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively

- A.  $n(n-1)/2, n-1$
- B.  $n, n-1$
- C.  $n-1, n$
- D.  $n-1, n(n-1)/2$

isro2017 computer-networks network-topologies

Answer key 

## 5.26

### Out of Gatecse Syllabus (1)

#### 5.26.1 Out of Gatecse Syllabus: GATE CSE 1993 | Question: 6.4, ISRO2008-14



Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through an asynchronous serial line at 2400 baud rate, and with two stop bits is

- A. 109
- B. 216
- C. 218
- D. 219

gate1993 computer-networks serial-communication normal isro2008 out-of-gatecse-syllabus

Answer key 

## 5.27

### Routers Bridge Hubs Switches (2)

#### 5.27.1 Routers Bridge Hubs Switches: ISRO CSE 2011 | Question: 71



One SAN switch has 24 ports. All 24 supports 8 Gbps Fiber Channel technology. What is the aggregate bandwidth of that SAN switch?

- A. 96 Gbps
- B. 192 Mbps
- C. 512 Gbps
- D. 192 Gbps

isro2011 computer-networks routers-bridge-hubs-switches

Answer key 

#### 5.27.2 Routers Bridge Hubs Switches: ISRO2015-48



A T-switch is used to

- A. Control how messages are passed between computers
- B. Echo every character that is received
- C. Transmit characters one at a time
- D. Rearrange the connections between computing equipments

isro2015 computer-networks routers-bridge-hubs-switches

Answer key 

## 5.28

### Routing (2)

#### 5.28.1 Routing: ISRO CSE 2014 | Question: 16



What is routing algorithm used by OSPF routing protocol?

- A. Distance vector
- B. Flooding
- C. Path vector
- D. Link state

Answer key **5.28.2 Routing: ISRO CSE 2016 | Question: 68**

Dynamic routing protocol enable routers to

- A. Dynamically discover and maintain routes
- B. Distribute routing updates to other routers
- C. Reach agreement with other routers about the network topology
- D. All of the above

isro2016 computer-networks routing

Answer key **5.29****Sliding Window (3)****5.29.1 Sliding Window: GATE CSE 2009 | Question: 57, ISRO2016-75**

Frames of 1000 bits are sent over a  $10^6$  bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

What is the minimum number of bits ( $I$ ) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

- A.  $I = 2$
- B.  $I = 3$
- C.  $I = 4$
- D.  $I = 5$

gatecse-2009 computer-networks sliding-window normal isro2016

Answer key **5.29.2 Sliding Window: ISRO CSE 2008 | Question: 8**

The TCP sliding window

- A. can be used to control the flow of information
- B. always occurs when the field value is 0
- C. always occurs when the field value is 1
- D. occurs horizontally

isro2008 computer-networks tcp sliding-window

Answer key **5.29.3 Sliding Window: ISRO CSE 2018 | Question: 44**

Station  $A$  uses 32 byte packets to transmit messages to Station  $B$  using a sliding window protocol. The round trip delay between  $A$  and  $B$  is 80 ms and the bottleneck bandwidth on the path between  $A$  and  $B$  is 128 kbps. What is the optimal window size that  $A$  should use?

- a. 20
- b. 40
- c. 160
- d. 320

isro2018 computer-networks sliding-window

Answer key **5.30****Slotted Aloha (1)****5.30.1 Slotted Aloha: ISRO2015-55**

A certain population of ALOHA users manages to generate 70 request/sec. If the time is slotted in units of 50 msec, then channel load would be

- A. 4.25
- B. 3.5
- C. 450
- D. 350

Answer key 

## 5.31

## Stop and Wait (1)

## 5.31.1 Stop and Wait: ISRO-2013-41



What will be the efficiency of a Stop and Wait protocol, if the transmission time for a frame is 20ns and the propagation time is 30ns?

- A. 20%                      B. 25%                      C. 40%                      D. 66%

isro2013 computer-networks stop-and-wait

Answer key 

## 5.32

## Subnetting (11)

## 5.32.1 Subnetting: GATE CSE 2003 | Question: 82, ISRO2009-1



The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?

- A. 172.57.88.62 and 172.56.87.23                      B. 10.35.28.2 and 10.35.29.4  
C. 191.203.31.87 and 191.234.31.88                      D. 128.8.129.43 and 128.8.161.55

gatecse-2003 computer-networks subnetting normal isro2009

Answer key 

## 5.32.2 Subnetting: GATE CSE 2007 | Question: 67, ISRO2016-72



The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

- A. 62 subnets and 262142 hosts.                      B. 64 subnets and 262142 hosts.  
C. 62 subnets and 1022 hosts.                      D. 64 subnets and 1024 hosts.

gatecse-2007 computer-networks subnetting easy isro2016

Answer key 

## 5.32.3 Subnetting: GATE CSE 2012 | Question: 34, ISRO-DEC2017-32



An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

- A. 245.248.136.0/21 and 245.248.128.0/22  
B. 245.248.128.0/21 and 245.248.128.0/22  
C. 245.248.132.0/22 and 245.248.132.0/21  
D. 245.248.136.0/24 and 245.248.132.0/21

gatecse-2012 computer-networks subnetting normal isrodec2017

Answer key 

## 5.32.4 Subnetting: GATE IT 2006 | Question: 63, ISRO2015-57



A router uses the following routing table:

| Destination  | Mask            | Interface |
|--------------|-----------------|-----------|
| 144.16.0.0   | 255.255.0.0     | eth0      |
| 144.16.64.0  | 255.255.224.0   | eth1      |
| 144.16.68.0  | 255.255.255.0   | eth2      |
| 144.16.68.64 | 255.255.255.224 | eth3      |

Packet bearing a destination address 144.16.68.117 arrives at the router. On which interface will it be forwarded?

- A. eth0                      B. eth1                      C. eth2                      D. eth3

gateit-2006   computer-networks   subnetting   normal   isro2015

Answer key 

### 5.32.5 Subnetting: ISRO CSE 2007 | Question: 73



Range of IP Address from 224.0.0.0 to 239.255.255.255 are

- A. Reserved for loopback                      B. Reserved for broadcast  
C. Used for multicast packets                D. Reserved for future addressing

isro2007   computer-networks   subnetting

Answer key 

### 5.32.6 Subnetting: ISRO CSE 2008 | Question: 3



The subnet mask 255.255.255.192

- A. extends the network portion to 16 bits  
B. extends the network portion to 26 bits  
C. extends the network portion to 36 bits  
D. has no effect on the network portion of an IP address

isro2008   computer-networks   subnetting

Answer key 

### 5.32.7 Subnetting: ISRO CSE 2011 | Question: 45



The broadcast address for IP network 172.16.0.0 with subnet mask 255.255.0.0 is

- A. 172.16.0.255                      B. 172.16.255.255  
C. 255.255.255.255                D. 172.255.255.255

isro2011   computer-networks   subnetting

Answer key 

### 5.32.8 Subnetting: ISRO CSE 2014 | Question: 75



An organization is granted the block 130.34.12.64/26. It needs to have 4 subnets. Which of the following is not an address of this organization?

- A. 130.34.12.124                      B. 130.34.12.89  
C. 130.34.12.70                      D. 130.34.12.132

computer-networks   subnetting   isro2014

Answer key 

### 5.32.9 Subnetting: ISRO CSE 2017 | Question: 29



The default subnet mask for a class B network can be

- A. 255.255.255.0      B. 255.0.0.0      C. 255.255.192.0      D. 255.255.0.0

isro2017   computer-networks   subnetting

Answer key 

### 5.32.10 Subnetting: ISRO-2013-43



What is IP class and number of sub-networks if the subnet mask is 255.224.0.0?

- A. Class A, 3      B. Class A, 8      C. Class B, 3      D. Class B, 32

isro2013   computer-networks   subnetting

Answer key 

### 5.32.11 Subnetting: ISRO2015-54



In a class B subnet, we know the IP address of one host and the mask as given below:

IP address : 125.134.112.66

Mask : 255.255.224.0

What is the first address(Network address)?

- A. 125.134.96.0      B. 125.134.112.0  
C. 125.134.112.66      D. 125.134.0.0

isro2015   computer-networks   subnetting

Answer key 

## 5.33

### Supernetting (1)

#### 5.33.1 Supernetting: ISRO CSE 2014 | Question: 57



A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives 4 packets with the following destination addresses. Which packet belongs to this supernet?

- A. 205.16.42.56      B. 205.17.32.76      C. 205.16.31.10      D. 205.16.39.44

isro2014   computer-networks   supernetting

Answer key 

## 5.34

### TCP (6)

#### 5.34.1 TCP: ISRO CSE 2007 | Question: 66



Silly Window Syndrome is related to

- A. Error during transmission      B. File transfer protocol  
C. Degrade in TCP performance      D. Interface problem

isro2007   computer-networks   tcp

Answer key 

#### 5.34.2 TCP: ISRO CSE 2014 | Question: 60



Suppose you are browsing the world wide web using a web browser and trying to access the web servers. What is the underlying protocol and port number that are being used?

- A. UDP, 80      B. TCP, 80      C. TCP, 25      D. UDP, 25

isro2014   computer-networks   tcp

Answer key 



### 5.34.3 TCP: ISRO CSE 2020 | Question: 53



The persist timer is used in TCP to

- A. To detect crashes from the other end of the connection
- B. To enable retransmission
- C. To avoid deadlock condition
- D. To timeout `FIN_Wait1` condition

isro-2020 computer-networks tcp normal

Answer key

### 5.34.4 TCP: ISRO CSE 2020 | Question: 54



Checksum field in TCP header is

- A. ones complement of sum of header and data in bytes
- B. ones complement of sum of header, data and pseudo header in 16 bit words
- C. dropped from IPv6 header format
- D. better than md5 or sh1 methods

isro-2020 computer-networks tcp normal

Answer key

### 5.34.5 TCP: ISRO2015-50



How many bits internet address is assigned to each host on a TCP/IP internet which is used in all communication with the host?

- A. 16 bits
- B. 32 bits
- C. 48 bits
- D. 64 bits

isro2015 computer-networks tcp

Answer key

### 5.34.6 TCP: ISRO2015-53



An ACK number of 1000 in TCP always means that

- A. 999 bytes have been successfully received
- B. 1000 bytes have been successfully received
- C. 1001 bytes have been successfully received
- D. None of the above

isro2015 computer-networks tcp

Answer key

## 5.35

### Token Ring (1)

#### 5.35.1 Token Ring: GATE CSE 2007 | Question: 66, ISRO2016-71



In a token ring network the transmission speed is  $10^7$  bps and the propagation speed is 200 meters/ $\mu$ s. The 1-bit delay in this network is equivalent to:

- A. 500 meters of cable.
- B. 200 meters of cable.
- C. 20 meters of cable.
- D. 50 meters of cable.

gatecse-2007 computer-networks token-ring out-of-syllabus-now isro2016

Answer key

## 5.36

### Transport Layer (1)

#### 5.36.1 Transport Layer: ISRO CSE 2018 | Question: 45



Assuming that for a given network layer implementation, connection establishment overhead is 100 bytes and disconnection overhead is 28 bytes. What would be the minimum size of the packet the

transport layer needs to keep up, if it wishes to implement a datagram service above the network layer and needs to keep its overhead to a minimum of 12.5%. (ignore transport layer overhead)

- a. 512 bytes                      b. 768 bytes                      c. 1152 bytes                      d. 1024 bytes

isro2018    transport-layer

Answer key 

## 5.37 Wifi (1)

### 5.37.1 Wifi: ISRO CSE 2017 | Question: 38



What is WPA?

- A. wired protected access                      B. wi-fi protected access  
C. wired process access                      D. wi-fi process access

isro2017    computer-networks    wifi

Answer key 

## 5.38 Wireless Networks (1)

### 5.38.1 Wireless Networks: ISRO CSE 2007 | Question: 74



IEEE 802.11 is standard for

- A. Ethernet  
B. Bluetooth  
C. Broadband Wireless  
D. Wireless LANs

isro2007    computer-networks    wireless-networks

Answer key 

## Answer Keys

|        |     |        |     |        |     |        |     |        |     |
|--------|-----|--------|-----|--------|-----|--------|-----|--------|-----|
| 5.0.1  | A   | 5.0.2  | C   | 5.0.3  | D   | 5.0.4  | D   | 5.0.5  | C   |
| 5.0.6  | A   | 5.0.7  | Q-Q | 5.0.8  | Q-Q | 5.0.9  | Q-Q | 5.0.10 | Q-Q |
| 5.0.11 | Q-Q | 5.0.12 | Q-Q | 5.0.13 | Q-Q | 5.0.14 | Q-Q | 5.0.15 | Q-Q |
| 5.0.16 | Q-Q | 5.0.17 | Q-Q | 5.0.18 | Q-Q | 5.0.19 | Q-Q | 5.0.20 | Q-Q |
| 5.0.21 | Q-Q | 5.0.22 | Q-Q | 5.0.23 | Q-Q | 5.0.24 | Q-Q | 5.0.25 | Q-Q |
| 5.0.26 | Q-Q | 5.0.27 | Q-Q | 5.0.28 | Q-Q | 5.0.29 | Q-Q | 5.0.30 | Q-Q |
| 5.0.31 | Q-Q | 5.0.32 | Q-Q | 5.0.33 | D   | 5.0.34 | A   | 5.0.35 | D   |
| 5.0.36 | C   | 5.0.37 | B   | 5.0.38 | C   | 5.0.39 | Q-Q | 5.0.40 | N/A |
| 5.1.1  | A   | 5.2.1  | B   | 5.2.2  | B   | 5.2.3  | C   | 5.2.4  | B   |
| 5.3.1  | A   | 5.3.2  | C   | 5.4.1  | D   | 5.4.2  | C   | 5.4.3  | A   |
| 5.4.4  | C   | 5.4.5  | D   | 5.4.6  | D   | 5.4.7  | D   | 5.5.1  | B   |
| 5.5.2  | B   | 5.5.3  | C   | 5.5.4  | A   | 5.5.5  | B   | 5.5.6  | A   |
| 5.5.7  | C   | 5.6.1  | A   | 5.7.1  | C   | 5.7.2  | C   | 5.8.1  | D   |
| 5.8.2  | B   | 5.9.1  | C   | 5.9.2  | Q-Q | 5.10.1 | A   | 5.10.2 | A   |

|         |   |
|---------|---|
| 5.10.3  | 6 |
| 5.11.5  | A |
| 5.14.2  | B |
| 5.17.1  | C |
| 5.20.3  | D |
| 5.22.2  | D |
| 5.23.4  | B |
| 5.23.9  | C |
| 5.24.5  | D |
| 5.24.10 | B |
| 5.25.2  | A |
| 5.28.2  | D |
| 5.31.1  | B |
| 5.32.5  | C |
| 5.32.10 | B |
| 5.34.3  | C |
| 5.36.1  | D |

|         |     |
|---------|-----|
| 5.11.1  | A   |
| 5.11.6  | A   |
| 5.15.1  | B   |
| 5.18.1  | C   |
| 5.20.4  | B   |
| 5.22.3  | A   |
| 5.23.5  | B   |
| 5.24.1  | D   |
| 5.24.6  | D   |
| 5.24.11 | A   |
| 5.26.1  | C   |
| 5.29.1  | D   |
| 5.32.1  | D   |
| 5.32.6  | B   |
| 5.32.11 | A   |
| 5.34.4  | B;C |
| 5.37.1  | B   |

|         |   |
|---------|---|
| 5.11.2  | B |
| 5.12.1  | A |
| 5.15.2  | C |
| 5.19.1  | B |
| 5.21.1  | B |
| 5.23.1  | D |
| 5.23.6  | A |
| 5.24.2  | D |
| 5.24.7  | A |
| 5.24.12 | D |
| 5.27.1  | B |
| 5.29.2  | A |
| 5.32.2  | C |
| 5.32.7  | B |
| 5.33.1  | D |
| 5.34.5  | B |
| 5.38.1  | D |

|         |     |
|---------|-----|
| 5.11.3  | D   |
| 5.13.1  | Q-Q |
| 5.15.3  | D   |
| 5.20.1  | D   |
| 5.21.2  | C   |
| 5.23.2  | C   |
| 5.23.7  | D   |
| 5.24.3  | B   |
| 5.24.8  | A   |
| 5.24.13 | C   |
| 5.27.2  | D   |
| 5.29.3  | B   |
| 5.32.3  | A   |
| 5.32.8  | D   |
| 5.34.1  | C   |
| 5.34.6  | D   |

|        |   |
|--------|---|
| 5.11.4 | D |
| 5.14.1 | C |
| 5.16.1 | C |
| 5.20.2 | A |
| 5.22.1 | B |
| 5.23.3 | D |
| 5.23.8 | A |
| 5.24.4 | C |
| 5.24.9 | C |
| 5.25.1 | B |
| 5.28.1 | D |
| 5.30.1 | B |
| 5.32.4 | C |
| 5.32.9 | D |
| 5.34.2 | B |
| 5.35.1 | C |

## 6.1.1 Knowledge Representation: ISRO-2013-3



Opportunistic reasoning is addressed by which of the following knowledge representation?

- A. Script                      B. Blackboard                      C. Production Rules                      D. Fuzzy Logic

isro2013    knowledge-representation

Answer key

## 6.1.2 Knowledge Representation: ISRO-DEC2017-78



Consider the results of a medical experiment that aims to predict whether someone is going to develop myopia based on some physical measurements and heredity. In this case, the input dataset consists of the person's medical characteristics and the target variable is binary: 1 for those who are likely to develop myopia and 0 for those who aren't. This can be best classified as

- A. Regression                                              B. Decision Tree  
C. Clustering                                              D. Association Rules

isrodec2017    non-gatecse    knowledge-representation

Answer key

## Answer Keys

6.1.1

B

6.1.2

B



### 7.0.1 ISRO CSE 2009 | Question: 69



The 'command' used to change contents of one database using the contents of another database by linking them on a common key field?

- A. Replace                      B. Join                      C. Change                      D. Update

isro2009   databases   bad-question

Answer key

### 7.0.2 ISRO-DEC2017-12



Consider a table that describes the customers:

Customers(custid, name, gender, rating)

The rating value is an integer in the range 1 to 5 and only two values (male and female) are recorded for *gender*. Consider the query "how many male customers have a rating of 5" ? The best indexing mechanism appropriate for the query is

- A. Linear hashing                      B. Extendible hashing  
C. B+ tree                      D. Bit-mapped index

isrodec2017

Answer key

### 7.0.3 ISRO-DEC2017-13



Consider the following schema:

Sailors(sid,sname,rating,age)

Boats(bid,bname,colour)

Reserves(sid,bid,day)

Two boats can have the same name but the colour differentiates them.

The two relations

$\rho(Tempside, (\pi_{sid,bid} Reserves) / (\pi_{bid} (\sigma_{bname='Ganga'} Boats))),$   
 $\pi_{sname} (Tempside \bowtie Sailors)$

If / is division operation, the above set of relations represents the query

- A. Names of sailors who have reserved all boats called Ganga.  
B. Names of sailors who have not reserved any Ganga boat.  
C. Names of sailors who have reserved at least one Ganga boat.  
D. Names of sailors who have reserved at most one Ganga boat.

isrodec2017

Answer key

### 7.0.4 ISRO-DEC2017-15



Consider the following table: Faculty(facName, dept, office, rank, dateHired)

| facName | dept | office | rank      | dateHired |
|---------|------|--------|-----------|-----------|
| Ravi    | Art  | A101   | Professor | 1975      |
| Murali  | Math | M201   | Assistant | 2000      |

| facName   | dept | office | rank       | dateHired |
|-----------|------|--------|------------|-----------|
| Narayanan | Art  | A101   | Associate  | 1992      |
| Lakshmi   | Math | M201   | Professor  | 1982      |
| Mohan     | CSC  | C101   | Professor  | 1980      |
| Sreeni    | Math | M203   | Associate  | 1990      |
| Tanuja    | CSC  | C101   | Instructor | 2001      |
| Ganesh    | CSC  | C105   | Associate  | 1995      |

(Assume that no faculty member within a single department has same name. Each faculty member has only one office identified in *office*). *3NF* refers to third normal form and *BCNF* refers to Boyce-Codd Normal Form

Then *Faculty* is

- A. Not in 3NF, in BCNF
- B. In 3NF, not in BCNF
- C. In 3NF, in BCNF
- D. Not in 3NF, not in BCNF

isrodec2017

Answer key 

## 7.0.5 ISRO-DEC2017-16



Consider the following query :

```
SELECT E.eno, COUNT(*)
FROM Employees E
GROUP BY E.eno
```

If an index on *eno* is available, the query can be answered by scanning only the index if

- A. the index is only hash and clustered
- B. the index is only *B+* tree and clustered
- C. index can be hash or *B+* tree and clustered or non-clustered
- D. index can be hash or *B+* tree and clustered

isrodec2017

Answer key 

## 7.0.6 ISRO CSE 2023 | Question: 7



Consider a database table *T* containing two columns *X* and *Y* each of type Integer. After the creation of the table, one record ( $X = 1, Y = 1$ ) is inserted in the table. Let *MX* and *MY* denote the respective maximum values of *X* and *Y* among all records in the table at any point in time. Using *MX* and *MY*, new records are inserted in the table 128 times with *X* and *Y* values being  $MX + 1, 2 \times MY + 1$  respectively. It may be noted that each time after the insertion, values of *MX* and *MY* change. What will be the output of the following SQL query after the steps mentioned above are carried out?

**SELECT Y FROM T WHERE X = 7;**

- A. 127
- B. 15
- C. 129
- D. 257

isro-cse-2023 databases

Answer key 

## 7.1

## B Tree (6)

### 7.1.1 B Tree: GATE CSE 2007 | Question: 63, ISRO2016-59



The order of a leaf node in a  $B^+$  - tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is  $1K$  bytes, data record pointer is  $7$  bytes long, the value field is  $9$  bytes long and a block pointer is  $6$  bytes long, what is the order of the leaf node?

- A. 63                      B. 64                      C. 67                      D. 68

gatecse-2007   databases   b-tree   normal   isro2016

Answer key

### 7.1.2 B Tree: GATE IT 2005 | Question: 23, ISRO2017-67



A B-Tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process are

- A. 5                      B. 4                      C. 3                      D. 2

gateit-2005   databases   b-tree   normal   isro2017

Answer key

### 7.1.3 B Tree: ISRO CSE 2018 | Question: 49



in a file which contains  $1$  million records and the order of the tree is  $100$ , then what is the maximum number of nodes to be accessed if  $B^+$  tree index is used?

- a. 5                      b. 4                      c. 3                      d. 10

isro2018   databases   b-tree

Answer key

### 7.1.4 B Tree: ISRO-2013-26



Calculate the order of leaf ( $P_{leaf}$ ) and non leaf ( $P$ ) nodes of a  $B^+$  tree based on the information given below.

Search key field =  $12$  field

Record pointer =  $10$  bytes

Block pointer =  $8$  bytes

Block size =  $1KB$

- A.  $P_{leaf} = 51$  &  $p = 46$                       B.  $P_{leaf} = 47$  &  $p = 52$   
C.  $P_{leaf} = 46$  &  $p = 51$                       D.  $P_{leaf} = 52$  &  $p = 47$

isro2013   databases   b-tree

Answer key

### 7.1.5 B Tree: ISRO-2013-27



The physical location of a record determined by a formula that transforms a file key into a record location is

- A. Hashed file                      B.  $B$ -Tree file                      C. Indexed file                      D. Sequential file

isro2013   databases   b-tree

Answer key

### 7.1.6 B Tree: ISRO2015-39



If a node has  $K$  children in  $B$  tree, then the node contains exactly \_\_\_\_\_ keys.

A.  $K^2$ B.  $K - 1$ C.  $K + 1$ D.  $\sqrt{K}$ 

isro2015 databases b-tree

Answer key 

## 7.2

## Candidate Key (1)

## 7.2.1 Candidate Key: ISRO CSE 2014 | Question: 42



Let  $x, y, z, a, b, c$  be the attributes of an entity set E. If  $\{x\}, \{x, y\}, \{a, b\}, \{a, b, c\}, \{x, y, z\}$  are superkeys then which of the following are the candidate keys?

- A.  $\{x, y\}$  and  $\{a, b\}$
- B.  $\{x\}$  and  $\{a, b\}$
- C.  $\{x, y, z\}$  and  $\{a, b, c\}$
- D.  $\{z\}$  and  $\{c\}$

databases candidate-key isro2014

Answer key 

## 7.3

## Database Normalization (12)

## 7.3.1 Database Normalization: ISRO CSE 2007 | Question: 51



Armstrong's inference rule doesnot determine

- A. Reflexivity
- B. Augmentation
- C. Transitivity
- D. Mutual dependency

isro2007 databases database-normalization

Answer key 

## 7.3.2 Database Normalization: ISRO CSE 2007 | Question: 54



BCNF is not used for cases where a relation has

- A. Two (or more) candidate keys
- B. Two candidate keys and composite
- C. The candidate key overlap
- D. Two mutually exclusive foreign keys

isro2007 databases database-normalization

Answer key 

## 7.3.3 Database Normalization: ISRO CSE 2011 | Question: 29



Which normal form is based on the concept of 'full functional dependency' is

- A. First Normal Form
- B. Second Normal Form
- C. Third Normal Form
- D. Fourth Normal Form

isro2011 databases database-normalization

Answer key 

## 7.3.4 Database Normalization: ISRO CSE 2011 | Question: 53



In functional dependency Armstrong inference rules refers to

- A. Reflexive, Augmentation and Decomposition
- B. Transitive, Augmentation and Reflexive
- C. Augmentation, Transitive, Reflexive and Decomposition
- D. Reflexive, Transitive and Decomposition



Answer key **7.3.5 Database Normalization: ISRO CSE 2014 | Question: 22**

Consider the schema  $R(A, B, C, D)$  and the functional dependencies  $A \rightarrow B$  and  $C \rightarrow D$ . If the decomposition is made as  $R_1(A, B)$  and  $R_2(C, D)$ , then which of the following is TRUE?

- A. Preserves dependency but cannot perform lossless join
- B. Preserves dependency and performs lossless join
- C. Does not perform dependency and cannot perform lossless join
- D. Does not preserve dependency but perform lossless join

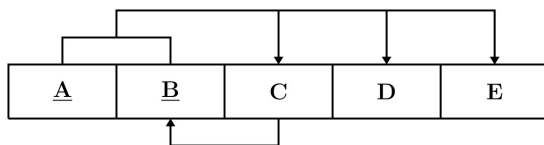
Answer key **7.3.6 Database Normalization: ISRO CSE 2014 | Question: 30**

Every time the attribute A appears, it is matched with the same value of attribute B but not the same value of attribute C. Which of the following is true?

- A.  $A \rightarrow (B, C)$
- B.  $A \rightarrow B, A \twoheadrightarrow C$
- C.  $A \rightarrow B, C \twoheadrightarrow A$
- D.  $A \twoheadrightarrow B, B \rightarrow C$

Answer key **7.3.7 Database Normalization: ISRO CSE 2014 | Question: 63**

Consider the following table



The table is in which normal form?

- A. First Normal Form
- B. Second Normal Form
- C. Third Normal Form but not BCNF
- D. Third Normal Form and BCNF

Answer key **7.3.8 Database Normalization: ISRO CSE 2018 | Question: 41**

For a database relation  $R(a, b, c, d)$  where the domain of  $a, b, c$  and  $d$  include only atomic value, only the following functions dependencies and those that can be inferred from them hold

$$a \rightarrow c$$

$$b \rightarrow d$$

The relation is in:

- a. First normal form but not in second normal form
- b. Second normal form but not in third normal form
- c. Third normal form
- d. None of the above

Answer key 

### 7.3.9 Database Normalization: ISRO CSE 2018 | Question: 5



Considering the following table in a relational database :

| Last Name | Rank      | Room | shift     |
|-----------|-----------|------|-----------|
| Smith     | Manager   | 234  | Morning   |
| Jones     | Custodian | 33   | Afternoon |
| Smith     | Custodian | 33   | Evening   |
| Doe       | Clerical  | 222  | Morning   |

According to the data shown in the table, which of the following could be a candidate key of the table?

- a. {Last Name}      b. {Room}      c. {Shift}      d. {Room, Shift}

isro2018   databases   database-normalization

Answer key

### 7.3.10 Database Normalization: ISRO CSE 2018 | Question: 54



The set of attributes  $X$  will be fully functionally dependent on the set of attributes  $Y$  if the following conditions are satisfied.

- a.  $X$  is functionally dependent on  $Y$       b.  $X$  is not functionally dependent on any subset of  $Y$   
c. Both (a) and (b)      d. None of these

isro2018   databases   database-normalization

Answer key

### 7.3.11 Database Normalization: ISRO CSE 2020 | Question: 50



If every non-key attribute functionally dependent on the primary key, then the relation will be in

- A. First normal form      B. Second normal form  
C. Third normal form      D. Fourth Normal form

isro-2020   databases   database-normalization   easy

Answer key

### 7.3.12 Database Normalization: ISRO-2013-25



Consider the following dependencies and the BOOK table in a relational database design. Determine the normal form of the given relation.

ISBN  $\rightarrow$  Title

ISBN  $\rightarrow$  Publisher

Publisher  $\rightarrow$  Address

- A. First Normal Form      B. Second Normal Form  
C. Third Normal Form      D. BCNF

isro2013   databases   database-normalization

Answer key

## 7.4

### ER Diagram (5)

#### 7.4.1 ER Diagram: ISRO CSE 2016 | Question: 56



Goals for the design of the logical scheme include

- A. avoiding data inconsistency      B. being able to construct query easily  
C. being able to access data efficiently      D. All of the above

Answer key **7.4.2 ER Diagram: ISRO CSE 2017 | Question: 8**

Which symbol denote derived attributes in ER Model?

- A. Double ellipse  
 B. Dashed ellipse  
 C. Squared ellipse  
 D. Ellipse with attribute name underlined

Answer key **7.4.3 ER Diagram: ISRO CSE 2018 | Question: 52**

In E-R model, Y is the dominant entity and X is subordinate entity

- a. If X is deleted, then Y is also deleted  
 b. If Y is deleted, then X is also deleted  
 c. If Y is deleted, then X is not deleted  
 d. None of the above

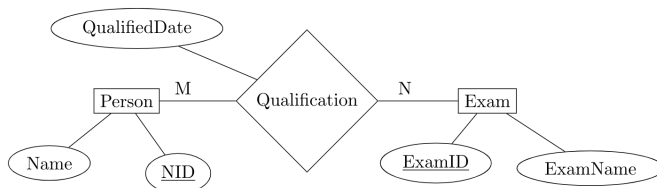
Answer key **7.4.4 ER Diagram: ISRO CSE 2023 | Question: 62**

Entity Relationship model shows the

- A. Static view  
 B. Functional view  
 C. Dynamic view  
 D. All of the above

Answer key **7.4.5 ER Diagram: ISRO2015-16**

Consider the following Relationship Entity Diagram(ERP)



Which of the following possible relations will not hold if the above ERD is mapped into a relation model?

- A. Person (NID, Name)  
 B. Qualification (NID, ExamID, QualifiedDate)  
 C. Exam (ExamID, NID, ExamName)  
 D. Exam (ExamID, ExamName)

Answer key **7.5****File System (6)****7.5.1 File System: ISRO CSE 2009 | Question: 70**

A locked database file can be

- A. Accessed by only one user  
 B. Modified by users with the correct password

C. Used to hide sensitive information

D. Updated by more than one user

isro2009 databases file-system

Answer key 

### 7.5.2 File System: ISRO CSE 2011 | Question: 46



Which RAID level gives block level striping with double distributed parity?

A. RAID 10

B. RAID 2

C. RAID 6

D. RAID 5

isro2011 file-system

Answer key 

### 7.5.3 File System: ISRO CSE 2017 | Question: 4



What does a data dictionary will identify?

A. Field name

B. Field format

C. Field types

D. All of the above

isro2017 databases file-system

Answer key 

### 7.5.4 File System: ISRO CSE 2017 | Question: 45



Which of these is characteristic of RAID 5?

A. Dedicated parity

B. Double parity

C. Hamming code parity

D. Distributed parity

isro2017 databases file-system non-gatecse

Answer key 

### 7.5.5 File System: ISRO CSE 2018 | Question: 53



Immunity of the external schemas (or application programs) to changes in the conceptual scheme is referred to as:

a. Physical Data Independence

b. Logical Data Independence

c. Both (a) and (b)

d. None of the above

isro2018 databases file-system

Answer key 

### 7.5.6 File System: ISRO2015-13



Six files F1, F2, F3, F4, F5 and F6 have 100, 200, 50, 80, 120, 150 records respectively. In what order should they be stored so as to optimize act. Assume each file is accessed with the same frequency

A. F3, F4, F1, F5, F6, F2

B. F2, F6, F5, F1, F4, F3

C. F1, F2, F3, F4, F5, F6

D. Ordering is immaterial as all files are accessed with the same frequency.

isro2015 file-system

Answer key 

## 7.6

## Functional Dependency (1)

### 7.6.1 Functional Dependency: ISRO2015-20



Let  $R = (A, B, C, D, E, F)$  be a relation scheme with the following dependencies  $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$ . Which of the following is a key of  $R$ ?

- A. CD                      B. EC                      C. AE                      D. AC

isro2015   databases   database-normalization   functional-dependency

Answer key

### 7.7 Indexing (6)

#### 7.7.1 Indexing: GATE CSE 2008 | Question: 16, ISRO2016-60



A clustering index is defined on the fields which are of type

- A. non-key and ordering                      B. non-key and non-ordering  
C. key and ordering                      D. key and non-ordering

gatecse-2008   easy   databases   indexing   isro2016

Answer key

#### 7.7.2 Indexing: ISRO CSE 2018 | Question: 51



Which of the following is dense index?

- A. Primary index                      B. Clustered index  
C. Secondary index                      D. Secondary non-key index

isro2018   indexing

Answer key

#### 7.7.3 Indexing: ISRO-2013-61, ISRO CSE 2008 | Question: 35



Embedded pointer provides

- A. A secondary access path                      B. A physical record key  
C. An inverted index                      D. A primary key

isro2013   databases   indexing   isro2008

Answer key

#### 7.7.4 Indexing: ISRO-2017



Q.Consider the following query :

```
Select E.no, COUNT (*)  
FROM Employee E  
GROUP BY E.no
```

If an index on *eno* is available, the query can be answered by scanning only the index if

- A) the index is only hash and clustered  
B) the index is only B+ tree and clustered  
C) index can be hash or B+ tree and clustered or non-clustered  
D) index can be hash or B+ tree and clustered

isro2017   databases   indexing   non-gatecse

Answer key

### 7.7.5 Indexing: ISRO-2017



Q.Consider a table that describes the customers:

*Customers*(custid,name,gender,rating)

The rating value is an integer in the range 1 to 5 and only two values(male and female) are recorded for gender.Consider the query "how many male customers have a rating of 5" ? The best indexing mechanism appropriate for the query is

- A) Linear hashing
- B)Extensible Hashing
- C)B+tree
- D)Bit-mapped index

isro-2017 indexing

Answer key

### 7.7.6 Indexing: ISRO2015-18



Given a block can hold either 3 records or 10 key pointers. A database contains n records, then how many blocks do we need to hold the data file and the dense index

- A.  $13n/30$
- B.  $n/3$
- C.  $n/10$
- D.  $n/30$

isro2015 databases indexing

Answer key

## 7.8

### Rdbms (2)

#### 7.8.1 Rdbms: ISRO CSE 2007 | Question: 50



A view of database that appears to an application program is known as

- A. Schema
- B. Subschema
- C. Virtual table
- D. None of these

isro2007 databases rdbms

Answer key

#### 7.8.2 Rdbms: ISRO CSE 2011 | Question: 21



Which type of DBMS provides support for maintaining several versions of the same entity?

- A. Relational Database Management System
- B. Hierarchical
- C. Object Oriented Database Management System
- D. Network

isro2011 databases rdbms

Answer key

## 7.9

### Referential Integrity (2)

#### 7.9.1 Referential Integrity: GATE CSE 1997 | Question: 6.10, ISRO2016-54



Let  $R(a,b,c)$  and  $S(d,e,f)$  be two relations in which  $d$  is the foreign key of  $S$  that refers to the primary key of  $R$ . Consider the following four operations  $R$  and  $S$

- I. Insert into  $R$
- II. Insert into  $S$
- III. Delete from  $R$
- IV. Delete from  $S$

Which of the following can cause violation of the referential integrity constraint above?

- A. Both I and IV      B. Both II and III      C. All of these      D. None of these

gate1997   databases   referential-integrity   easy   isro2016

Answer key 

### 7.9.2 Referential Integrity: ISRO CSE 2009 | Question: 72



Purpose of 'Foreign Key' in a table is to ensure

- A. Null Integrity      B. Referential Integrity  
C. Domain Integrity      D. Null and Domain Integrity

isro2009   databases   referential-integrity

Answer key 

## 7.10 Relational Algebra (4)

### 7.10.1 Relational Algebra: GATE CSE 1999 | Question: 1.18, ISRO2016-53



Consider the join of a relation  $R$  with a relation  $S$ . If  $R$  has  $m$  tuples and  $S$  has  $n$  tuples then the maximum and minimum sizes of the join respectively are

- A.  $m + n$  and 0  
B.  $mn$  and 0  
C.  $m + n$  and  $|m - n|$   
D.  $mn$  and  $m + n$

gate1999   databases   relational-algebra   easy   isro2016

Answer key 

### 7.10.2 Relational Algebra: GATE CSE 2000 | Question: 1.23, ISRO2016-57



Given the relations

- employee (name, salary, dept-no), and
- department (dept-no, dept-name, address),

Which of the following queries cannot be expressed using the basic relational algebra operations ( $\sigma, \pi, \times, \bowtie, \cup, \cap, -$ )?

- A. Department address of every employee  
B. Employees whose name is the same as their department name  
C. The sum of all employees' salaries  
D. All employees of a given department

gatecse-2000   databases   relational-algebra   easy   isro2016

Answer key 

### 7.10.3 Relational Algebra: ISRO CSE 2007 | Question: 52



Which operation is used to extract specified columns from a table?

- A. Project      B. Join      C. Extract      D. Substitute

isro2007   databases   relational-algebra

Answer key 

#### 7.10.4 Relational Algebra: ISRO CSE 2008 | Question: 33



The join operation can be defined as

- A. a cartesian product of two relations followed by a selection
- B. a cartesian product of two relations
- C. a union of two relations followed by cartesian product of the two relations
- D. a union of two relations

isro2008 databases relational-algebra

Answer key

#### 7.11

#### Relational Model (2)

##### 7.11.1 Relational Model: ISRO CSE 2014 | Question: 11



An aggregation association is drawn using which symbol?

- A. A line which loops back on to the same table
- B. A small open diamond at the end of a line connecting two tables
- C. A small closed diamond at the end of a line connecting two tables
- D. A small closed triangle at the end of a line connecting two tables

isro2014 databases relational-model

Answer key

##### 7.11.2 Relational Model: ISRO2015-21



If  $D_1, D_2, \dots, D_n$  are domains in a relational model, then the relation is a table, which is a subset of

- A.  $D_1 \oplus D_2 \oplus \dots \oplus D_n$
- B.  $D_1 \times D_2 \times \dots \times D_n$
- C.  $D_1 \cup D_2 \cup \dots \cup D_n$
- D.  $D_1 \cap D_2 \cap \dots \cap D_n$

databases isro2015 relational-model

Answer key

#### 7.12

#### SQL (13)

##### 7.12.1 SQL: GATE CSE 2005 | Question: 77, ISRO2016-55



The relation **book** (title, price) contains the titles and prices of different books. Assuming that no two books have the same price, what does the following SQL query list?

```
select title
from book as B
where (select count(*)
      from book as T
      where T.price > B.price) < 5
```

- A. Titles of the four most expensive books
- B. Title of the fifth most inexpensive book
- C. Title of the fifth most expensive book
- D. Titles of the five most expensive books

gatecse-2005 databases sql easy isro2016

Answer key



### 7.12.2 SQL: ISRO CSE 2007 | Question: 47



Which commands are used to control access over objects in relational database?

- A. CASCADE & MVD
- B. GRANT & REVOKE
- C. QUE & QUIST
- D. None of these

isro2007 databases sql

Answer key

### 7.12.3 SQL: ISRO CSE 2007 | Question: 48



Which of the following is aggregate function in SQL?

- A. Avg
- B. Select
- C. Ordered by
- D. distinct

isro2007 databases sql

Answer key

### 7.12.4 SQL: ISRO CSE 2017 | Question: 7



| Employee | Department | OT_allowance |
|----------|------------|--------------|
| RAMA     | Mechanical | 5000         |
| GOPI     | Electrical | 2000         |
| SINDHU   | Computer   | 4000         |
| MAHESH   | Civil      | 1500         |

What is the output of the following SQL query?

```
select count(*) from
((select Employee, Department from Overtime_allowance)
natural join
(select Department, OT_allowance from Overtime_allowance)
as T);
```

- A. 16
- B. 4
- C. 8
- D. None of the above

isro2017 databases sql

Answer key

### 7.12.5 SQL: ISRO CSE 2018 | Question: 40



Given relations  $R(w,x)$  and  $S(y,z)$ , the result of

```
SELECT DISTINCT w, x
FROM R, S
```

Is guaranteed to be same as  $R$ , if

- a.  $R$  has no duplicates and  $S$  is non-empty
- b.  $R$  and  $S$  have no duplicates
- c.  $S$  has no duplicates and  $R$  is non-empty
- d.  $R$  and  $S$  have the same number of tuples

isro2018 sql databases

Answer key

### 7.12.6 SQL: ISRO CSE 2018 | Question: 42



Consider the set of relations given below and the SQL query that follows:

```
Students: (Roll_number, Name, Date_of_birth)
Coursed: (Course_number, Course_name, Instructor)
Grades: (Roll_number, Course_number, Grade)
```

```
SELECT DISTINCT Name
```

```
FROM Students, Courses, Grades
WHERE Students.Roll_number = Grades.Roll_number
      AND Courses.Instructor =Sriram
      AND Courses.Course_number = Grades.Course_number
      AND Grades.Grade = A
```

Which of the following sets is computed by the above query?

- Names of Students who have got an A grade in all courses taught by Sriram
- Names of Students who have got an A grade in all courses
- Names of Students who have got an A grade in at least one of the courses taught by Sriram
- None of the above

isro2018 sql databases

Answer key 

### 7.12.7 SQL: ISRO CSE 2020 | Question: 48



Properties of 'DELETE' and 'TRUNCATE' commands indicate that

- After the execution of 'TRUNCATE' operation, COMMIT, and ROLLBACK statements cannot be performed to retrieve the lost data, while 'DELETE' allow it
- After the execution of 'DELETE' and 'TRUNCATE' operation retrieval is easily possible for the lost data
- After the execution of 'DELETE' operation, COMMIT and ROLLBACK statements can be performed to retrieve the lost data, while TRUNCATE do not allow it
- After the execution of 'DELETE' and 'TRUNCATE' operation no retrieval is possible for the lost data

isro-2020 databases sql normal

Answer key 

### 7.12.8 SQL: ISRO CSE 2020 | Question: 51



The SQL query

```
SELECT columns
FROM TableA
RIGHT OUTER JOIN TableB
ON A.columnName = B.columnName
WHERE A.columnName IS NULL
```

returns the following:

- All rows in Table B, which meets equality condition above and, none from Table A which meets the condition.
- All rows in Table A, which meets equality condition above and none from Table B, which meets the condition.
- All rows in Table B, which meets the equality condition
- All rows in Table A, which meets the equality condition

isro-2020 databases sql normal

Answer key 

### 7.12.9 SQL: ISRO-2013-24



Consider the following relational schema:

- Suppliers (sid:integer, sname:string, saddress:string)
- Parts (pid:integer, pname:string, pcolor:string)
- Catalog (sid:integer, pid:integer, pcost:real)

What is the result of the following query?

```
(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sid = Catalog.sid)
MINUS
(SELECT Catalog.pid from Suppliers, Catalog
WHERE Suppliers.sname <> 'Sachin' and Suppliers.sid = Catalog.sid)
```

- A. Pid of parts supplied by all except Sachin
- B. Pid of parts supplied only by Sachin
- C. Pid of parts available in catalog supplied by Sachin
- D. Pid of parts available in catalog supplied by all except Sachin

isro2013 sql databases

Answer key 

### 7.12.10 SQL: ISRO-DEC2017-11



Consider the schema

Sailors(sid,sname,rating,age) with the following data

| sid | sname  | rating | age |
|-----|--------|--------|-----|
| 22  | Dustin | 7      | 45  |
| 29  | Borg   | 1      | 33  |
| 31  | Pathy  | 8      | 55  |
| 32  | Robert | 8      | 25  |
| 58  | Raghu  | 10     | 17  |
| 64  | Herald | 7      | 35  |
| 71  | Vishnu | 10     | 16  |
| 74  | King   | 9      | 35  |
| 85  | Archer | 3      | 26  |
| 84  | Bob    | 3      | 64  |
| 96  | Flinch | 3      | 17  |

For the query

```
SELECT S.rating, AVG(S.age) AS avgage FROM Sailors S
Where S.age >= 18
GROUP BY S.rating
HAVING 1<(SELECT COUNT(*) FROM$ Sailors S2 Where S.rating=S2.rating)
```

The number of rows returned is

- A. 6
- B. 5
- C. 4
- D. 3

isrodec2017 sql

Answer key 

### 7.12.11 SQL: ISRO2015-19



The maximum length of an attribute of type text is

- A. 127
- B. 255
- C. 256
- D. It is variable

isro2015 databases sql

Answer key 

### 7.12.12 SQL: ISRO2015-22



Consider the following relational query on the above database:

```
SELECT S.name FROM Suppliers S
Where S.sid NOT IN (SELECT C.sid FROM Catalog C
WHERE C.pid NOT IN (SELECT P.pid FROM Parts P WHERE P.color <>'blue'))
```

- A. Find the names of all suppliers who have supplied non-blue part.
- B. Find the names of all suppliers who have not supplied non-blue part.
- C. Find the names of all suppliers who have supplied only non-blue parts
- D. Find the names of all suppliers who have not supplied only non-blue part.

isro2015 databases sql ambiguous

Answer key

### 7.12.13 SQL: ISRO2015-23



Consider the following schema:

**Emp** (Empcode, Name, Sex, Salary, Deptt)

A simple SQL query is executed as follows:

```
SELECT Deptt FROM Emp
GROUP by Deptt
Having avg (Salary) > {select avg (Salary) from Emp}
```

The output will be

- A. Average salary of male employee is the average salary of the organization
- B. Average salary of male employee is less than the average salary of the organization
- C. Average salary of male employee is equal to the average salary of the organization
- D. Average salary of male employee is more than the average salary of the organization

isro2015 databases sql

Answer key

## 7.13

### Transaction and Concurrency (12)

#### 7.13.1 Transaction and Concurrency: GATE CSE 2003 | Question: 29, ISRO2009-73



Which of the following scenarios may lead to an irrecoverable error in a database system?

- A. A transaction writes a data item after it is read by an uncommitted transaction
- B. A transaction reads a data item after it is read by an uncommitted transaction
- C. A transaction reads a data item after it is written by a committed transaction
- D. A transaction reads a data item after it is written by an uncommitted transaction

gatecse-2003 databases transaction-and-concurrency easy isro2009

Answer key

#### 7.13.2 Transaction and Concurrency: GATE CSE 2006 | Question: 20, ISRO2015-17



Consider the following log sequence of two transactions on a bank account, with initial balance 12000, that transfer 2000 to a mortgage payment and then apply a 5% interest.

1. T1 start
2. T1 B old = 12000 new = 10000
3. T1 M old = 0 new = 2000

- Suppose the database system crashes just before log record 7 is written. When the system is restarted, which one statement is true of the recovery procedure?

- gatecse-2006 databases transaction-and-concurrency normal isro2015

### 7.13.3 Transaction and Concurrency: ISRO 2017-5



- isro2017 databases transaction-and-concurrency

#### 7.13.4 Transaction and Concurrency: ISRO CSE 2007 | Question: 60



- isro2007   databases   transaction-and-concurrency

### 7.13.5 Transaction and Concurrency: ISRO CSE 2009 | Question: 71



- isro2009 databases transaction-and-concurrency

### 7.13.6 Transaction and Concurrency: ISRO CSE 2011 | Question: 15



|           |           |           |
|-----------|-----------|-----------|
| <b>T1</b> | <b>T2</b> | <b>T3</b> |
|-----------|-----------|-----------|

|      |      |      |
|------|------|------|
|      |      | R(Y) |
|      |      | R(Z) |
| R(X) |      |      |
| W(X) |      |      |
|      |      | W(Y) |
|      |      | W(Z) |
|      | W(Z) |      |
| R(Y) |      |      |
| W(Y) |      |      |
|      | R(Y) |      |
|      | W(Y) |      |
|      | R(X) |      |
|      |      |      |

|  |      |  |
|--|------|--|
|  | W(X) |  |
|--|------|--|

- A.  $T_1 - T_2 - T_3$   
C.  $T_2 - T_1 - T_3$

- B.  $T_3 - T_1 - T_2$   
D.  $T_1 - T_3 - T_2$

isro2011 databases transaction-and-concurrency

Answer key 

### 7.13.7 Transaction and Concurrency: ISRO CSE 2016 | Question: 58



Trigger is

- A. Statement that enables to start any DBMS  
B. Statement that is executed by the user when debugging an application program  
C. The condition that the system tests for the validity of the database user  
D. Statement that is executed automatically by the system as a side effect of a modification of the database

isro2016 databases transaction-and-concurrency

Answer key 

### 7.13.8 Transaction and Concurrency: ISRO CSE 2017 | Question: 6



ACID properties of a transactions are

- A. Atomicity, consistency, isolation, database  
B. Atomicity, consistency, isolation, durability  
C. Atomicity, consistency, integrity, durability  
D. Atomicity, consistency, integrity, database

isro2017 databases transaction-and-concurrency

Answer key 

### 7.13.9 Transaction and Concurrency: ISRO CSE 2018 | Question: 55



Let us assume that transaction  $T_1$  has arrived before transaction  $T_2$ . Consider the schedule  
 $s = r_1(A); r_2(B); w_2(A); w_1(B)$

Which of the following is true?

- a. Allowed under basic timestamp protocol.  
b. Not allowed under basic timestamp protocols because  $T_1$  is rolled back.  
c. Not allowed under basic timestamp protocols because  $T_2$  is rolled back.  
d. None of these

isro2018 databases transaction-and-concurrency

Answer key 

### 7.13.10 Transaction and Concurrency: ISRO CSE 2020 | Question: 22



Raymonds tree based algorithm ensures

- A. no starvation, but deadlock may occur in rare cases  
B. no deadlock, but starvation may occur

- C. neither deadlock nor starvation can occur  
D. deadlock may occur in cases where the process is already starved

isro-2020 databases transaction-and-concurrency normal

Answer key 

### 7.13.11 Transaction and Concurrency: ISRO CSE 2023 | Question: 49



Amongst the ACID properties of a transaction, the 'Durability' property requires that the changes made to the database by a successful transaction persist

- A. Except in case of an operating system crash  
B. Except in case of a disk crash  
C. Except in case of a power failure  
D. Always, even if there is a failure of any kind

isro-cse-2023 databases transaction-and-concurrency

Answer key 

### 7.13.12 Transaction and Concurrency: ISRO-2013-23



Which of the following is the highest isolation level in transaction management?

- A. Serializable  
B. Repeated Read  
C. Committed Read  
D. Uncommitted Read

isro2013 transaction-and-concurrency

Answer key 

## Answer Keys

|         |     |         |     |         |   |         |     |         |     |
|---------|-----|---------|-----|---------|---|---------|-----|---------|-----|
| 7.0.1   | B   | 7.0.2   | D   | 7.0.3   | A | 7.0.4   | B   | 7.0.5   | C   |
| 7.0.6   | Q-Q | 7.1.1   | A   | 7.1.2   | A | 7.1.3   | B   | 7.1.4   | C   |
| 7.1.5   | A   | 7.1.6   | B   | 7.2.1   | B | 7.3.1   | D   | 7.3.2   | C   |
| 7.3.3   | B   | 7.3.4   | B   | 7.3.5   | A | 7.3.6   | B   | 7.3.7   | C   |
| 7.3.8   | A   | 7.3.9   | D   | 7.3.10  | C | 7.3.11  | C   | 7.3.12  | B   |
| 7.4.1   | D   | 7.4.2   | B   | 7.4.3   | B | 7.4.4   | Q-Q | 7.4.5   | C   |
| 7.5.1   | A   | 7.5.2   | C   | 7.5.3   | D | 7.5.4   | D   | 7.5.5   | B   |
| 7.5.6   | A   | 7.6.1   | B   | 7.7.1   | A | 7.7.2   | C   | 7.7.3   | A   |
| 7.7.4   | Q-Q | 7.7.5   | Q-Q | 7.7.6   | A | 7.8.1   | B   | 7.8.2   | C   |
| 7.9.1   | B   | 7.9.2   | B   | 7.10.1  | B | 7.10.2  | C   | 7.10.3  | A   |
| 7.10.4  | A   | 7.11.1  | B   | 7.11.2  | B | 7.12.1  | D   | 7.12.2  | B   |
| 7.12.3  | A   | 7.12.4  | B   | 7.12.5  | A | 7.12.6  | C   | 7.12.7  | A;C |
| 7.12.8  | A   | 7.12.9  | B   | 7.12.10 | D | 7.12.11 | D   | 7.12.12 | X   |
| 7.12.13 | D   | 7.13.1  | D   | 7.13.2  | B | 7.13.3  | A   | 7.13.4  | A   |
| 7.13.5  | A   | 7.13.6  | B   | 7.13.7  | D | 7.13.8  | B   | 7.13.9  | B   |
| 7.13.10 | B   | 7.13.11 | Q-Q | 7.13.12 | A |         |     |         |     |





### 8.0.1 ISRO CSE 2008 | Question: 48



Which of the following is termed as minimum error code ?

- A. Binary code      B. Gray code      C. Excess-3 code      D. Octal code

isro2008    digital-logic    binary-codes

Answer key

### 8.0.2 ISRO CSE 2014 | Question: 26



The output of a tristate buffer when the enable input is 0 is

- A. Always 0      B. Always 1  
C. Retains the last value when enable input was high      D. Disconnected state

isro2014    digital-logic    translation-lookaside-buffer

Answer key

### 8.0.3 ISRO CSE 2016 | Question: 15



The Excess-3 code is also called

- A. Cyclic Redundancy Code      B. Weighted Code  
C. Self-Complementing Code      D. Algebraic Code

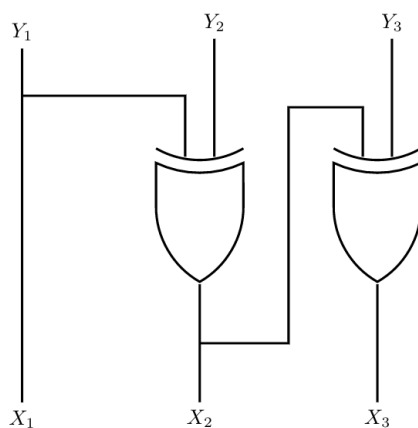
isro2016    digital-logic    binary-codes

Answer key

### 8.0.4 ISRO CSE 2016 | Question: 12



The logic circuit given below converts a binary code  $Y_1Y_2Y_3$  into



- A. Excess-3 code      B. Gray code      C. BCD code      D. Hamming code

digital-logic    binary-codes    isro2016

Answer key

### 8.0.5 ISRO 2016-ECE Digital Logic



A 1 MHz clock signal is applied to a J-K Flip Flop with  $J=K=1$ .

What is the frequency of the Flip-Flop O/P signal?

- (a) 2 MHz

- (b) 500 kHz
- (c) 250 kHz
- (d) 500 MHz

isro2016-ece

Answer key 

### 8.0.6 ISRO ECE 2016 | Digital Logic



For a 10-bit digital ramp ADC using 500 kHz clock, the maximum conversion time is

- a.  $2048 \mu S$
- b.  $2046 \mu S$
- c.  $2064 \mu S$
- d.  $2084 \mu S$

isro2016-ece digital-logic

Answer key 

### 8.0.7 ISRO ECE 2016 | Digital Logic



A 10-bit DAC has a step size of 10mV.

What is its Full scale O/P voltage and the percentage resolution?

- a. 10.24V, 0.2%
- b. 10.23V, 0.5%
- c. 10.23V, 0.1%
- d. 10.24V, 0.1%

isro2016-ece digital-logic

Answer key 

### 8.0.8 ISRO ECE 2016 | Digital logic



How is the status of the carry, auxiliary carry and parity flag affected if write instruction?

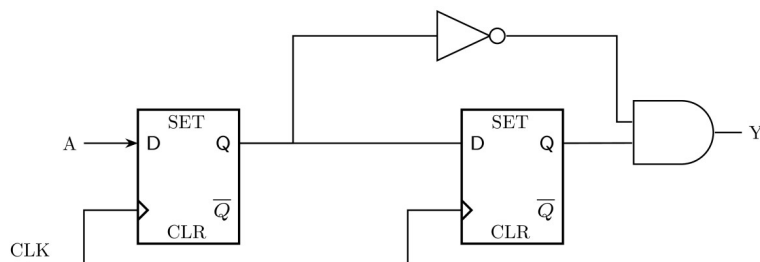
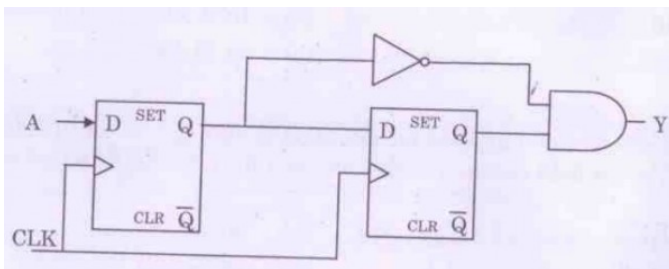
```
MOV A, #9C
ADD A, #64H
```

- a. CY = 0, AC = 0, P = 0
- b. CY = 1, AC = 1, P = 0
- c. CY = 0, AC = 1, P = 0
- d. CY = 1, AC = 1, P = 1

isro2016-ece digital-logic isro-ece

Answer key 

### 8.0.9 ISRO ECE 2015 | Flipflop



What is the functionality of following digital circuit? A is input data, CLK is system clock and Y is output.

- a. Falling edge detection of input A
- c. Rising edge detection of input A

- b. Clock division by 2
- d. Clock division by 4

isro2015-ece isro-ece digital-logic

Answer key 

### 8.0.10 ISRO2012- ECE Digital logic: ADC



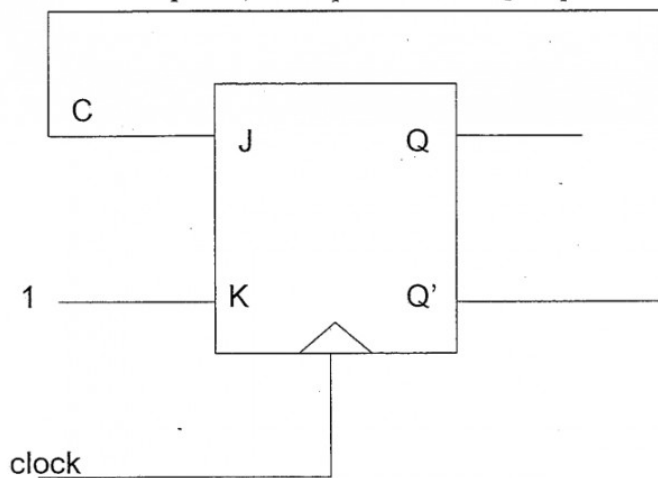
The number of comparators needed in a 8 bit flash type ADC is:

- a. 8
- b. 16
- c. 255
- d. 256

digital-logic isro-ece isro2012-ece

Answer key 

### 8.0.11 ISRO 2011-ECE Digital Logic



In a JK Flipflop we have  $J = Q'$  and  $K = 1$ . assuming that the flipflop was initially cleared and clocked for 6 pulses, the sequence at the Q output will be

- a) 010000
- b) 011001
- c) 010010
- d) 010101

isro-ece isro2011-ece digital-logic

Answer key 

### 8.0.12 ISRO 2011- ECE Digital Logic



\_\_\_\_\_ has the maximum fan out capacity

- a. MOS
- b. CMOS
- c. ECL
- d. RTL

isro-ece isro2011-ece digital-logic

### 8.0.13 ISRO 2011- ECE Digital Logic

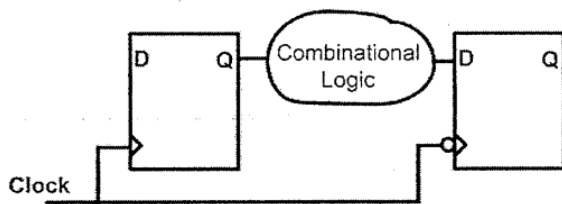


The function of a strobe function in digital system is

- a. To reset memory register.
- b. To check the functioning of a logic gate
- c. To avoid race problem
- d. To tri-state the output of the register

isro-ece isro2011-ece digital-logic

### 8.0.14 ISRO 2010-ECE Digital logic



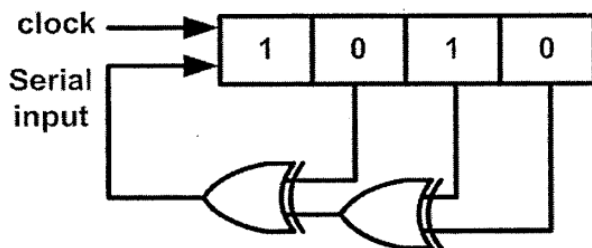
What is the maximum clock frequency at the given circuit can be operated without timing violations? Assume that the Combinational logic delay is 10 ns and the clock duty cycle varies from 40% to 60 %.

- a) 100 MHz
- b) 50 MHz
- c) 40 MHz
- d) 25 MHz

isro-ece digital-logic

Answer key

### 8.0.15 ISRO 2010-ECE Shift register



Given shift register is initially loaded with the bit pattern "1010". After how many clock cycles will the content of shift register be "1010" again?

- a) 5
- b) 9
- c) 7
- d) 15

isro-ece digital-logic

Answer key 

### 8.0.16 ISRO 2009-ECE MOD Counter



A certain JK FF has  $t_{pd}=12$  ns. The largest MOD counter that can be constructed from such FFs and still operate up to 10 MHz is

- a) 16
- b) 256
- c) 8
- d) 128

isro-ece digital-logic

Answer key 

### 8.0.17 ISRO 2009- ECE Digital Logic



Which of the following types of devices is not field programmable?

- a) FPGA
- b) ASIC
- c) CPLD
- d) PLD

isro-ece digital-logic

### 8.0.18 ISRO 2009 -ECE Logic Family



Which of the following technology results in least power dissipation ?

- a) CMOS
- b) ECL
- c) TTL
- d) NMOS

isro-ece digital-logic

Answer key 

### 8.0.19 ISRO 2009- ECE Digital Logic



A 5 bit DAC has a current output. For a digital input of 10100, an output current of 10 mA is produced. What will be the output current for a digital input of 11101?

- a) 14.5 mA
- b) 10 mA
- c) 100 mA
- d) Not possible to calculate

**8.0.20 ISRO 2008- ECE Digital Logic**

The fastest ADC among the following is

- (a) Successive approximation type
- (b) Dual slope type
- (c) Sigma - Delta ADC
- (d) Flash converter

**8.0.21 ISRO 2008- ECE Odd parity**

Odd parity generator uses \_\_\_\_\_ logic

- (a) XNOR
- (b) XOR
- (c) Sequential
- (d) OR

Answer key

**8.0.22 ISRO 2008-ECE A/D Converter**

The number of comparators required in an 8-bit flash-type A/D converter is

- (a) 256
- (b) 255
- (c) 9
- (d) 8

**8.0.23 ISRO 2008- ECE minimum number of NAND Gates**

Minimum number of 2-input NAND gates that will be required to implement the function:  $Y = AB + CD + EF$  is

- (a) 4
- (b) 5
- (c) 6
- (d) 7

Answer key

### 8.0.24 ISRO 2008-ECE - Ripple counter



In a ripple counter how many changes in state happen when count changes from 7 to 8?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

isro-ece digital-logic

Answer key

### 8.0.25 ISRO 2008- ECE PROM



A programmable device (**PROM**) is

- (a) programmable OR and fixed AND array
- (b) programmable AND and fixed OR array
- (c) programmable AND and programmable OR array
- (d) none

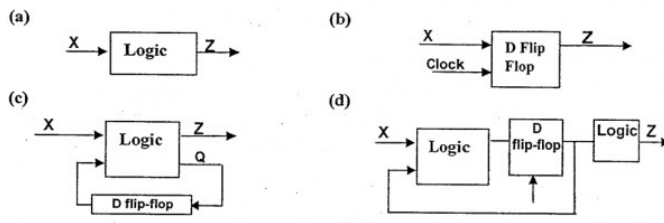
isro-ece digital-logic

Answer key

### 8.0.26 ISRO 2008-ECE Moore model for sequential circuits



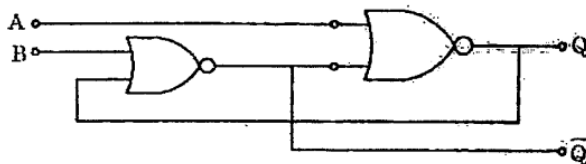
Which of the following represents the Moore model for sequential circuits?



isro-ece digital-logic

Answer key

### 8.0.27 ISRO 2007- ECE Digital logic



The circuit is a

- a) Monostable MV
- b) Astable MV
- c) Adder
- d) SR FF

digital-logic isro-ece

Answer key 

### 8.0.28 ISRO 2007-ECE Up counter



A 4-bit presetable UP counter has preset input 0101. The preset operation takes place as soon as the counter reaches 1111. The modulus of the counter is

- a) 5
- b) 10
- c) 11
- d) 15

isro-ece digital-logic

Answer key 

### 8.0.29 ISRO 2007-ECE Counters



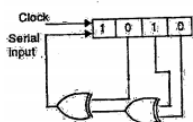
A 4-bit synchronous counter uses flip-flops with propagation delay time of 25 ns each. The maximum possible time required for change of state will be

- a) 25 ns
- b) 50 ns
- c) 75 ns
- d) 100 ns

isro-ece digital-logic

Answer key 

### 8.0.30 ISRO 2007- ECE Shift Register



The shift register shown in the given figure is initially loaded with the bit pattern 1010. Subsequently the shift register is clocked, and with each clock pulse the pattern gets shifted by one bit position to the right. With each shift, the bit at the serial input is pushed to the left most position (MSB). After how many clock pulses will the content of the shift register become 1010 again?

- a) 3
- b) 7
- c) 11
- d) 15

isro-ece

Answer key 

### 8.0.31 ISRO 2007- ECE combinational logic circuit







d)  $X_0 X_1 X_3 \dots X_{n-1} + X_2 + X_3 + X_5 \dots X_{n-1} + \dots + X_{n-2} + X_{n-1} + X_n$

isro-ece digital-logic

### 8.0.35 ISRO 2006-ECE Ripple counter



A Pulse train with a frequency of 1MHz is counted using a modulo 1024 ripple-counter built with J-K flip-flops. For proper operation of the counter the maximum permissible propagation delay per flip-flop stage is

- a) 100 n sec
- b) 50 n sec
- c) 20 n sec
- d) 10 n sec

isro-ece digital-logic

Answer key

### 8.0.36 ISRO 2006-ECE A/D Converter



The A/D converter used in a digital voltmeter could be (1) successive approximation type (2) Flash converter type (3) Dual slope converter type. The correct sequence in the increasing order of their conversion times is

- a) 1,2,3
- b) 2,1,3
- c) 3,2,1
- d) 3,1,2

isro-ece digital-logic

### 8.0.37 ISRO 2006-ECE A/D converter



10 bit A/D converters, the quantization error is given by (in percent)

- a) 1
- b) 2
- c) 0.1
- d) 0.2

isro-ece digital-logic

Answer key

### 8.0.38 ISRO 2013 - Boolean algebra - [EE]



Which of the following Boolean algebra rules is correct?

- a)  $A.A' = 1$
- b)  $A+AB = A+B$
- c)  $A+A'B = A+B$

d)  $A(A+B) = B$

engineering-mathematics isro-ee

Answer key 

### 8.0.39 ISRO 2013- Combinational circuit [EE]



In an all NOR gate realization of a combinational circuit all EVEN and ODD level gates behave like

- a) OR and AND
- b) AND and OR
- c) OR and NOT
- d) NOR and AND

isro-ee digital-logic

Answer key 

### 8.0.40 ISRO 2015 Truth table [EE]



The truth table

| X | Y | F |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

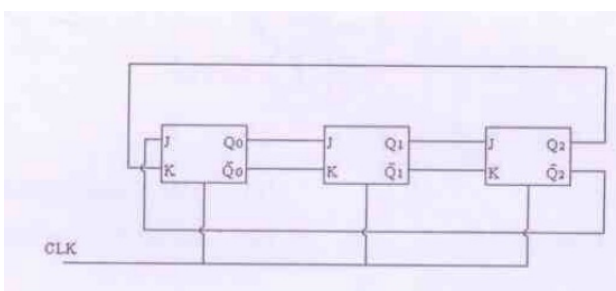
Represent the Boolean function

- (a) X
- (b)  $X - Y$
- (c)  $X + Y$
- (d) Y

isro-ee digital-logic

Answer key 

### 8.0.41 ISRO 2015 - Johnson counter [EE]



A three stage Johnson counter ring in figure is clocked at a constant frequency of  $f_c$  from starting state of  $Q_0Q_1Q_2 = 101$ . The frequency of output  $Q_0Q_1Q_2$  will be

- (a)  $f_c / 2$

- (b)  $f_c / 6$
- (c)  $f_c / 3$
- (d)  $f_c / 8$

digital-logic isro-ee

Answer key 

#### 8.0.42 ISRO 2015- Ripple Counter [EE]



A 4 bit module -16 ripple counter uses JK F/F. If the propagation delay of each F/F is 50 nano seconds, the maximum clock frequency that can be used is equal to

- (a) 20 MHz
- (b) 5 MHz
- (c) 10 MHz
- (d) 4 MHz

isro-ee digital-logic

Answer key 

#### 8.0.43 ISRO 2017-25



The 2-input XOR has a high output only when the input values are

- (a) low
- (b) high
- (c) same
- (d) different

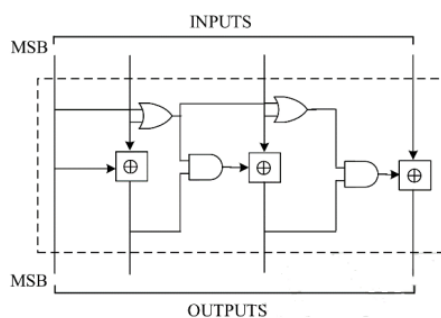
isro2017

Answer key 

#### 8.0.44 ISRO 2017-ECE Computational circuits



The circuit shown in the figure converts



- (a) BCD to binary code
- (b) Binary to Excess-3 code
- (c) Excess-3 to Gray code
- (d) Gray to Binary code

isro2017-ece

Answer key 

### 8.0.45 ISRO 2017-ECE T- Flip flops



If input to T flip flop is **200** Hz signal, then what will be the output signal frequency if **four** T flip flops are connected in cascade

- (a) 200 Hz
- (b) 50 Hz
- (c) 800 Hz
- (d) None of the above

isro2017-ece

Answer key

### 8.0.46 ISRO 2017-ECE Minterms



Simplify the below function represented in sum of minterms

$$F(A, B, C, D, E) = \sum(0, 1, 2, 3, 8, 9, 16, 17, 20, 21, 24, 25, 28, 29, 30, 31)$$

- (a)  $A\bar{D} + (\bar{C} + \bar{D}) + \bar{A}\bar{B}\bar{C} + (\bar{A} + \bar{B} + \bar{C})$
- (b)  $A\bar{D}E + (\bar{C} + \bar{E})D + \bar{A}\bar{B}\bar{C}D + AE\bar{C}$
- (c)  $A\bar{D}B + (\bar{C} + \bar{B}) + \bar{C}\bar{D}\bar{E} + (\bar{B} + \bar{C} + \bar{D})$
- (d)  $A\bar{C} + (\bar{D} + \bar{E}) + \bar{A}\bar{E} + (\bar{C} + \bar{D} + \bar{E})$

isro2017-ece

Answer key

### 8.0.47 ISRO 2017-ECE Digital Logic



Which of the following statement is **true** for Programmable Logic array (**PLA**)?

- (a) Fixed AND array and Fused programmable OR array
- (b) Fused programmable AND array and Fixed OR array
- (c) Fused programmable AND array and Fused programmable OR array
- (d) None of the above

isro2017-ece

Answer key

### 8.0.48 ISRO 2017-ECE Digital logic



Consider following 8085 microprocessor program

```
MVI A, DATA1
ORA A
JM DISPLAY
OUT PORT1
CMA
DISPLAY : ADI 01H
OT PORT1
HLT
```

If DATA1 = A7H, the output at PORT1 is

- (a) A7H
- (b) 58H
- (c) 00H

(d) 59H

isro2017-ece

#### 8.0.49 ISRO-DEC2017-40



Let  $M = 11111010$  and  $N = 00001010$  be two 8-bit two's complement number. Their product in two's complement is

- A. 11000100      B. 10011100      C. 10100101      D. 11010101

isrodec2017

Answer key

#### 8.0.50 ISRO-DEC2017-39



A computer with 32-bit word size uses 2's complement to represent numbers, The range of integers that can be represented by this computer is

- A.  $-2^{32}$  to  $2^{32}$       B.  $-2^{31}$  to  $2^{32} - 1$   
C.  $-2^{31}$  to  $2^{31} - 1$       D.  $-2^{31}$  to  $2^{32} - 1$

isrodec2017

Answer key

#### 8.0.51 ISRO-DEC2017-75



In IEEE floating point representation, the hexadecimal number 0xC0000000 corresponds to

- A. -3.0      B. -1.0      C. -4.0      D. -2.0

isrodec2017

Answer key

#### 8.0.52 ISRO-DEC2017-71



A 32-bit adder is formed by cascading 4-bit CLA adder. The gate delays (latency) for getting the sum bits is

- A. 16      B. 18      C. 17      D. 19

isrodec2017

Answer key

#### 8.0.53 ISRO-DEC2017-72



We consider the addition of two 2's complement numbers  $b_{n-1}b_{n-2} \dots b_0$  and  $a_{n-1}a_{n-2} \dots a_0$ . A binary adder for adding two unsigned binary numbers is used to add two binary numbers. The sum is denoted by  $c_{n-1}c_{n-2} \dots c_0$ . The carry out is denoted by  $c_{out}$ . The overflow condition is identified by

- A.  $c_{out}(\overline{a_{n-1}} \oplus \overline{b_{n-1}})$   
B.  $\overline{a_{n-1}}b_{n-1}\overline{c_{n-1}} + \overline{a_{n-1}}b_{n-1}c_{n-1}$   
C.  $c_{out} \oplus c_{n-1}$   
D.  $a_{n-1} \oplus b_{n-1} \oplus c_{n-1}$

Answer key **8.0.54 ISRO CSE 2023 | Question: 10**

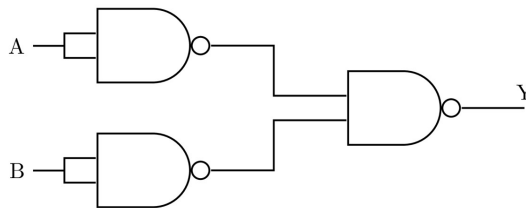
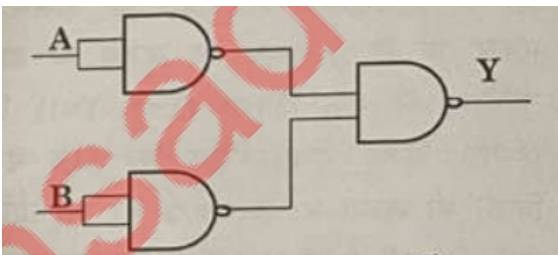
A logic circuit that provides a LOW output when both inputs are HIGH or both inputs are LOW is

- A. AND                      B. NAND                      C. XNOR                      D. XOR

isro-cse-2023   digital-logic   easy

Answer key **8.0.55 ISRO CSE 2023 | Question: 9**

The logical operation of the following circuit is



- A. XOR                      B. NAND                      C. AND                      D. OR

isro-cse-2023   digital-logic

Answer key **8.0.56 ISRO CSE 2023 | Question: 8**

A full adder circuit requires

- A. two inputs and two outputs                      B. two inputs and three outputs  
C. three inputs and two outputs                      D. three inputs and one output

isro-cse-2023   digital-logic

Answer key **8.1 Adder (7)****8.1.1 Adder: ISRO 2007-ECE Half Adder Circuit**

A half adder can be constructed using two 2-input logic gates. One of them is an AND-gate, the other is

- A. OR                      B. NAND                      C. NOR                      D. EX-OR

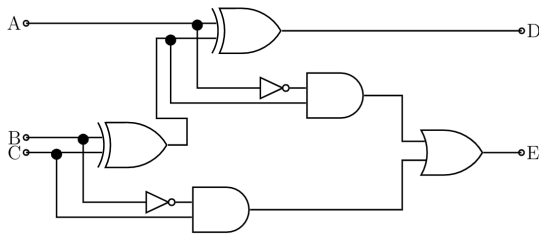
isro-ece   digital-logic   adder

Answer key 

### 8.1.2 Adder: ISRO CSE 2007 | Question: 03



The circuit shown in the given figure is a



- A. full adder  
B. full subtractor  
C. shift register  
D. decade counter

isro2007 digital-logic digital-circuits adder

Answer key 

### 8.1.3 Adder: ISRO CSE 2016 | Question: 9



For a binary half-subtractor having two inputs A and B, the correct set of logical outputs D (= A minus B) and X (= borrow) are

- A.  $D = AB + \bar{A}B, X = \bar{A}B$   
B.  $D = \bar{A}B + A\bar{B}, X = A\bar{B}$   
C.  $D = \bar{A}B + A\bar{B}, X = \bar{A}B$   
D.  $D = AB + \bar{A}B, X = A\bar{B}$

isro2016 digital-logic adder

Answer key 

### 8.1.4 Adder: ISRO CSE 2017 | Question: 24



When two  $n$ -bit binary numbers are added the sum will contain at the most

- A.  $n$  bits  
B.  $n + 2$  bits  
C.  $n + 3$  bits  
D.  $n + 1$  bits

isro2017 digital-logic adder

Answer key 

### 8.1.5 Adder: ISRO CSE 2020 | Question: 9



In a 8-bit ripple carry adder using identical full adders, each full adder takes 34 ns for computing sum. If the time taken for 8-bit addition is 90 ns, find time taken by each full adder to find carry.

- A. 6 ns  
B. 7 ns  
C. 10 ns  
D. 8 ns

isro-2020 digital-logic combinational-circuit adder normal

Answer key 

### 8.1.6 Adder: ISRO ECE 2016 | Digital Logic



How many inputs & outputs does a full adder have?

- a. 3,2  
b. 2,3  
c. 3,3  
d. 2,2

isro2016-ece digital-logic combinational-circuit adder

Answer key 



### 8.1.7 Adder: ISRO2015-7



If half adders and full adders are implements using gates, then for the addition of two 17 bit numbers (using minimum gates) the number of half adders and full adders required will be

- A. 0,17                      B. 16,1                      C. 1,16                      D. 8,8

isro2015    digital-logic    adder

Answer key

## 8.2 Asynchronous Circuit (1)

### 8.2.1 Asynchronous Circuit: ISRO 2008- ECE Dynamic hazard



In asynchronous circuits, which of the following best explains a dynamic hazard

- A. Output changes several times for a single change in an input  
B. Output changes to a different state for a single change in an input  
C. Output changes momentarily when it is supposed to remain constant for a single change in an input  
D. None of the above

isro-ece    digital-logic    asynchronous-circuit

Answer key

## 8.3 Bcd (1)

### 8.3.1 Bcd: ISRO-2013-11



When two BCD numbers  $0 \times 14$  and  $0 \times 08$  are added what is the binary representation of the resultant number ?

- A.  $0 \times 22$                       B.  $0 \times 1c$                       C.  $0 \times 16$                       D. results in overflow

isro2013    number-representation    bcd

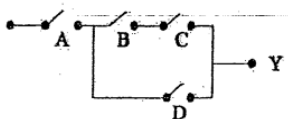
Answer key

## 8.4 Boolean Algebra (20)

### 8.4.1 Boolean Algebra: ISRO 2006- ECE Boolean algebra



For the switch circuit, taking open as 0 and closed as 1, the expression for the circuit is Y.



- A.  $A + (B + C)D$                       B.  $A + BC + D$                       C.  $A(BC + D)$                       D. None of these

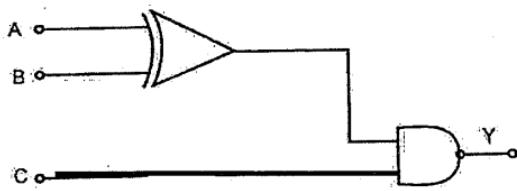
isro-ece    digital-logic    boolean-algebra

Answer key

### 8.4.2 Boolean Algebra: ISRO 2007- ECE Combinational circuit



The Boolean expression for the output of the logic circuit shown in the figure is



A.  $Y = AB + A'B' + C$

C.  $Y = AB' + A'B + C$

digital-logic isro-ece boolean-algebra

Answer key

B.  $Y = AB + A'B' + C'$

D.  $Y = AB + A'B + C'$

#### 8.4.3 Boolean Algebra: ISRO 2007-ECE Boolean Dual



For the identity  $AB + A'C + BC = AB + A'C$ , the dual form is

A.  $(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

B.  $(A'+B')(A'+C')(B'+C') = (A'+B')(A+C')$

C.  $(A+B)(A'+C)(B+C) = (A'+B')(A+C')$

D.  $A'B' + AC' + B'C' = A'B' + AC'$

isro-ece digital-logic boolean-algebra

Answer key

#### 8.4.4 Boolean Algebra: ISRO CSE 2007 | Question: 01



The Boolean expression  $Y = (A + \bar{B} + \bar{A}B)\bar{C}$  is given by

A.  $A\bar{C}$

B.  $B\bar{C}$

C.  $\bar{C}$

D.  $AB$

isro2007 digital-logic boolean-algebra

Answer key

#### 8.4.5 Boolean Algebra: ISRO CSE 2008 | Question: 11



The Boolean theorem  $AB + \bar{A}C + BC = AB + \bar{A}C$  corresponds to

A.  $(A+B) \cdot (\bar{A} + C) \cdot (B+C) = (A+B) \cdot (\bar{A} + C)$

B.  $AB + \bar{A}C + BC = AB + BC$

C.  $AB + \bar{A}C + BC = (A+B) \cdot (\bar{A} + C) \cdot (B+C)$

D.  $(A+B) \cdot (\bar{A} + C) \cdot (B+C) = AB + \bar{A}C$

isro2008 digital-logic boolean-algebra

Answer key

#### 8.4.6 Boolean Algebra: ISRO CSE 2008 | Question: 24



The Boolean expression  $(A + \bar{C})(\bar{B} + \bar{C})$  simplifies to

A.  $\bar{C} + A\bar{B}$

B.  $\bar{C}(\bar{A} + B)$

C.  $\bar{B}\bar{C} + A\bar{B}$

D. None of these

isro2008 digital-logic boolean-algebra

Answer key

#### 8.4.7 Boolean Algebra: ISRO CSE 2008 | Question: 25



In the expression  $\bar{A}(\bar{A} + \bar{B})$  by writing the first term  $A$  as  $A + 0$ , the expression is best simplified as

- A.  $A+AB$
- B.  $AB$
- C.  $A$
- D.  $A+B$

isro2008 digital-logic boolean-algebra

Answer key

#### 8.4.8 Boolean Algebra: ISRO CSE 2008 | Question: 28



Which of the following is not a valid rule of XOR?

- A.  $0 \text{ XOR } 0 = 0$
- B.  $1 \text{ XOR } 1 = 1$
- C.  $1 \text{ XOR } 0 = 1$
- D.  $B \text{ XOR } B = 0$

isro2008 digital-logic boolean-algebra

Answer key

#### 8.4.9 Boolean Algebra: ISRO CSE 2009 | Question: 19



The switching expression corresponding to  $f(A, B, C, D) = \Sigma(1, 4, 5, 9, 11, 12)$  is

- A.  $BC'D' + A'C'D + AB'D$
- B.  $ABC' + ACD + B'C'D$
- C.  $ACD' + A'BC' + AC'D'$
- D.  $A'BD + ACD' + BCD'$

isro2009 digital-logic boolean-algebra

Answer key

#### 8.4.10 Boolean Algebra: ISRO CSE 2009 | Question: 20



Consider the following boolean function of four variables  $f(w, x, y, z) = \Sigma(1, 3, 4, 6, 9, 11, 12, 14)$ , the function is

- A. Independent of one variable
- B. Independent of two variables
- C. Independent of three variables
- D. Dependent on all variables

isro2009 digital-logic boolean-algebra

Answer key

#### 8.4.11 Boolean Algebra: ISRO CSE 2011 | Question: 30



In Boolean algebra, rule  $(X + Y)(X + Z) =$

- A.  $Y + XZ$
- B.  $X + YZ$
- C.  $XY + Z$
- D.  $XZ + Y$

isro2011 digital-logic boolean-algebra

Answer key

#### 8.4.12 Boolean Algebra: ISRO CSE 2011 | Question: 6



Evaluate  $(X \text{ xor } Y) \text{ xor } Y$ ?

- A. All 1's
- B. All 0's
- C.  $X$
- D.  $Y$

isro2011 digital-logic boolean-algebra

Answer key

#### 8.4.13 Boolean Algebra: ISRO CSE 2014 | Question: 56



Which of the following is not valid Boolean algebra rule?

- A.  $X.X = X$
- B.  $(X+Y).X = X$
- C.  $\overline{X} + XY = Y$
- D.  $(X+Y).(X+Z) = X + YZ$

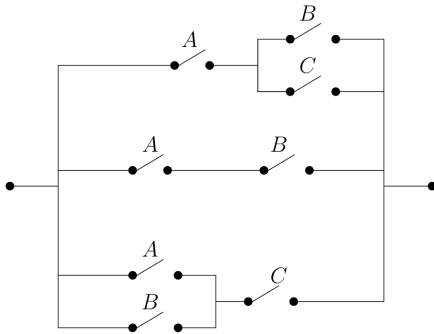
digital-logic boolean-algebra isro2014

Answer key

#### 8.4.14 Boolean Algebra: ISRO CSE 2016 | Question: 8



The minimum Boolean expression for the following circuit is



- A.  $AB + AC + BC$
- B.  $A + BC$
- C.  $A + B$
- D.  $A + B + C$

isro2016 digital-logic boolean-algebra

Answer key

#### 8.4.15 Boolean Algebra: ISRO CSE 2018 | Question: 62



Any set of Boolean operation that is sufficient to represent all Boolean expression is said to be complete. Which of the following is not complete ?

- A. {AND, OR}
- B. {AND, NOT}
- C. {NOT, OR}
- D. {NOR}

isro2018 digital-logic boolean-algebra

Answer key

#### 8.4.16 Boolean Algebra: ISRO ECE 2012 | Boolean Logic



| C | B | A | F |
|---|---|---|---|
| 0 | 0 | 0 | d |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | d |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

The truth table for implementing a Boolean variable F is given where d represents don't care states. The minimized expression for F is:

- A.  $BC' + AC' + AB$
- B.  $AB + C'$
- C.  $A'BC' + AB'C' + ABC$
- D. None of the above

isro-ece isro2012-ece digital-logic boolean-algebra

Answer key 

#### 8.4.17 Boolean Algebra: ISRO ECE 2013 | Digital Logic



Which of the following is the Boolean function for majority voting, assuming  $A, B, C$  are inputs and  $Y$  is output?

- a.  $Y = AB + AC + BC$
- b.  $Y = A + B + C$
- c.  $Y = ABC$
- d.  $Y = AB + BC$

isro-ece isro2013-ece digital-logic boolean-algebra

Answer key 

#### 8.4.18 Boolean Algebra: ISRO ECE 2016 | Digital Logic



Simplify Boolean function represented in sum of product of min-terms,  $F(x, y, z) = \Sigma(0, 2, 4, 5, 6)$

- a.  $z' + xy'$
- b.  $x'y'z' + xyz + xy'z'$
- c.  $xyz + x' + y' + z'$
- d.  $xy + yz + zx$

isro2016-ece digital-logic boolean-algebra

Answer key 

#### 8.4.19 Boolean Algebra: ISRO2015-10



The boolean expression  $AB + AB' + A'C + AC$  is independent of the boolean variable

- A.  $A$
- B.  $B$
- C.  $C$
- D. None of these

isro2015 digital-logic boolean-algebra

Answer key 

#### 8.4.20 Boolean Algebra: ISRO2015-5



The complement of the Boolean expression  $AB(\overline{B}C + AC)$  is

- A.  $(\overline{A} + \overline{B}) + (B + \overline{C}).(\overline{A} + \overline{C})$
- B.  $(\overline{A}.\overline{B}) + (B\overline{C} + \overline{A}.\overline{C})$
- C.  $(\overline{A} + \overline{B}).(B + \overline{C}) + (A + \overline{C})$
- D.  $(A+B).(\overline{B} + C)(A+C)$

isro2015 digital-logic boolean-algebra

Answer key 

### 8.5

#### Booths Algorithm (1)

#### 8.5.1 Booths Algorithm: ISRO CSE 2009 | Question: 40 , GATE2008-IT\_42



The two numbers given below are multiplied using the Booth's algorithm

Multiplicand : 0101 1010 1110 1110

Multiplier : 0111 0111 1011 1101

How many additions/subtractions are required for the multiplication of the above two numbers?

- A. 6
- B. 8
- C. 10
- D. 12

isro2009 digital-logic booths-algorithm

Answer key 

## 8.6.1 CO and Architecture: ISRO 2009-ECE RISC vs CISC



The theoretical dividing line between Reduced Instruction Set computing (RISC) microprocessor and Complex Instructions Set Computing (CISC) microprocessor is

- a) Instruction execution rate to be one instruction per clock cycle
- b) Number of address and data lines
- c) Number of pins in the chip
- d) None of the above

isro-ece co-and-architecture

Answer key

## 8.7.1 Canonical Normal Form: ISRO CSE 2016 | Question: 16



The simplified SOP (Sum of Product) from the Boolean expression

$$(P + \bar{Q} + \bar{R}). (P + Q + R). (P + Q + \bar{R})$$

is

- A.  $(\bar{P}.Q + \bar{R})$
- B.  $(P + Q.\bar{R})$
- C.  $(P.\bar{Q} + R)$
- D.  $(P.Q + R)$

digital-logic canonical-normal-form isro2016

Answer key

## 8.7.2 Canonical Normal Form: ISRO-2013-28



The most simplified form of the Boolean function

$$x(A, B, C, D) = \sum(7, 8, 9, 10, 11, 12, 13, 14, 15)$$

(expressed in sum of minterms) is?

- A.  $A + A'BCD$
- B.  $AB + CD$
- C.  $A + BCD$
- D.  $ABC + D$

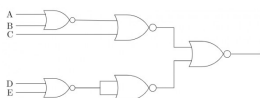
isro2013 digital-logic canonical-normal-form

Answer key

## 8.8.1 Circuit Output: ISRO CSE 2007 | Question: 02



The circuit shown in the following figure realizes the function.



- A.  $(\overline{A+B+C})(\overline{D} \overline{E})$
- B.  $(\overline{A+B+C})(D\overline{E})$
- C.  $(A+B+C)(\overline{D} \overline{E})$
- D.  $(A+B+C)(D\overline{E})$

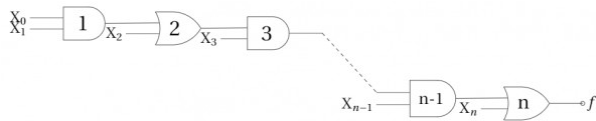
isro2007 digital-logic digital-circuits circuit-output

Answer key

### 8.8.2 Circuit Output: ISRO CSE 2008 | Question: 12



In the given network of AND and OR gates  $f$  can be written as



- A.  $X_0X_1X_2 \dots X_n + X_1X_2 \dots X_n + X_2X_3 \dots X_n + \dots + X_n$
- B.  $X_0X_1 + X_2X_3 + \dots X_{n-1}X_n$
- C.  $X_0 + X_1 + X_2 + \dots + X_n$
- D.  $X_0X_1 + X_3 \dots X_{n-1} + X_2X_3 + X_5 \dots X_{n-1} + \dots + X_{n-2}X_{n-1} + X_n$

isro2008 digital-logic circuit-output

Answer key

### 8.8.3 Circuit Output: ISRO CSE 2008 | Question: 26



The logic operations of two combinational circuits in Figure-I and Figure -II are

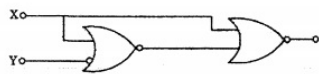


Figure -I



Figure -II

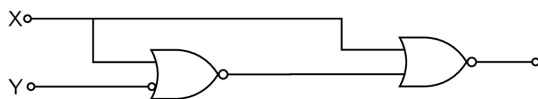


Figure -I

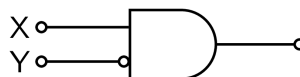


Figure -II

- A. entirely different
- B. identical
- C. complementary
- D. dual

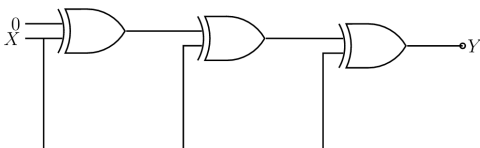
isro2008 digital-logic circuit-output

Answer key

### 8.8.4 Circuit Output: ISRO CSE 2008 | Question: 27



The output  $Y$  of the given circuit



- A. 1
- B. 0
- C. X
- D.  $X'$

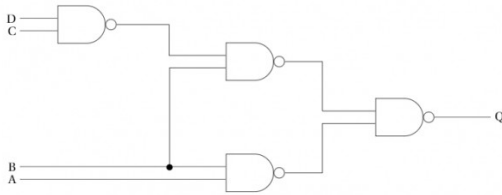
isro2008 digital-logic circuit-output

Answer key 

8.8.5 Circuit Output: ISRO CSE 2014 | Question: 15



Consider the logic circuit given below:



$Q =$  \_\_\_\_\_?

- A.  $\bar{A}C + B\bar{C} + CD$   
C.  $AB + B\bar{C} + B\bar{D}$

- B.  $ABC + \bar{C}D$   
D.  $\bar{A}\bar{B} + A\bar{C} + \bar{C}D$

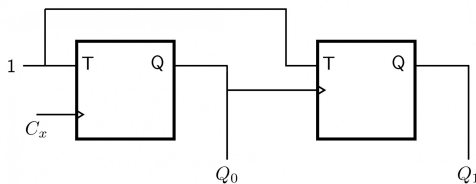
isro2014 digital-logic circuit-output

Answer key 

8.8.6 Circuit Output: ISRO CSE 2014 | Question: 21, UGCNET-Dec2012-III: 23, UGCNET-Dec2013-III: 22



What are the final values of  $Q_1$  and  $Q_0$  after 4 clock cycles, if initial values are 00 in the sequential circuit shown below:



A. 11

B. 01

C. 10

D. 00

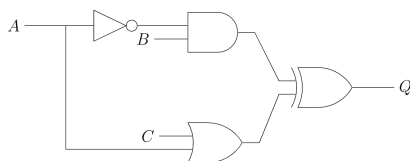
isro2014 digital-logic circuit-output ugcnetcse-dec2012-paper3 ugcnetcse-dec2013-paper3

Answer key 

8.8.7 Circuit Output: ISRO CSE 2014 | Question: 53



Consider the logic circuit given below.



The inverter, AND and OR gates have delays of 6, 10 and 11 nanoseconds respectively. Assuming that wire delays are negligible, what is the duration of glitch for Q before it becomes stable?

A. 5

B. 11

C. 16

D. 27

isro2014 digital-logic circuit-output

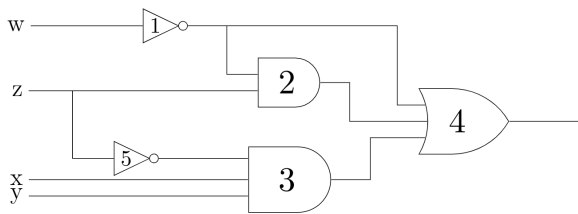
Answer key 

8.8.8 Circuit Output: ISRO CSE 2016 | Question: 10



Consider the following gate network





Which one of the following gates is redundant?

- A. Gate No. 1                      B. Gate No. 2                      C. Gate No. 3                      D. Gate No. 4

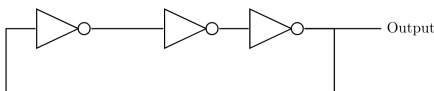
isro2016    digital-logic    circuit-output

Answer key

### 8.8.9 Circuit Output: ISRO CSE 2016 | Question: 13



The circuit given in the figure below is

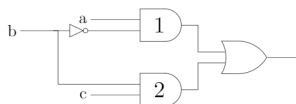


- A. An oscillating circuit and its output is square wave  
 B. The one whose output remains stable in '1' state  
 C. The one having output remains stable in '0' state  
 D. has a single pulse of three times propagation delay

isro2016    digital-logic    circuit-output

Answer key

### 8.8.10 Circuit Output: ISRO CSE 2018 | Question: 9



In the diagram above, the inverter (NOT gate) and the AND-gates labeled 1 and 2 have delays of 9, 10 and 12 nanoseconds (ns), respectively. Wire delays are negligible. For certain values  $a$  and  $c$ , together with certain transition of  $b$ , a glitch (spurious output) is generated for a short time, after which the output assumes its correct value. The duration of glitch is:

- a. 7 ns                      b. 9 ns                      c. 11 ns                      d. 13 ns

isro2018    digital-logic    circuit-output

Answer key

### 8.8.11 Circuit Output: ISRO CSE 2020 | Question: 11



Minimum number of NAND gates required to implement the following binary equation

$$Y = (\bar{A} + \bar{B})(C + D)$$

- A. 4                      B. 5                      C. 3                      D. 6

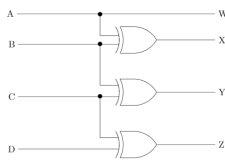
isro-2020    digital-logic    combinational-circuit    circuit-output    normal

Answer key

### 8.8.12 Circuit Output: ISRO CSE 2020 | Question: 12



If  $ABCD$  is a 4-bit binary number, then what is the code generated by the following circuit?



- A. BCD code      B. Gray code      C. 8421 code      D. Excess-3 code

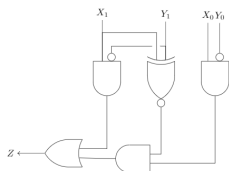
isro-2020   digital-logic   combinational-circuit   circuit-output   normal

Answer key

### 8.8.13 Circuit Output: ISRO CSE 2020 | Question: 66



The following circuit compares two 2-bit binary numbers,  $X$  and  $Y$  represented by  $X_1X_0$  and  $Y_1Y_0$  respectively. ( $X_0$  and  $Y_0$  represent Least Significant Bits)



Under what conditions  $Z$  will be 1?

- A.  $X > Y$       B.  $X < Y$       C.  $X = Y$       D.  $X \neq Y$

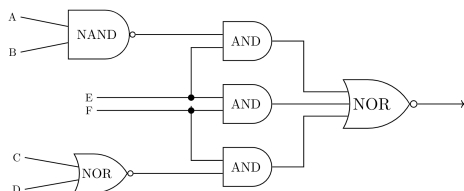
isro-2020   digital-logic   digital-circuits   circuit-output   normal

Answer key

### 8.8.14 Circuit Output: ISRO CSE 2020 | Question: 77



Consider the following circuit



The function by the network above is

- A.  $\overline{A}BE + EF + \overline{C}DF$   
 B.  $(\overline{E} + AB\overline{F})(C + D + \overline{F})$   
 C.  $(\overline{A}\overline{B} + E)(\overline{E} + \overline{F})(C + D + \overline{F})$   
 D.  $(A + B)\overline{E} + \overline{E}F + CDF$

isro-2020   digital-logic   combinational-circuit   circuit-output   normal

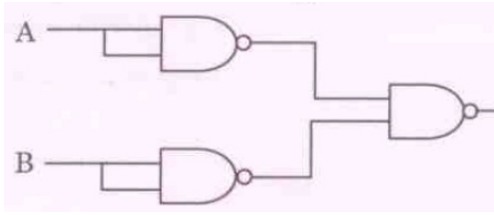
Answer key

## 8.9

### Combinational Circuit (2)

#### 8.9.1 Combinational Circuit: ISRO 2016-EC Combinational circuit





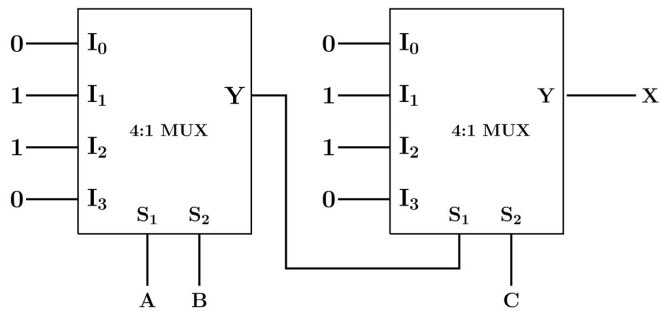
Given circuit represents

- (a) INVERTER
- (b) AND
- (c) OR
- (d) NOR

digital-logic isro2016-ece isro-ece combinational-circuit

Answer key

## 8.9.2 Combinational Circuit: ISRO ECE 2016 | combinational logic



in given combinational logic  $X = ?$

- a.  $X = AB'C' + A'BC' + A'B'C + ABC$
- b.  $X = A'BC + A'BC' + AB'C + A'B'C'$
- c.  $X = AB + BC + AC$
- d.  $X = A'B' + B'C' + A'C'$

isro2016-ece isro-ece digital-logic combinational-circuit

Answer key

## 8.10 Decoder (1)

### 8.10.1 Decoder: GATE CSE 2007 | Question: 8, ISRO2011-31



How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

- A. 7
- B. 8
- C. 9
- D. 10

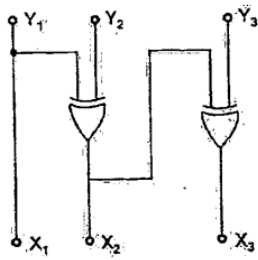
gatecse-2007 digital-logic normal isro2011 decoder

Answer key

## 8.11 Digital Circuits (3)

### 8.11.1 Digital Circuits: ISRO 2007-ECE : Binary Codes





The logic circuit given below converts a binary code  $Y_1 Y_2 Y_3$  into

- a) Excess -3 code
- b) Gray code
- c) BCD code
- d) Hamming code

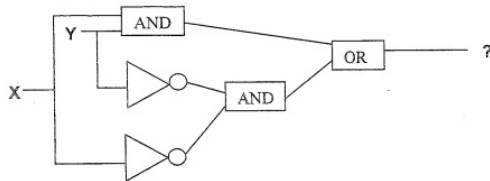
isro-ece digital-logic digital-circuits

Answer key

### 8.11.2 Digital Circuits: ISRO CSE 2011 | Question: 47



The output expression of the following gate network is



- A.  $X.Y + \bar{X}.\bar{Y}$
- B.  $X.Y + X.\bar{Y}$
- C.  $X.Y$
- D.  $X + Y$

isro2011 digital-logic combinational-circuit digital-circuits

Answer key

### 8.11.3 Digital Circuits: ISRO CSE 2017 | Question: 26



Advantage of synchronous sequential circuits over asynchronous one is :

- A. Lower hardware requirement
- B. Better noise immunity
- C. Faster operation
- D. All of the above

isro2017 digital-logic digital-circuits bad-question

Answer key

## 8.12

### Digital Counter (3)

### 8.12.1 Digital Counter: ISRO CSE 2007 | Question: 34



Ring counter is analogous to

- A. Toggle Switch
- B. Latch
- C. Stepping Switch
- D. S-R flip flop

isro2007 digital-logic digital-counter

Answer key 

### 8.12.2 Digital Counter: ISRO ECE 2013 | Digital Logic



If a counter having 10 flip flops is initially at 0, what count will it hold after 2060 pulses?

- a. 000 000 1100
- b. 000 001 1100
- c. 000 001 1000
- d. 000 000 1110

digital-logic isro-ece isro2013-ece sequential-circuit digital-counter

Answer key 

### 8.12.3 Digital Counter: ISRO2015-4



A modulus -12 ring counter requires a minimum of

- A. 10 flip-flops
- B. 12 flip-flops
- C. 8 flip-flops
- D. 6 flip-flops

isro2015 digital-logic digital-counter

Answer key 

## 8.13 Flip Flop (10)

### 8.13.1 Flip Flop: GATE CSE 2004 | Question: 18, ISRO2007-31



In an  $SR$  latch made by cross-coupling two NAND gates, if both  $S$  and  $R$  inputs are set to 0, then it will result in

- A.  $Q = 0, Q' = 1$
- B.  $Q = 1, Q' = 0$
- C.  $Q = 1, Q' = 1$
- D. Indeterminate states

gatecse-2004 digital-logic easy isro2007 flip-flop

Answer key 

### 8.13.2 Flip Flop: ISRO 2006-ECE Flipflops



Which of the following statements are correct ?

1. A flip-flop is used to store 1 bit of information.
2. Race-around condition occurs in a  $J-K$  flip-flop when both the inputs are 1.
3. Master-slave configuration is used in flip-flops to store 2 bits of information.
4. A transparent latch consists of a  $D$ -type flip-flop.

- A. 1,2 and 3
- B. 1,3 and 4
- C. 1,2 and 4
- D. 2,3 and 4

isro-ece digital-logic flip-flop

Answer key 

### 8.13.3 Flip Flop: ISRO 2007-ECE JK flipflop as Divide by 2 circuit



For one of the following conditions, clocked  $J-K$  flip-flop can be used as divided by 2 circuit where the pulse train to be divided is applied at clock input :

- A.  $J = 1, K = 1$  and the flip-flop should have active *HIGH* inputs.
- B.  $J = 1, K = 1$  and the flip-flop should have active *LOW* inputs.
- C.  $J = 0, K = 0$  and the flip-flop should have active *HIGH* inputs.
- D.  $J = 1, K = 1$  and the flip-flop should be a negative edge triggered one.

isro-ece digital-logic flip-flop

Answer key 

#### 8.13.4 Flip Flop: ISRO 2008- ECE Flip flop counter



A counter is designed with six stages of flip flops. Determine the output frequency at the last (sixth) stage, when input frequency is  $1\text{ MHz}$ .

- A.  $1\text{ MHz}$                       B.  $166\text{ KHz}$                       C.  $15.625\text{ KHz}$                       D. 0

digital-logic   isro-ece   flip-flop

#### 8.13.5 Flip Flop: ISRO 2008- ECE Johnson Counter



The mod number of a Johnson counter will be always equal to the number of flip flops used :

- A. same                                      B. twice  
C.  $2N$  where  $N$  is the number of flip flops.                      D. None of the these

isro-ece   digital-logic   flip-flop

Answer key

#### 8.13.6 Flip Flop: ISRO 2008- ECE SR flipflop to D flip flop



A  $S$ - $R$  flip flop with a clock input can be converted to a ' $D$ ' flip flop using :

- A. Two inverters  
B. the flip flop outputs ( $Q$  &  $\bar{Q}$ ) connected to its inputs ( $S$  &  $R$ )  
C. One inverter  
D. Not possible

isro-ece   digital-logic   flip-flop

Answer key

#### 8.13.7 Flip Flop: ISRO CSE 2007 | Question: 05



The characteristic equation of an  $SR$  flip-flop is given by :

- A.  $Q_{n+1} = S + RQ_n$                                       B.  $Q_{n+1} = R\bar{Q}_n + \bar{S}Q_n$   
C.  $Q_{n+1} = \bar{S} + RQ_n$                                       D.  $Q_{n+1} = S + \bar{R}Q_n$

isro2007   digital-logic   flip-flop

Answer key

#### 8.13.8 Flip Flop: ISRO CSE 2011 | Question: 74



In an  $RS$  flip-flop, if the  $S$  line (Set line) is set high (1) and the  $R$  line (Reset line) is set low (0), then the state of the flip-flop is :

- A. Set to 1                                      B. Set to 0  
C. No change in state                                      D. Forbidden

isro2011   digital-logic   flip-flop

Answer key

#### 8.13.9 Flip Flop: ISRO CSE 2016 | Question: 18



The functional difference between  $SR$  flip-flop and  $J$ - $K$  flip-flop is that :

- A.  $J$ - $K$  flip-flop is faster than  $SR$  flip-flop                                      B.  $J$ - $K$  flip-flop has a feedback path  
C.  $J$ - $K$  flip-flop accepts both inputs 1                                      D. None of them

digital-logic   flip-flop   isro2016

Answer key

### 8.13.10 Flip Flop: ISRO-2013-30



In a three stage counter, using *RS* flip flops what will be the value of the counter after giving 9 pulses to its input? Assume that the value of counter before giving any pulses is 1:

- A. 1                      B. 2                      C. 9                      D. 10

isro2013   digital-logic   flip-flop

Answer key

### 8.14 Floating Point Representation (3)

#### 8.14.1 Floating Point Representation: ISRO CSE 2007 | Question: 30



0.75 decimal system is equivalent to \_\_\_\_ in octal system

- A. 0.60  
B. 0.52  
C. 0.54  
D. 0.50

isro2007   digital-logic   number-representation   floating-point-representation

Answer key

#### 8.14.2 Floating Point Representation: ISRO CSE 2007 | Question: 36



Consider a computer system that stores a floating-point numbers with 16-bit mantissa and an 8-bit exponent, each in two's complement. The smallest and largest positive values which can be stored are:

- A.  $1 \times 10^{-128}$  and  $2^{15} \times 10^{128}$                       B.  $1 \times 10^{-256}$  and  $2^{15} \times 10^{255}$   
C.  $1 \times 10^{-128}$  and  $2^{15} \times 10^{127}$                       D.  $1 \times 10^{-128}$  and  $2^{15} - 1 \times 10^{127}$

isro2007   digital-logic   number-representation   floating-point-representation

Answer key

#### 8.14.3 Floating Point Representation: ISRO CSE 2008 | Question: 23



A computer uses 8 digit mantissa and 2 digit exponent. If  $a = 0.052$  and  $b = 28E + 11$  then  $b + a - b$  will:

- A. result in an overflow error                      B. result in an underflow error  
C. be 0                      D. be  $5.28E+11$

isro2008   digital-logic   number-representation   floating-point-representation

Answer key

### 8.15 Functional Completeness (2)

#### 8.15.1 Functional Completeness: ISRO-2013-22



Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is not complete?

- A.  $\{NOT, OR\}$                       B.  $\{NOR\}$                       C.  $\{AND, OR\}$                       D.  $\{AND, NOT\}$

isro2013   functional-completeness   digital-logic   digital-circuits   combinational-circuit

Answer key

#### 8.15.2 Functional Completeness: ISRO-DEC2017-76



Which of the following set of components is sufficient to implement any arbitrary Boolean function?

- A. *XOR* gates, *NOT* gates
- B. *AND* gates, *XOR* gates and 1
- C. 2 to 1 multiplexer
- D. Three input gates that output  $(A.B) + C$  for the inputs  $A, B, C$

isrodec2017 functional-completeness digital-logic

Answer key 

## 8.16 Gray Code (1)

### 8.16.1 Gray Code: ISRO 2017-ECE Number system



The Gray code for (A5) 16 is equivalent to

- A. 10010101
- B. 11010101
- C. 11011111
- D. 11011011

isro2017-ece digital-logic number-system gray-code

Answer key 

## 8.17 Hazards (1)

### 8.17.1 Hazards: ISRO CSE 2016 | Question: 11



The dynamic hazard problem occurs in

- A. combinational circuit alone
- B. sequential circuit only
- C. Both (a) and (b)
- D. None of the above

isro2016 digital-logic digital-circuits hazards

Answer key 

## 8.18 IEEE Representation (3)

### 8.18.1 IEEE Representation: ISRO CSE 2011 | Question: 8



What is the decimal value of the floating-point number  $C1D00000$  (hexadecimal notation)? (Assume 32-bit, single precision floating point IEEE representation)

- A. 28
- B. -15
- C. -26
- D. -28

isro2011 digital-logic number-representation floating-point-representation ieee-representation

Answer key 

### 8.18.2 IEEE Representation: ISRO CSE 2014 | Question: 8



In the standard IEEE 754 single precision floating point representation, there is 1 bit for sign, 23 bits for fraction and 8 bits for exponent. What is the precision in terms of the number of decimal digits?

- A. 5
- B. 6
- C. 7
- D. 8

number-representation ieee-representation isro2014

Answer key 



### 8.18.3 IEEE Representation: ISRO2015-1



Which of the given number has its IEEE - 754 32-bit floating point representation as (0 10000000 110 0000 0000 0000 0000 0000)

- A. 2.5                      B. 3.0                      C. 3.5                      D. 4.5

isro2015   digital-logic   number-representation   floating-point-representation   ieee-representation

Answer key

### 8.19

#### K Map (1)

#### 8.19.1 K Map: ISRO 2006- ECE K Map



| CD \ AB | AB |    |    |    |
|---------|----|----|----|----|
|         | 00 | 01 | 11 | 10 |
| 00      | 1  | 0  | 1  | 0  |
| 01      | 0  | 1  | 0  | 0  |
| 11      | 1  | 0  | 1  | 0  |
| 10      | 0  | 1  | 0  | 1  |

The function shown in the figure when simplified will yield a result with \_\_\_\_\_ terms

- A. 2                      B. 4                      C. 7                      D. 14

digital-logic   isro-ece   k-map

Answer key

### 8.20

#### Matrix Chain Ordering (1)

#### 8.20.1 Matrix Chain Ordering: ISRO CSE 2023 | Question: 72



The complexity of matrix multiplication of two matrices A and B whose orders are  $m \times n$  and  $n \times p$  respectively is

- A.  $O(m \times p)$   
B.  $O(m \times n^2 \times p)$   
C.  $O(m \times n \times p^2)$   
D.  $O(m \times n \times p)$

isro-cse-2023   algorithms   time-complexity   matrix-chain-ordering

Answer key

### 8.21

#### Memory Interfacing (2)

#### 8.21.1 Memory Interfacing: GATE CSE 2009 | Question: 7, ISRO2015-3



How many  $32K \times 1$  RAM chips are needed to provide a memory capacity of  $256K$  bytes?

- A. 8                      B. 32                      C. 64                      D. 128

gatecse-2009   digital-logic   memory-interfacing   easy   isro2015

Answer key

#### 8.21.2 Memory Interfacing: ISRO CSE 2014 | Question: 25



Suppose you want to build a memory with 4 byte words and a capacity of  $2^{21}$  bits. What is type of decoder

required if the memory is built using  $2K \times 8$  RAM chips?

- A. 5 to 32                      B. 6 to 64                      C. 4 to 16                      D. 7 to 128

digital-logic   memory-interfacing   isro2014

Answer key 

## 8.22

### Min No Gates (2)

#### 8.22.1 Min No Gates: ISRO CSE 2016 | Question: 7



The minimum number of NAND gates required to implement the Boolean function  $A + A\bar{B} + A\bar{B}C$  is equal to

- A. 0 (Zero)                      B. 1                      C. 4                      D. 7

digital-logic   min-no-gates   isro2016

Answer key 

#### 8.22.2 Min No Gates: ISRO CSE 2017 | Question: 23



What is the minimum number of two-input NAND gates used to perform the function of two-input OR gate?

- A. One                      B. Two                      C. Three                      D. Four

isro2017   digital-logic   min-no-gates

Answer key 

## 8.23

### Multiplexer (5)

#### 8.23.1 Multiplexer: ISRO CSE 2008 | Question: 22



How many 2-input multiplexers are required to construct a  $2^{10}$ -input multiplexer?

- A. 1023                      B. 31                      C. 10                      D. 127

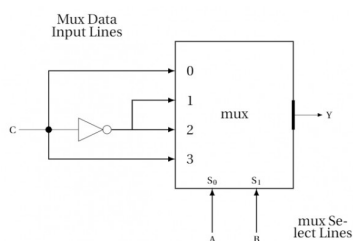
isro2008   digital-logic   multiplexer

Answer key 

#### 8.23.2 Multiplexer: ISRO CSE 2020 | Question: 10



Following Multiplexer circuit is equivalent to



- A. Sum equation of full adder                      B. Carry equation of full adder  
C. Borrow equation for full subtractor                      D. Difference equation of a full subtractor

isro-2020   digital-logic   combinational-circuit   multiplexer   normal

Answer key 

### 8.23.3 Multiplexer: ISRO CSE 2023 | Question: 70



Which of the following definitions is true

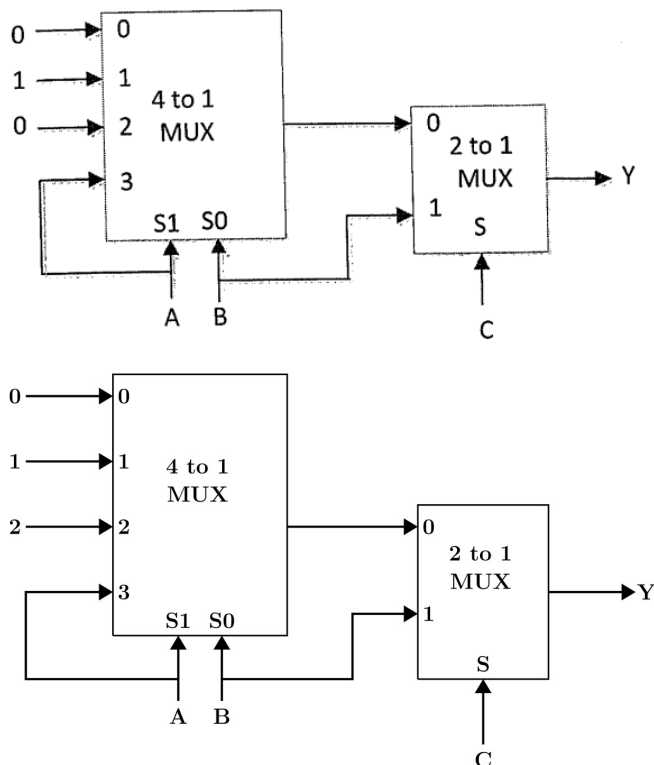
- i. A digital multiplexer takes one input from many inputs and outputs the selected one
- ii. Four select lines are required for 1-16 multiplexer
- iii. Eight OR gates are required to implement an octal to binary encoder

A. (i) and (ii)      B. (ii) and (iii)      C. (i) alone      D. none of the above

isro-cse-2023   digital-logic   digital-circuits   multiplexer

Answer key

### 8.23.4 Multiplexer: ISRO ECE 2014 | Digital logic



In the following circuit, Y can be expressed as :

A.  $Y = BC + A$       B.  $Y = AC' + BC$       C.  $Y = C$       D.  $Y = B$

isro2014-ece   isro-ece   digital-logic   combinational-circuit   multiplexer

Answer key

### 8.23.5 Multiplexer: ISRO2015-8



Minimum number of  $2 \times 1$  multiplexers required to realize the following function,  $f = \bar{A} \bar{B} C + \bar{A} B \bar{C}$   
Assume that inputs are available only in true form and Boolean a constant 1 and 0 are available.

A. 1      B. 2      C. 3      D. 7

isro2015   digital-logic   multiplexer

Answer key

### 8.24.1 Number Representation: GATE CSE 1995 | Question: 2.12, ISRO2015-9



The number of 1's in the binary representation of  $(3 * 4096 + 15 * 256 + 5 * 16 + 3)$  are:

- A. 8                      B. 9                      C. 10                      D. 12

gate1995   digital-logic   number-representation   normal   isro2015

Answer key

### 8.24.2 Number Representation: GATE CSE 2005 | Question: 16, ISRO2009-18, ISRO2015-2



The range of integers that can be represented by an  $n$  bit  $2$ 's complement number system is:

- A.  $-2^{n-1}$  to  $(2^{n-1} - 1)$                       B.  $-(2^{n-1} - 1)$  to  $(2^{n-1} - 1)$   
C.  $-2^{n-1}$  to  $2^{n-1}$                       D.  $-(2^{n-1} + 1)$  to  $(2^{n-1} - 1)$

gatecse-2005   digital-logic   number-representation   easy   isro2009   isro2015

Answer key

### 8.24.3 Number Representation: GATE CSE 2009 | Question: 5, ISRO2017-57



$(1217)_8$  is equivalent to

- A.  $(1217)_{16}$                       B.  $(028F)_{16}$                       C.  $(2297)_{10}$                       D.  $(0B17)_{16}$

gatecse-2009   digital-logic   number-representation   isro2017

Answer key

### 8.24.4 Number Representation: GATE IT 2006 | Question: 7, ISRO2009-41



The addition of 4 – *bit*, two's complement, binary numbers 1101 and 0100 results in

- A. 0001 and an overflow                      B. 1001 and no overflow  
C. 0001 and no overflow                      D. 1001 and an overflow

gateit-2006   digital-logic   number-representation   normal   isro2009

Answer key

### 8.24.5 Number Representation: ISRO 2006- ECE Gray code



Gray code for number 7 is

- a) 1100  
b) 1001  
c) 0110  
d) 0100

isro-ece   digital-logic   number-representation

Answer key

### 8.24.6 Number Representation: ISRO 2006-ECE



Given the decimal number -19 an eight bit two's complement representation is given by

- a) 11101110  
b) 11101101  
c) 11101100

d) none of these

isro-ece digital-logic number-representation

Answer key 

#### 8.24.7 Number Representation: ISRO CSE 2007 | Question: 04



When two numbers are added in excess-3 code and the sum is less than 9, then in order to get the correct answer it is necessary to

- A. subtract 0011 from the sum
- B. add 0011 to the sum
- C. subtract 0110 from the sum
- D. add 0110 to the sum

isro2007 digital-logic number-representation

Answer key 

#### 8.24.8 Number Representation: ISRO CSE 2007 | Question: 18



The number of digit 1 present in the binary representation of  $3 \times 512 + 7 \times 64 + 5 \times 8 + 3$  is

- A. 8
- B. 9
- C. 10
- D. 12

isro2007 digital-logic number-representation

Answer key 

#### 8.24.9 Number Representation: ISRO CSE 2007 | Question: 38



The Hexadecimal equivalent of 01111100110111100011 is

- A. CD73E
- B. ABD3F
- C. 7CDE3
- D. FA4CD

isro2007 digital-logic number-representation

Answer key 

#### 8.24.10 Number Representation: ISRO CSE 2007 | Question: 49



One approach to handling fuzzy logic data might be to design a computer using ternary (base-3) logic so that data could be stored as “true,” “false,” and “unknown.” If each ternary logic element is called a flit, how many flits are required to represent at least 256 different values?

- A. 4
- B. 5
- C. 6
- D. 7

isro2007 digital-logic number-representation

Answer key 

#### 8.24.11 Number Representation: ISRO CSE 2008 | Question: 13



If  $N^2 = (7601)_8$  where  $N$  is a positive integer, then the value of  $N$  is

- A.  $(241)_5$
- B.  $(143)_6$
- C.  $(165)_7$
- D.  $(39)_{16}$

isro2008 digital-logic number-representation

Answer key 

#### 8.24.12 Number Representation: ISRO CSE 2008 | Question: 20



If  $(12x)_3 = (123)_x$ , then the value of  $x$  is

- A. 3
- B. 3 or 4
- C. 2
- D. None of these

Answer key **8.24.13 Number Representation: ISRO CSE 2009 | Question: 39 , GATE2008-IT\_15**

A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively will be

- A. 1,1,0                      B. 1,0,0                      C. 0,1,0                      D. 1,0,1

isro2009 number-representation digital-logic

Answer key **8.24.14 Number Representation: ISRO CSE 2014 | Question: 27**

How many different BCD numbers can be stored in 12 switches ? (Assume two position or on-off switches).

- A.  $2^{12}$                       B.  $2^{12} - 1$                       C.  $10^{12}$                       D.  $10^3$

isro2014 digital-logic number-representation

Answer key **8.24.15 Number Representation: ISRO CSE 2016 | Question: 14**

If  $12A7C_{16} = X_8$  then the value of  $X$  is

- A. 224174                      B. 425174                      C. 6173                      D. 225174

digital-logic number-representation isro2016

Answer key **8.24.16 Number Representation: ISRO CSE 2016 | Question: 17**

Which of the following binary number is the same as its 2's complement ?

- A. 1010                      B. 0101                      C. 1000                      D. 1001

digital-logic number-representation isro2016

Answer key **8.24.17 Number Representation: ISRO CSE 2018 | Question: 19**

Given  $\sqrt{224_r} = 13_r$ , the value of radix  $r$  is

- a. 10                      b. 8                      c. 6                      d. 5

isro2018 number-representation digital-logic

Answer key **8.24.18 Number Representation: ISRO CSE 2018 | Question: 3**

If a variable can take only integral values from 0 to  $n$ , where  $n$  is an integer, then the variable can be represented as a bit-field whose width is (the log in the answer are to the base 2, and  $\lceil \log n \rceil$  means the floor of  $\log n$ )

- a.  $\lceil \log(n) \rceil + 1$  bits  
b.  $\lceil \log(n - 1) \rceil + 1$  bits  
c.  $\lceil \log(n + 1) \rceil + 1$  bits

d. None of the above

isro2018 number-representation digital-logic

Answer key 

#### 8.24.19 Number Representation: ISRO CSE 2018 | Question: 32



A computer uses ternary system instead of the traditional system, An  $n$  bit string in the binary system will occupy

- a.  $3 + n$  ternary digits
- b.  $2n/3$  ternary digits
- c.  $n \log_2 3$  ternary digits
- d.  $n \log_3 2$  ternary digits

isro2018 digital-logic number-representation

Answer key 

#### 8.24.20 Number Representation: ISRO ECE 2012 | Number system



If  $(146)_x + (313)_{x-2} = (246)_8$  then the value of  $x$  will be

- a. 5
- b. 6
- c. 7
- d. 9

isro-ece isro2012-ece digital-logic number-representation

Answer key 

#### 8.24.21 Number Representation: ISRO ECE 2013 | 2s complement number



The two numbers represented in signed 2s complement form are  $P = 11101101$  and  $Q = 11100110$ . If  $Q$  is subtracted from  $P$ , the value obtained in signed 2s complement form is:

- a. 1000001111
- b. 00000111
- c. 11111001
- d. 111111001

isro-ece isro2013-ece digital-logic number-representation

Answer key 

#### 8.24.22 Number Representation: ISRO-2013-13



The number 1102 in base 3 is equivalent to 123 in which base system?

- A. 4
- B. 5
- C. 6
- D. 8

isro2013 number-representation

Answer key 

#### 8.24.23 Number Representation: ISRO-2013-17



Two eight bit bytes 11000011 and 01001100 are added. What are the values of the overflow, carry and zero flags respectively, if the arithmetic unit of the CPU uses 2s complement form?

- A. 0,1,1'
- B. 1,1,0
- C. 1,0,1
- D. 0,1,0

isro2013 digital-logic number-representation

Answer key 

#### 8.24.24 Number Representation: ISRO-2013-77



The binary equivalent of the decimal number 42.75 is

- A. 101010.110
- B. 100110.101
- C. 101010.101
- D. 100110.110

Answer key **8.24.25 Number Representation: ISRO-2013-8**

Which logic gate is used to detect overflow in 2's complement arithmetic?

- A. OR gate                      B. AND gate                      C. NAND gate                      D. XOR gate

Answer key **8.24.26 Number Representation: ISRO2015-6**

The code which uses 7 bits to represent a character is :

- A. ASCII                      B. BCD                      C. EBCDIC                      D. Gray

Answer key **8.24.27 Number Representation: ISRO2015-78**

The decimal number has 64 digits. The number of bits needed for its equivalent binary representation is?

- A. 200                      B. 213                      C. 246                      D. 277

Answer key **8.25****Number System (1)****8.25.1 Number System: ISRO 2007-ECE Conversion to binary number**

Which of the following binary number is equal to octal number 66.3

- a) 101101.100  
b) 1101111.111  
c) 111111.1111  
d) 110110.011

Answer key **8.26****Pla (1)****8.26.1 Pla: ISRO-2013-29**

How many programmable fuses are required in a PLA which takes 16 inputs and gives 8 outputs? It has to use 8 OR gates and 32 AND gates.

- A. 1032                      B. 776                      C. 1284                      D. 1536



8.27.1 Sequential Circuit: ISRO CSE 2020 | Question: 80



A new flipflop with inputs  $X$  and  $Y$ , has the following property

| $X$ | $Y$ | Current state | Next state     |
|-----|-----|---------------|----------------|
| 0   | 0   | $Q$           | 1              |
| 0   | 1   | $Q$           | $\overline{Q}$ |
| 1   | 1   | $Q$           | 0              |
| 1   | 0   | $Q$           | $Q$            |

Which of the following expresses the next state in terms of  $X, Y$ , current state?

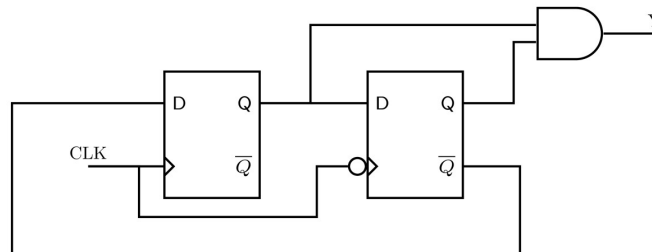
- A.  $(\overline{X} \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$
- B.  $(\overline{X} \wedge Q) \vee (\overline{Y} \wedge \overline{Q})$
- C.  $(X \wedge \overline{Q}) \vee (Y \wedge Q)$
- D.  $(X \wedge \overline{Q}) \vee (\overline{Y} \wedge Q)$

isro-2020 digital-logic sequential-circuit flip-flop normal

8.27.2 Sequential Circuit: ISRO ECE 2013 | Digital Logic



What is the frequency and duty cycle of output  $Y$ , when CLK frequency is 1MHz @ 50% duty cycle?

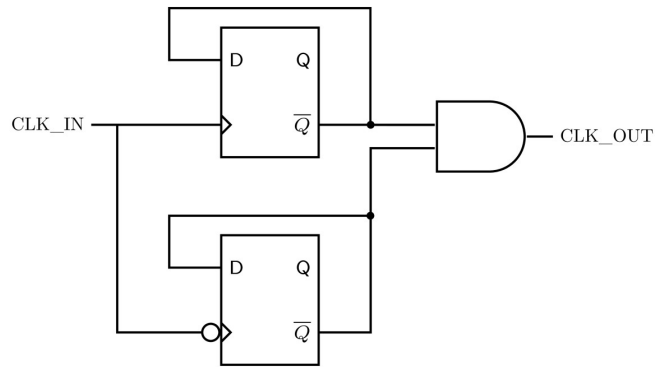


- a. 500 KHz @ 50% duty cycle
- b. 500 KHz @ 25% duty cycle
- c. 250 KHz @ 50% duty cycle
- d. 250 KHz @ 25% duty cycle

isro2013-ece isro-ece digital-logic sequential-circuit flip-flop

8.27.3 Sequential Circuit: ISRO ECE 2013 | Digital Logic





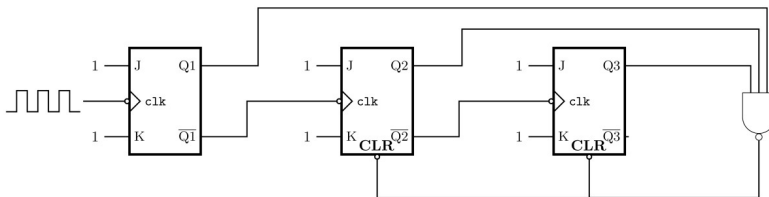
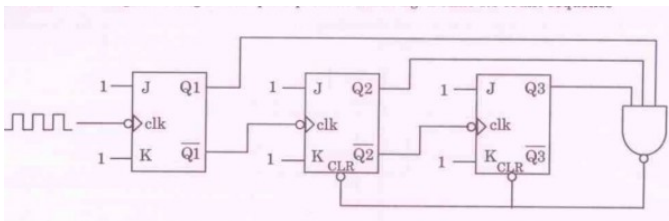
What is the division factor of the given clock divider circuit?

- a. 2                      b. 3                      c. 1.5                      d. 2.5

isro2013-ece   isro-ece   digital-logic   sequential-circuit   flip-flop

Answer key

#### 8.27.4 Sequential Circuit: ISRO2016 -EC Digital Logic



- a. 000,001,010,011,100 & repeats  
b. 100,011,010,001,000 & repeat  
c. 010,011,100,000,001 & repeats  
d. 101,110,111,000,001,010,011,100 & repeats

isro2016-ece   isro-ece   digital-logic   sequential-circuit   flip-flop

### 8.28

#### Shift Registers (2)

#### 8.28.1 Shift Registers: ISRO CSE 2023 | Question: 71



The time delay obtained through an 8-bit serial register with 400 MHz clock is:

- a. 20 ns  
b. 2.5  $\mu$ s  
c. 20  $\mu$ s  
d. 2.5  $\mu$ s

isro-cse-2023   shift-registers   digital-circuits



Which shift register counter requires the most decoding circuitry?

- A. Johnson Counter  
 B. Ring Counter  
 C. Ripple Counter  
 D. MOD counter

isro2016-ece digital-logic sequential-circuit digital-counter shift-registers

Answer key

## Answer Keys

|        |     |         |     |        |     |        |     |        |     |
|--------|-----|---------|-----|--------|-----|--------|-----|--------|-----|
| 8.0.1  | B   | 8.0.2   | D   | 8.0.3  | C   | 8.0.4  | X   | 8.0.5  | Q-Q |
| 8.0.6  | Q-Q | 8.0.7   | Q-Q | 8.0.8  | Q-Q | 8.0.9  | Q-Q | 8.0.10 | Q-Q |
| 8.0.11 | Q-Q | 8.0.12  | Q-Q | 8.0.13 | Q-Q | 8.0.14 | Q-Q | 8.0.15 | Q-Q |
| 8.0.16 | Q-Q | 8.0.17  | Q-Q | 8.0.18 | Q-Q | 8.0.19 | Q-Q | 8.0.20 | Q-Q |
| 8.0.21 | Q-Q | 8.0.22  | Q-Q | 8.0.23 | Q-Q | 8.0.24 | Q-Q | 8.0.25 | Q-Q |
| 8.0.26 | Q-Q | 8.0.27  | Q-Q | 8.0.28 | Q-Q | 8.0.29 | Q-Q | 8.0.30 | Q-Q |
| 8.0.31 | Q-Q | 8.0.32  | Q-Q | 8.0.33 | Q-Q | 8.0.34 | Q-Q | 8.0.35 | Q-Q |
| 8.0.36 | Q-Q | 8.0.37  | Q-Q | 8.0.38 | Q-Q | 8.0.39 | Q-Q | 8.0.40 | Q-Q |
| 8.0.41 | Q-Q | 8.0.42  | Q-Q | 8.0.43 | D   | 8.0.44 | Q-Q | 8.0.45 | Q-Q |
| 8.0.46 | Q-Q | 8.0.47  | Q-Q | 8.0.48 | Q-Q | 8.0.49 | A   | 8.0.50 | C   |
| 8.0.51 | D   | 8.0.52  | B   | 8.0.53 | X   | 8.0.54 | Q-Q | 8.0.55 | Q-Q |
| 8.0.56 | Q-Q | 8.1.1   | Q-Q | 8.1.2  | B   | 8.1.3  | C   | 8.1.4  | D   |
| 8.1.5  | D   | 8.1.6   | Q-Q | 8.1.7  | C   | 8.2.1  | A   | 8.3.1  | A   |
| 8.4.1  | C   | 8.4.2   | Q-Q | 8.4.3  | A   | 8.4.4  | C   | 8.4.5  | A   |
| 8.4.6  | A   | 8.4.7   | X   | 8.4.8  | B   | 8.4.9  | A   | 8.4.10 | B   |
| 8.4.11 | B   | 8.4.12  | C   | 8.4.13 | C   | 8.4.14 | A   | 8.4.15 | A   |
| 8.4.16 | Q-Q | 8.4.17  | Q-Q | 8.4.18 | Q-Q | 8.4.19 | B   | 8.4.20 | A   |
| 8.5.1  | B   | 8.6.1   | Q-Q | 8.7.1  | B   | 8.7.2  | C   | 8.8.1  | A   |
| 8.8.2  | X   | 8.8.3   | A   | 8.8.4  | C   | 8.8.5  | C   | 8.8.6  | D   |
| 8.8.7  | A   | 8.8.8   | B   | 8.8.9  | A   | 8.8.10 | A   | 8.8.11 | A   |
| 8.8.12 | B   | 8.8.13  | A   | 8.8.14 | B   | 8.9.1  | Q-Q | 8.9.2  | Q-Q |
| 8.10.1 | C   | 8.11.1  | Q-Q | 8.11.2 | A   | 8.11.3 | X   | 8.12.1 | C   |
| 8.12.2 | Q-Q | 8.12.3  | B   | 8.13.1 | C   | 8.13.2 | Q-Q | 8.13.3 | Q-Q |
| 8.13.4 | Q-Q | 8.13.5  | Q-Q | 8.13.6 | Q-Q | 8.13.7 | D   | 8.13.8 | A   |
| 8.13.9 | C   | 8.13.10 | B   | 8.14.1 | A   | 8.14.2 | D   | 8.14.3 | C   |
| 8.15.1 | C   | 8.15.2  | B   | 8.16.1 | Q-Q | 8.17.1 | C   | 8.18.1 | C   |
| 8.18.2 | C   | 8.18.3  | C   | 8.19.1 | Q-Q | 8.20.1 | Q-Q | 8.21.1 | C   |
| 8.21.2 | A   | 8.22.1  | A   | 8.22.2 | C   | 8.23.1 | A   | 8.23.2 | A;D |
| 8.23.3 | Q-Q | 8.23.4  | D   | 8.23.5 | B   | 8.24.1 | C   | 8.24.2 | A   |
| 8.24.3 | B   | 8.24.4  | C   | 8.24.5 | Q-Q | 8.24.6 | Q-Q | 8.24.7 | A   |

|         |     |
|---------|-----|
| 8.24.8  | B   |
| 8.24.13 | B   |
| 8.24.18 | A   |
| 8.24.23 | D   |
| 8.25.1  | Q-Q |
| 8.27.4  | Q-Q |

|         |     |
|---------|-----|
| 8.24.9  | C   |
| 8.24.14 | D   |
| 8.24.19 | D   |
| 8.24.24 | A   |
| 8.26.1  | C   |
| 8.28.1  | Q-Q |

|         |     |
|---------|-----|
| 8.24.10 | C   |
| 8.24.15 | D   |
| 8.24.20 | C   |
| 8.24.25 | D   |
| 8.27.1  | A   |
| 8.28.2  | Q-Q |

|         |     |
|---------|-----|
| 8.24.11 | B   |
| 8.24.16 | C   |
| 8.24.21 | Q-Q |
| 8.24.26 | A   |
| 8.27.2  | Q-Q |

|         |     |
|---------|-----|
| 8.24.12 | D   |
| 8.24.17 | D   |
| 8.24.22 | B   |
| 8.24.27 | B   |
| 8.27.3  | Q-Q |

**9.1.1 Counting: ISRO CSE 2014 | Question: 19**

The number of bit strings of length 8 that will either start with 1 or end with 00 is?

- A. 32
- B. 128
- C. 160
- D. 192

isro2014 combinatory counting

Answer key 

**9.2.1 Process Scheduling: ISRO-2013-53**

A CPU scheduling algorithm determines an order for the execution of its scheduled processes. Given 'n' processes to be scheduled on one processor, how many possible different schedules are there?

- A.  $n$
- B.  $n^2$
- C.  $n!$
- D.  $2^n$

isro2013 process-scheduling combinatory

Answer key 

## Answer Keys

9.1.1

C

9.2.1

C



## 10.0.1 ISRO-2013-76

The number of edges in a 'n' vertex complete graph is?

- A.  $n*(n-1)/2$                       B.  $n^2$   
 C.  $n*(n+1)/2$                       D.  $n*(n+1)$

isro2013 graph-theory

Answer key

## 10.0.2 ISRO CSE 2023 | Question: 63

Maximum number of Simple graphs possible with  $n$  vertices

- A.  $2^{n(n-1)/2}$                       B.  $2^{(n-1)/2}$                       C.  $2^{(n(n+1)/2)}$                       D.  $2^{n(n+1)}$

isro-cse-2023 graph-theory discrete-mathematics

Answer key

## 10.0.3 ISRO CSE 2023 | Question: 51

Which of the following are true?

- i. In a graph  $G$  with 'n' vertices and 'e' edges, sum of degrees of vertices =  $2*e$ .
  - ii. Eccentricity of a connected graph can never be equal to radius of the graph
  - iii. Girth of a graph is the shortest cycle of the graph
  - iv. Graph with equal degree for all vertices is multigraph
- a. (i), (ii), (iii)
  - b. (ii), (iii), (iv)
  - c. (i), (iii), (iv)
  - d. None of the above

isro-cse-2023 discrete-mathematics graph-theory

Answer key

## 10.1

## Degree of Graph (1)

## 10.1.1 Degree of Graph: ISRO CSE 2009 | Question: 52

If  $G$  is a graph with  $e$  edges and  $n$  vertices the sum of the degrees of all vertices in  $G$  is

- A.  $e$                       B.  $e/2$                       C.  $e^2$                       D.  $2e$

isro2009 graph-theory degree-of-graph

Answer key

## 10.2

## Euler Graph (2)

## 10.2.1 Euler Graph: ISRO CSE 2007 | Question: 06

A graph with  $n$  vertices and  $n - 1$  edges that is not a tree, is



- A. Connected      B. Disconnected      C. Euler      D. A circuit

isro2007 graph-theory euler-graph

Answer key 

### 10.2.2 Euler Graph: ISRO CSE 2016 | Question: 5



A given connected graph  $G$  is a Euler Graph if and only if all vertices of  $G$  are of

- A. same degree  
B. even degree  
C. odd degree  
D. different degree

graph-theory euler-graph isro2016

Answer key 

## 10.3

### Graph Coloring (1)

#### 10.3.1 Graph Coloring: ISRO CSE 2007 | Question: 07



If a graph requires  $k$  different colours for its proper colouring, then the chromatic number of the graph is

- A. 1      B.  $k$       C.  $k - 1$       D.  $k/2$

isro2007 graph-theory graph-coloring

Answer key 

## 10.4

### Graph Connectivity (10)

#### 10.4.1 Graph Connectivity: GATE CSE 1994 | Question: 1.6, ISRO2008-29



The number of distinct simple graphs with up to three nodes is

- A. 15      B. 10      C. 7      D. 9

gate1994 graph-theory graph-connectivity combinatory normal isro2008 counting

Answer key 

#### 10.4.2 Graph Connectivity: GATE CSE 2002 | Question: 1.25, ISRO2008-30, ISRO2016-6



The maximum number of edges in a  $n$ -node undirected graph without self loops is

- A.  $n^2$       B.  $\frac{n(n-1)}{2}$       C.  $n - 1$       D.  $\frac{(n+1)(n)}{2}$

gatecse-2002 graph-theory easy isro2008 isro2016 graph-connectivity

Answer key 

#### 10.4.3 Graph Connectivity: GATE CSE 2003 | Question: 8, ISRO2009-53



Let  $G$  be an arbitrary graph with  $n$  nodes and  $k$  components. If a vertex is removed from  $G$ , the number of components in the resultant graph must necessarily lie down between

- A.  $k$  and  $n$       B.  $k - 1$  and  $k + 1$   
C.  $k - 1$  and  $n - 1$       D.  $k + 1$  and  $n - k$

gatecse-2003 graph-theory graph-connectivity normal isro2009

Answer key 

#### 10.4.4 Graph Connectivity: ISRO CSE 2007 | Question: 62



Let  $X$  be the adjacency matrix of a graph  $G$  with no self loops. The entries along the principal diagonal of  $X$  are

- A. all zeros
- B. all ones
- C. both zeros and ones
- D. different

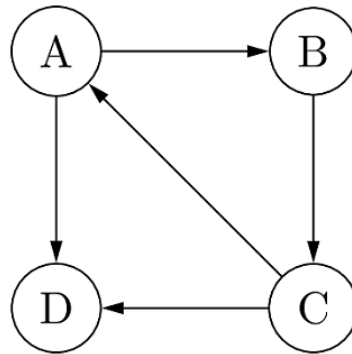
isro2007 graph-theory graph-connectivity

Answer key

#### 10.4.5 Graph Connectivity: ISRO CSE 2008 | Question: 78



Consider the graph shown in the figure below:



Which of the following is a valid strong component?

- A.  $a, c, d$
- B.  $a, b, d$
- C.  $b, c, d$
- D.  $a, b, c$

isro2008 graph-theory graph-connectivity

Answer key

#### 10.4.6 Graph Connectivity: ISRO CSE 2009 | Question: 54



A graph in which all nodes are of equal degree, is known as

- A. Multigraph
- B. Non regular graph
- C. Regular graph
- D. Complete graph

isro2009 graph-theory graph-connectivity

Answer key

#### 10.4.7 Graph Connectivity: ISRO CSE 2009 | Question: 55



In a graph  $G$  there is one and only one path between every pair of vertices then  $G$  is a

- A. Path
- B. Walk
- C. Tree
- D. Circuit

isro2009 graph-theory graph-connectivity

Answer key

#### 10.4.8 Graph Connectivity: ISRO CSE 2009 | Question: 56



A simple graph ( a graph without parallel edge or loops) with  $n$  vertices and  $k$  components can have at most

- A.  $n$  edges
- B.  $n - k$  edges
- C.  $(n - k)(n - k + 1)$  edges
- D.  $(n - k)(n - k + 1)/2$  edges

isro2009 graph-theory graph-connectivity



Answer key 

### 10.4.9 Graph Connectivity: ISRO CSE 2011 | Question: 35



How many edges are there in a forest with  $v$  vertices and  $k$  components?

- A.  $(v+1)-k$   
 B.  $(v+1)/2-k$   
 C.  $v-k$   
 D.  $v+k$

isro2011 graph-theory graph-connectivity

Answer key 

#### 10.4.10 Graph Connectivity: ISRO CSE 2018 | Question: 38



The number of edges in a regular graph of degree:  $d$  and  $n$  vertices is:

- A. maximum of  $n$  and  $d$   
 B.  $n + d$   
 C.  $nd$   
 D.  $nd/2$

isro2018   graph-theory   graph-connectivity

Answer key 

10.5 Graph Planarity (1)

### 10.5.1 Graph Planarity: ISRO CSE 2023 | Question: 52



If there are five faces and nine vertices in an undirected planar graph, then number of edges is

- A. 14                      B. 16                      C. 12                      D. None of the above

isro-cse-2023 graph-theory graph-planarity

Answer key 

## Answer Keys

|        |   |         |     |        |     |        |   |        |   |
|--------|---|---------|-----|--------|-----|--------|---|--------|---|
| 10.0.1 | A | 10.0.2  | Q-Q | 10.0.3 | Q-Q | 10.1.1 | D | 10.2.1 | B |
| 10.2.2 | B | 10.3.1  | B   | 10.4.1 | C   | 10.4.2 | B | 10.4.3 | C |
| 10.4.4 | A | 10.4.5  | D   | 10.4.6 | C   | 10.4.7 | C | 10.4.8 | D |
| 10.4.9 | C | 10.4.10 | D   | 10.5.1 | Q-Q |        |   |        |   |



## 11.0.1 ISRO-DEC2017-6



The proposition  $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$  is a

- A. Tautology                      B. Contradiction                      C. Contingency                      D. Absurdity

isrodec2017

Answer key

## 11.0.2 ISRO-DEC2017-7



If  $T(x)$  denotes  $x$  is a trigonometric function,  $P(x)$  denotes  $x$  is a periodic function and  $C(x)$  denotes  $x$  is a continuous function then the statement "It is not the case that some trigonometric functions are not periodic" can be logically represented as

- A.  $\neg \exists x [T(x) \wedge \neg P(x)]$   
 B.  $\neg \exists x [T(x) \vee \neg P(x)]$   
 C.  $\neg \exists x [\neg T(x) \wedge \neg P(x)]$   
 D.  $\neg \exists x [T(x) \wedge P(x)]$

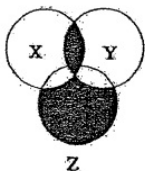
isrodec2017

Answer key

## 11.1

## Boolean Algebra (1)

## 11.1.1 Boolean Algebra: ISRO 2006 -ECE Venn Diagrams



The Boolean expression for the shaded area in the Venn diagram is

- a)  $X' + Y' + Z$   
 b)  $XY'Z + X'YZ$   
 c)  $X + Y + Z$   
 d)  $X'Y'Z + XY$

isro-ece    digital-logic    boolean-algebra

Answer key

## 11.2

## First Order Logic (2)

## 11.2.1 First Order Logic: GATE CSE 2004 | Question: 23, ISRO2007-32



Identify the correct translation into logical notation of the following assertion.

Some boys in the class are taller than all the girls

Note:  $\text{taller}(x, y)$  is true if  $x$  is taller than  $y$ .

- A.  $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
- B.  $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \wedge \text{taller}(x, y)))$
- C.  $(\exists x)(\text{boy}(x) \rightarrow (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$
- D.  $(\exists x)(\text{boy}(x) \wedge (\forall y)(\text{girl}(y) \rightarrow \text{taller}(x, y)))$

gatecse-2004 mathematical-logic easy isro2007 first-order-logic

Answer key 

### 11.2.2 First Order Logic: ISRO CSE 2020 | Question: 73



Given that

$B(a)$  means “ $a$  is a bear”

$F(a)$  means “ $a$  is a fish” and

$E(a, b)$  means “ $a$  eats  $b$ ”

Then what is the best meaning of

$\forall x[F(x) \rightarrow \forall y(E(y, x) \rightarrow b(y))]$

- A. Every fish is eaten by some bear
- B. Bears eat only fish
- C. Every bear eats fish
- D. Only bears eat fish

isro-2020 mathematical-logic first-order-logic normal

Answer key 

## 11.3 Propositional Logic (1)

### 11.3.1 Propositional Logic: ISRO CSE 2017 | Question: 22



Which one of the following Boolean expressions is NOT a tautology?

- A.  $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$
- B.  $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$
- C.  $(a \wedge b \wedge c) \rightarrow (c \vee a)$
- D.  $a \rightarrow (b \rightarrow a)$

isro2017 mathematical-logic propositional-logic

Answer key 

## Answer Keys

|        |   |        |   |        |     |        |   |        |   |
|--------|---|--------|---|--------|-----|--------|---|--------|---|
| 11.0.1 | C | 11.0.2 | A | 11.1.1 | Q-Q | 11.2.1 | D | 11.2.2 | D |
| 11.3.1 | B |        |   |        |     |        |   |        |   |



## 12.0.1 ISRO-DEC2017-4



If  $x = -1$  and  $x = 2$  are extreme points of  $f(x) = \alpha \log |x| + \beta x^2 + x$  then

A.  $\alpha = -6, \beta = \frac{-1}{2}$

B.  $\alpha = 2, \beta = \frac{-1}{2}$

C.  $\alpha = 2, \beta = \frac{1}{2}$

D.  $\alpha = -6, \beta = \frac{1}{2}$

isrodec2017

Answer key

## 12.0.2 ISRO-DEC2017-8



The number of elements in the power set of  $\{\{1, 2\}, \{2, 1, 1\}, \{2, 1, 1, 2\}\}$  is

A. 3

B. 8

C. 4

D. 2

isrodec2017

Answer key

## 12.0.3 ISRO-DEC2017-9



The function  $f : [0, 3] \rightarrow [1, 29]$  defined by  $f(x) = 2x^3 - 15x^2 + 36x + 1$  is

A. injective and surjective

B. surjective but not injective

C. injective but not surjective

D. neither injective nor surjective

isrodec2017

Answer key

## 12.1

## Abelian Group (1)

## 12.1.1 Abelian Group: ISRO CSE 2018 | Question: 17



$(G, *)$  is an abelian group. Then

a.  $x = x^{-1}$  for any  $x$  belonging to  $G$

b.  $x = x^2$  for any  $x$  belonging to  $G$

c.  $(x * y)^2 = x^2 * y^2$ , for any  $x, y$  belonging to  $G$

d.  $G$  is of finite order

isro2018 group-theory set-theory&amp;algebra abelian-group

Answer key

## 12.2

## Equivalence Class (1)

## 12.2.1 Equivalence Class: ISRO CSE 2007 | Question: 29



The set of all Equivalence Classes of a set  $A$  of Cardinality  $C$

A. is of cardinality  $2^C$

B. have the same cardinality as  $A$

C. forms a partition of  $A$

D. is of cardinality  $C^2$

Answer key **12.3 Functions (2)****12.3.1 Functions: ISRO CSE 2014 | Question: 50**

Let  $A$  be a finite set having  $x$  elements and let  $B$  be a finite set having  $y$  elements. What is the number of distinct functions mapping  $B$  into  $A$ .

- A.  $x^y$                       B.  $2^{(x+y)}$                       C.  $y^x$                       D.  $y!/(y-x)!$

isro2014 set-theory&amp;algebra functions combinatory

Answer key **12.3.2 Functions: ISRO-DEC2017-5**

Let  $f(x) = \log|x|$  and  $g(x) = \sin x$ . If  $A$  is the range of  $f(g(x))$  and  $B$  is the range of  $g(f(x))$  then  $A \cap B$  is

- A.  $[-1, 0]$                       B.  $[-1, 0)$                       C.  $[-\infty, 0]$                       D.  $[-\infty, 1]$

isrodec2017 set-theory&amp;algebra functions

Answer key **12.4 Group Theory (1)****12.4.1 Group Theory: ISRO CSE 2016 | Question: 4**

If  $(G, \cdot)$  is a group such that  $(ab)^{-1} = a^{-1}b^{-1}, \forall a, b \in G$ , then  $G$  is a/an

- A. Commutative semi group                      B. Abelian group  
C. Non-abelian group                      D. None of these

set-theory&amp;algebra group-theory isro2016

Answer key **12.5 Relations (2)****12.5.1 Relations: ISRO CSE 2018 | Question: 56**

The time complexity of computing the transitive closure of binary relation on a set of  $n$  elements is known to be

- a.  $O(n)$                       b.  $O(n * \log(n))$                       c.  $O(n^{\frac{3}{2}})$                       d.  $O(n^3)$

isro2018 set-theory&amp;algebra relations time-complexity

Answer key **12.5.2 Relations: ISRO-DEC2017-2**

Consider the set of integers  $I$ . Let  $D$  denote "divides with an integer quotient" (e.g.  $4D8$  but not  $4D7$ ). Then  $D$  is

- A. Reflexive, Not Symmetric, Transitive  
B. Not Reflexive, Not Anti-symmetric, Transitive  
C. Reflexive, Anti-symmetric, Transitive  
D. Not Reflexive, Not Anti-symmetric, Not Transitive

isrodec2017 set-theory&amp;algebra relations

Answer key 

12.6

Set Theory (4)

12.6.1 Set Theory: ISRO CSE 2011 | Question: 27



Which one of the following is true?

- A.  $R \cap S = (R \cup S) - [(R - S) \cup (S - R)]$
- B.  $R \cup S = (R \cap S) - [(R - S) \cup (S - R)]$
- C.  $R \cap S = (R \cup S) - [(R - S) \cap (S - R)]$
- D.  $R \cap S = (R \cup S) \cup (R - S)$

isro2011 set-theory&algebra set-theory

Answer key 

12.6.2 Set Theory: ISRO CSE 2017 | Question: 9



The symmetric difference of sets  $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$  and  $B = \{1, 3, 5, 6, 7, 8, 9\}$  is:

- A.  $\{1, 3, 5, 6, 7, 8\}$
- B.  $\{2, 4, 9\}$
- C.  $\{2, 4\}$
- D.  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

isro2017 set-theory&algebra set-theory

Answer key 

12.6.3 Set Theory: ISRO CSE 2020 | Question: 76



If  $A = \{x, y, z\}$  and  $B = \{u, v, w, x\}$ , and the universe is  $\{s, t, u, v, w, x, y, z\}$ . Then  $(A \cup \bar{B}) \cap (A \cap B)$  is equal to

- A.  $\{u, v, w, x\}$
- B.  $\{\}$
- C.  $\{u, v, w, x, y, z\}$
- D.  $\{u, v, w\}$

isro-2020 discrete-mathematics set-theory&algebra set-theory easy

Answer key 

12.6.4 Set Theory: ISRO-2013-74



The number of elements in the power set of the set  $\{\{A, B\}, C\}$  is

- A. 7
- B. 8
- C. 3
- D. 4

isro2013 set-theory&algebra set-theory easy

Answer key 

## Answer Keys

|        |   |        |   |        |   |        |   |        |   |
|--------|---|--------|---|--------|---|--------|---|--------|---|
| 12.0.1 | B | 12.0.2 | D | 12.0.3 | B | 12.1.1 | C | 12.2.1 | C |
| 12.3.1 | A | 12.3.2 | A | 12.4.1 | B | 12.5.1 | D | 12.5.2 | B |
| 12.6.1 | A | 12.6.2 | B | 12.6.3 | X | 12.6.4 | D |        |   |



## 13.0.1 ISRO CSE 2009 | Question: 49



$x = a \cos(t), y = b \sin(t)$  is the parametric form of

- A. Ellipse                      B. Hyperbola                      C. Circle                      D. Parabola

isro2009 calculus

Answer key

## 13.0.2 ISRO 2011-ECE Mathematics



$$\int_0^{\pi/2} (\cos^3 x) dx =$$

- a) 3/2  
b) 2/3  
c) 8/9  
d) 8/13

isro-ece isro2011-ece engineering-mathematics calculus

Answer key

## 13.0.3 ISRO 2010-ECE Calculus



What does the following integral evaluate to?

$$\int_0^{\pi/2} \sin^6 \theta d\theta$$

- a)  $5\pi/16$   
b)  $5\pi/8$   
c) 0  
d)  $5\pi/32$

isro-ece engineering-mathematics calculus

Answer key

## 13.0.4 ISRO 2010-ECE Continuity in functions



Which of the following functions have a jump discontinuity at  $x=0$

- a)  $g(x) = \ln |x|$                       b)  $g(x) = 1/x^2$   
c)  $\begin{cases} \frac{\sin(x)}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$                       d) None of the above

isro-ece engineering-mathematics calculus

Answer key

### 13.0.5 ISRO 2009- ECE Calculus



$f(x) = [x^2]$  ( $[ ]$  stands for greatest integer function)

- a)  $f(x)$  continuous at  $x = 0$
- b) limit  $f(x)$  does not exist as  $x$  tend to 0
- c)  $f'(0) = 1$
- d)  $f(x)$  not derivable at  $x = 0$

isro-ece engineering-mathematics calculus

Answer key

### 13.0.6 ISRO 2009- ECE Calculus



$\lim_{x \rightarrow 0} (x \log \sin x)$  is

- a) 0
- b)  $1/2$
- c) 1
- d) 2

isro-ece engineering-mathematics calculus

Answer key

### 13.0.7 ISRO 2009-ECE Engineering Mathematics



There is a function  $f(x)$ , such that  $f(0) = 1$  and  $f'(0) = -1$  and  $f(x)$  is positive for all values of  $x$ . Then,

- a)  $f''(x) < 0$  for all  $x$
- b)  $-1 < f''(x) < 0$  for all  $x$
- c)  $-2 < f''(x) < -1$  for all  $x$
- d) None of the above

isro-ece engineering-mathematics calculus

Answer key

### 13.0.8 ISRO 2008-ECE Integration



Evaluate the function  $\int_0^{\infty} t^2 \delta(t-3) dt$

- (a)  $e^{-t^2}$
- (b)  $\frac{t^3}{3}$
- (c)  $t^2(t-3)$
- (d) 9

isro-ece engineering-mathematics calculus

### 13.0.9 ISRO 2007- ECE Integration





The value of  $\int_0^1 \int_0^1 \frac{dx dy}{\sqrt{(1-x^2)(1-y^2)}}$  is

a)  $\frac{\pi^2}{2}$                       b)  $\frac{\pi}{4}$                       c)  $\frac{\pi^2}{4}$                       d)  $\frac{\pi}{2}$

isro-ece   engineering-mathematics   calculus

### 13.0.10 ISRO 2012- [Mech] Calculus



The approximate value of  $y(0.1)$  from  $dy/dx = x^2y-1$ ,  $y(0)=1$  is

- (a) 0.900
- (b) 1.001
- (c) 0.802
- (d) 0.994

engineering-mathematics   isro-mech   calculus

Answer key 

### 13.0.11 ISRO 2013 Area under curve [Mech]



Area bounded by the parabola  $2y = x^2$  and the line  $x = y-4$  is equal to

- (a) 4.5
- (b) 9
- (c) 18
- (d) 36

engineering-mathematics   isro-mech   calculus

Answer key 

### 13.0.12 ISRO 2013- Calculus [Mech]



$\lim_{x \rightarrow 0} \frac{x^2 + x - \sin x}{x^2}$

- (a) 0
- (b)  $\infty$
- (c) 1
- (d) None of these

engineering-mathematics   isro-mech   calculus

Answer key 

### 13.0.13 ISRO 2013 Differentiation - [EE]



The derivative of a ideal step function is:

- a) Zero
- b) undefined
- c) Impulse function
- d) Sine function

Answer key **13.0.14 ISRO 2013- Functions [EE]**The minimum point of the function  $(x^3 / 3) - x$  is at

- a)  $x = 1$
- b)  $x = -1$
- c)  $x = 0$
- d)  $x = 1/\sqrt{3}$

Answer key **13.0.15 ISRO 2013- Integration [EE]**The integrating factor of equation  $\sec^2 y \, dy/dx + x \tan y = x^3$  is

- a)  $[e]^{x^2/2}$
- b)  $[e]^{-x^2/2}$
- c)  $[e]^{x/2}$
- d)  $[e]^{-x/2}$

Answer key **13.0.16 ISRO 2015- Differential equation [EE]**The solution of the differential equation  $(d y / dx) = ky$ ,  $Y(0) = C$  is

- (a)  $x = ce^{-xy}$
- (b)  $y = cekx$
- (c)  $x = ke^{cy}$
- (d)  $y = ce^{-kx}$

Answer key **13.1****Area (1)****13.1.1 Area: ISRO ECE 2012 |: Engineering Mathematics**The area bounded by the curves  $y^2 = x$ ,  $y = x$  is given by

- a.  $2/3$                       b.  $1/2$                       c.  $1/6$                       d.  $1/3$

Answer key 

## 13.2

## Definite Integral (1)

## 13.2.1 Definite Integral: ISRO CSE 2016 | Question: EC Calculus



Evaluate  $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dx \cdot dy}{(1+x^2+y^2)}$

A.  $\frac{\pi}{2} [\log(1 + \sqrt{2})]$   
 C.  $\frac{\pi}{2} [\log(1 - \sqrt{2})]$

B.  $\frac{\pi}{4} [\log(1 + \sqrt{2})]$   
 D.  $\frac{\pi}{4} [\log(1 - \sqrt{2})]$

isro2016-ece isro-ece calculus definite-integral

Answer key

## 13.3

## Differentiation (2)

## 13.3.1 Differentiation: ISRO CSE 2011 | Question: 59



$n$ -th derivative of  $x^n$  is

A.  $nx^{n-1}$

B.  $n^n \cdot n!$

C.  $nx^n!$

D.  $n!$

isro2011 calculus differentiation

Answer key

## 13.3.2 Differentiation: ISRO ECE 2012 | Engineering Mathematics



If  $x = e^{y+e^{y+e^{y+\dots}}}$   
 then  $\frac{dy}{dx}$  is

a.  $1-x$

b.  $(1-x)/x$

c.  $1/x$

d.  $x/(1-x)$

engineering-mathematics isro2012-ece isro-ece calculus differentiation

Answer key

## 13.4

## Functions (1)

## 13.4.1 Functions: ISRO CSE 2018 | Question: 15



The domain of the function  $\log(\log \sin(x))$  is:

a.  $0 < x < \pi$

c. Empty set

b.  $2n\pi < x < (2n+1)\pi$ , for  $n$  in  $N$

d. None of the above

isro2018 calculus functions

Answer key

## 13.5

## Integration (2)

## 13.5.1 Integration: ISRO 2013- Area under curve [EE]



The area bounded by the curves  $y^2 = 9x$ ,  $x - y + 2 = 0$  is given by

a) 1

b) 1/2

C) 3/2

d) 5/4

Answer key **13.5.2 Integration: ISRO2006-ECE calculus**

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} \text{ is}$$

a)  $\frac{1}{a} \operatorname{cosec}^{-1}\left(\frac{x}{a}\right)$

b)  $\frac{1}{a} \sin^{-1}\left(\frac{x}{a}\right)$

c)  $\frac{1}{a} \cos^{-1}\left(\frac{x}{a}\right)$

d)  $\frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right)$

Answer key **13.6 Limits (2)****13.6.1 Limits: ISRO CSE 2016 | Question: 3**

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x} \text{ is given by}$$

A. 0

B. -1

C. 1

D.  $\frac{1}{2}$

Answer key **13.6.2 Limits: ISRO ECE 2016 | Mathematics**

$$\text{The value of } \lim_{x \rightarrow 8} \left( \frac{x^{1/3} - 2}{x - 8} \right) \text{ is}$$

a.  $1/4$

b.  $1/8$

c.  $1/12$

d.  $1/16$

Answer key **13.7 Maxima Minima (2)****13.7.1 Maxima Minima: ISRO CSE 2009 | Question: 50**

The value of  $x$  at which  $y$  is minimum for  $y = x^2 - 3x + 1$  is

A.  $-3/2$

B.  $3/2$

C. 0

D.  $-5/4$

Answer key **13.7.2 Maxima Minima: ISRO-2013-49**

What is the least value of the function  $f(x) = 2x^2 - 8x - 3$  in the interval  $[0, 5]$ ?

A. -15

B. 7

C. -11

D. -3

Answer key **13.8 Vector Calculus (1)**



Which of the following statement is correct

- A.  $\Delta(U_k V_k) = U_k \Delta V_k + V_k \Delta U_k$   
 B.  $\Delta(U_k V_k) = U_{k+1} \Delta V_k + V_{k+1} \Delta U_k$   
 C.  $\Delta(U_k V_k) = V_{k+1} \Delta U_k + U_k \Delta V_k$   
 D.  $\Delta(U_k V_k) = U_{k+1} \Delta V_k + V_k \Delta U_k$

isro2009 calculus vector-calculus non-gatecse

Answer key

## Answer Keys

|         |     |
|---------|-----|
| 13.0.1  | A   |
| 13.0.6  | Q-Q |
| 13.0.11 | Q-Q |
| 13.0.16 | Q-Q |
| 13.4.1  | C   |
| 13.7.1  | B   |

|         |     |
|---------|-----|
| 13.0.2  | Q-Q |
| 13.0.7  | Q-Q |
| 13.0.12 | Q-Q |
| 13.1.1  | Q-Q |
| 13.5.1  | Q-Q |
| 13.7.2  | C   |

|         |     |
|---------|-----|
| 13.0.3  | Q-Q |
| 13.0.8  | Q-Q |
| 13.0.13 | Q-Q |
| 13.2.1  | Q-Q |
| 13.5.2  | Q-Q |
| 13.8.1  | B   |

|         |     |
|---------|-----|
| 13.0.4  | Q-Q |
| 13.0.9  | Q-Q |
| 13.0.14 | Q-Q |
| 13.3.1  | D   |
| 13.6.1  | C   |

|         |     |
|---------|-----|
| 13.0.5  | Q-Q |
| 13.0.10 | Q-Q |
| 13.0.15 | Q-Q |
| 13.3.2  | Q-Q |
| 13.6.2  | Q-Q |



#### 14.0.1 ISRO 2012- ECE : Engineering mathematics



If  $A$ ,  $B$  &  $C$  are vectors and  $A = 1.u_x + 2.u_y + 3.u_z$ ,  $B = 1.u_x + 1.u_y + 1.u_z$ , and  $C = 3.u_x + 2.u_y + 1.u_z$  then  $(A \times B) \cdot C$  is :

- a) 2
- b) 1
- c) 0
- d) -1

isro-ece engineering-mathematics linear-algebra

#### 14.0.2 ISRO 2011- ECE Mathematics



If  $\alpha$  and  $\beta$  are the roots of the equation  $x^2 - px + q = 0$ , then  $\sum \alpha^2$  is

- a.  $p^2 + 2q$
- b.  $p + 2q$
- c.  $p^2 - 2q$
- d.  $p - 2q$

isro-ece isro2011-ece engineering-mathematics linear-algebra

Answer key

#### 14.0.3 ISRO 2011-ECE Linear Algebra



Which of the following system is linear?

- a)  $y(n) = e^{x(n)}$
- b)  $y(n) = Ax(n) + B$
- c)  $y(n) = x(n^2)$
- d)  $y(n) = x^2(n)$

isro-ece isro2011-ece engineering-mathematics linear-algebra

Answer key

#### 14.0.4 ISRO 2011- Engineering mathematics



The solution of differential equation  $dy/dx = e^{x-y} + x^2e^{-y}$  is

- a.  $e^y = e^x + x^3/3 + c$
- b.  $e^y - e^x = c$
- c.  $x - e^y = c$
- d.  $e^y + e^x + x^3/3 + y = 0$

isro-ece engineering-mathematics isro2011-ece calculus

Answer key 

#### 14.0.5 ISRO 2011-ECE : Complex Numbers



$(3 + i)/(5 + 5i)$  is same as

- a.  $(2 - i)/5$
- b.  $3 - i$
- c.  $5 - 5i$
- d.  $(2 + i)/5$

isro-ece isro2011-ece engineering-mathematics linear-algebra

Answer key 

#### 14.0.6 ISRO 2011-ECE Mathematics



$(\cos 5\theta - i \sin 5\theta)^2$  is same as:

- a.  $\cos 10\theta + i \sin 10\theta$
- b.  $\cos 25\theta - \sin 25\theta$
- c.  $(\cos \theta + i \sin \theta)^{-10}$
- d.  $(\cos \theta - i \sin \theta)^{-10}$

isro-ece isro2011-ece engineering-mathematics

#### 14.0.7 ISRO 2011- ECE Linear algebra



The value of x at which y has a minimum for  $y = x^2 - 3x + 1$  is

- a.  $-3/2$
- b.  $3/2$
- c. 0
- d. None of these

isro-ece isro2011-ece engineering-mathematics linear-algebra

Answer key 

#### 14.0.8 ISRO 2010-ECE matrices



If every minor of order 'r' of a matrix 'A' is zero, then rank of 'A' is

- a) greater than 'r'
- b) equal to 'r'
- c) less than or equal to 'r'
- d) less than 'r'

isro-ece engineering-mathematics linear-algebra

Answer key 

#### 14.0.9 ISRO 2010-ECE Matrices



Which of the following is **true**?

- a) The product of the eigenvalues of a matrix is equal to the trace of the matrix
- b) The eigenvalues of a skew-symmetric matrix are real
- c) A is a nonzero column matrix and B is a nonzero row matrix, then the rank of AB is one
- d) A system of linear non-homogeneous equations is consistent if and only if the rank of the coefficient matrix is less than or equal to the rank of the augmented matrix

isro-ece   engineering-mathematics   linear-algebra

Answer key 

#### 14.0.10 ISRO 2009-ECE Matrices



$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Evaluate  $A^3 - 6A^2 + 11A - 10I$

- a) Null matrix
- b) Identity matrix
- c)  $-4I$
- d) None of the above

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#### 14.0.11 ISRO 2009- ECE Linear algebra



If  $\alpha$  and  $\beta$  are the roots of the equation  $x^2 + x + 1 = 0$ . Then  $\alpha^{3n} + \beta^{3n}$  is

- a) 1
- b) 2
- c) 0
- d) 3

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Answer key 

#### 14.0.12 ISRO 2009 -ECE Complex Numbers



Number of real values of  $(a + ib)^{1/n} + (a - ib)^{1/n}$  is

- a) 0
- b) 1
- c) n
- d) None of the above

isro-ece   engineering-mathematics



#### 14.0.13 ISRO 2009- Engineering Mathematics



$$T_r = r/(1 + r^2 + r^4)$$

$$\sum_1^{\infty} T_r =$$

- a) 1/4
- b) 1/2
- c) 1
- d) None of above

isro-ece engineering-mathematics linear-algebra

Answer key

#### 14.0.14 ISRO 2008-ECE Maths / Vectors



The value of c which makes the angle  $45^\circ$  between  $a = i + cj$  and  $b = i + j$  is

- a) 1
- b)  $1/\sqrt{2}$
- c)  $-1/\sqrt{2}$
- d) 0

isro-ece engineering-mathematics

Answer key

#### 14.0.15 ISRO 2008- ECE adjoint Matrix



$$A = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & x \end{bmatrix}$$

The matrix is its own adjoint. The value of x will be

- (a) 5 (b) 3 (c) -3 (d) -5

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Answer key

#### 14.0.16 ISRO 2007- ECE Vectors



A unit vector perpendicular to the vectors  $a = 2i - 3j + k$  and  $b = i + j - 2k$ , is

- a)  $(1/\sqrt{3})(-i + j + k)$
- b)  $(1/\sqrt{3})(i + j - k)$
- c)  $(1/\sqrt{3})(i + j + k)$
- d)  $(i + j + k)$

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Answer key

#### 14.0.17 ISRO 2007- ECE Z plane



The region of the z plane for which  $|z-a|/|z+a| = 1$  ( $\operatorname{Re}(a) \neq 0$ ) is:

- a) x-axis
- b) y axis
- c) The straight line  $z = |a|$
- d) None of the above

isro-ece engineering-mathematics

Answer key

#### 14.0.18 ISRO 2007-ECE Roots of equation



If a,b,c are the roots of equations  $x^3 + px^2 + qx + p = 0$ , Then the value of

$$\tan^{-1}a + \tan^{-1}b + \tan^{-1}c$$

is

- a)  $n\pi/4$
- b)  $n\pi/2$
- c)  $n\pi$
- d)  $2n\pi$

isro-ece engineering-mathematics

Answer key

#### 14.0.19 ISRO 2007- ECE Determinant



The value of determinant  $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$  is:

- a) 0
- b) 1
- c) 3
- d)  $a+b+c$

isro-ece engineering-mathematics linear-algebra

Answer key

#### 14.0.20 ISRO 2007- ECE Matrices



The Algebraic multiplicity of the matrix  $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$  is

- a) 1
- b) 2
- c) 3
- d) 4

isro-ece engineering-mathematics linear-algebra

Answer key 

#### 14.0.21 ISRO 2006-ECE Matrix determinant



The determinant of  $\begin{vmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{vmatrix}$  is:

- a) 0
- b) -1
- c) 1
- d) 2

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Answer key 

#### 14.0.22 ISRO 2006-ECE Concurrent Lines



The value of k for which the lines  $2x + y - 1 = 0$ ,  $4x + 3y - 3 = 0$  and  $3x + ky - 2 = 0$  are concurrent is

- a) -2
- b) 3
- c) 2
- d) -3

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Answer key 

#### 14.0.23 ISRO 2012 - [Mech]



$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$$

rank of given matrix is

- a) 0
- b) 1
- c) 2
- d) 3

engineering-mathematics   isro-mech   linear-algebra

Answer key 

#### 14.0.24 ISRO 2012- [Mech] Vectors



The value of t for which  $A + tB$  is perpendicular to C where  $A = i + 2j + 3k$ ,  $B = -i + 2j + k$  and  $C = 3i + j$

- (a) 5
- (b) 4
- (c) 12
- (d) 0

isro-mech   engineering-mathematics

#### 14.0.25 ISRO 2012- [Mech] Engineering mathematics Cross product



$A \times B$  is a vector

- (a) Parallel to A, but perpendicular to B
- (b) Parallel to B, but perpendicular to A.
- (c) Perpendicular to both A & B
- (d) Parallel to both A & B

isro-mech engineering-mathematics

Answer key

#### 14.0.26 ISRO 2012: [Mech] Complex Numbers



$1 + i$  is equivalent to

- (a)  $\sqrt{2} e^{-i\pi/4}$
- (b)  $\sqrt{2} e^{i\pi/4}$
- (c)  $2 e^{-i\pi/4}$
- (d)  $2 e^{i\pi/4}$

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Answer key

#### 14.0.27 ISRO 2012 - [Mech]



In a class of 45 students, the mean mark of 25 girls is 32 and the mean mark of 20 boys is 27.5. What is the class mean?

- (a) 32
- (b) 27.5
- (c) 29.75
- (d) 30

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Answer key

#### 14.0.28 ISRO 2016- Vectors [Mech]



The sine of the angle between the two vectors  $a = 3i + j + k$  and  $b = 2i - 2j + k$  is

- (a)  $\sqrt{74/99}$
- (b)  $\sqrt{25/99}$
- (c)  $\sqrt{37/99}$
- (d)  $\sqrt{5/99}$

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Answer key

#### 14.0.29 ISRO 2016 Coordinate geometry [Mech]



Equation of the line normal to function  $f(x) = (x - 8)^{2/3} + 1$  at  $P(0, 5)$  is

- (a)  $y = 3x - 5$

(b)  $3y = x - 15$

(c)  $3y = x + 15$

(d)  $y = 3x + 5$

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Answer key 

#### 14.0.30 ISRO 2015- matrices [Mech]



For which value of  $x$ , will the matrix  $\begin{bmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$  become singular ? 12 6 0

- (a) 4 (b) 6 (c) 8 (d) 12

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Answer key 

#### 14.0.31 ISRO 2015- Conic sections [Mech]



Circular shapes appear in this fashion when viewed at an angle other than 90 degrees

- (a) Circular (b) Elliptical (c) Lengthened (d) Angular

engineering-mathematics isro-mech linear-algebra

Answer key 

#### 14.0.32 ISRO 2014- Matrices [mech]



Let  $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

The only correct statement about the matrix  $A$  is

- (a)  $A$  is a ZERO matrix  
(b)  $A^2 = I$   
(c)  $A^{-1}$  does not exist  
(d)  $A = (-1)I$ , where  $I$  is a Unit matrix

engineering-mathematics isro-mech linear-algebra

Answer key 

#### 14.0.33 ISRO 2014- vector {mech}



Angle made between vector  $C = 2i - 3j + 4k$  and the  $Z$ -axis is

- a)  $4/\sqrt{29}$   
b)  $\sin^{-1} 4/\sqrt{29}$

c)  $\cos^{-1} 4/\sqrt{29}$

d)  $\tan^{-1} 4/\sqrt{29}$

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Answer key 

#### 14.0.34 ISRO 2013 - matrices [Mech]



If  $A = \begin{bmatrix} 8 & 5 \\ 7 & 6 \end{bmatrix}$  then  $|A^{121} - A^{120}|$  is:

- a) 0
- b) 1
- c) 120
- d) 121

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Answer key 

#### 14.0.35 ISRO 2013- Angle between vectors [Mech]



Vector  $a = 3i + 2j - 6k$ , vector  $b = 4i - 3j + k$ , angle between above vectors is

- (a)  $90^\circ$
- (b)  $0^\circ$
- (c)  $45^\circ$
- (d)  $60^\circ$

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Answer key 

#### 14.0.36 ISRO 2013- Vectors [EE]



The vectors  $x_1 = (1, 2, 4)$ ,  $x_2 = (2, -1, 3)$ ,  $x_3 = (0, 1, 2)$ ,  $x_4 = (-3, 7, 2)$  are

- a) Linearly independent
- b) Linearly dependent
- c) No relation
- d) Exponentially dependent

engineering-mathematics isro-ee

Answer key 

#### 14.0.37 ISRO 2013- Newton Raphson [EE]



The equation  $x^3 - x^2 + 4x - 4 = 0$  is to be solved using the Newton - Raphson method. If  $x = 2$  is taken as the initial approximation of the solution, then the next approximation using the method will be

- a)  $2/3$
- b)  $4/3$

c)  $1/3$

d)  $5/3$

isro-ee engineering-mathematics

Answer key 

#### 14.0.38 ISRO 2013- Polynomials [EE]



The unique polynomial  $P(x)$  of degree 2 such that:  $P(1) = 1$ ,  $P(3) = 27$ ,  $P(4) = 64$  is

a)  $8x^2 - 19x + 12$

b)  $8x^2 + 19x + 12$

c)  $-8x^2 - 19x + 12$

d)  $-8x^2 - 19x - 12$

isro-ee engineering-mathematics

Answer key 

#### 14.0.39 ISRO 2013- Quadratic equation [EE]



If the sum of the roots of the quadratic equation  $ax^2 + bx + c = 0$  is equal to the sum of the squares of their reciprocals, then  $a/c$ ,  $b/a$  and  $c/b$  are in:

(A) arithmetic progression

(B) geometric progression

(C) arithmetic-geometric-progression

(D) harmonic progression

isro-ee engineering-mathematics linear-algebra

Answer key 

#### 14.0.40 ISRO 2014 -Complex Numbers [EE]



If  $\left[\frac{1+i}{1-i}\right]^x = 1$  then

(A)  $x = 2n + 1$ , where  $n$  is any positive integer

(B)  $x = 2n$ , where  $n$  is any positive integer

(C)  $x = 4n + 1$ , where  $n$  is any positive integer

(D)  $x = 4n$ , where  $n$  is any positive integer

isro-ee engineering-mathematics linear-algebra

Answer key 

#### 14.0.41 ISRO 2014- Geometry [EE]



If a vertex of a triangle is  $(1, 1)$  and the mid-points of two sides through this vertex are  $(-1, 2)$  and  $(3, 2)$ , then the centroid of the triangle is

(A)  $(-1, 7/3)$

(B)  $(-1/3, 7/3)$

(C)  $(1, 7/3)$

(D)  $(1/3, 7/3)$

engineering-mathematics isro-ee

Answer key 

#### 14.0.42 ISRO 2014- Quadratic Equation [EE]



If one root of the equation  $x^2 + px + 12 = 0$  is 4, while the equation  $x^2 + px + q = 0$  has equal roots, then the value of 'q' is

(A)  $49/4$

(B) 4

(C)  $49/16$

(D) 12

isro-ee engineering-mathematics

Answer key 

#### 14.0.43 ISRO 2014 -Trigonometry [EE]



If  $\tan \Theta = 8/15$  and  $\Theta$  is acute, then  $\operatorname{cosec} \Theta$

(A)  $8/17$

(B)  $8/15$

(C)  $17/8$

(D)  $17/15$

isro-ee engineering-mathematics

Answer key 

#### 14.0.44 ISRO 2015- Complex Numbers [EE]



If  $x = \sqrt{-1}$ , then the value of  $x^x$  is

(a)  $e^{-\pi/2}$

(b)  $x$

(c)  $e^{\pi/2}$

(d) 1

isro-ee engineering-mathematics

Answer key 

#### 14.0.45 ISRO-DEC2017-1



Suppose  $A$  is a finite set with  $n$  elements. The number of elements and the rank of the largest equivalence relation on  $A$  are



A.  $\{n, 1\}$

B.  $\{n, n\}$

C.  $\{n^2, 1\}$

D.  $\{1, n^2\}$

isrodec2017

Answer key 

## 14.1

## Determinant (4)

## 14.1.1 Determinant: ISRO CSE 2008 | Question: 31



If the two matrices  $\begin{bmatrix} 1 & 0 & x \\ 0 & x & 1 \\ 0 & 1 & x \end{bmatrix}$  and  $\begin{bmatrix} x & 1 & 0 \\ x & 0 & 1 \\ 0 & x & 1 \end{bmatrix}$  have the same determinant, then the value of  $x$  is

A.  $\frac{1}{2}$

B.  $\sqrt{2}$

C.  $\pm \frac{1}{2}$

D.  $\pm \frac{1}{\sqrt{2}}$

isro2008 linear-algebra matrix determinant

Answer key 

## 14.1.2 Determinant: ISRO CSE 2009 | Question: 60



If two adjacent rows of a determinant are interchanged, the value of the determinant

A. becomes zero

B. remains unaltered

C. becomes infinitive

D. becomes negative of its original value

isro2009 linear-algebra matrix determinant

Answer key 

## 14.1.3 Determinant: ISRO CSE 2009 | Question: 61



If  $\begin{vmatrix} 3 & 3 \\ x & 5 \end{vmatrix} = 3$  then the value of  $x$  is

A. 2

B. 3

C. 4

D. 5

isro2009 linear-algebra matrix determinant

Answer key 

## 14.1.4 Determinant: ISRO CSE 2009 | Question: 63



$$\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix} =$$

A. 779

B. 679

C. 0

D. 256

isro2009 linear-algebra matrix determinant

Answer key 

## 14.2

## Eigen Value (5)

## 14.2.1 Eigen Value: ISRO 2006-ECE Eigen Values



The Eigen values of  $\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$  are:

a) 6, 1

b) 6, -1

- c) -6, 1  
d) -6,-1

isro-ece engineering-mathematics linear-algebra eigen-value

Answer key 

### 14.2.2 Eigen Value: ISRO 2012- ECE Matrices



The Eigen values of matrix are:

$$\begin{bmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

- a)  $\pm \cos \alpha$   
b)  $\pm \sin \alpha$   
c)  $\tan \alpha$  &  $\cot \alpha$   
d)  $\cos \alpha \pm \sin \alpha$

engineering-mathematics isro-ece isro2012-ece linear-algebra eigen-value

Answer key 

### 14.2.3 Eigen Value: ISRO 2015- Eigen Values [mech]



$$\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

Eigen values of the matrix are:

- a) 3,-3  
b) -3, -5  
c) 3, 5  
d) 5 ,0

engineering-mathematics isro-mech linear-algebra eigen-value

Answer key 

### 14.2.4 Eigen Value: ISRO CSE 2007 | Question: 09



Eigen vectors of  $\begin{bmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{bmatrix}$  are

- A.  $\begin{bmatrix} a^n & 1 \\ 0 & a^n \end{bmatrix}$   
B.  $\begin{bmatrix} a^n & n \\ 0 & a^n \end{bmatrix}$   
C.  $\begin{bmatrix} a^n & na^{n-1} \\ 0 & a^n \end{bmatrix}$   
D.  $\begin{bmatrix} a^n & na^{n-1} \\ -n & a^n \end{bmatrix}$

isro2007 linear-algebra matrix eigen-value

Answer key 

#### 14.2.5 Eigen Value: ISRO-2013-33



What is the matrix transformation which takes the independent vectors  $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$  and  $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$  and transforms them to  $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$  and  $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$  respectively?

A.  $\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$   
C.  $\begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix}$

B.  $\begin{pmatrix} 0 & 0 \\ 0.5 & 0.5 \end{pmatrix}$   
D.  $\begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$

isro2013 linear-algebra eigen-value

Answer key 

#### 14.3

#### Functions (1)

##### 14.3.1 Functions: ISRO 2006-ECE Functions



$x^3 + x \sin x$  is

- a) Constant function
- b) Odd function
- c) Even function
- d) Periodic function

isro-ece engineering-mathematics linear-algebra functions

Answer key 

#### 14.4

#### Geometry (2)

##### 14.4.1 Geometry: ISRO 2007- ECE Geometry



Equation of a straight line passing through the point (-1,2) and making equal intercepts on the axes is

- a)  $x - y = 1$
- b)  $x - 2y = 1$
- c)  $x + y = 1$
- d)  $x - y = 2$

isro-ece engineering-mathematics geometry

Answer key 

##### 14.4.2 Geometry: ISRO 2012- Coordinate geometry [Mech]



The equation of the plane through the line  $(x - 1)/3 = (y - 4)/2 = (z - 4)/-2$  and parallel to the line  $(x + 1)/2 = (y - 1)/-4 = (z + 2)/1$  is

- (a)  $3x + 2y - 2z = 101$
- (b)  $2x - 4y + z = 98$
- (c)  $6x + 7y + 16z = 98$
- (d)  $6x - 8y - 2z = 101$

isro-mech engineering-mathematics geometry

## 14.5.1 Matrix: ISRO 2013 - Matrices [Mech]



If  $A$  is Square Matrix of order 3, then product of  $A$  and its transpose is

- (a) Unit Matrix
- (b) Zero Matrix
- (c) Identity Matrix
- (d) Symmetric Matrix

engineering-mathematics isro-mech linear-algebra matrix

Answer key

## 14.5.2 Matrix: ISRO 2013- Matrices /Mech



The matrix  $\begin{bmatrix} 0 & -4 & 1 \\ 4 & 0 & -5 \\ -1 & 5 & 0 \end{bmatrix}$  is

- (a) Orthogonal Matrix
- (b) Skew Symmetric
- (c) Symmetric
- (d) Idempotent

engineering-mathematics isro-mech matrix linear-algebra

Answer key

## 14.5.3 Matrix: ISRO 2013- Matrices [EE]



Characteristic roots of matrix  $A$  and  $A^T$  will be

- a) Different
- b) Same
- c) Cannot say about roots
- d) None of these

isro-ee matrix

Answer key

## 14.5.4 Matrix: ISRO 2014 -Matrices [Mech]



Let  $M$  be a  $3 \times 3$  matrix satisfying:

$$M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad \text{and} \quad M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

Then the sum of the diagonal entries of  $M$  is:

- (a) 9

- (b) 12  
(c) 6  
(d) 0

engineering-mathematics isro-mech linear-algebra matrix

Answer key 

#### 14.5.5 Matrix: ISRO 2016- matrices [Mech]



What is the value of

$$\begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix}$$

- a)  $2 a^2 b^2 c^2$   
b)  $4 a^2 b^2 c^2$   
c)  $8 a^2 b^2 c^2$   
d) 0

matrix engineering-mathematics isro-mech linear-algebra

Answer key 

#### 14.5.6 Matrix: ISRO CSE 2008 | Question: 34



If a square matrix  $A$  satisfies  $A^T A = I$ , then the matrix  $A$  is

- A. Idempotent      B. Symmetric      C. Orthogonal      D. Hermitian

isro2008 linear-algebra matrix

Answer key 

#### 14.5.7 Matrix: ISRO CSE 2009 | Question: 59



A square matrix  $A$  is called orthogonal if  $A' A =$

- A.  $I$       B.  $A$       C.  $-A$       D.  $-I$

isro2009 linear-algebra matrix

Answer key 

#### 14.5.8 Matrix: ISRO CSE 2009 | Question: 62



If  $A, B, C$  are any three matrices, then  $A' + B' + C'$  is equal to

- A. a null matrix      B.  $A + B + C$       C.  $(A + B + C)'$       D.  $-(A + B + C)$

isro2009 linear-algebra matrix

Answer key 

#### 14.5.9 Matrix: ISRO CSE 2011 | Question: 36



If  $A$  and  $B$  are square matrices with same order and  $A$  is symmetric, then  $B^T A B$  is

- A. Skew symmetric  
B. Symmetric  
C. Orthogonal  
D. Idempotent

Answer key **14.5.10 Matrix: ISRO CSE 2017 | Question: 1**If  $A$  is a skew symmetric matrix then  $A^t$  is

- A. Diagonal matrix      B.  $A$       C.  $0$       D.  $-A$

Answer key **14.5.11 Matrix: ISRO-DEC2017-17**If  $C$  is a skew-symmetric matrix of order  $n$  and  $X$  is  $n \times 1$  column matrix, then  $X^T C X$  is a

- A. scalar matrix      B. null matrix      C. unit matrix      D. matrix will all elements 1

Answer key **14.6 Rank of Matrix (1)****14.6.1 Rank of Matrix: ISRO CSE 2014 | Question: 72**

The rank of the matrix  $A = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 9 & 5 & 2 & 2 \\ 7 & 1 & 0 & 4 \end{pmatrix}$  is \_\_\_\_\_.

- A. 0      B. 1      C. 2      D. 3

Answer key **14.7 System of Equations (1)****14.7.1 System of Equations: ISRO ECE 2012 | Engineering Mathematics**The system of equations  $x + y + z = 6, 2x + y + z = 7, x + 2y + z = 8$  has

- a. A unique solution      b. No solution  
c. An infinite number of solutions      d. None of these

Answer key **14.8 Vector Space (1)****14.8.1 Vector Space: ISRO-DEC2017-10**If vectors  $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$  and  $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$  are perpendicular to each other, then value of  $\lambda$  is

- A.  $\frac{2}{5}$       B. 2      C. 3      D.  $\frac{5}{2}$

Answer key 

## Answer Keys

|         |     |
|---------|-----|
| 14.0.1  | Q-Q |
| 14.0.6  | Q-Q |
| 14.0.11 | Q-Q |
| 14.0.16 | Q-Q |
| 14.0.21 | Q-Q |
| 14.0.26 | Q-Q |
| 14.0.31 | Q-Q |
| 14.0.36 | Q-Q |
| 14.0.41 | Q-Q |
| 14.1.1  | A   |
| 14.2.2  | Q-Q |
| 14.4.1  | Q-Q |
| 14.5.4  | Q-Q |
| 14.5.9  | B   |
| 14.8.1  | D   |

|         |     |
|---------|-----|
| 14.0.2  | Q-Q |
| 14.0.7  | Q-Q |
| 14.0.12 | Q-Q |
| 14.0.17 | Q-Q |
| 14.0.22 | Q-Q |
| 14.0.27 | Q-Q |
| 14.0.32 | Q-Q |
| 14.0.37 | Q-Q |
| 14.0.42 | Q-Q |
| 14.1.2  | D   |
| 14.2.3  | Q-Q |
| 14.4.2  | Q-Q |
| 14.5.5  | Q-Q |
| 14.5.10 | D   |

|         |     |
|---------|-----|
| 14.0.3  | Q-Q |
| 14.0.8  | Q-Q |
| 14.0.13 | Q-Q |
| 14.0.18 | Q-Q |
| 14.0.23 | Q-Q |
| 14.0.28 | Q-Q |
| 14.0.33 | Q-Q |
| 14.0.38 | Q-Q |
| 14.0.43 | Q-Q |
| 14.1.3  | C   |
| 14.2.4  | Q-Q |
| 14.5.1  | Q-Q |
| 14.5.6  | C   |
| 14.5.11 | B   |

|         |     |
|---------|-----|
| 14.0.4  | Q-Q |
| 14.0.9  | Q-Q |
| 14.0.14 | Q-Q |
| 14.0.19 | Q-Q |
| 14.0.24 | Q-Q |
| 14.0.29 | Q-Q |
| 14.0.34 | Q-Q |
| 14.0.39 | Q-Q |
| 14.0.44 | Q-Q |
| 14.1.4  | C   |
| 14.2.5  | D   |
| 14.5.2  | Q-Q |
| 14.5.7  | A   |
| 14.6.1  | C   |

|         |     |
|---------|-----|
| 14.0.5  | Q-Q |
| 14.0.10 | Q-Q |
| 14.0.15 | Q-Q |
| 14.0.20 | Q-Q |
| 14.0.25 | Q-Q |
| 14.0.30 | Q-Q |
| 14.0.35 | Q-Q |
| 14.0.40 | Q-Q |
| 14.0.45 | C   |
| 14.2.1  | Q-Q |
| 14.3.1  | Q-Q |
| 14.5.3  | Q-Q |
| 14.5.8  | C   |
| 14.7.1  | Q-Q |



## 15.0.1 ISRO 2009- ECE Engineering Mathematics



A straight line and a circle of radius  $a$  are given. A chord is drawn at random to this circle parallel to the given line. Expected length of the chord is

- a)  $2a$
- b)  $\pi a$
- c)  $\pi a/2$
- d) None of the above

isro-ece engineering-mathematics

Answer key

## 15.0.2 ISRO-DEC2017-3



A bag contains 19 red balls and 19 black balls. Two balls are removed at a time repeatedly and discarded if they are of the same colour, but if they are different, black ball is discarded and red ball is returned to the bag. The probability that this process will terminate with one red ball is

- A. 1
- B.  $\frac{1}{21}$
- C. 0
- D. 0.5

isrodec2017

Answer key

## 15.1

## Arithmetic Mean (1)

## 15.1.1 Arithmetic Mean: ISRO CSE 2011 | Question: 67



The arithmetic mean of attendance of 49 students of class A is 40 and that of 53 students of class B is 35. Then the percentage of arithmetic mean of attendance of class A and B is

- A. 27.2%
- B. 50.25%
- C. 51.13%
- D. 37.4%

isro2011 probability arithmetic-mean

Answer key

## 15.2

## Conditional Probability (4)

## 15.2.1 Conditional Probability: GATE CSE 1994 | Question: 1.4, ISRO2017-2



Let  $A$  and  $B$  be any two arbitrary events, then, which one of the following is TRUE?

- A.  $P(A \cap B) = P(A)P(B)$
- B.  $P(A \cup B) = P(A) + P(B)$
- C.  $P(A | B) = P(A \cap B)P(B)$
- D.  $P(A \cup B) \leq P(A) + P(B)$

gate1994 probability conditional-probability normal isro2017

Answer key

## 15.2.2 Conditional Probability: ISRO CSE 2007 | Question: 33



Company X shipped 5 computer chips, 1 of which was defective. and company Y shipped 4 computer chips, 2 of which were defective. One computer chip is to be chosen uniformly at a random from the 9 chips shipped by the companies. If the chosen chip is found to be defective, what is the probability that the chip came from the company Y?



A. 2/9

B. 4/9

C. 2/3

D. 1/2

isro2007 probability conditional-probability

Answer key **15.2.3 Conditional Probability: ISRO-2013-71**

Let  $P(E)$  denote the probability of the occurrence of event  $E$ . If  $P(A) = 0.5$  and  $P(B) = 1$  then the values of  $P(A|B)$  and  $P(B|A)$  respectively are

A. 0.5, 0.25

B. 0.25, 0.5

C. 0.5, 1

D. 1, 0.5

isro2013 probability conditional-probability

Answer key **15.2.4 Conditional Probability: ISRO2016 EC Probability**

A person on a trip has a choice between a private car and public transport. The probability of using a private car is 0.45. While using public transport, the further choice available are bus and metro. Out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (rounded up to two decimals) of using a car, bus and metro respectively would be

a. 0.45, 0.30 and 0.25

b. 0.45, 0.25 and 0.30

c. 0.45, 0.55 and 0

d. 0.45, 0.35 and 0.20

isro2016-ece isro-ece probability conditional-probability

Answer key **15.3****Expectation (1)****15.3.1 Expectation: ISRO 2009-ECE Probability**

A man with  $n$  keys wants to open a lock. He tries his keys at random.

The expected number of attempts for his success is (keys are replaced after every attempt)

a)  $n/2$ b)  $n$ c)  $\sqrt{n}$ 

d) None of the above

isro-ece engineering-mathematics expectation

**15.4****Mean Mode Median (1)****15.4.1 Mean Mode Median: ISRO CSE 2014 | Question: 74**

What is the median of data if its mode is 15 and the mean is 30?

A. 30

B. 25

C. 22.5

D. 27.5

probability statistics mean-mode-median isro2014

Answer key **15.5****Normal Distribution (1)**

**15.5.1 Normal Distribution: ISRO CSE 2009 | Question: 65**

If the mean of a normal frequency distribution of 1000 items is 25 and its standard deviation is 2.5, then its maximum ordinate is

- A.  $\frac{1000}{\sqrt{2\pi}} e^{-25}$  B.  $\frac{1000}{\sqrt{2\pi}}$   
 C.  $\frac{1000}{\sqrt{2\pi}} e^{-2.5}$  D.  $\frac{400}{\sqrt{2\pi}}$

isro2009 statistics normal-distribution

Answer key

**15.6****Poisson Distribution (1)****15.6.1 Poisson Distribution: ISRO CSE 2009 | Question: 66**

If the pdf of a Poisson distribution is given by  $f(x) = \frac{e^{-2} 2^x}{x!}$  then its mean is

- A.  $2^x$  B. 2 C. -2 D. 1

isro2009 probability poisson-distribution

Answer key

**15.7****Probability (20)****15.7.1 Probability: ISRO 2006- ECE Probability**

A box contains 5 black and 5 red balls. Two balls are one after another from the box, without replacement. The probability of both balls being red is

- a) 1/90  
 b) 1/5  
 c) 19/90  
 d) 2/9

isro-ece engineering-mathematics probability

Answer key

**15.7.2 Probability: ISRO 2007- ECE Probability**

A bag contains eight white and six red marbles. The probability of drawing two marbles of same colour is

- a)  $8C_2 \cdot 6C_2 / 14C_2$   
 b)  $8C_2 / 14C_2 + 6C_2 / 14C_2$   
 c)  $8C_2 \cdot 6C_2 / 14C_2 \cdot 14C_2$   
 d)  $8C_2 / 14C_2 + 6C_2 / 12C_2$

isro-ece engineering-mathematics probability

Answer key

**15.7.3 Probability: ISRO 2008- ECE Probability**

A husband and wife appear in an interview for two vacancies for same post. The probability of husband

getting selected is  $1/5$  while the probability of wife getting selected is  $1/7$ . Then the probability that anyone of them getting selected is

- a)  $1/35$
- b)  $11/35$
- c)  $12/35$
- d)  $34/35$

isro-ece probability engineering-mathematics

Answer key 

#### 15.7.4 Probability: ISRO 2012- ECE Probability

(79) A random variable  $X$  has  $\bar{X}=0$  &  $\sigma_X^2=1$ . Form a new random variable  $Y=2X+1$ . The values of  $\bar{Y}$  &  $\sigma_Y^2$  are :

- a) 0 & 1
- b) 1 & 2
- c) 1 & 4
- d) None of these

engineering-mathematics isro-ece isro2012-ece probability

Answer key 

#### 15.7.5 Probability: ISRO 2012- Probability [Mech]

The probability that A happens is  $1/3$ . the odds against happening A are:

- a) 2:1
- b) 3:2
- c) 1:2
- d) 2:3

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.6 Probability: ISRO 2012: [Mech] Probability

Let E and F be any two events with  $P(E \cup F)=0.8$ ,  $P(E)=0.4$  and  $P(E/F)=0.3$ . Then  $P(F)$  is

- (a)  $3/7$
- (b)  $4/7$
- (c)  $3/5$
- (d)  $2/5$

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.7 Probability: ISRO 2013 Probability [EE]

An urn contains 5 black and 5 white balls. The probability of drawing two balls of the same colour

- a)  $2/9$
- b)  $4/9$
- c)  $1/9$
- d)  $5/9$

isro-ee engineering-mathematics probability

Answer key 

#### 15.7.8 Probability: ISRO 2013- Probability [Mech]



If the probability for A to fail an examination is 0.2 and that for B is 0.3, then probability that either A or B fail is

- (a) 0.5
- (b) 0.06
- (c) 0.44
- (d) 0.38

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.9 Probability: ISRO 2013- Probability/Leap year [Mech]



Chance that a leap year selected at random will contain 53 Sundays is

- (a)  $3/7$
- (b)  $7/2$
- (c)  $7/3$
- (d)  $2/7$

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.10 Probability: ISRO 2013- probability [EE]



Ten percent of screws produced in a certain factory turn out to be defective. Find the probability that in a sample of 10 screws chosen at random, exactly two will be defective.

- a) 0.2
- b) 0.25
- c) 0.8
- d) 0.379.

isro-ee probability

Answer key 

#### 15.7.11 Probability: ISRO 2014- Probability [Mech]



A purse contains 4 copper coins and 3 silver coins. A second purse contains 6 copper coins and 4 silver coins. A purse is chosen randomly and a coin is taken out of it. What is the probability that it is a copper coin?

- a)  $70/82$
- b)  $35/70$

c)  $41/82$

d)  $41/70$

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.12 Probability: ISRO 2014- probability [EE]



A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is

(A) 140

(B) 196

(C) 280

(D) 346

isro-ee engineering-mathematics probability

Answer key 

#### 15.7.13 Probability: ISRO 2015- probability [Mech]



The probability that a teacher will give an unannounced test during any class is  $1/5$ . If a student is absent twice, then probability that misses at least one test is

(a)  $24/25$

(b)  $16/25$

(c)  $7/25$

(d)  $9/25$

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.14 Probability: ISRO 2016 Mech: Probability



A and B are two candidates appearing for an interview by a company. The probability that A is selected is 0.5 and the probability that both A and B are selected is at most 0.3. The probability of B getting selected is

(a) 0.9

(b)  $\leq 0.6$

(c)  $\leq 0.3$

(d) 0.5

probability isro-mech2016 engineering-mathematics

Answer key 

#### 15.7.15 Probability: ISRO 2016 Number of trials [Mech]



There are 20 locks and 20 matching keys. Maximum number of trials required to match all the locks is

- (a) 190
- (b) 210
- (c) 400
- (d) 40

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.16 Probability: ISRO 2016 Probability [Mech]



A and B are two candidates appearing for an interview by a company. The probability that A is selected is 0.5 and the probability that both A and B are selected is at most 0.3. The probability of B getting selected is

- (a) 0.9
- (b) 0.3
- (c) 0.6
- (d) 0.5

engineering-mathematics isro-mech probability

Answer key 

#### 15.7.17 Probability: ISRO CSE 2011 | Question: 61



Three coins are tossed simultaneously. The probability that they will fall two heads and one tail is

- A.  $\frac{5}{8}$                       B.  $\frac{1}{8}$                       C.  $\frac{2}{3}$                       D.  $\frac{3}{8}$

isro2011 probability

Answer key 

#### 15.7.18 Probability: ISRO CSE 2014 | Question: 37



The probability that two friends are born in the same month is \_\_\_\_ ?

- A.  $\frac{1}{6}$                       B.  $\frac{1}{12}$                       C.  $\frac{1}{144}$                       D.  $\frac{1}{24}$

probability isro2014

Answer key 

#### 15.7.19 Probability: ISRO CSE 2018 | Question: 14



A class of 30 students occupy a classroom containing 5 rows of seats, with 8 seats in each row. If the students seat themselves at random, the probability that sixth seat in the fifth row will be empty is:

- a.  $\frac{1}{5}$                       b.  $\frac{1}{3}$                       c.  $\frac{1}{4}$                       d.  $\frac{2}{5}$

isro2018 probability

Answer key 

### 15.7.20 Probability: ISRO ECE 2012 | Probability



Person X can solve 80% of the ISRO question paper and Person Y can solve 60%. The probability that at least one of them will solve a problem from the question paper, selected at random is :

- a. 0.48
- b. 0.70
- c. 0.88
- d. 0.92

isro2012-ece isro-ece engineering-mathematics probability

Answer key

## 15.8

### Probability Density Function (1)

#### 15.8.1 Probability Density Function: GATE CSE 2003 | Question: 60, ISRO2007-45



A program consists of two modules executed sequentially. Let  $f_1(t)$  and  $f_2(t)$  respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

- A.  $f_1(t) + f_2(t)$
- B.  $\int_0^t f_1(x)f_2(x)dx$
- C.  $\int_0^t f_1(x)f_2(t-x)dx$
- D.  $\max\{f_1(t), f_2(t)\}$

gatecse-2003 probability normal isro2007 probability-density-function

Answer key

## 15.9

### Random Variable (1)

#### 15.9.1 Random Variable: GATE CSE 2005 | Question: 12, ISRO2009-64



Let  $f(x)$  be the continuous probability density function of a random variable  $x$ , the probability that  $a < x \leq b$  is :

- A.  $f(b-a)$
- B.  $f(b) - f(a)$
- C.  $\int_a^b f(x)dx$
- D.  $\int_a^b xf(x)dx$

gatecse-2005 probability random-variable easy isro2009

Answer key

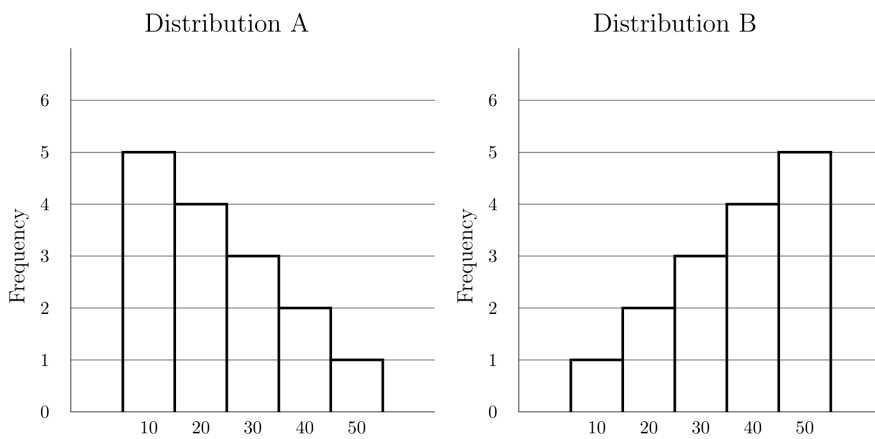
## 15.10

### Standard Deviation (1)

#### 15.10.1 Standard Deviation: ISRO CSE 2020 | Question: 56



For the distributions given below:



Which of the following is correct for the above distributions?

- A. Standard deviation of  $A$  is significantly lower than standard deviation of  $B$
- B. Standard deviation of  $A$  is slightly lower than standard deviation of  $B$
- C. Standard deviation of  $A$  is same as standard deviation of  $B$
- D. Standard deviation of  $A$  is significantly higher than standard deviation of  $B$

isro-2020 probability standard-deviation normal

Answer key

## Answer Keys

|         |     |         |     |         |     |         |     |         |     |
|---------|-----|---------|-----|---------|-----|---------|-----|---------|-----|
| 15.0.1  | Q-Q | 15.0.2  | A   | 15.1.1  | D   | 15.2.1  | D   | 15.2.2  | C   |
| 15.2.3  | C   | 15.2.4  | A   | 15.3.1  | Q-Q | 15.4.1  | B   | 15.5.1  | D   |
| 15.6.1  | B   | 15.7.1  | Q-Q | 15.7.2  | B   | 15.7.3  | Q-Q | 15.7.4  | Q-Q |
| 15.7.5  | Q-Q | 15.7.6  | Q-Q | 15.7.7  | Q-Q | 15.7.8  | Q-Q | 15.7.9  | Q-Q |
| 15.7.10 | Q-Q | 15.7.11 | Q-Q | 15.7.12 | Q-Q | 15.7.13 | Q-Q | 15.7.14 | B   |
| 15.7.15 | Q-Q | 15.7.16 | Q-Q | 15.7.17 | D   | 15.7.18 | B   | 15.7.19 | C   |
| 15.7.20 | Q-Q | 15.8.1  | C   | 15.9.1  | C   | 15.10.1 | C   |         |     |





## 16.1.1 Sequence Series: ISRO CSE 2007 | Question: 61



The Fibonacci sequence is the sequence of integers

A. 1, 3, 5, 7, 9, 11, 13

B. 0, 1, 1, 2, 3, 5, 8, 13, 21, 34

C. 0, 1, 3, 4, 7, 11, 18, 29, 47

D. 0, 1, 3, 7, 15

isro2007 analytical-aptitude sequence-series

Answer key 

## Answer Keys

|        |   |
|--------|---|
| 16.1.1 | B |
|--------|---|

17

## General Aptitude: Quantitative Aptitude (4)



17.1

## Geometry (1)

## 17.1.1 Geometry: ISRO-2013-72



How many diagonals can be drawn by joining the angular points of an octagon?

A. 14

B. 20

C. 21

D. 28

isro2013 quantitative-aptitude geometry

Answer key

17.2

## Inequality (1)

## 17.2.1 Inequality: ISRO CSE 2016 | Question: 1



Which of the following is true ?

A.  $\sqrt{3} + \sqrt{7} = \sqrt{10}$

B.  $\sqrt{3} + \sqrt{7} \leq \sqrt{10}$

C.  $\sqrt{3} + \sqrt{7} < \sqrt{10}$

D.  $\sqrt{3} + \sqrt{7} > \sqrt{10}$

quantitative-aptitude isro2016 inequality

Answer key

17.3

## LCM HCF (1)

## 17.3.1 LCM HCF: ISRO CSE 2020 | Question: 55



If  $x + 2y = 30$ , then  $\left(\frac{2y}{5} + \frac{x}{3}\right) + \left(\frac{x}{5} + \frac{2y}{3}\right)$  will be equal to

A. 8

B. 16

C. 18

D. 20

isro-2020 quantitative-aptitude easy lcm-hcf

Answer key

17.4

## Summation (1)

## 17.4.1 Summation: ISRO CSE 2016 | Question: 2



What is the sum to infinity of the series,

$$3 + 6x^2 + 9x^4 + 12x^6 + \dots \text{ given } |x| < 1$$

A.  $\frac{3}{(1+x^2)}$

B.  $\frac{3}{(1+x^2)^2}$

C.  $\frac{3}{(1-x^2)^2}$

D.  $\frac{3}{(1-x^2)}$

quantitative-aptitude summation isro2016

Answer key

## Answer Keys

17.1.1

B

17.2.1

D

17.3.1

B

17.4.1

C

## 18.1.1 Cloud Computing: ISRO-2013-78



Which of the following is not provided as a service in cloud computing?

- A. Infrastructure as a service
- B. Architecture as a service
- C. Software as a service
- D. Platform as a service

isro2013 cloud-computing non-gatecse

Answer key 

## Answer Keys

18.1.1

B

## 19.1.1 Computer Graphics: ISRO CSE 2008 | Question: 2



The term Phong associated with

- A. Ray tracing
- B. shading
- C. Hiddenline removal
- D. a game

isro2008 computer-graphics non-gatecse

Answer key

## 19.1.2 Computer Graphics: ISRO CSE 2008 | Question: 74



A Steiner patch is

- A. Biquadratic Bezeir patch
- B. Bicubic patch
- C. Circular patch only
- D. Bilinear Bezier patch

isro2008 non-gatecse computer-graphics

Answer key

## 19.1.3 Computer Graphics: ISRO CSE 2011 | Question: 7



Which of the following is true about  $z$ -buffer algorithm?

- A. It is a depth sort algorithm
- B. No limitation on total number of objects
- C. Comparisons of objects is done
- D.  $z$ -buffer is initialized to background colour at start of algorithm

isro2011 non-gatecse computer-graphics

Answer key

## 19.1.4 Computer Graphics: ISRO CSE 2014 | Question: 34



How much memory is required to implement  $z$ -buffer algorithm for a  $512 \times 512 \times 24$  bit-plane image?

- A. 768 KB
- B. 1 MB
- C. 1.5 MB
- D. 2 MB

non-gatecse computer-graphics isro2014

Answer key

## 19.1.5 Computer Graphics: ISRO CSE 2014 | Question: 48



A frame buffer array is addressed in row major order for a monitor with pixel locations starting from  $(0,0)$  and ending with  $(100,100)$ . What is address of the pixel  $(6,10)$ ? Assume one bit storage per pixel and starting pixel location is at 0.

- A. 1016
- B. 1006
- C. 610
- D. 616

isro2014 non-gatecse computer-graphics

Answer key

## 19.1.6 Computer Graphics: ISRO CSE 2018 | Question: 39



Perform window to viewport transformation for the point  $(20,15)$ . Assume that  $(X_{wmin}, Y_{wmin})$  is  $(0,0)$ ;  $(X_{wmax}, Y_{wmax})$  is  $(100,100)$ ;  $(X_{vmin}, Y_{vmin})$  is  $(5,5)$ ;  $(X_{vmax}, Y_{vmax})$  is  $(20,20)$ . The value of  $x$  and  $y$

in the viewport is

- A.  $x = 4, y = 4$
- B.  $x = 3, y = 3$
- C.  $x = 8, y = 7.25$
- D.  $x = 3, y = 4$

isro2018 non-gatecse normal computer-graphics

Answer key 

### 19.1.7 Computer Graphics: ISRO-2013-80



In graphics, the number of vanishing points depends on

- A. The number of axes cut by the projection plane
- B. The center of projection
- C. The number of axes which are parallel to the projection plane
- D. The perspective projections of any set of parallel lines that are not parallel to the projection plane

isro2013 computer-graphics non-gatecse

Answer key 

## 19.2

### Shading (1)

#### 19.2.1 Shading: ISRO-2013-31



In which of the following shading models of polygons, the interpolation of intensity values is done along the scan line?

- A. Gourard shading
- B. Phong shading
- C. Constant shading
- D. Flat shading

isro2013 shading non-gatecse

Answer key 

## 19.3

### Unix (1)

#### 19.3.1 Unix: ISRO CSE 2011 | Question: 57



Which of the following UNIX command allows scheduling a program to be executed at the specifies time?

- A. cron
- B. nice
- C. date and time
- D. schedule

isro2011 unix non-gatecse

Answer key 

## Answer Keys

|        |   |        |     |        |   |        |     |        |   |
|--------|---|--------|-----|--------|---|--------|-----|--------|---|
| 19.1.1 | B | 19.1.2 | A   | 19.1.3 | D | 19.1.4 | Q-Q | 19.1.5 | A |
| 19.1.6 | C | 19.1.7 | Q-Q | 19.2.1 | A | 19.3.1 | A   |        |   |

## 20.1.1 Computer Peripherals: ISRO CSE 2008 | Question: 44



The device which is used to connect a peripheral to bus is known as

- A. control register
- B. interface
- C. communication protocol
- D. none of these

isro2008 non-gatecse computer-peripherals

Answer key

## 20.1.2 Computer Peripherals: ISRO CSE 2011 | Question: 22



A system is having 8 M bytes of video memory for bit-mapped graphics with 64-bit colour. What is the maximum resolution it can support?

- A.  $800 \times 600$
- B.  $1024 \times 768$
- C.  $1280 \times 1024$
- D.  $1920 \times 1440$

isro2011 computer-peripherals

Answer key

## 20.1.3 Computer Peripherals: ISRO CSE 2011 | Question: 23



What is the meaning of  $\overline{RD}$  signal in Intel 8151A?

- A. Read (when it is low)
- B. Read (when it is high)
- C. Write (when it is low)
- D. Read and Write (when it is high)

isro2011 non-gatecse computer-peripherals

Answer key

## 20.1.4 Computer Peripherals: ISRO CSE 2011 | Question: 9



What is the raw throughput of USB 2.0 technology?

- A. 480 Mbps
- B. 400 Mbps
- C. 200 Mbps
- D. 12 Mbps

isro2011 non-gatecse computer-peripherals

Answer key

## 20.1.5 Computer Peripherals: ISRO CSE 2017 | Question: 46



SATA is the abbreviation of

- A. Serial Advanced Technology Attachment
- B. Serial Advanced Technology Architecture
- C. Serial Advanced Technology Adapter
- D. Serial Advanced Technology Array

isro2017 computer-peripherals non-gatecse

Answer key

## 20.2.1 Ieee1394: ISRO-2013-20



IEEE 1394 is related to

- A. RS-232
- B. USB
- C. Firewire
- D. PCI

isro2013 ieee1394

Answer Keys

|        |   |
|--------|---|
| 20.1.1 | B |
| 20.2.1 | C |

|        |   |
|--------|---|
| 20.1.2 | B |
|--------|---|

|        |   |
|--------|---|
| 20.1.3 | A |
|--------|---|

|        |   |
|--------|---|
| 20.1.4 | A |
|--------|---|

|        |   |
|--------|---|
| 20.1.5 | A |
|--------|---|

## 21.1.1 Digital Image Processing: ISRO CSE 2014 | Question: 7



What does a pixel mask mean?

- A. string containing only 1's  
 B. string containing only 0's  
 C. string containing two 0's  
 D. string containing 1's and 0's

non-gatecse digital-image-processing isro2014

Answer key

## 21.1.2 Digital Image Processing: ISRO-2013-10



If the frame buffer has 8 bits per pixel and 8 bits are allocated for each of the  $R, G, B$  components, what would be the size of the lookup table?

- A. 24 bytes  
 B. 1024 bytes  
 C. 768 bytes  
 D. 256 bytes

isro2013 digital-image-processing

Answer key

## 21.1.3 Digital Image Processing: ISRO2015-65



Which of the following is not an image type used in MPEG?

- A. A frame  
 B. B frame  
 C. D frame  
 D. P frame

isro2015 digital-image-processing non-gatecse

Answer key

## 21.2.1 Image Compression: ISRO2015-63



Which of the following compression algorithms is used to generate a .png file?

- A. LZ78  
 B. Deflate  
 C. LZW  
 D. Huffman

isro2015 digital-image-processing image-compression non-gatecse

Answer key

## Answer Keys

21.1.1

D

21.1.2

C

21.1.3

A

21.2.1

B



## 22.1.1 Semiconductor: ISRO CSE 2011 | Question: 64



When n-type semiconductor is heated?

- A. number of electrons increases while that of holes decreases
- B. number of holes increases while that of electrons decreases
- C. number of electrons and holes remain the same
- D. number of electron and holes increases equally

isro2011 semiconductor non-gatecse

Answer key 

## Answer Keys

22.1.1

B

## 23.1.1 Distributed Computing: ISRO CSE 2014 | Question: 80



A computing architecture, which allows the user to use computers from multiple administrative domains to reach a common goal is called as

- A. Grid Computing
- B. Neutral Networks
- C. Parallel Processing
- D. Cluster Computing

isro2014 non-gatecse distributed-computing

Answer key

## Answer Keys

23.1.1

A

## 24.1

## Circle (1)

## 24.1.1 Circle: ISRO CSE 2014 | Question: 9



Let  $R$  be the radius of the circle. What is the angle subtended by an arc of length  $R$  at the center of the circle?

- A. 1 degree  
C. 90 degrees  
B. 1 radian  
D.  $\pi$  radians

isro2014 circle geometry

Answer key

## 24.2

## Geometry (3)

## 24.2.1 Geometry: ISRO CSE 2011 | Question: 76



What is the matrix that represents rotation of an object by  $\theta^\circ$  about the origin in 2D?

- A.  

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$
B.  

$$\begin{pmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{pmatrix}$$
C.  

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \cos \theta & \sin \theta \end{pmatrix}$$
D.  

$$\begin{pmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{pmatrix}$$

isro2011 geometry

Answer key

## 24.2.2 Geometry: ISRO CSE 2014 | Question: 54



The conic section that is obtained when a right circular cone is cut through a plane that is parallel to the side of the cone is called \_\_\_\_\_

- A. parabola  
B. hyperbola  
C. circle  
D. ellipse

isro2014 non-gatecse geometry

Answer key

## 24.2.3 Geometry: ISRO CSE 2014 | Question: 65



A cube of side 1 unit is placed in such a way that the origin coincides with one of its top vertices and the three axes along three of its edges. What are the co-ordinates of the vertex which is diagonally opposite to the vertex whose co-ordinates are  $(1, 0, 1)$ ?

- A.  $(0, 0, 0)$   
B.  $(0, -1, 0)$   
C.  $(0, 1, 0)$   
D.  $(1, 1, 1)$

isro2014 geometry non-gatecse

Answer key

**Answer Keys**

|        |   |
|--------|---|
| 24.1.1 | B |
|--------|---|

|        |   |
|--------|---|
| 24.2.1 | A |
|--------|---|

|        |   |
|--------|---|
| 24.2.2 | A |
|--------|---|

|        |   |
|--------|---|
| 24.2.3 | B |
|--------|---|

### 25.1.1 Integrated Circuits: ISRO CSE 2007 | Question: 35



The output 0 and 1 level for TTL Logic family is approximately

- A. 0.1 and 5 V  
B. 0.6 and 3.5 V  
C. 0.9 and 1.75 V  
D. -1.75 and 0.9 V

isro2007   digital-logic   integrated-circuits   non-gatecse

Answer key 

### 25.1.2 Integrated Circuits: ISRO CSE 2008 | Question: 21



The advantage of MOS devices over bipolar devices is that

- A. it allows higher bit densities and also cost effective      B. it is easy to fabricate  
C. it is higher-impedance and operational speed                D. all of these

isro2008 non-gatecse integrated-circuits

Answer key 

## 25.1.3 Integrated Circuits: ISRO CSE 2011 | Question: 80



Logic family popular for low power dissipation

- A. CMOS                      B. ECL                      C. TTL                      D. DTL

isro2011 non-gatecse integrated-circuits

Answer key 

## 25.1.4 Integrated Circuits: ISRO CSE 2014 | Question: 46



If the maximum output voltage of a DAC is  $V$  volts and if the resolution is  $R$  bits then the weight of the most significant bit is \_\_\_\_\_

- A.  $V/(2^R - 1)$   
 B.  $(2^{R-1}).V/(2^R - 1)$   
 C.  $(2^{R-1}).V$   
 D.  $V/(2^{R-1})$

isro2014 non-gatecse integrated-circuits

Answer key 

## Answer Keys

|        |   |        |   |        |   |        |   |
|--------|---|--------|---|--------|---|--------|---|
| 25.1.1 | B | 25.1.2 | D | 25.1.3 | A | 25.1.4 | B |
|--------|---|--------|---|--------|---|--------|---|

## 26.1.1 Cmm Model: ISRO CSE 2017 | Question: 47



Capability maturity Model (CMM) is the methodology to

- A. develop and refine an organization's software development process
- B. develop the software
- C. test the software
- D. All of the above

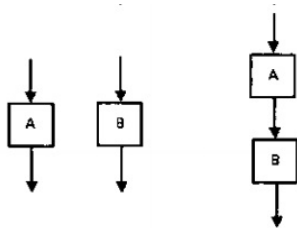
isro2017 is&software-engineering non-gatecse cmm-model

Answer key

## 26.2.1 Cyclomatic Complexity: GATE CSE 2010 | Question: 21,ISRO2017-60



The cyclomatic complexity of each of the modules A and B shown below is 10. What is the cyclomatic complexity of the sequential integration shown on the right hand side?



- A. 19
- B. 21
- C. 20
- D. 10

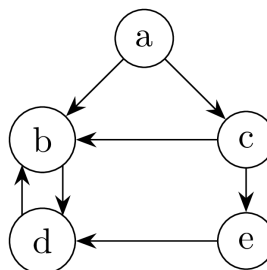
gatecse-2010 is&software-engineering cyclomatic-complexity easy isro2017

Answer key

## 26.2.2 Cyclomatic Complexity: ISRO CSE 2011 | Question: 65



The Cyclomatic Complexity metric  $V(G)$  of the following control flow graph



- A. 3
- B. 4
- C. 5
- D. 6

isro2011 is&software-engineering cyclomatic-complexity

Answer key

### 26.2.3 Cyclomatic Complexity: ISRO CSE 2014 | Question: 4



Consider the following pseudo- code

```
while (m<n)
  if (x>y ) and (a<b) then
    a=a+1
    y=y-1
  end if
m=m+1 end while
```

What is cyclomatic complexity of the above pseudo -code?

- A. 2                      B. 3                      C. 4                      D. 5

isro2014   is&software-engineering   cyclomatic-complexity   non-gatecse

Answer key

### 26.2.4 Cyclomatic Complexity: ISRO CSE 2020 | Question: 69



Consider the following pseudo-code

```
I=0; J=0; K=8;
while (I<K-1) //while-1
{
  J=J+1;
  while (J<K) //while-2
  {
    if (x[I]<x[J])
    {
      temp = x[I];
      x[I]=x[J];
      x[J]=temp;
    }
  } // end of while-2
  I=I+1;
} // end of while-1
```

The cyclomatic complexity of the above is

- A. 3                      B. 2                      C. 4                      D. 1

isro-2020   is&software-engineering   cyclomatic-complexity   normal

Answer key

### 26.2.5 Cyclomatic Complexity: ISRO CSE 2023 | Question: 61



McCabe's cyclomatic complexity number of a program control graph 'G' with  $e$  edges,  $n$  nodes and  $p$  disconnected paths is defined as

- A.  $n - e + 2p$                       B.  $e - n + 2 + p$   
C.  $e - n + 2p$                       D.  $n - e + p$

isro-cse-2023   is&software-engineering   non-gatecse   cyclomatic-complexity

Answer key

### 26.2.6 Cyclomatic Complexity: ISRO-2013-66



What is the cyclomatic complexity of a module which has seventeen edges and thirteen nodes?

- A. 4                      B. 5                      C. 6                      D. 7

isro2013   is&software-engineering   cyclomatic-complexity   non-gatecse

Answer key

**26.3.1 Is&software Engineering: ISRO 2008- ECE Software Engineering**

Which of the following relation is valid?

Where

MTBF = Mean Time Between Failures

MTTF = Mean Time To Failures

MTTR = Mean Time To Repair

- (a)  $MTBF = MTTF + MTTR$   
 (b)  $MTTR + MTTF + MTBF = 1$   
 (c)  $1/MTTR + 1/MTTF = 1/MTBF$   
 (d)  $MTBF \cdot MTTF \cdot MTTR = 1$

isro-ece is&software-engineering

Answer key

**26.3.2 Is&software Engineering: ISRO 2017-39**

Estimation at software development effort for organic software in basic COCOMO is:

- A.  $E = 2.0 \text{ (KLOC)}^{1.05} \text{ PM}$  B.  $E = 3.4 \text{ (KLOC)}^{1.06} \text{ PM}$   
 C.  $E = 2.4 \text{ (KLOC)}^{1.05} \text{ PM}$  D.  $E = 2.4 \text{ (KLOC)}^{1.07} \text{ PM}$

isro2017 is&software-engineering non-gatecse

Answer key

**26.3.3 Is&software Engineering: ISRO CSE 2007 | Question: 28**

The level of aggregation of information required for operational control is

- A. Detailed B. Aggregate  
 C. Qualitative D. None of the above

isro2007 non-gatecse is&software-engineering

Answer key

**26.3.4 Is&software Engineering: ISRO CSE 2007 | Question: 76**

Which of the following is not an approach to Software Process Assessment?

- A. SPICE(ISO/IEC15504) B. Standard CMMI Assessment Method for process improvement  
 C. ISO 9001:2000 D. IEEE 2000:2001

isro2007 is&software-engineering

Answer key

**26.3.5 Is&software Engineering: ISRO CSE 2007 | Question: 77**

A physical DFD specifies

- A. what processes will be used B. who generates data and who processes it  
 C. what each person in an organization does D. which data will be generated

isro2007 is&software-engineering

Answer key



### 26.3.6 Is&software Engineering: ISRO CSE 2011 | Question: 17



A context model of a software system can be shown by drawing a

- A. LEVEL-0 DFD      B. LEVEL-1 DFD      C. LEVEL-2 DFD      D. LEVEL-3 DFD

isro2011 is&software-engineering

Answer key

### 26.3.7 Is&software Engineering: ISRO CSE 2011 | Question: 51



Warnier Diagram enables the analyst to represent

- A. Class structure      B. Information Hierarchy  
C. Data Flow      D. State Transition

isro2011 is&software-engineering non-gatecse

Answer key

### 26.3.8 Is&software Engineering: ISRO CSE 2014 | Question: 20



Which of the following is not a maturity level as per Capability Maturity Model?

- A. Initial      B. Measurable      C. Repeatable      D. Optimized

is&software-engineering non-gatecse isro2014

Answer key

### 26.3.9 Is&software Engineering: ISRO CSE 2014 | Question: 69



For a software project, the spiral model was employed. When will the spiral stop?

- A. When the software product is retired  
B. When the software product is released after Beta testing  
C. When the risk analysis is completed  
D. After completing five loops

isro2014 is&software-engineering non-gatecse

Answer key

### 26.3.10 Is&software Engineering: ISRO CSE 2016 | Question: 61



The extent to which the s/w can continue to operate correctly despite the introduction of invalid inputs is called as

- A. Reliability      B. Robustness      C. Fault tolerance      D. Portability

is&software-engineering isro2016

Answer key

### 26.3.11 Is&software Engineering: ISRO CSE 2016 | Question: 62



Which one of the following is a functional requirement ?

- A. Maintainability      B. Portability  
C. Robustness      D. None of the mentioned

is&software-engineering isro2016

Answer key

### 26.3.12 Is&software Engineering: ISRO CSE 2016 | Question: 63



Configuration management is not concerned with

- A. controlling changes to the source code
- B. choice of hardware configuration for an application
- C. controlling documentation changes
- D. maintaining versions of software

non-gatecse is&software-engineering isro2016

Answer key

### 26.3.13 Is&software Engineering: ISRO CSE 2016 | Question: 65



A company needs to develop a digital signal processing software for one of its newest inventions. The software is expected to have 20000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.2 for the software development on embedded systems, while the exponentiation factor is given as 1.5. What is the estimated effort in person-months?

- A. 196.77
- B. 206.56
- C. 199.56
- D. 210.68

non-gatecse is&software-engineering isro2016

Answer key

### 26.3.14 Is&software Engineering: ISRO CSE 2017 | Question: 58



Which of the following is not a life cycle model?

- A. Spiral model
- B. Prototyping model
- C. Waterfall model
- D. Capability maturity model

isro2017 is&software-engineering non-gatecse

Answer key

### 26.3.15 Is&software Engineering: ISRO CSE 2017 | Question: 61



In software maintenance tackling the changes in the hardware or software environment where the software works, is

- A. Corrective maintenance
- B. Perfective maintenance
- C. Adaptive maintenance
- D. Preventive maintenance

isro2017 is&software-engineering non-gatecse

Answer key

### 26.3.16 Is&software Engineering: ISRO CSE 2017 | Question: 65



Which product metric gives the measure of the average length of words and sentence in documents?

- A. SCI number
- B. Cyclomatic complexity
- C. LOC
- D. Fog index

isro2017 is&software-engineering non-gatecse

Answer key

### 26.3.17 Is&software Engineering: ISRO CSE 2018 | Question: 47



In a particular program, it is found that 1% of the code account for 50% of the execution time. To code a program in C++, it takes 100 man-days. Coding in assembly language is 10 times harder than coding in C++, but runs 5 times faster. Converting an existing C++ program into an assembly language program is 4 times

faster.

To completely write the program in C++ and rewrite the 1% code in assembly language, if a project team needs 13 days, the team consist of

- a. 13 programmers
- b. 10 programmers
- c. 8 programmers
- d. 100/13 programmers

isro2018 is&software-engineering

Answer key 

### 26.3.18 Is&software Engineering: ISRO CSE 2018 | Question: 59



The lower degree of cohesion is kind of:

- a. Logical Cohesion
- b. Coincidental Cohesion
- c. Procedural Cohesion
- d. Communicational Cohesion

isro2018 is&software-engineering

Answer key 

### 26.3.19 Is&software Engineering: ISRO CSE 2018 | Question: 76



Cloaking is a search engine optimization (*SEO*) technique. During cloaking:

- a. Content presented to search engine spider is different from that presented to user's browser
- b. Content present to search engine spider and browser is same
- c. Contents of user's requested website are changed
- d. None of the above

isro2018 is&software-engineering

Answer key 

### 26.3.20 Is&software Engineering: ISRO CSE 2020 | Question: 64



Regression testing is primarily related to

- A. Functional testing
- B. Development testing
- C. Data flow testing
- D. Maintenance testing

isro-2020 is&software-engineering normal

Answer key 

### 26.3.21 Is&software Engineering: ISRO CSE 2020 | Question: 67



What is the availability of the software with the following reliability figures

Mean Time Between Failures (MTBF) is 20 days

Mean Time To Repair (MTTR) is 20 hours

- A. 90%
- B. 96%
- C. 24%
- D. 50%

isro-2020 is&software-engineering normal

Answer key 

### 26.3.22 Is&software Engineering: ISRO CSE 2020 | Question: 68



What is the defect rate for Six sigma?

- A. 1.0 defect per million lines of code
- B. 1.4 defects per million lines of code
- C. 3.0 defects per million lines of code
- D. 3.4 defects per million lines of code

isro-2020 is&software-engineering normal

Answer key 

### 26.3.23 Is&software Engineering: ISRO CSE 2020 | Question: 70



In a class definition with 10 methods, to make the class maximally cohesive, number of direct and indirect connections required among the methods are

- A. 90,0                      B. 45,0                      C. 10,10                      D. 45,45

isro-2020   is&software-engineering   normal

Answer key 

## 26.4

### Out of Gatecse Syllabus (1)

### 26.4.1 Out of Gatecse Syllabus: GATE CSE 2014 Set 3 | Question: 19, ISRO2017-78



In the context of modular software design, which one of the following combinations is desirable?

- A. High cohesion and high coupling                      B. High cohesion and low coupling  
C. Low cohesion and high coupling                      D. Low cohesion and low coupling

gatecse-2014-set3   is&software-engineering   easy   isro2017   out-of-gatecse-syllabus

Answer key 

## 26.5

### Project Cost (1)

### 26.5.1 Project Cost: GATE CSE 2011 | Question: 5, ISRO2016-64



A company needs to develop a strategy for software product development for which it has a choice of two programming languages L1 and L2. The number lines of code (LOC) developed using L2 is estimated to be twice of the LOC developed with L1. The product will have to be maintained for five years. Various parameters for the company are given in the table below.

| Parameter                        | Language L1   | Language L2  |
|----------------------------------|---------------|--------------|
| Man years needed for development | LOC/10000     | LOC/10000    |
| Development Cost per man year    | ₹ 10, 00, 000 | ₹ 7, 50, 000 |
| Maintenance Time                 | 5 years       | 5 years      |
| Cost of maintenance per year     | ₹ 1,00,000    | ₹ 50, 000    |

Total cost of the project includes cost of development and maintenance. What is the LOC for L1 for which of the cost of the project using L1 is equal to the cost of the project using L2?

- A. 4000                      B. 5000                      C. 4333                      D. 4667

gatecse-2011   is&software-engineering   project-cost   normal   isro2016

Answer key 

## 26.6

### Software Coupling (1)

### 26.6.1 Software Coupling: ISRO-2013-67



Which of the following types of coupling has the weakest coupling?

- A. Pathological coupling                      B. Control coupling  
C. Data coupling                      D. Message coupling

isro2013   is&software-engineering   software-coupling   non-gatecse

Answer key 

## 26.7

## Software Metrics (1)

## 26.7.1 Software Metrics: ISRO CSE 2018 | Question: 58



The Functions Point (FP) metric is

- A. Calculated from user requirement
- B. Calculated from lines of code
- C. Calculated from software complexity assessment
- D. None of the above

isro2018 software-metrics is&software-engineering non-gatecse

Answer key

## 26.8

## Software Productivity (1)

## 26.8.1 Software Productivity: ISRO2015-60



If in a software project the number of user input, user output, enquiries, files and external interfaces are (15, 50, 24, 12, 8), respectively, with complexity average weighing factor. The productivity if effort = 70 percent-month is

- A. 110.54
- B. 408.74
- C. 304.78
- D. 220.14

isro2015 is&software-engineering software-productivity non-gatecse

Answer key

## 26.9

## Software Reliability (1)

## 26.9.1 Software Reliability: ISRO CSE 2007 | Question: 79



Which of the following models used for software reliability

- A. Waterfall
- B. Musa
- C. COCOMO
- D. Rayleigh

isro2007 is&software-engineering software-reliability

Answer key

## 26.10

## Software Testing (10)

## 26.10.1 Software Testing: ISRO CSE 2007 | Question: 67



To execute all loops at their boundaries and within their operational bounds is an example of

- A. Black Box Testing
- B. Alpha Testing
- C. Recovery Testing
- D. White Box Testing

isro2007 is&software-engineering software-testing

Answer key

## 26.10.2 Software Testing: ISRO CSE 2009 | Question: 67



Activities which ensure that the software that has been built, is traceable to customer requirement is covered as part of

- A. Verification
- B. Validation
- C. Maintenance
- D. Modeling

isro2009 is&software-engineering software-testing

Answer key

## 26.10.3 Software Testing: ISRO CSE 2009 | Question: 68



A testing method which is normally used as the acceptance test for a software system, is

- A. Regression Testing
- C. Unit Testing

- B. Integration Testing
- D. System Testing

isro2009 is&software-engineering software-testing

Answer key 

#### 26.10.4 Software Testing: ISRO CSE 2011 | Question: 34



Black Box Software Testing method focuses on the

- A. Boundary condition of the software
- C. Functional requirement of the software
- B. Control structure of the software
- D. Independent paths of the software

isro2011 non-gatecse software-testing is&software-engineering

Answer key 

#### 26.10.5 Software Testing: ISRO CSE 2014 | Question: 23



The test suite (set of test input) used to perform unit testing on a module could cover 70% of the code. What is the reliability of the module if the probability of success is 0.95 during testing?

- A. 0.665 to 0.95
- C. At the most 0.95
- B. At the most 0.665
- D. At least 0.665

isro2014 is&software-engineering software-testing

Answer key 

#### 26.10.6 Software Testing: ISRO CSE 2018 | Question: 48



In unit testing of a module, it is found that for a set of test data, at the maximum 90% of the code alone were tested with the probability of success 0.9. The reliability of module is

- a. Greater than 0.9
- c. At most 0.81
- b. Equal to 0.9
- d. At least 0.81

isro2018 is&software-engineering software-testing

Answer key 

#### 26.10.7 Software Testing: ISRO CSE 2023 | Question: 39



The MCDC software testing technique stands for

- A. Multiple Conditions Decision Coverage
- C. Modified Condition Decision Coverage
- B. Multiple coverage Decision conditions
- D. None of the above

isro-cse-2023 non-gatecse software-testing

Answer key 

#### 26.10.8 Software Testing: ISRO-2013-68



Which of the following testing methods uses fault simulation technique?

- A. Unit testing
- C. Stress testing
- B. Beta testing
- D. Mutation testing

isro2013 is&software-engineering software-testing non-gatecse

Answer key 

#### 26.10.9 Software Testing: ISRO-2013-69, UGCNET-Dec2012-III: 62



If a program  $P$  calls two subprograms  $P_1$  and  $P_2$  and  $P_1$  can fail 50% of the time and  $P_2$  can fail 40% of the time, what is the failure rate of program  $P$ ?

- A. 50%                      B. 60%                      C. 70%                      D. 10%

isro2013   is&software-engineering   software-testing   non-gatecse   ugcnetcse-dec2012-paper3

Answer key 

## 26.10.10 Software Testing: ISRO2015-59



Alpha and Beta testing are forms of

- A. Acceptance testing                      B. Integration testing  
C. System testing                      D. Unit testing

isro2015   is&software-engineering   software-testing

Answer key 

## 26.11

### Spiral Model (1)

#### 26.11.1 Spiral Model: GATE IT 2007 | Question: 4, ISRO2016-66



In the Spiral model of software development, the primary determinant in selecting activities in each iteration is

- A. Iteration size  
B. Cost  
C. Adopted process such as Rational Unified Process or Extreme Programming  
D. Risk

gateit-2007   is&software-engineering   spiral-model   normal   isro2016

Answer key 

## 26.12

### Uml (3)

#### 26.12.1 Uml: ISRO CSE 2007 | Question: 65



Which of the following is not a UML DIAGRAM?

- A. Use Case                      B. Class Diagram  
C. Analysis Diagram                      D. Swimlane Diagram

isro2007   is&software-engineering   uml

Answer key 

#### 26.12.2 Uml: ISRO CSE 2007 | Question: 78



In UML diagram of a class

- A. state of object cannot be represented                      B. state is irrelevant  
C. state is represented as an attribute                      D. state is represented as a result of an operation

isro2007   is&software-engineering   uml

Answer key 

#### 26.12.3 Uml: ISRO CSE 2011 | Question: 28



The above figure represents which one of the following UML diagram for a single send session of an online chat system?

A. Package diagram  
C. Class diagram

isro2011 is&software-engineering uml

Answer key 

B. Activity diagram  
D. Sequence diagram

## Answer Keys

|          |     |
|----------|-----|
| 26.1.1   | A   |
| 26.2.5   | Q-Q |
| 26.3.4   | D   |
| 26.3.9   | A   |
| 26.3.14  | D   |
| 26.3.19  | A   |
| 26.4.1   | B   |
| 26.9.1   | D   |
| 26.10.5  | B   |
| 26.10.10 | C   |

|         |   |
|---------|---|
| 26.2.1  | A |
| 26.2.6  | C |
| 26.3.5  | B |
| 26.3.10 | B |
| 26.3.15 | C |
| 26.3.20 | D |
| 26.5.1  | B |
| 26.10.1 | D |
| 26.10.6 | C |
| 26.11.1 | D |

|         |     |
|---------|-----|
| 26.2.2  | B   |
| 26.3.1  | Q-Q |
| 26.3.6  | A   |
| 26.3.11 | D   |
| 26.3.16 | D   |
| 26.3.21 | B   |
| 26.6.1  | D   |
| 26.10.2 | B   |
| 26.10.7 | Q-Q |
| 26.12.1 | C   |

|         |   |
|---------|---|
| 26.2.3  | C |
| 26.3.2  | C |
| 26.3.7  | B |
| 26.3.12 | B |
| 26.3.17 | C |
| 26.3.22 | D |
| 26.7.1  | C |
| 26.10.3 | D |
| 26.10.8 | D |
| 26.12.2 | C |

|         |   |
|---------|---|
| 26.2.4  | C |
| 26.3.3  | A |
| 26.3.8  | B |
| 26.3.13 | A |
| 26.3.18 | B |
| 26.3.23 | B |
| 26.8.1  | B |
| 26.10.4 | C |
| 26.10.9 | C |
| 26.12.3 | B |



## 27.1.1 Java: ISRO CSE 2011 | Question: 44



In Java, after executing the following code what are the values of x, y and z?

```
int x,y=10; z=12;
```

```
x=y++ + z++;
```

A. x=22, y=10, z=12

B. x=24, y=10, z=12

C. x=24, y=11, z=13

D. x=22, y=11, z=13

isro2011 non-gatecse java

Answer key

## 27.1.2 Java: ISRO CSE 2011 | Question: 72



Find the output of the following Java code line

```
System.out.println(Math.floor(-7.4))
```

A. -7

B. -8

C. -7.4

D. -7.0

isro2011 non-gatecse java

Answer key

## 27.1.3 Java: ISRO CSE 2014 | Question: 29



Consider the following Java code fragment:

```
public class While
{
    public void loop()
    {
        int x = 0;
        while(1)
        {
            System.out.println("x plus one is" +(x+1));
        }
    }
}
```

A. There is syntax error in line no. 1

B. There is syntax errors in line nos. 1 & 6

C. There is syntax error in line no. 8

D. There is syntax error in line no. 6

isro2014 java non-gatecse

Answer key

## 27.1.4 Java: ISRO CSE 2017 | Question: 74



Which of these is a super class of all errors and exceptions in the Java language?

A. Runtime Exceptions

B. Throwable

C. Catchable

D. None of the above

isro2017 java

Answer key

### 27.1.5 Java: ISRO-2013-64



What is the output of the following program?

```
Class Test
{
    public static void main (String [] args)
    {
        int x = 0;
        int y = 0
        for (int z = 0; z < 5; z++)
        {
            if ((++x > 2) || (++y > 2))
            {
                x++;
            }
        }
        System.out.println (x+ " " + y);
    }
}
```

A. 8 2

B. 8 5

C. 8 3

D. 5 3

isro2013 java non-gatecse

Answer key

### 27.1.6 Java: ISRO-2013-79



The built-in base class in java, which is used to handle all exceptions is

A. Raise

B. Exception

C. Error

D. Throwable

isro2013 java non-gatecse

Answer key

## Answer Keys

|        |   |        |   |        |   |        |   |        |   |
|--------|---|--------|---|--------|---|--------|---|--------|---|
| 27.1.1 | D | 27.1.2 | B | 27.1.3 | D | 27.1.4 | B | 27.1.5 | A |
| 27.1.6 | D |        |   |        |   |        |   |        |   |

## 28.1.1 Multimedia: ISRO2015-66



Consider an uncompressed stereo audio signal of CD quality which is sampled at 44.1 kHz and quantized using 16 bits. What is required storage space if a compression ratio of 0.5 is achieved for 10 seconds of this audio?

- A. 172 KB      B. 430 KB      C. 860 KB      D. 1720 KB

isro2015   multimedia   non-gatecse   audio-compression

Answer key

## 28.1.2 Multimedia: ISRO2015-67



What is compression ratio in a typical mp3 audio files?

- A. 4:1      B. 6:1      C. 8:1      D. 10:1

isro2015   multimedia   audio-compression

Answer key

## Answer Keys

28.1.1

C

28.1.2

D

## 29.0.1 ISRO CSE 2009 | Question: 46



The shift operator  $E$  is defined as  $E[f(x_i)] = f(x_i + h)$  and  $E'[f(x_i)] = f(x_i - h)$  then  $\Delta$  (forward difference) in terms of  $E$  is

- A.  $E - 1$                       B.  $E$                       C.  $1 - E^{-1}$                       D.  $1 - E$

isro2009

Answer key

## 29.1

## Guass Seidal Iterative Method (1)

## 29.1.1 Guass Seidal Iterative Method: ISRO-2013-48



The Guass-Seidal iterative method can be used to solve which of the following sets?

- A. Linear algebraic equations                      B. Linear and non-linear algebraic equations  
C. Linear differential equations                      D. Linear and non-linear differential equations

isro2013   numerical-methods   guass-seidal-iterative-method

Answer key

## 29.2

## Interpolation (1)

## 29.2.1 Interpolation: ISRO CSE 2011 | Question: 52



Given

X: 0   10   16

Y: 6   16   28

The interpolated value  $X=4$  using piecewise linear interpolation is

- A. 11                      B. 4                      C. 22                      D. 10

isro2011   interpolation   non-gatecse

Answer key

## 29.3

## Newton Raphson (1)

## 29.3.1 Newton Raphson: ISRO CSE 2017 | Question: 3



Using Newton-Raphson method, a root correct to 3 decimal places of  $x^3 - 3x - 5 = 0$

- A. 2.222                      B. 2.275                      C. 2.279                      D. None of the above

isro2017   newton-raphson   non-gatecse

Answer key

## 29.4

## Numerical Methods (4)

## 29.4.1 Numerical Methods: GATE CSE 2006 | Question: 1, ISRO2009-57



Consider the polynomial  $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$ , where  $a_i \neq 0, \forall i$ . The minimum number of multiplications needed to evaluate  $p$  on an input  $x$  is:

- A. 3                      B. 4                      C. 6                      D. 9

gatecse-2006   numerical-methods   normal   isro2009

Answer key 

29.4.2 Numerical Methods: ISRO CSE 2009 | Question: 44



A root  $\alpha$  of equation  $f(x) = 0$  can be computed to any degree of accuracy if a 'good' initial approximation  $x_0$  is chosen for which

- A.  $f(x_0) > 0$   
 B.  $f(x_0)f''(x_0) > 0$   
 C.  $f(x_0)f''(x_0) < 0$   
 D.  $f''(x_0) > 0$

isro2009 numerical-methods

Answer key 

29.4.3 Numerical Methods: ISRO CSE 2009 | Question: 47



The formula  $\int_{x_0}^{x_n} y(n)dx \simeq h/2(y_0 + 2y_1 + \dots + 2y_{n-1} + y_n) - h/12(\nabla y_n - \Delta y_0) - h/24(\nabla^2 y_n + \Delta^2 y_0) - 19h/720(\nabla^3 y_n - \Delta^3 y_0) \dots$  is called

- A. Simpson rule  
 B. Trapezoidal rule  
 C. Romberg's rule  
 D. Gregory's formula

isro2009 numerical-methods non-gatecse

Answer key 

29.4.4 Numerical Methods: ISRO CSE 2009 | Question: 51



The formula  $P_k = y_0 + k\nabla y_0 + \frac{k(k+1)}{2}\nabla^2 y_0 + \dots + \frac{k \dots (k+n-1)}{n!}\nabla^n y_0$  is

- A. Newton's backward formula  
 B. Gauss forward formula  
 C. Gauss backward formula  
 D. Stirling's formula

isro2009 numerical-methods

Answer key 

29.5 Polynomials (1)

29.5.1 Polynomials: ISRO CSE 2009 | Question: 48



The cubic polynomial  $y(x)$  which takes the following values:  $y(0) = 1, y(1) = 0, y(2) = 1$  and  $y(3) = 10$  is

- A.  $x^3 + 2x^2 + 1$   
 B.  $x^3 + 3x^2 - 1$   
 C.  $x^3 + 1$   
 D.  $x^3 - 2x^2 + 1$

isro2009 polynomials

Answer key 

## Answer Keys

|        |     |        |   |        |   |        |   |        |   |
|--------|-----|--------|---|--------|---|--------|---|--------|---|
| 29.0.1 | A   | 29.1.1 | A | 29.2.1 | D | 29.3.1 | C | 29.4.1 | A |
| 29.4.2 | Q-Q | 29.4.3 | B | 29.4.4 | A | 29.5.1 | D |        |   |

## 30.1

## Copy Constructor (1)

## 30.1.1 Copy Constructor: ISRO-2013-75



What is the right way to declare a copy constructor of a class if the name of the class is MyClass?

- A. MyClass (constant MyClass \* arg)
- B. MyClass (constant MyClass & arg)
- C. MyClass (MyClass arg)
- D. MyClass (MyClass \* arg)

isro2013 copy-constructor non-gatecse

Answer key

## 30.2

## Java (1)

## 30.2.1 Java: ISRO CSE 2008 | Question: 43



Consider the following code segment:

```
for (int k=0; k<20; k=k+2)
{
    if (k % 3 == 1)
        system.out.print(k+ " ");
}
```

What is printed as a result of executing the code segment?

- A. 4 16
- B. 4 10 16
- C. 0 6 12 18
- D. 1 4 7 10 13 16 19

isro2008 java

Answer key

## 30.3

## Object Oriented Programming (6)

## 30.3.1 Object Oriented Programming: ISRO CSE 2009 | Question: 33



The feature in object-oriented programming that allows the same operation to be carried out differently, depending on the object, is

- A. Inheritance
- B. Polymorphism
- C. Overfunctioning
- D. Overriding

isro2009 object-oriented-programming non-gatecse

Answer key

## 30.3.2 Object Oriented Programming: ISRO CSE 2016 | Question: 78



If a class  $C$  is derived from class  $B$ , which is derived from class  $A$ , all through public inheritance, then a class  $C$  member function can access

- A. only protected and public data of  $C$  and  $B$
- B. Only protected and public data of  $C$
- C. all data of  $C$  and private data of  $A$  and  $B$
- D. public and protected data of  $A$  and  $B$  and all data of  $C$

non-gatecse object-oriented-programming isro2016

Answer key

## 30.3.3 Object Oriented Programming: ISRO CSE 2017 | Question: 41



What is the output of the C++ program?

```
#include <iostream>
using namespace std;
```

```

void square(int *x) {
    *x = (*x)++ * (*x);
}

void square(int *x, int *y) {
    *x = (*x) * --(*y);
}

int main()
{
    int number = 30;
    square(&number, &number);
    cout<<number;
    return 0;
}

```

- A. 910                      B. 920                      C. 870                      D. 900

isro2017   object-oriented-programming

Answer key 

### 30.3.4 Object Oriented Programming: ISRO CSE 2017 | Question: 43



Which of the following UML 2.0 diagrams capture behavioral aspects of a system?

- A. Use case diagram, Object diagram, Activity diagram and state machine diagram  
 B. Use case diagram, Activity diagram and state machine diagram  
 C. Object diagram, Communication Diagram, Timing diagram and Interaction diagram  
 D. Object diagram, Composite structure diagram, package diagram and Deployment diagram

isro2017   object-oriented-programming   non-gatecse

Answer key 

### 30.3.5 Object Oriented Programming: ISRO CSE 2017 | Question: 44



Which of the following is associated with objects?

- A. State                      B. Behavior                      C. Identity                      D. All of the above

isro2017   object-oriented-programming   non-gatecse

Answer key 

### 30.3.6 Object Oriented Programming: ISRO CSE 2018 | Question: 43



Consider the following C++ program

```

int a (int m)
{return ++m;}
int b(int&m)
{return ++m;}
int{char &m}
{return ++m;}
void main()
{
    int p = 0, q=0, r = 0;
    p += a(b(p)) ;
    q+= b(a(q));
    r+=a(c(r));
    cout<<p<<q << r;
}

```

Assuming the required header first are already included, the above program

- a. results in compilation error                      b. print 123  
 c. print 111                      d. print 322

Answer key 

## 30.4

## Operator Overloading (1)

## 30.4.1 Operator Overloading: ISRO CSE 2017 | Question: 42



Which of the following operator(s) cannot be overloaded?

- A. . (member Access or Dot operator)                      B. ?: (ternary or Conditional Operator)  
 C. :: ( Scope Resolution Operator)                      D. All of the above

isro2017 operator-overloading

Answer key 

## 30.5

## Virtual Function (1)

## 30.5.1 Virtual Function: ISRO CSE 2008 | Question: 73



We can make a class abstract by

- A. Declaring it abstract using the virtual keyword  
 B. Making at least one member function as virtual function  
 C. Making at least one member function as pure virtual function  
 D. Making all member function const.

isro2008 non-gatecse virtual-function

Answer key 

## Answer Keys

|        |   |        |   |        |   |        |   |        |   |
|--------|---|--------|---|--------|---|--------|---|--------|---|
| 30.1.1 | B | 30.2.1 | B | 30.3.1 | B | 30.3.2 | D | 30.3.3 | C |
| 30.3.4 | B | 30.3.5 | D | 30.3.6 | A | 30.4.1 | D | 30.5.1 | C |



## 31.0.1 ISRO-DEC2017-77



Consider the following :

|           |                                |           |                     |
|-----------|--------------------------------|-----------|---------------------|
| <b>A.</b> | Condition Coverage             | <b>p.</b> | Black box testing   |
| <b>B.</b> | Equivalence Class partitioning | <b>q.</b> | System testing      |
| <b>C.</b> | Volume Testing                 | <b>r.</b> | White box testing   |
| <b>D.</b> | Beta Testing                   | <b>s.</b> | Performance testing |

Matching *A, B, C, D* in the same order gives,

- A. r,p,s,q                      B. p,r,q,s                      C. s,r,q,p                      D. q,r,s,p

isrodec2017

Answer key

## 31.0.2 ISRO-DEC2017-79



Which of the following related to snowflake schema is true?

- A. Each dimension is represented by a single dimensional table  
 B. Maintenance efforts are less  
 C. Dimension tables are normalized  
 D. It is not an extension of star schema

isrodec2017

Answer key

## 31.1

## Rs232 (1)

## 31.1.1 Rs232: ISRO-2013-37



The voltage ranges for a logic high and a logic low in RS- 232 C standard is

- A. Low is  $0.0V$  to  $1.8V$ , High is  $2.0V$  to  $5.0V$   
 B. Low is  $-15.0V$  to  $-3.0V$ , High is  $3.0V$  to  $15.0V$   
 C. Low is  $3.0V$  to  $15.0V$ , High is  $-3.0V$  to  $-15.0V$   
 D. Low is  $2.0V$  to  $5.0V$ , High is  $0.0V$  to  $1.8V$

isro2013 rs232

Answer key

## Answer Keys

31.0.1

A

31.0.2

C

31.1.1

C

## 32.1

## Html (4)

## 32.1.1 Html: GATE IT 2008 | Question: 19, ISRO2016-76



Which of the following is TRUE only of XML but NOT HTML?

- A. It is derived from SGML
- B. It describes content and layout
- C. It allows user defined tags
- D. It is restricted only to be used with web browsers

gateit-2008 web-technologies html normal isro2016

Answer key

## 32.1.2 Html: ISRO CSE 2011 | Question: 75



In HTML, which of the following can be considered a container?

- A. <SELECT>
- B. <Value>
- C. <INPUT>
- D. <BODY>

isro2011 non-gatecse web-technologies html

Answer key

## 32.1.3 Html: ISRO CSE 2017 | Question: 75



Choose the most appropriate HTML tag in the following to create a numbered list

- A. <dl>
- B. <list>
- C. <ul>
- D. <ol>

isro2017 web-technologies html non-gatecse

Answer key

## 32.1.4 Html: ISRO2015-45



To add a background color for all <h1> elements, which of the following HTML syntax is used

- A. h1 { background-color :#FFFFFF }
- B. { background-color :#FFFFFF } . h1
- C. h1 { background-color :#FFFFFF } . h1(all)
- D. h1. all{bgcolor= #FFFFFF }

isro2015 html non-gatecse

Answer key

## 32.2

## Javascript (1)

## 32.2.1 Javascript: ISRO2015-46



The correct syntax to write "Hi there" in Javascript is

- A. jsript.write ("Hi There")
- B. response.write ("Hi There")
- C. print ("Hi There")
- D. print.jsript ("Hi There")

javascript non-gatecse isro2015

Answer key

## 32.3

## Web Technologies (3)

## 32.3.1 Web Technologies: ISRO CSE 2007 | Question: 72



An email contains a textual birthday greeting, a picture of a cake and a song. The order is not important. What is the content-type?

- A. Multipart/mixed
- C. Multipart/digest

isro2007 web-technologies

Answer key 

- B. Multipart/parallel
- D. Multipart/alternative

### 32.3.2 Web Technologies: ISRO CSE 2014 | Question: 76



A web client sends a request to a web server. The web server transmits a program to that client and is executed at client. It creates a web document. What are such web documents called?

- A. Active
- B. Static
- C. Dynamic
- D. Passive

isro2014 web-technologies non-gatecse

Answer key 

### 32.3.3 Web Technologies: ISRO CSE 2023 | Question: 38



CAPTCHA is used to provide protection from

- A. Zero day attack
- B. Buffer overflow
- C. Automated scripted attack
- D. Man in the middle attack

isro-cse-2023 non-gatecse network-security web-technologies

Answer key 

## 32.4

## Xml (2)

### 32.4.1 Xml: ISRO CSE 2017 | Question: 40



XPath is used to navigate through elements and attributes in

- A. XSL document
- B. XML document
- C. XHTML document
- D. XQuery document

isro2017 web-technologies xml non-gatecse

Answer key 

### 32.4.2 Xml: ISRO2015-47



To declare the version of XML, the correct syntax is

- A. `<?xml version='1.0'/>`
- B. `<*xml version='1.0'/>`
- C. `<?xml version="1.0"/>`
- D. `<xml version='1.0'/>`

isro2015 xml non-gatecse

Answer key 

## Answer Keys

|        |   |        |   |        |     |        |   |        |   |
|--------|---|--------|---|--------|-----|--------|---|--------|---|
| 32.1.1 | C | 32.1.2 | D | 32.1.3 | D   | 32.1.4 | A | 32.2.1 | X |
| 32.3.1 | B | 32.3.2 | A | 32.3.3 | Q-Q | 32.4.1 | B | 32.4.2 | C |

**33.0.1 ISRO CSE 2008 | Question: 53**

Thrashing

- A. always occurs on large computers
- B. is a natural consequence of virtual memory systems
- C. can always be avoided by swapping
- D. can be caused by poor paging algorithms

isro2008 operating-system paging

Answer key

**33.0.2 ISRO CSE 2008 | Question: 52**

Dynamic address translation

- A. is part of the operating system paging algorithm
- B. is useless when swapping is used
- C. is the hardware necessary to implement paging
- D. storage pages at a specific location on disk

isro2008 operating-system paging

Answer key

**33.0.3 ISRO CSE 2009 | Question: 8**

The primary purpose of an operating system is

- A. To make most efficient use of the computer hardware
- B. To allow people to use the computer
- C. To keep systems programmers employed
- D. To make computers easier to use

isro2009 operating-system

Answer key

**33.0.4 ISRO CSE 2007 | Question: 25**

What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

- A. Partitioning
- B. Multi-tasking
- C. Windowing
- D. Paging

isro2007 operating-system paging

Answer key

**33.0.5 ISRO CSE 2008 | Question: 54**

What is the name of the operating system that reads and reacts in terms of operating system?

- A. Batch system
- B. Quick response time
- C. real time system
- D. Time sharing system

isro2008 operating-system

Answer key 

### 33.0.6 ISRO CSE 2008 | Question: 58



Checkpointing a job

- A. allows it to be completed successfully
- B. allows it to continue executing later
- C. prepares it for finishing
- D. occurs only when there is an error in it

isro2008 operating-system

Answer key 

### 33.0.7 ISRO CSE 2009 | Question: 11



A page fault

- A. Occurs when a program accesses an available page on memory
- B. is an error in a specific page
- C. is a reference to a page belonging to another program
- D. occurs when a program accesses a page not currently in memory

isro2009 operating-system page-fault

Answer key 

### 33.0.8 ISRO-2013-52 - Need clarification



Suppose we have variable logical records of lengths of 55 bytes, 1010 bytes and 2525 bytes while the physical block size in disk is 1515 bytes. What is the maximum and minimum fragmentation seen in bytes?

1. 2525 and 55
2. 1515 and 55
3. 1515 and 00
4. 1010 and 5

what is the answer c or D?

i guess it should be C. Here they haven't mentioned that we have 2 blocks. All those who is supporting answer D uses 2 blocks.

Some please give the concrete solution.

isro2013 operating-system

### 33.0.9 ISRO 2011- ECE Operating System



\_\_\_\_\_ is a primitive that can execute code. It contains an instruction pointer (=program counter) and sometimes has its own stack.

- a) *Process*
- b) *Task*
- c) *Kernel*
- d) *Thread*

isro-ece isro2011-ece operating-system

Answer key 

### 33.0.10 ISRO 2011-ECE Operating System



A run-time stack cannot be used in a round-robin scheduling system because of the \_\_\_\_\_ nature of scheduling.

- a. LIFO (Last in First out)
- b. FIFO (First in First out)
- c. FILO (First in Last out)
- d. None of the above

isro-ece isro2011-ece operating-system

Answer key

### 33.0.11 ISRO 2011-ECE Operating system



In a real time system, the simplest scheme that allows the operating system to allocate memory to two processes simultaneously is \_\_\_\_\_

- a. Over lays
- b. Pipeline
- c. Swapping
- d. None of the above

isro-ece operating-system isro2011-ece

Answer key

### 33.0.12 ISRO 2015- OS [EE]



OS that permits multiples programs to run simultaneously using **single** processor is referred as

- (a) Multitasking
- (b) Multi user
- (c) Multithreading
- (d) Multiprocessing

isro-ee operating-system

Answer key

### 33.0.13 ISRO CSE 2017 | Question: 52



Given reference to the following pages by a program

**0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3**

How many page faults will occur if the program has three page frames available to it and uses an optimal replacement?

- A. 7
- B. 8
- C. 9
- D. None of the above

isro2017 operating-system paging

Answer key

### 33.0.14 ISRO-DEC2017-58



A virtual memory system uses *FIFO* page replacement policy and allocates a fixed number of frames to the process. Consider the following statements

Increasing the number of page frames  
**M:** allocated to a process sometimes increases the page fault rate.

**N:** Some programs do not exhibit locality of reference.

Which one of the following is true?

- A. Both *M* and *N* are true and *N* is the reason for *M*.
- B. Both *M* and *N* are true but *N* is not the reason for *M*.
- C. Both *M* and *N* are false.
- D. *M* is false, but *N* is true.

isrodec2017

Answer key

### 33.0.15 ISRO-DEC2017-63



Which of the following is false?

- A. User level threads are not scheduled by the kernel.
- B. Context switching between user-level threads is faster than context switching between kernel-level threads.
- C. When a user thread is blocked all other threads of its processes are blocked.
- D. Kernel-level threads cannot utilize multiprocessor systems by splitting threads on different processors or cores.

isrodec2017

Answer key

### 33.0.16 ISRO-DEC2017-61



Consider the following scheduling:

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>A.</b> Gang scheduling           | <b>s.</b> Guaranteed scheduling |
| <b>B.</b> Rate monotonic scheduling | <b>t.</b> Thread scheduling     |
| <b>C.</b> Fair share scheduling     | <b>u.</b> Real-time scheduling  |

Matching the table in the order *A, B, C* gives

- A. t,u,s
- B. s,t,u
- C. u,s,t
- D. u,t,s

isrodec2017

Answer key

### 33.0.17 ISRO-DEC2017-62



A system uses *FIFO* policy for page replacement. It has 4 page frames with no pages loaded, to begin with. The system first accesses 50 distinct pages in some order and then accesses the same 50 pages in reverse order. How many page faults will occur?

- A. 96
- B. 100
- C. 97
- D. 92

isrodec2017

Answer key 

### 33.0.18 ISRO-DEC2017-60



A process executes the following code

```
for (i=0; i<n; i++)  
    fork();
```

The total number of child processes created is

- A.  $n^2$                       B.  $2^{n+1} - 1$                       C.  $2^n$                       D.  $2^n - 1$

isrodec2017

### 33.0.19 ISRO-DEC2017-69



Consider a disk sequence with 100 cylinders. The request to access the cylinder occur in the following sequence :

4, 34, 10, 7, 19, 73, 2, 15, 6, 20

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 2ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

- A. 190                      B. 238                      C. 233                      D. 276

isrodec2017

Answer key 

### 33.0.20 ISRO-DEC2017-64



Which of the following is not true with respect to deadlock prevention and deadlock avoidance schemes?

- A. In deadlock prevention, the request for resources is always granted if resulting state is safe.
- B. In deadlock avoidance, the request for resources is always granted. if the resulting state is safe.
- C. Deadlock avoidance requires knowledge of resource requirements a priori.
- D. Deadlock prevention is more restrictive than deadlock avoidance.

isrodec2017

Answer key 

### 33.0.21 ISRO CSE 2018 | Question: 20



Determine the number of page faults when references to pages occur in the order - 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 having brought earlier than page 2. (assume LRU i.e. Least-Recently-Used algorithm is applied)

- a. 3                      b. 4                      c. 5                      d. None of the above

isro2018    operating-system    page-fault

Answer key 

### 33.0.22 ISRO CSE 2023 | Question: 57



State whether true or false:

- i. Virtualization facilitates multiple OS simultaneously on the same hardware
- ii. In dual boot, multiple OS can run simultaneously

- A. (i) True (ii) False                      B. (i) False (ii) True  
C. (i) True (ii) True                      D. (i) False (ii) False



Answer key 

## 33.1

## Bankers Algorithm (1)

## 33.1.1 Bankers Algorithm: ISRO CSE 2017 | Question: 48



What problem is solved by Dijkstra banker' algorithm?

- A. Mutual exclusion
- B. Deadlock recovery
- C. Deadlock avoidance
- D. Cache coherence

isro2017 operating-system bankers-algorithm

Answer key 

## 33.2

## Concurrency (1)

## 33.2.1 Concurrency: ISRO-2013-62



A particular parallel program computation requires 100 seconds when executed on a single CPU. If 20% of this computation is strictly sequential, then theoretically the best possible elapsed times for this program running on 2 CPUs and 4 CPUs respectively are

- A. 55 and 45 seconds
- B. 80 and 20 seconds
- C. 75 and 25 seconds
- D. 60 and 40 seconds

isro2013 operating-system concurrency

Answer key 

## 33.3

## Context Switch (1)

## 33.3.1 Context Switch: GATE CSE 2000 | Question: 1.20, ISRO2008-47



Which of the following need not necessarily be saved on a context switch between processes?

- A. General purpose registers
- B. Translation look-aside buffer
- C. Program counter
- D. All of the above

gatecse-2000 operating-system easy isro2008 context-switch

Answer key 

## 33.4

## Critical Section (1)

## 33.4.1 Critical Section: ISRO CSE 2017 | Question: 68



A critical region

- A. is a piece of code which only one process executes at a time
- B. is a region prone to deadlock
- C. is a piece of code which only a finite number of processes execute
- D. is found only in windows NT operating system

isro2017 operating-system critical-section process-synchronization

Answer key 

## 33.5

## Deadlock Prevention Avoidance Detection (3)

## 33.5.1 Deadlock Prevention Avoidance Detection: ISRO CSE 2009 | Question: 77



When a process is rolled back as a result of deadlock the difficulty which arises is

- A. Starvation
- C. Low device utilization

- B. System throughput
- D. Cycle stealing

isro2009 deadlock-prevention-avoidance-detection

Answer key 

### 33.5.2 Deadlock Prevention Avoidance Detection: ISRO CSE 2018 | Question: 21



Consider a system having  $m$  resources of the same type. These resources are shared by 3 processes  $A, B, C$ , which have peak time demands of 3, 4, 6 respectively. The minimum value of  $m$  that ensures that deadlock will never occur is:

- a. 11
- b. 12
- c. 13
- d. 14

isro2018 deadlock-prevention-avoidance-detection operating-system

Answer key 

### 33.5.3 Deadlock Prevention Avoidance Detection: ISRO CSE 2020 | Question: 29



An aid to determine the deadlock occurrence is

- A. resource allocation graph
- B. starvation graph
- C. inversion graph
- D. none of the above

isro-2020 operating-system deadlock-prevention-avoidance-detection easy

Answer key 

## 33.6 Disk (11)

### 33.6.1 Disk: GATE CSE 1999 | Question: 2-18, ISRO2008-46



Raid configurations of the disks are used to provide

- A. Fault-tolerance
- B. High speed
- C. High data density
- D. (A) & (B)

gate1999 operating-system disk easy isro2008

Answer key 

### 33.6.2 Disk: GATE CSE 2003 | Question: 25, ISRO2009-12



Using a larger block size in a fixed block size file system leads to

- A. better disk throughput but poorer disk space utilization
- B. better disk throughput and better disk space utilization
- C. poorer disk throughput but better disk space utilization
- D. poorer disk throughput and poorer disk space utilization

gatecse-2003 operating-system disk normal isro2009

Answer key 

### 33.6.3 Disk: GATE CSE 2007 | Question: 11, ISRO2009-36, ISRO2016-21



Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

- A. 256 Mbyte, 19 bits
- B. 256 Mbyte, 28 bits
- C. 512 Mbyte, 20 bits
- D. 64 Gbyte, 28 bits

gatecse-2007 operating-system disk normal isro2016

Answer key 

#### 33.6.4 Disk: GATE IT 2007 | Question: 44, ISRO2015-34



A hard disk system has the following parameters :

- Number of tracks = 500
- Number of sectors/track = 100
- Number of bytes /sector = 500
- Time taken by the head to move from one track to adjacent track =  $1\text{ ms}$
- Rotation speed =  $600\text{ rpm}$ .

What is the average time taken for transferring 250 bytes from the disk ?

- A.  $300.5\text{ ms}$       B.  $255.5\text{ ms}$       C.  $255\text{ ms}$       D.  $300\text{ ms}$

gateit-2007   operating-system   disk   normal   isro2015

Answer key 

#### 33.6.5 Disk: ISRO CSE 2007 | Question: 39



Disk requests are received by a disk drive for cylinder 5, 25, 18, 3, 39, 8 and 35 in that order. A seek takes 5 msec per cylinder moved. How much seek time is needed to serve these requests for a Shortest Seek First (SSF) algorithm? Assume that the arm is at cylinder 20 when the last of these requests is made with none of the requests yet served

- A. 125 msec      B. 295 msec      C. 575 msec      D. 750 msec

isro2007   operating-system   disk

Answer key 

#### 33.6.6 Disk: ISRO CSE 2007 | Question: 56



Which of the following RAID level provides the highest Data Transfer Rate (Read/Write)

- A. RAID 1      B. RAID 3      C. RAID 4      D. RAID 5

isro2007   operating-system   disk

Answer key 

#### 33.6.7 Disk: ISRO CSE 2008 | Question: 49



The total time to prepare a disk drive mechanism for a block of data to be read from it is

- A. seek time      B. latency  
C. latency plus seek time      D. transmission time

isro2008   operating-system   disk

Answer key 

#### 33.6.8 Disk: ISRO CSE 2009 | Question: 36



Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stores in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively

- A. 256 Mbyte, 19 bits      B. 256 Mbute, 28 bits  
C. 512 Mbyte, 20 bits      D. 64 Gbyte, 28 bits

isro2009   operating-system   disk

Answer key 

### 33.6.9 Disk: ISRO CSE 2011 | Question: 38



A fast wide SCSI-II disk drive spins at 7200 RPM, has a sector size of 512 bytes, and holds 160 sectors per track. Estimate the sustained transfer rate of this drive

- A. 576000 Kilobytes / sec
- B. 9600 Kilobytes / sec
- C. 4800 Kilobytes / sec
- D. 19200 Kilobytes / sec

isro2011 operating-system disk

Answer key

### 33.6.10 Disk: ISRO CSE 2018 | Question: 23



Disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38 at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms/cylinder. The total seek time, if the disk arm scheduling algorithms is first-come-first-served is

- a. 360
- b. 850
- c. 900
- d. None of the above

isro2018 disk operating-system

Answer key

### 33.6.11 Disk: ISRO CSE 2018 | Question: 50



A particular disk unit uses a bit string to record the occupancy or vacancy of its tracks, with 0 denoting vacant and 1 for occupied. A 32-bit segment of this string has hexadecimal value D4FE2003. The percentage of occupied tracks for the corresponding part of the disk, to the nearest percentage is:

- a. 12
- b. 25
- c. 38
- d. 44

isro2018 disk operating-system

Answer key

## 33.7

### Disk Scheduling (2)

#### 33.7.1 Disk Scheduling: ISRO CSE 2014 | Question: 14



There are 200 tracks on a disc platter and the pending requests have come in the order - 36, 69, 167, 76, 42, 51, 126, 12 and 199. Assume the arm is located at the 100<sup>th</sup> track and moving towards track 200. If sequence of disc access is 126, 167, 199, 12, 36, 42, 51, 69 and 76 then which disc access scheduling policy is used?

- A. Elevator
- B. Shortest seek-time first
- C. C-SCAN
- D. First Come First Served

disk-scheduling isro2014 operating-system

Answer key

#### 33.7.2 Disk Scheduling: ISRO CSE 2017 | Question: 66



Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence.

**4, 37, 10, 7, 19, 73, 2, 15, 6, 20**

Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first algorithm is used.

- A. 95 ms
- B. 119 ms
- C. 233 ms
- D. 276 ms

isro2017 operating-system disk-scheduling

Answer key

## 33.8

## File System (1)

## 33.8.1 File System: ISRO CSE 2014 | Question: 66



Consider a system where each file is associated with a 16 bit number. For each file, each user should have the read and write capability. How much memory is needed to store each user's access data?

- A. 16 KB                      B. 32 KB                      C. 64 KB                      D. 128 KB

isro2014   operating-system   file-system

Answer key

## 33.9

## Fork System Call (4)

## 33.9.1 Fork System Call: ISRO CSE 2008 | Question: 63



Fork is

- A. the creation of a new job                      B. the dispatching of a task  
C. increasing the priority of a task              D. the creation of a new process

isro2008   operating-system   fork-system-call

Answer key

## 33.9.2 Fork System Call: ISRO CSE 2017 | Question: 51



What is the output of the following program?

```
main()
{
    int a = 10;
    if(fork() == 0)
        a++;
    printf("%d\n",a);
}
```

- A. 10 and 11                      B. 10                      C. 11                      D. 11 and 11

isro2017   operating-system   fork-system-call

Answer key

## 33.9.3 Fork System Call: ISRO CSE 2018 | Question: 4



The following C program:

```
{
    fork(); fork(); printf("yes");
}
```

If we execute this code segment, how many times the string yes will be printed?

- a. Only once                      b. 2 times                      c. 4 times                      d. 8 times

isro2018   fork-system-call   operating-system

Answer key

## 33.9.4 Fork System Call: ISRO2015-37



Consider the following program.

```
main()
{
    fork();
    fork();
}
```

```
} fork();
```

How many new processes will be created?

- A. 8                      B. 6                      C. 7                      D. 5

isro2015   operating-system   fork-system-call

Answer key 

### 33.10

#### Fragmentation (1)

##### 33.10.1 Fragmentation: ISRO CSE 2018 | Question: 22



A computer has  $1000K$  of main memory. The jobs arrive and finish in the following sequence.

Job 1 requiring  $200 K$  arrives

Job 2 requiring  $350 K$  arrives

Job 3 requiring  $300 K$  arrives

Job 1 finishes

Job 4 requiring  $120 K$  arrives

Job 5 requiring  $150 K$  arrives

Job 6 requiring  $80 K$  arrives

Among best fit and first fit, which performs better for this sequence?

- a. First fit                                              b. Best fit  
c. Both perform the same                      d. None of the above

isro2018   memory-management   fragmentation   operating-system

Answer key 

### 33.11

#### IO Devices (1)

##### 33.11.1 IO Devices: ISRO-2013-51



In case of a DVD, the speed of data transfer is mentioned in multiples of?

- A.  $150 \text{ KB/s}$                       B.  $1.38 \text{ MB/s}$                       C.  $300 \text{ KB/s}$                       D.  $2.40 \text{ MB/s}$

isro2013   io-devices   non-gatecse   out-of-syllabus-now

Answer key 

### 33.12

#### IO Handling (3)

##### 33.12.1 IO Handling: GATE CSE 1996 | Question: 1.20, ISRO2008-56



Which of the following is an example of spooled device?

- A. A line printer used to print the output of a number of jobs  
B. A terminal used to enter input data to a running program  
C. A secondary storage device in a virtual memory system  
D. A graphic display device

gate1996   operating-system   io-handling   normal   isro2008

Answer key 

### 33.12.2 IO Handling: GATE IT 2004 | Question: 11, ISRO2011-33



What is the bit rate of a video terminal unit with 80 characters/line, 8 bits/character and horizontal sweep time of 100  $\mu$ s (including 20  $\mu$ s of retrace time)?

- A. 8 Mbps                      B. 6.4 Mbps                      C. 0.8 Mbps                      D. 0.64 Mbps

gateit-2004   operating-system   io-handling   easy   isro2011

Answer key

### 33.12.3 IO Handling: ISRO-2013-52



Suppose we have variable logical records of lengths of 5 bytes, 10 bytes and 25 bytes while the physical block size in disk is 15 bytes. What is the maximum and minimum fragmentation seen in bytes?

- A. 25 and 5                      B. 15 and 5                      C. 15 and 0                      D. 10 and 5

isro2013   operating-system   io-handling   disk

Answer key

## 33.13

### Interrupts (1)

#### 33.13.1 Interrupts: ISRO 2011 -ECE Operating System



Which of the following is **not true** regarding a preemptive kernel

- a. If a high priority thread becomes ready to run, low priority thread is preempted
- b. The kernel checks for the high priority ready to run threads when ever called
- c. The executing thread is never interrupted
- d. There are special demands on communication between threads and handling common resources

isro-ece   operating-system   threads   interrupts

Answer key

## 33.14

### Macros (1)

#### 33.14.1 Macros: ISRO CSE 2018 | Question: 67



In multi-programmed systems, it is advantageous if some programs such as editors and compilers can be shared by several users.

Which of the following must be true of multi-programmed systems in order that a single copy of a program can be shared by several users?

- I. The program is a macro
- II. The program is recursive
- III. The program is reentrant

- a. I only                      b. II only                      c. III only                      d. I, II and III

isro2018   operating-system   macros

Answer key

## 33.15

### Memory Management (6)

#### 33.15.1 Memory Management: ISRO CSE 2018 | Question: 61



The Operating System of a computer may periodically collect all the free memory space to form

contiguous block of free space. This is called:

- a. Concatenation
- b. Garbage Collection
- c. Collision
- d. Dynamic Memory Allocation

isro2018 memory-management operating-system

Answer key 

### 33.15.2 Memory Management: ISRO CSE 2020 | Question: 25



What is compaction refers to

- A. a technique for overcoming internal fragmentation
- B. a paging technique
- C. a technique for overcoming external fragmentation
- D. a technique for compressing the data

isro-2020 operating-system memory-management easy

Answer key 

### 33.15.3 Memory Management: ISRO CSE 2023 | Question: 2



Which of the following heap memory *E* allocation strategies is likely to exploit spatial locality of memory accesses in a program, the most?

- A. Best Fit
- B. First Fit
- C. Next Fit
- D. Last Fit

isro-cse-2023 operating-system memory-management

Answer key 

### 33.15.4 Memory Management: ISRO CSE 2023 | Question: 76



Any attempt by a process to access memory allocated to OS or process of other user results into

- A. Trap to OS
- B. Context Switching
- C. Page fault
- D. Invocation to Scheduler Despatch

isro-cse-2023 operating-system memory-management

Answer key 

### 33.15.5 Memory Management: ISRO-2013-56



In a 64-bit machine, with 2 GB RAM, and 8 KB page size, how many entries will be there in the page table if its is inverted?

- A.  $2^{18}$
- B.  $2^{20}$
- C.  $2^{33}$
- D.  $2^{51}$

isro2013 operating-system memory-management paging

Answer key 

### 33.15.6 Memory Management: ISRO2013-55



Consider a logical address space of 8 pages of 1024 words each, mapped onto a physical memory of 32 frames. How many bits are there in the physical address and logical address respectively?

- A. 5,3
- B. 10,10
- C. 15,13
- D. 15,15

isro2013 operating-system memory-management



Answer key 

33.16

### Multilevel Paging (1)

#### 33.16.1 Multilevel Paging: ISRO CSE 2011 | Question: 49



Consider a 32-bit machine where four-level paging scheme is used. If the hit ratio to TLB is 98%, and it takes 20 nanosecond to search the TLB and 100 nanoseconds to access the main memory what is effective memory access time in nanoseconds?

- A. 126                      B. 128                      C. 122                      D. 120

isro2011   operating-system   multilevel-paging

Answer key 

33.17

### Mutual Exclusion (2)

#### 33.17.1 Mutual Exclusion: ISRO CSE 2017 | Question: 56



Mutual exclusion problem occurs

- A. between two disjoint processes that do not interact  
B. among processes that share resources  
C. among processes that do not use the same resource  
D. between two processes that uses different resources of different machine

isro2017   operating-system   process-synchronization   mutual-exclusion

Answer key 

#### 33.17.2 Mutual Exclusion: ISRO CSE 2020 | Question: 57



The hardware implementation which provides mutual exclusion is

- A. Semaphores                      B. Test and set instructions  
C. Both options                      D. None of the options

isro-2020   operating-system   semaphore   mutual-exclusion   normal

Answer key 

33.18

### Page Replacement (12)

#### 33.18.1 Page Replacement: GATE CSE 1997 | Question: 3.10, ISRO2008-57, ISRO2015-64



Dirty bit for a page in a page table

- A. helps avoid unnecessary writes on a paging device      B. helps maintain LRU information  
C. allows only read on a page                      D. None of the above

gate1997   operating-system   page-replacement   easy   isro2008   isro2015

Answer key 

#### 33.18.2 Page Replacement: GATE CSE 2004 | Question: 21, ISRO2007-44



The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

- A. the instruction set architecture                      B. page size  
C. number of processes in memory                      D. physical memory size

gatecse-2004   operating-system   virtual-memory   page-replacement   normal   isro2007

Answer key 

### 33.18.3 Page Replacement: GATE CSE 2005 | Question: 22, ISRO2015-36



Increasing the RAM of a computer typically improves performance because:

- A. Virtual Memory increases
- B. Larger RAMs are faster
- C. Fewer page faults occur
- D. Fewer segmentation faults occur

gatecse-2005 operating-system page-replacement easy isro2015

Answer key

### 33.18.4 Page Replacement: GATE CSE 2009 | Question: 9, ISRO2016-52



In which one of the following page replacement policies, Belady's anomaly may occur?

- A. FIFO
- B. Optimal
- C. LRU
- D. MRU

gatecse-2009 operating-system page-replacement normal isro2016

Answer key

### 33.18.5 Page Replacement: ISRO CSE 2007 | Question: 10



The term 'aging' refers to

- A. booting up the priority of the process in multi-level of queue without feedback.
- B. gradually increasing the priority of jobs that wait in the system for a long time to remedy infinite blocking
- C. keeping track of the following a page has been in memory for the purpose of LRU replacement
- D. letting job reside in memory for a certain amount of time so that the number of pages required can be estimated accurately.

isro2007 operating-system memory-management page-replacement

Answer key

### 33.18.6 Page Replacement: ISRO CSE 2008 | Question: 67



The page replacement algorithm which gives the lowest page fault rate is

- A. LRU
- B. FIFO
- C. Optimal page replacement
- D. Second chance algorithm

isro2008 operating-system page-replacement

Answer key

### 33.18.7 Page Replacement: ISRO CSE 2011 | Question: 73



Belady's anomaly means

- A. Page fault rate is constant even on increasing the number of allocated frames
- B. Page fault rate may increase on increasing the number of allocated frames
- C. Page fault rate may increase on decreasing the number of allocated frames
- D. Page fault rate may decrease on increasing the number of allocated frames

isro2011 operating-system page-replacement page-fault

Answer key

### 33.18.8 Page Replacement: ISRO CSE 2016 | Question: 48



Determine the number of page faults when references to pages occur in the following order:

1, 2, 4, 5, 2, 1, 2, 4

Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2,

with page one having brought earlier than page 2. (LRU page replacement algorithm is used)

- A. 3                      B. 5                      C. 4                      D. None of these

page-replacement   page-fault   isro2016   operating-system

Answer key 

### 33.18.9 Page Replacement: ISRO CSE 2016 | Question: 49



Working Set  $(t, k)$  at an instant of time  $t$  is

- A. the set of  $k$  future references that the OS will make  
B. the set of future references that the OS will make in next  $t$  unit of time  
C. the set of  $k$  references with high frequency  
D. the  $k$  set of pages that have been referenced in the last  $t$  time units

operating-system   page-replacement   isro2016

Answer key 

### 33.18.10 Page Replacement: ISRO CSE 2018 | Question: 20



Determine the number of page faults when references to pages occur in the order 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 brought earlier than page 2. (assume LRU i.e., Least-Recently-Used algorithm is applied)

- A. 3                      B. 4                      C. 5                      D. None of the above

isro2018   operating-system   page-replacement

Answer key 

### 33.18.11 Page Replacement: ISRO CSE 2020 | Question: 30



Consider the following page reference string.

1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6

What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU replacement strategy?

- A. 7                      B. 4                      C. 6                      D. 5

isro-2020   operating-system   memory-management   page-replacement   page-fault   normal

Answer key 

### 33.18.12 Page Replacement: ISRO-2013-65



Consider the list of page references in the time line as below:

9 6 2 3 4 4 4 4 3 4 4 2 5 8 6 8 5 5 3 2 3 3 9 6 2 7

What is the working set at the penultimate page reference if  $\Delta$  is 5?

- A. {8, 5, 3, 2, 9, 6}                      B. {4, 3, 6, 2, 5}  
C. {3, 9, 6, 2, 7}                      D. {3, 9, 6, 2}

isro2013   operating-system   memory-management   page-replacement

Answer key 

## 33.19

## Pipes (1)

### 33.19.1 Pipes: ISRO CSE 2018 | Question: 13



The difference between a named pipe and a regular file in Unix is that

- a. Unlike a regular file, named pipe is a special file
- b. The data in a pipe is transient, unlike the content of a regular file
- c. Pipes forbid random accessing, while regular files do allow this.
- d. All of the above

isro2018 operating-system pipes non-gatecse

Answer key

## 33.20

### Process (7)

### 33.20.1 Process: ISRO CSE 2007 | Question: 41



A task in a blocked state

- A. is executable
- B. is running
- C. must still be placed in the run queues
- D. is waiting for some temporarily unavailable resources

isro2007 operating-system process

Answer key

### 33.20.2 Process: ISRO CSE 2008 | Question: 64



Which of the following need not necessarily be saved on a Context Switch between processes?

- A. General purpose registers
- B. Translation look-aside buffer
- C. Program counter
- D. Stack pointer

isro2008 operating-system process

Answer key

### 33.20.3 Process: ISRO CSE 2009 | Question: 75



Special software to create a job queue is called a

- A. Driver
- B. Spooler
- C. Interpreter
- D. Linkage editor

isro2009 operating-system process

Answer key

### 33.20.4 Process: ISRO CSE 2009 | Question: 76



Process is

- A. A program in high level language kept on disk
- B. Contents of main memory
- C. A program in execution
- D. A job in secondary memory

isro2009 operating-system process

Answer key

### 33.20.5 Process: ISRO CSE 2009 | Question: 9



Which is the correct definition of a valid process transition in an operating system?

- A. Wake up: ready  $\rightarrow$  running
- B. Dispatch: ready  $\rightarrow$  running
- C. Block: ready  $\rightarrow$  running
- D. Timer runoff:  $\rightarrow$  ready  $\rightarrow$  running

isro2009 operating-system process

Answer key

### 33.20.6 Process: ISRO CSE 2011 | Question: 55



There are three processes in the ready queue. When the currently running process requests for I/O how many process switches take place?

- A. 1                      B. 2                      C. 3                      D. 4

isro2011   operating-system   process

Answer key

### 33.20.7 Process: ISRO-2013-60



The state of a process after it encounters an *I/O* instruction is?

- A. Ready                      B. Blocked                      C. Idle                      D. Running

isro2013   operating-system   process

Answer key

## 33.21

## Process Scheduling (22)

### 33.21.1 Process Scheduling: GATE CSE 1996 | Question: 2.20, ISRO2008-15



Four jobs to be executed on a single processor system arrive at time 0 in the order *A, B, C, D*. Their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of *A* under round robin scheduling with time slice of one time unit is

- A. 10                      B. 4                      C. 8                      D. 9

gate1996   operating-system   process-scheduling   normal   isro2008

Answer key

### 33.21.2 Process Scheduling: GATE CSE 2006 | Question: 06, ISRO2009-14



Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

- A. 1                      B. 2                      C. 3                      D. 4

gatecse-2006   operating-system   process-scheduling   normal   isro2009

Answer key

### 33.21.3 Process Scheduling: ISRO CSE 2007 | Question: 11, GATE CSE 2001 | Question: 1.19



Consider a set of  $n$  tasks with known runtimes  $r_1, r_2, \dots, r_n$  to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?

- A. Round Robin                      B. Shortest job first  
C. Highest response ratio next                      D. first come first served

isro2007   operating-system   process-scheduling   gatecse-2001

Answer key

### 33.21.4 Process Scheduling: ISRO CSE 2007 | Question: 12



Consider a job scheduling problem with 4 jobs  $J_1, J_2, J_3$  and  $J_4$  with corresponding deadlines:  $(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$ . Which of the following is not a feasible schedule without violating any job schedule?

- A.  $J_2, J_4, J_1, J_3$                       B.  $J_4, J_1, J_2, J_3$

C.  $J_4, J_2, J_1, J_3$

D.  $J_4, J_2, J_3, J_1$

isro2007 operating-system process-scheduling

Answer key 

### 33.21.5 Process Scheduling: ISRO CSE 2007 | Question: 17



Round Robin schedule is essentially the pre-emptive version of

A. FIFO

B. Shortest job first

C. Shortest remaining time

D. Longest remaining time

isro2007 operating-system process-scheduling

Answer key 

### 33.21.6 Process Scheduling: ISRO CSE 2007 | Question: 43



On a system using non-preemptive scheduling, processes with expected run times of 5, 18, 9 and 12 are in the ready queue. In what order should they be run to minimize wait time?

A. 5, 12, 9, 18

B. 5, 9, 12, 18

C. 12, 18, 9, 5

D. 9, 12, 18, 5

isro2007 operating-system process-scheduling

Answer key 

### 33.21.7 Process Scheduling: ISRO CSE 2007 | Question: 64, ISRO CSE 2008 | Question: 50



Feedback queues

A. are very simple to implement

B. dispatch tasks according to execution characteristics

C. are used to favour real time tasks

D. require manual intervention to implement properly

isro2007 operating-system isro2008 process-scheduling

Answer key 

### 33.21.8 Process Scheduling: ISRO CSE 2008 | Question: 51



With Round-Robin CPU scheduling in a time shared system

A. using very large time slices (quantas) degenerates into First-Come First served (FCFS) algorithm.

B. using extremely small time slices improves performance

C. using very small time slices degenerates into Last-In First-Out (LIFO) algorithm.

D. using medium sized times slices leads to shortest Request time First (SRTF) algorithm

isro2008 operating-system process-scheduling

Answer key 

### 33.21.9 Process Scheduling: ISRO CSE 2008 | Question: 66, ISRO CSE 2009 | Question: 15



The performance of Round Robin algorithm depends heavily on

A. size of the process

B. the I/O bursts of the process

C. the CPU bursts of the process

D. the size of the time quantum

isro2008 isro2009 operating-system process-scheduling

Answer key 

**33.21.10 Process Scheduling: ISRO CSE 2009 | Question: 10**

The correct matching of the following pairs is

|    |                  |      |             |
|----|------------------|------|-------------|
| A. | Disk check       | i.   | Round robin |
| B. | Batch processing | ii.  | Scan        |
| C. | Time sharing     | iii. | LIFO        |
| D. | Stack operation  | iv.  | FIFO        |

1. a-iii, b-iv, c-ii, d-i
2. a-iv, b-iii, c-ii, d-i
3. a-iii, b-iv, c-i, d-ii
4. a-ii, b-iv, c-i, d-iii

isro2009 operating-system process-scheduling

Answer key

**33.21.11 Process Scheduling: ISRO CSE 2009 | Question: 17**

Consider a set of 5 processes whose arrival time, CPU time needed and the priority are given below:

| Process<br>Priority | Arrival Time<br>(in ms) | CPU Time<br>Needed | Priority |
|---------------------|-------------------------|--------------------|----------|
| P1                  | 0                       | 10                 | 5        |
| P2                  | 0                       | 5                  | 2        |
| P3                  | 2                       | 3                  | 1        |
| P4                  | 5                       | 20                 | 4        |
| P5                  | 10                      | 2                  | 3        |

(smaller the number, higher the priority)

If the CPU scheduling policy is priority scheduling without pre-emption, the average waiting time will be

- A. 12.8 ms                      B. 11.8 ms                      C. 10.8 ms                      D. 09.8 ms

isro2009 operating-system process-scheduling

Answer key

**33.21.12 Process Scheduling: ISRO CSE 2011 | Question: 4**

The following table shows the processes in the ready queue and time required for each process for completing its job.

| Process Time |    |
|--------------|----|
| $P_1$        | 10 |
| $P_2$        | 5  |
| $P_3$        | 20 |
| $P_4$        | 8  |
| $P_5$        | 15 |

If round-robin scheduling with 5 ms is used what is the average waiting time of the processes in the queue?

- A. 27 ms                      B. 26.2 ms                      C. 27.5 ms                      D. 27.2 ms

isro2011 operating-system process-scheduling

Answer key

### 33.21.13 Process Scheduling: ISRO CSE 2014 | Question: 78



Which of the following is not an optimization criterion in the design of a CPU scheduling algorithm?

- A. Minimum CPU utilization
- B. Maximum throughput
- C. Minimum turnaround time
- D. Minimum waiting time

isro2014 operating-system process-scheduling

Answer key

### 33.21.14 Process Scheduling: ISRO CSE 2016 | Question: 51



For the real time operating system, which of the following is the most suitable scheduling scheme?

- A. Round robin
- B. First come first serve
- C. Pre-emptive
- D. Random scheduling

isro2016 operating-system process-scheduling

Answer key

### 33.21.15 Process Scheduling: ISRO CSE 2020 | Question: 27



Which of the following algorithms defines time quantum?

- A. shortest job scheduling algorithm
- B. round robin scheduling algorithm
- C. priority scheduling algorithm
- D. multilevel queue scheduling algorithm

isro-2020 operating-system process-scheduling easy

Answer key

### 33.21.16 Process Scheduling: ISRO CSE 2020 | Question: 31



Three CPU-bound tasks, with execution times of 15, 12 and 5 time units respectively arrive at times 0,  $t$  and 8, respectively. If the operating system implements a shortest remaining time first scheduling algorithm, what should be the value of  $t$  to have 4 context switches? Ignore the context switches at time 0 and at the end.

- A.  $0 < t < 3$
- B.  $t = 0$
- C.  $t \leq 3$
- D.  $3 < t < 8$

isro-2020 operating-system process-scheduling normal

Answer key

### 33.21.17 Process Scheduling: ISRO CSE 2023 | Question: 54



|       | Process Scheduler Type |     | Activity                                           |
|-------|------------------------|-----|----------------------------------------------------|
| (i)   | Long Term Scheduler    | (P) | executes much faster to reduce wastage of CPU time |
| (ii)  | Medium term scheduler  | (Q) | controls degree of multi programming               |
| (iii) | Short Term scheduler   | (R) | is associated with swapping                        |

with respect to Process Schedulers of an operating system mentioned above in the table, which of the following statements is correct?

- a. (i) – (R), (ii) – (P), (iii) – (Q)
- b. (i) – (Q), (ii) – (R), (iii) – (P)
- c. (i) – (P), (ii) – (R), (iii) – (Q)
- d. (i) – (R), (ii) – (Q), (iii) – (P)

isro-cse-2023 operating-system process-scheduling

Answer key



### 33.21.18 Process Scheduling: ISRO-2013-50



Consider the following set of processes, with arrival times and the required CPU-burst times given in milliseconds.

| Process | Arrival time | Burst Time |
|---------|--------------|------------|
| P1      | 0            | 4          |
| P2      | 2            | 2          |
| P3      | 3            | 1          |

What is the sequence in which the processes are completed? Assume round robin scheduling with a time quantum of 2 milliseconds?

- A. P1, P2, P3      B. P2, P1, P3      C. P3, P2, P1      D. P2, P3, P1

isro2013 process-scheduling round-robin-scheduling

Answer key

### 33.21.19 Process Scheduling: ISRO-2013-59



A starvation free job scheduling policy guarantees that no job indefinitely waits for a service. Which of the following job scheduling policies is starvation free?

- A. Priority queuing      B. Shortest job first  
C. Youngest job first      D. Round robin

isro2013 operating-system process-scheduling

Answer key

### 33.21.20 Process Scheduling: ISRO-2013-70



Which of the following strategy is employed for overcoming the priority inversion problem?

- A. Temporarily raise the priority of lower priority level process  
B. Have a fixed priority level scheme.  
C. Implement Kernel pre-emption scheme.  
D. Allow lower priority process to complete its job.

isro2013 operating-system process-scheduling

Answer key

### 33.21.21 Process Scheduling: ISRO2015-33



Suppose a system contains  $n$  processes and system uses the round-robin algorithm for CPU scheduling then which data structure is best suited ready queue of the process

- A. stack      B. queue      C. circular queue      D. tree

isro2015 operating-system process-scheduling

Answer key

### 33.21.22 Process Scheduling: ISRO2015-38



Suppose two jobs, each of which needs 10 minutes of CPU time, start simultaneously. Assume 50% I/O wait time. How long will it take for both to complete, if they run sequentially?

- A. 10      B. 20      C. 30      D. 40

process-scheduling isro2015

**33.22.1 Process Synchronization: GATE CSE 1996 | Question: 1.19, ISRO2008-61**

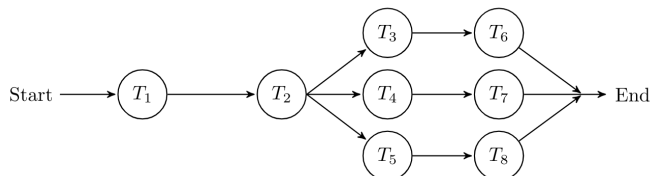
A critical section is a program segment

- A. which should run in a certain amount of time
- B. which avoids deadlocks
- C. where shared resources are accessed
- D. which must be enclosed by a pair of semaphore operations,  $P$  and  $V$

gate1996 operating-system process-synchronization easy isro2008

**33.22.2 Process Synchronization: ISRO CSE 2011 | Question: 10**

Below is the precedence graph for a set of tasks to be executed on a parallel processing system  $S$ .



What is the efficiency of this precedence graph on  $S$  if each of the tasks  $T_1, \dots, T_8$  takes the same time and the system  $S$  has five processors?

- A. 25%
- B. 40%
- C. 50%
- D. 90%

isro2011 operating-system process-synchronization

**33.22.3 Process Synchronization: ISRO CSE 2011 | Question: 77**

In a system using single processor, a new process arrives at the rate of six processes per minute and each such process requires seven seconds of service time. What is the CPU utilization?

- A. 70%
- B. 30%
- C. 60%
- D. 64%

isro2011 operating-system process-synchronization

**33.22.4 Process Synchronization: ISRO CSE 2018 | Question: 67**

Procedures  $P1$  and  $P2$  have a producer-consumer relationship, communicating by the use of a set of shared buffers.

$P1$ : repeat

Obtain an empty buffer

Fill it

Return a full buffer

forever

$P2$ : repeat

Obtain a full buffer

Empty it

Return an empty buffer  
forever

Increasing the number of buffers is likely to do which of the following?

- I. Increase the rate at which requests are satisfied (throughput)
- II. Decrease the likelihood of deadlock
- III. Increase the ease of achieving a correct implementation

- a. III only                      b. II only                      c. I only                      d. II and III only

isro2018   process-synchronization   operating-system

Answer key 

### 33.22.5 Process Synchronization: ISRO CSE 2020 | Question: 26



The operating system and the other processes are protected from being modified by an already running process because

- A. they run at different time instants and not in parallel
- B. they are in different logical addresses
- C. they use a protection algorithm in the scheduler
- D. every address generated by the CPU is being checked against the relocation and limit parameters

isro-2020   operating-system   process-synchronization   easy

Answer key 

### 33.22.6 Process Synchronization: ISRO CSE 2020 | Question: 28



Dispatch latency is defined as

- A. the speed of dispatching a process from running to the ready state
- B. the time of dispatching a process from running to ready state and keeping the CPU idle
- C. the time to stop one process and start running another one
- D. none of these

isro-2020   operating-system   process-synchronization   easy

Answer key 

### 33.22.7 Process Synchronization: ISRO CSE 2023 | Question: 75



In a producer-consumer scenario also known as Bounded-Buffer Problem, what would be the most appropriate synchronization primitive to ensure that the consumer waits when the buffer is empty?

- A. Spinlock                      B. Mutex lock                      C. Semaphore                      D. Monitors

isro-cse-2023   operating-system   process-synchronization

Answer key 

## 33.23

## Realtime Systems (1)

### 33.23.1 Realtime Systems: ISRO CSE 2017 | Question: 73



Which of the following statement is true?

- A. Hard real time OS has less jitter than soft real time OS

- B. Hard real time OS has more jitter than soft real time OS
- C. Hard real time OS has equal jitter as soft real time OS
- D. None of the above

isro2017 operating-system realtime-systems

Answer key 

### 33.24

#### Resource Allocation (10)

##### 33.24.1 Resource Allocation: ISRO CSE 2007 | Question: 40



Consider a system having 'm' resources of the same type. The resources are shared by 3 processes A, B, C, which have peak time demands of 3, 4, 6 respectively. The minimum value of 'm' that ensures that deadlock will never occur is

- A. 11
- B. 12
- C. 13
- D. 14

isro2007 operating-system resource-allocation

Answer key 

##### 33.24.2 Resource Allocation: ISRO CSE 2008 | Question: 62



In which of the following four necessary conditions for deadlock processes claim exclusive control of the resources they require?

- A. no preemption
- B. mutual exclusion
- C. circular wait
- D. hold and wait

isro2008 operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key 

##### 33.24.3 Resource Allocation: ISRO CSE 2009 | Question: 16



Consider a system having "n" resources of same type. These resources are shared by 3 processes, A, B, C. These have peak demands of 3, 4, and 6 respectively. For what value of "n" deadlock won't occur

- A. 15
- B. 9
- C. 10
- D. 13

isro2009 operating-system resource-allocation deadlock-prevention-avoidance-detection

Answer key 

##### 33.24.4 Resource Allocation: ISRO CSE 2011 | Question: 60



A total of 9 units of a resource type available, and given the safe state shown below, which of the following sequence will be a safe state?

**Process Used Max**

|       |   |   |
|-------|---|---|
| $P_1$ | 2 | 7 |
| $P_2$ | 1 | 6 |
| $P_3$ | 2 | 5 |
| $P_4$ | 1 | 4 |

- A.  $\langle P_4, P_1, P_3, P_2 \rangle$
- B.  $\langle P_4, P_2, P_1, P_3 \rangle$
- C.  $\langle P_4, P_2, P_3, P_1 \rangle$
- D.  $\langle P_3, P_1, P_2, P_4 \rangle$

isro2011 operating-system resource-allocation

Answer key 

### 33.24.5 Resource Allocation: ISRO CSE 2014 | Question: 68



What is the minimum number of resources required to ensure that deadlock will never occur, if there are currently three processes  $P_1, P_2$  and  $P_3$  running in a system whose maximum demand for the resources of same type are 3, 4, and 5 respectively.

- A. 3                                      B. 7                                      C. 9                                      D. 10

isro2014   operating-system   resource-allocation

Answer key

### 33.24.6 Resource Allocation: ISRO CSE 2016 | Question: 46



With single resource, deadlock occurs

- A. if there are more than two processes competing for that resources  
B. if there are only two processes competing for that resources  
C. if there is a single process competing for that resources  
D. none of these

operating-system   resource-allocation   isro2016

Answer key

### 33.24.7 Resource Allocation: ISRO CSE 2016 | Question: 47



A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units, then

- A. Deadlock can never occur                                      B. Deadlock may occur  
C. Deadlock has to occur                                              D. None of these

operating-system   resource-allocation   isro2016

Answer key

### 33.24.8 Resource Allocation: ISRO-2013-57



Which of the following is not a necessary condition for deadlock?

- A. Mutual exclusion                      B. Reentrancy                      C. Hold and wait                      D. No pre-emption

isro2013   operating-system   resource-allocation   deadlock-prevention-avoidance-detection

Answer key

### 33.24.9 Resource Allocation: ISRO-2013-58



Consider the following process and resource requirement of each process.

| Process | Type 1 |     | Type 2 |     |
|---------|--------|-----|--------|-----|
|         | Used   | Max | Used   | Max |
| P1      | 1      | 2   | 1      | 3   |
| P2      | 1      | 3   | 1      | 2   |
| P3      | 2      | 4   | 1      | 4   |

Predict the state of this system, assuming that there are a total of 5 instances of resource type 1 and 4 instances of resource type 2.

- A. Can go to safe or unsafe state based on sequence  
 C. Unsafe state
- B. Safe state  
 D. Deadlock state

isro2013 operating-system resource-allocation

Answer key 

### 33.24.10 Resource Allocation: ISRO2015-32



In a lottery scheduler with 40 tickets, how we will distribute the tickets among 4 processes  $P_1, P_2, P_3$  and  $P_4$  such that each process gets 10%, 5%, 60% and 25% respectively?

|   | $P_1$ | $P_2$ | $P_3$ | $P_4$ |
|---|-------|-------|-------|-------|
| A | 12    | 4     | 70    | 30    |
| B | 7     | 5     | 20    | 10    |
| C | 4     | 2     | 24    | 10    |
| D | 8     | 5     | 30    | 40    |

isro2015 operating-system resource-allocation

Answer key 

### 33.25 Runtime Environment (1)

#### 33.25.1 Runtime Environment: ISRO-DEC2017-68



Consider the following table :

|           |                    |           |                    |
|-----------|--------------------|-----------|--------------------|
| <b>A.</b> | Activation record  | <b>p.</b> | Linking loader     |
| <b>B.</b> | Location counter   | <b>q.</b> | Garbage collection |
| <b>C.</b> | Reference counts   | <b>r.</b> | Subroutine call    |
| <b>D.</b> | Address relocation | <b>s.</b> | Assembler          |

- A.  $a - p, b - q, c - r, d - s$   
 C.  $a - r, b - s, c - q, d - p$
- B.  $a - q, b - r, c - s, d - p$   
 D.  $a - r, b - s, c - p, d - q$

isrodec2017 runtime-environment compiler-design

Answer key 

### 33.26 Segmentation (2)

#### 33.26.1 Segmentation: ISRO CSE 2014 | Question: 18



Consider the following segment table in segmentation scheme :

| Segment ID | Base | Limit |
|------------|------|-------|
| 0          | 200  | 200   |
| 1          | 5000 | 1210  |
| 2          | 1527 | 498   |
| 3          | 2500 | 50    |

What happens if the logical address requested is – Segment Id 2 and offset 1000?

- A. Fetches the entry at the physical address 2527 for segment Id 2  
 B. A trap is generated  
 C. Deadlock  
 D. Fetches the entry at offset 27 in Segment Id 3

segmentation operating-system isro2014

Answer key 

### 33.26.2 Segmentation: ISRO2015-31



If there are 32 segments, each size 1 k bytes, then the logical address should have

- A. 13 bits                      B. 14 bits                      C. 15 bits                      D. 16 bits

isro2015   operating-system   virtual-memory   segmentation

Answer key

### 33.27

#### Semaphore (6)

### 33.27.1 Semaphore: GATE CSE 1992 | Question: 02,x, ISRO2015-35



At a particular time of computation, the value of a counting semaphore is 7. Then 20  $P$  operations and 15  $V$  operations were completed on this semaphore. The resulting value of the semaphore is :

- A. 42                      B. 2                      C. 7                      D. 12

gate1992   operating-system   semaphore   easy   isro2015   process-synchronization

Answer key

### 33.27.2 Semaphore: ISRO CSE 2007 | Question: 42, UGCNET-June2010-II: 37



Semaphores

- A. synchronize critical resources to prevent deadlock  
B. synchronize critical resources to prevent contention  
C. are used to do I/O  
D. are used for memory management

isro2007   operating-system   process-synchronization   ugcnetcse-june2010-paper2   semaphore

Answer key

### 33.27.3 Semaphore: ISRO CSE 2016 | Question: 45



At a particular time of computation the value of a counting semaphore is 7. Then 20  $P$  operations and  $x$   $V$  operations were completed on this semaphore. If the new value of semaphore is 5,  $x$  will be

- A. 18                      B. 22                      C. 15                      D. 13

operating-system   process-synchronization   semaphore   isro2016

Answer key

### 33.27.4 Semaphore: ISRO CSE 2017 | Question: 71



At a particular time the value of counting semaphore is 10. It will become 7 after:

- A. 3  $V$  operations                      B. 3  $P$  operations  
C. 5  $V$  operations and 2  $P$  operations                      D. 2  $V$  operations and 5  $P$  operations

isro2017   operating-system   semaphore

Answer key

### 33.27.5 Semaphore: ISRO-DEC2017-70



A counting semaphore was initialized to 7. Then 20  $P$  (wait) operations and  $x$   $V$  (signal) operations were completed on this semaphore. If the final value of semaphore is 5, then the value  $x$  will be

- A. 0                      B. 13                      C. 18                      D. 5

Answer key **33.27.6 Semaphore: ISRO2015-30**

Semaphores are used to solve the problem of

- i. Race Condition
- ii. Process Synchronization
- iii. Mutual Exclusion
- iv. None of the above

- A. I and II
- B. II and III
- C. All of the above
- D. None of the above

semaphore isro2015 process-synchronization

Answer key **33.28****Thrashing (1)****33.28.1 Thrashing: ISRO-2013-54**

Which of the following are the likely causes of thrashing?

- A. Page size was very small.
- B. There are too many users connected to the system.
- C. Least recently used policy is used for page replacement.
- D. First in First out policy is used for page replacement.

isro2013 operating-system memory-management thrashing

Answer key **33.29****Threads (3)****33.29.1 Threads: ISRO CSE 2008 | Question: 60**

Overlaying

- |                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| A. requires use of a loader        | B. allows larger programs, but requires more effort |
| C. is most used on large computers | D. is transparent to the user                       |

isro2008 operating-system threads

Answer key **33.29.2 Threads: ISRO CSE 2023 | Question: 11**

Which of the following multithreading model is followed in Linux OS?

- A. One User Thread mapping to One Kernel Thread
- B. Many User Threads mapping to One Kernel Thread
- C. One User Thread mapping to Many Kernel Threads
- D. Many User Threads mapping to Many Kernel Threads

isro-cse-2023 operating-system threads



Answer key 

### 33.29.3 Threads: ISRO CSE 2023 | Question: 12



Purpose of priority inheritance protocol in synchronization mechanisms is

- A. To prevent priority inversion in systems having two or lesser priorities for a resource
- B. To provide mutual exclusion between threads and prevent data corruption
- C. To prevent priority inversion and ensure fairness in resource allocation
- D. To allow multiple threads to access any resource simultaneously

isro-cse-2023   operating-system   process-synchronization   threads

33.30

Unix (1)

### 33.30.1 Unix: ISRO CSE 2017 | Question: 72



The Linux command `mknod myfifo b 4 16`

- A. will create a character device if user is root
- B. will create a named pipe FIFO if user is root
- C. will create a block device if user is root
- D. None of these

isro2017   operating-system   unix   non-gatecse

Answer key 

33.31

Virtual Memory (9)

### 33.31.1 Virtual Memory: GATE CSE 2006 | Question: 62, ISRO2016-50



A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is:

- A. 11 bits
- B. 13 bits
- C. 15 bits
- D. 20 bits

gatecse-2006   operating-system   virtual-memory   normal   isro2016

Answer key 

### 33.31.2 Virtual Memory: ISRO CSE 2007 | Question: 27



Virtual memory is

- A. Part of Main Memory only used for swapping
- B. A technique to allow a program, of size more than the size of main memory, to run
- C. Part of secondary storage used in program execution
- D. None of these

isro2007   operating-system   virtual-memory

Answer key 

### 33.31.3 Virtual Memory: ISRO CSE 2008 | Question: 65



Consider a logical address space of 8 pages of 1024 words mapped into memory of 32 frames. How many bits are there in the logical address?

- A. 13 bits
- B. 15 bits
- C. 14 bits
- D. 12 bits

isro2008   operating-system   virtual-memory

Answer key 

### 33.31.4 Virtual Memory: ISRO CSE 2011 | Question: 24



If the page size in a 32-bit machine is 4K bytes then the size of page table is

- A. 1 M bytes      B. 2 M bytes      C. 4 M bytes      D. 4 K bytes

isro2011   operating-system   paging   virtual-memory

Answer key 

### 33.31.5 Virtual Memory: ISRO CSE 2014 | Question: 35



Using the page table shown below, translate the physical address 25 to virtual address. The address length is 16 bits and page size is 2048 words while the size of the physical memory is four frames.

| Page | Present (1-In, 0-Out) | Frame |
|------|-----------------------|-------|
| 0    | 1                     | 3     |
| 1    | 1                     | 2     |
| 2    | 1                     | 0     |
| 3    | 0                     | —     |

- A. 25      B. 6169      C. 2073      D. 4121

operating-system   virtual-memory   paging   isro2014

Answer key 

### 33.31.6 Virtual Memory: ISRO CSE 2014 | Question: 44



A computer has 16 pages of virtual address space but the size of main memory is only four frames. Initially the memory is empty. A program references the virtual pages in the order 0, 2, 4, 5, 2, 4, 3, 11, 2, 10. How many page faults occur if LRU page replacement algorithm is used?

- A. 3      B. 5      C. 7      D. 8

isro2014   operating-system   virtual-memory   paging

Answer key 

### 33.31.7 Virtual Memory: ISRO CSE 2014 | Question: 70



Dirty bit is used to indicate which of the following?

- A. A page fault has occurred      B. A page has corrupted data  
C. A page has been modified after being loaded into cache      D. An illegal access of page

isro2014   operating-system   virtual-memory   paging

Answer key 

### 33.31.8 Virtual Memory: ISRO CSE 2014 | Question: 77



What is the size of the physical address space in a paging system which has a page table containing 64 entries of 11 bit each (including valid and invalid bit) and a page size of 512 bytes?

- A.  $2^{11}$       B.  $2^{15}$       C.  $2^{19}$       D.  $2^{20}$

isro2014   virtual-memory   paging

Answer key 

**33.31.9 Virtual Memory: ISRO CSE 2016 | Question: 22**

Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every  $10^6$  memory accesses, what is the effective access time for the memory?

A. 21.4 ns

B. 29.9 ns

C. 23.5 ns

D. 35.1 ns

operating-system virtual-memory paging isro2016

Answer key

**Answer Keys**

|          |     |          |     |          |     |          |     |          |     |
|----------|-----|----------|-----|----------|-----|----------|-----|----------|-----|
| 33.0.1   | D   | 33.0.2   | C   | 33.0.3   | D   | 33.0.4   | B   | 33.0.5   | C   |
| 33.0.6   | B   | 33.0.7   | D   | 33.0.8   | Q-Q | 33.0.9   | Q-Q | 33.0.10  | Q-Q |
| 33.0.11  | Q-Q | 33.0.12  | Q-Q | 33.0.13  | A   | 33.0.14  | B   | 33.0.15  | D   |
| 33.0.16  | A   | 33.0.17  | A   | 33.0.18  | D   | 33.0.19  | B   | 33.0.20  | A   |
| 33.0.21  | B   | 33.0.22  | Q-Q | 33.1.1   | C   | 33.2.1   | D   | 33.3.1   | B   |
| 33.4.1   | A   | 33.5.1   | A   | 33.5.2   | A   | 33.5.3   | A   | 33.6.1   | D   |
| 33.6.2   | A   | 33.6.3   | A   | 33.6.4   | D   | 33.6.5   | B   | 33.6.6   | A   |
| 33.6.7   | C   | 33.6.8   | A   | 33.6.9   | B   | 33.6.10  | D   | 33.6.11  | D   |
| 33.7.1   | C   | 33.7.2   | B   | 33.8.1   | A   | 33.9.1   | D   | 33.9.2   | A   |
| 33.9.3   | C   | 33.9.4   | C   | 33.10.1  | A   | 33.11.1  | B   | 33.12.1  | A   |
| 33.12.2  | B   | 33.12.3  | D   | 33.13.1  | Q-Q | 33.14.1  | C   | 33.15.1  | B   |
| 33.15.2  | C   | 33.15.3  | Q-Q | 33.15.4  | Q-Q | 33.15.5  | A   | 33.15.6  | C   |
| 33.16.1  | B   | 33.17.1  | B   | 33.17.2  | B   | 33.18.1  | A   | 33.18.2  | A   |
| 33.18.3  | C   | 33.18.4  | A   | 33.18.5  | B   | 33.18.6  | C   | 33.18.7  | B   |
| 33.18.8  | C   | 33.18.9  | D   | 33.18.10 | B   | 33.18.11 | C   | 33.18.12 | D   |
| 33.19.1  | D   | 33.20.1  | D   | 33.20.2  | B   | 33.20.3  | B   | 33.20.4  | C   |
| 33.20.5  | B   | 33.20.6  | B   | 33.20.7  | B   | 33.21.1  | D   | 33.21.2  | B   |
| 33.21.3  | B   | 33.21.4  | B   | 33.21.5  | A   | 33.21.6  | B   | 33.21.7  | B   |
| 33.21.8  | A   | 33.21.9  | D   | 33.21.10 | D   | 33.21.11 | C   | 33.21.12 | B   |
| 33.21.13 | A   | 33.21.14 | C   | 33.21.15 | B   | 33.21.16 | A   | 33.21.17 | Q-Q |
| 33.21.18 | B   | 33.21.19 | D   | 33.21.20 | A   | 33.21.21 | C   | 33.21.22 | D   |
| 33.22.1  | C   | 33.22.2  | B   | 33.22.3  | A   | 33.22.4  | C   | 33.22.5  | D   |
| 33.22.6  | C   | 33.22.7  | Q-Q | 33.23.1  | A   | 33.24.1  | A   | 33.24.2  | B   |
| 33.24.3  | D   | 33.24.4  | D   | 33.24.5  | D   | 33.24.6  | D   | 33.24.7  | A   |
| 33.24.8  | B   | 33.24.9  | C   | 33.24.10 | C   | 33.25.1  | C   | 33.26.1  | B   |
| 33.26.2  | C   | 33.27.1  | B   | 33.27.2  | B   | 33.27.3  | A   | 33.27.4  | B   |
| 33.27.5  | C   | 33.27.6  | B   | 33.28.1  | A   | 33.29.1  | D   | 33.29.2  | Q-Q |
| 33.29.3  | Q-Q | 33.30.1  | C   | 33.31.1  | C   | 33.31.2  | B   | 33.31.3  | A   |

|         |   |
|---------|---|
| 33.31.4 | C |
| 33.31.9 | B |

|         |   |
|---------|---|
| 33.31.5 | D |
|---------|---|

|         |   |
|---------|---|
| 33.31.6 | C |
|---------|---|

|         |   |
|---------|---|
| 33.31.7 | C |
|---------|---|

|         |   |
|---------|---|
| 33.31.8 | C |
|---------|---|

**34.0.1 Need Guidance to work in ISRO**

What languages,Scripts should learn to serve best at ISRO and what kind of skills and experience are required to get ISRO/Scientist Post?

isro-preparation

Answer key 

## Answer Keys

|        |     |
|--------|-----|
| 34.0.1 | Q-Q |
|--------|-----|

**35.0.1 ISRO-DEC2017-28**

The number of structurally different possible binary trees with 4 nodes is

- A. 14                      B. 12                      C. 336                      D. 168

isrodec2017

Answer key

**35.0.2 ISRO-DEC2017-47**

Which of the following permutation can be obtained in the same order using a stack assuming that input is the sequence 5, 6, 7, 8, 9 in that order?

- A. 7, 8, 9, 5, 6                      B. 5, 9, 6, 7, 8                      C. 7, 8, 9, 6, 5                      D. 9, 8, 7, 5, 6

isrodec2017

Answer key

**35.0.3 ISRO-DEC2017-49**

A binary search tree is used to locate the number 43. Which one of the following probe sequence is not possible?

- A. 61, 52, 14, 17, 40, 43                      B. 10, 65, 31, 48, 37, 43  
C. 81, 61, 52, 14, 41, 43                      D. 17, 77, 27, 66, 18, 43

isrodec2017

Answer key

**35.0.4 ISRO-DEC2017-54**

A strictly binary tree with 10 leaves

- A. cannot have more than 19 nodes                      B. has exactly 19 nodes  
C. has exactly 17 nodes                      D. has exactly 20 nodes

isrodec2017

Answer key

**35.0.5 ISRO-DEC2017-56**

Which one of the following property is correct for a red-black tree?

- A. Every simple path from anode to a descendant leaf contains the same number of black nodes.  
B. If a node is red, then one children is red and another is black.  
C. If a node is red, then both its children are red.  
D. Every leaf node (sentinel node) is red.

isrodec2017

Answer key

**35.0.6 ISRO-DEC2017-52**

A priority queue is implemented as a *Max-heap*. Initially, it has 5 elements. The level order traversal of the heap is 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level order

traversal of the heap after the insertion of the elements is

1. 10, 8, 7, 5, 3, 2, 1
2. 10, 8, 7, 2, 3, 1, 5
3. 10, 8, 7, 1, 2, 3, 5
4. 10, 8, 7, 3, 2, 1, 5

isrodec2017

Answer key 

### 35.0.7 ISRO-DEC2017-53



The minimum number of stacks needed to implement a queue is

- A. 3                      B. 1                      C. 2                      D. 4

isrodec2017

Answer key 

## 35.1

### AVL Tree (3)

#### 35.1.1 AVL Tree: GATE CSE 2009 | Question: 37, ISRO-DEC2017-55



What is the maximum height of any AVL-tree with 7 nodes? Assume that the height of a tree with a single node is 0.

- A. 2                      B. 3                      C. 4                      D. 5

gatecse-2009   data-structures   binary-search-tree   normal   isrodec2017   avl-tree

Answer key 

#### 35.1.2 AVL Tree: ISRO CSE 2020 | Question: 20



The minimum height of an AVL tree with  $n$  nodes is

- A.  $\text{Ceil}(\log_2(n+1))$                       B.  $1.44 \log_2 n$   
C.  $\text{Floor}(\log_2(n+1))$                       D.  $1.64 \log_2 n$

isro-2020   data-structures   avl-tree   normal

Answer key 

#### 35.1.3 AVL Tree: ISRO-2013-2



The number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree is?

- A. 0                      B. 1                      C. 2                      D. 3

isro2013   data-structures   avl-tree

Answer key 

## 35.2

### Array (2)

#### 35.2.1 Array: ISRO-2013-1



Let  $A(1 : 8, -5 : 5, -10 : 5)$  be a three dimensional array. How many elements are there in the array  $A$ ?

- A. 1200                      B. 1408                      C. 33                      D. 1050

isro2013   array   easy

Answer key 

### 35.2.2 Array: ISRO-2013-9



In an array of  $2N$  elements that is both 2-ordered and 3-ordered, what is the maximum number of positions that an element can be from its position if the array were 1-ordered?

- A. 1                      B. 2                      C.  $N/2$                       D.  $2N - 1$

isro2013   array

Answer key 

## 35.3

### Binary Heap (2)

#### 35.3.1 Binary Heap: ISRO CSE 2018 | Question: 57



Given a binary-max heap. The elements are stored in an arrays as 25, 14, 16, 13, 10, 8, 12. What is the content of the array after two delete operations?

- a. 14, 13, 8, 12, 10                      b. 14, 12, 13, 10, 8  
c. 14, 13, 12, 8, 10                      d. 14, 13, 12, 10, 8

isro2018   data-structures   binary-heap

Answer key 

#### 35.3.2 Binary Heap: ISRO-DEC2017-45



Match the following and choose the correct answer in the order  $A, B, C$

|                                                |                  |
|------------------------------------------------|------------------|
| A. Heap Construction                           | p. $O(n \log n)$ |
| B. Hash Table construction with linear probing | q. $O(n^2)$      |
| C. AVL Tree Construction                       | r. $O(n)$        |

(Bounds given may or may not be asymptotically tight)

- A.  $a - q, b - r, c - p$                       B.  $a - p, b - q, c - r$   
C.  $a - q, b - p, c - r$                       D.  $a - r, b - q, c - p$

isrodec2017   binary-heap   data-structures

Answer key 

## 35.4

### Binary Search Tree (8)

#### 35.4.1 Binary Search Tree: GATE CSE 2003 | Question: 19, ISRO2009-24



Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree?

- A. 7510324689  
B. 0243165987  
C. 0123456789  
D. 9864230157

gatecse-2003   binary-search-tree   easy   isro2009

Answer key 



### 35.4.2 Binary Search Tree: GATE CSE 2003 | Question: 63, ISRO2009-25



A data structure is required for storing a set of integers such that each of the following operations can be done in  $O(\log n)$  time, where  $n$  is the number of elements in the set.

- I. Deletion of the smallest element
- II. Insertion of an element if it is not already present in the set

Which of the following data structures can be used for this purpose?

- A. A heap can be used but not a balanced binary search tree
- B. A balanced binary search tree can be used but not a heap
- C. Both balanced binary search tree and heap can be used
- D. Neither balanced search tree nor heap can be used

gatecse-2003 data-structures easy isro2009 binary-search-tree

Answer key

### 35.4.3 Binary Search Tree: GATE CSE 2004 | Question: 4, ISRO2009-26



The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?

- A. 2
- B. 3
- C. 4
- D. 6

gatecse-2004 data-structures binary-search-tree easy isro2009

Answer key

### 35.4.4 Binary Search Tree: ISRO CSE 2011 | Question: 11



How many distinct binary search trees can be created out of 4 distinct keys?

- A. 5
- B. 14
- C. 24
- D. 35

isro2011 data-structures binary-search-tree combinatory

Answer key

### 35.4.5 Binary Search Tree: ISRO CSE 2011 | Question: 62



The average depth of a binary search tree is

- A.  $O(n^{0.5})$
- B.  $O(n)$
- C.  $O(\log n)$
- D.  $O(n \log n)$

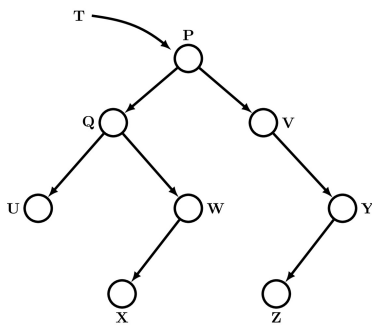
isro2011 data-structures binary-search-tree time-complexity

Answer key

### 35.4.6 Binary Search Tree: ISRO CSE 2014 | Question: 41



Consider the following binary search tree T given below: Which node contains the fourth smallest element in T?



- A. Q                      B. V                      C. W                      D. X

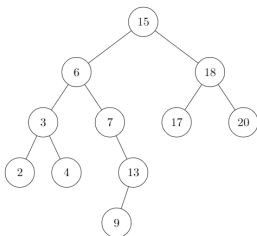
isro2014   data-structures   binary-search-tree

Answer key

### 35.4.7 Binary Search Tree: ISRO CSE 2020 | Question: 19



What is the in-order successor of 15 in the given binary search tree?



- A. 18                      B. 6                      C. 17                      D. 20

isro-2020   data-structures   binary-search-tree   easy

Answer key

### 35.4.8 Binary Search Tree: ISRO-DEC2017-51



Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the reversal ordering on natural numbers i.e. 9 is assumed to be smallest and 0 is assumed to be largest. The *in-order* traversal of the resultant binary search tree is

- A. 9, 8, 6, 4, 2, 3, 0, 1, 5, 7                      B. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9  
C. 0, 2, 4, 3, 1, 6, 5, 9, 8, 7                      D. 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

isrodec2017   binary-search-tree

Answer key

## 35.5 Binary Tree (5)

### 35.5.1 Binary Tree: ISRO CSE 2008 | Question: 75



A complete binary tree with the property that the value at each node is at least as large as the values at its children is known as

- A. binary search tree                      B. AVL tree  
C. completely balanced tree                      D. Heap

isro2008   data-structures   binary-tree

Answer key

### 35.5.2 Binary Tree: ISRO CSE 2009 | Question: 31



A full binary tree with  $n$  leaves contains

- A.  $n$  nodes      B.  $\log_2 n$  nodes      C.  $2n - 1$       D.  $2^n$  nodes

isro2009   data-structures   binary-tree

Answer key

### 35.5.3 Binary Tree: ISRO CSE 2016 | Question: 30



A complete binary tree with  $n$  non-leaf nodes contains

- A.  $\log_2 n$  nodes      B.  $n + 1$  nodes      C.  $2n$  nodes      D.  $2n + 1$  nodes

data-structures   binary-tree   isro2016

Answer key

### 35.5.4 Binary Tree: ISRO-2013-32



Which of the following number of nodes can form a full binary tree?

- A. 8      B. 15      C. 14      D. 13

isro2013   binary-tree   easy

Answer key

### 35.5.5 Binary Tree: ISRO-DEC2017-57



The *in-order* and *pre-order* traversal of a binary tree are  $db e a f c g$  and  $a b d e c f g$  respectively. The *post-order* traversal of a binary tree is

1.  $ed b g f c a$
2.  $ed b f g c a$
3.  $de b f g c a$
4.  $de f g b c a$

isrodec2017   data-structures   binary-tree

Answer key

## 35.6

## Data Structures (3)

### 35.6.1 Data Structures: ISRO 2011-ECE Stack



In what order the elements of a pushdown stack are accessed?

- a. First In-First Out (FIFO)
- b. Last In Last Out (LILO)
- c. Last In First Out (LIFO)
- d. None of the above

isro-ece   isro2011-ece   data-structures

Answer key

### 35.6.2 Data Structures: ISRO CSE 2020 | Question: 32



$G$  is an undirected graph with vertex set  $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$  and edge set  $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$ . A breadth first search of the graph is performed with  $v_1$  as the root node. Which of the following is a tree edge?

A.  $v2v4$ B.  $v1v4$ C.  $v4v5$ D.  $v3v4$ 

isro-2020 data-structures breadth-first-search normal

Answer key **35.6.3 Data Structures: ISRO CSE 2023 | Question: 4**In Reverse Polish notation, expression  $A * B + C * D$  is written asA.  $AB * CD * +$   
C.  $AB * CD + *$ B.  $A * BCD * +$   
D.  $A * B * CD +$ 

isro-cse-2023 data-structures

Answer key **35.7****Infix Prefix (5)****35.7.1 Infix Prefix: GATE CSE 2004 | Question: 38, ISRO2009-27**

Assume that the operators  $+$ ,  $-$ ,  $\times$  are left associative and  $^$  is right associative. The order of precedence (from highest to lowest) is  $^$ ,  $\times$ ,  $+$ ,  $-$ . The postfix expression corresponding to the infix expression  $a + b \times c - d ^ e ^ f$  is

A.  $abc \times + def ^ ^ -$   
 B.  $abc \times + de ^ f ^ -$   
 C.  $ab + c \times d - e ^ f ^$   
 D.  $- + a \times bc ^ ^ def$

gatecse-2004 stack isro2009 infix-prefix

Answer key **35.7.2 Infix Prefix: GATE CSE 2007 | Question: 38, ISRO2016-27**

The following postfix expression with single digit operands is evaluated using a stack:

$$823 ^ / 23 * + 51 * -$$

Note that  $^$  is the exponentiation operator. The top two elements of the stack after the first  $*$  is evaluated are

A. 6,1                      B. 5,7                      C. 3,2                      D. 1,5

gatecse-2007 data-structures stack normal infix-prefix isro2016

Answer key **35.7.3 Infix Prefix: ISRO CSE 2009 | Question: 28**The infix expression  $A + (B - C) * D$  is correctly represented in prefix notation as

A.  $A + B - C * D$   
 B.  $+ A * - BCD$   
 C.  $ABC - D * +$   
 D.  $A + BC - D *$

isro2009 data-structures stack infix-prefix

Answer key 

#### 35.7.4 Infix Prefix: ISRO CSE 2017 | Question: 69



Choose the equivalent prefix form of the following expression

$$(a+(b-c))*((d-e)/(f+g-h))$$

- A.  $*+a-bc/-de-+fgh$
- B.  $*+a-bc-/de-+fgh$
- C.  $*+a-bc/-ed-+fgh$
- D.  $*+ab-c/-de-+fgh$

isro2017 data-structures infix-prefix stack

Answer key

#### 35.7.5 Infix Prefix: ISRO CSE 2020 | Question: 4



Convert the pre-fix expression to in-fix  $-* + ABC*-DE + FG$

- A.  $(A - B)*C + (D*E) - (F + G)$
- B.  $(A + B)*C - (D - E)*(F + G)$
- C.  $(A + B - C)*(D - E)*(F + G)$
- D.  $(A + B)*C - (D*E) - (F + G)$

isro-2020 data-structures infix-prefix normal

Answer key

### 35.8

#### Linked List (9)

#### 35.8.1 Linked List: ISRO CSE 2008 | Question: 71



The time required to search an element in a linked list of length n is

- A.  $O(\log_2 n)$
- B.  $O(n)$
- C.  $O(1)$
- D.  $O(n^2)$

isro2008 data-structures linked-list time-complexity

Answer key

#### 35.8.2 Linked List: ISRO CSE 2008 | Question: 72



Which of the following operations is performed more efficiently by doubly linked list than by linear linked list?

- A. Deleting a node whose location is given
- B. Searching an unsorted list for a given item
- C. Inserting a node after the node with a given location
- D. Traversing the list to process each node

isro2008 data-structures linked-list

Answer key

#### 35.8.3 Linked List: ISRO CSE 2008 | Question: 76



The minimum number of fields with each node of doubly linked list is

- A. 1
- B. 2
- C. 3
- D. 4

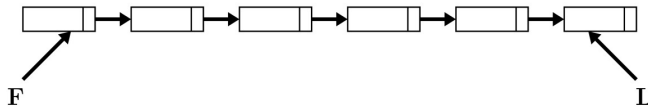
isro2008 data-structures linked-list

Answer key

#### 35.8.4 Linked List: ISRO CSE 2014 | Question: 49



Consider a single linked list where  $F$  and  $L$  are pointers to the first and last elements respectively of the linked list. The time for performing which of the given operations depends on the length of the linked list?



- A. Delete the first element of the list
- B. Interchange the first two elements of the list
- C. Delete the last element of the list
- D. Add an element at the end of the list

isro2014 data-structures linked-list

Answer key

#### 35.8.5 Linked List: ISRO CSE 2017 | Question: 16



Given two statements

- I. Insertion of an element should be done at the last node of the circular list
- II. Deletion of an element should be done at the last node of the circular list

- A. Both are true
- B. Both are false
- C. First is false and second is true
- D. None of the above

isro2017 data-structures linked-list bad-question

Answer key

#### 35.8.6 Linked List: ISRO CSE 2017 | Question: 53



In a doubly linked list the number of pointers affected for an insertion operation will be

- A. 4
- B. 0
- C. 1
- D. Depends on the nodes of doubly linked list

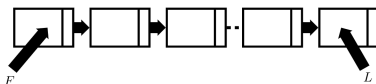
isro2017 data-structures linked-list bad-question

Answer key

#### 35.8.7 Linked List: ISRO CSE 2018 | Question: 63



Consider a singly linked list of the form where  $F$  is a pointer to the first element in the linked list and  $L$  is the pointer to the last element in the list. The time of which of the following operations depends on the length of the list?



- a. Delete the last element of the list
- b. Delete the first element of the list
- c. Add an element after the last element of the list
- d. Interchange the first two elements of the list

isro2018 linked-list data-structures

Answer key

### 35.8.8 Linked List: ISRO CSE 2018 | Question: 79



A doubly linked list is declared as:

```
struct Node {  
    int Value;  
    struct Node *Fwd;  
    struct Node *Bwd;  
};
```

Where Fwd and Bwd represent forward and backward link to the adjacent elements of the list. Which of the following segment of code deletes the node pointed to by  $X$  from the doubly linked list, if it is assumed that  $X$  points to neither the first nor the last node of the list?

- a.  $X \rightarrow Bwd \rightarrow Fwd = X \rightarrow Fwd$ ;  $X \rightarrow Fwd \rightarrow Bwd = X \rightarrow Bwd$ ;
- b.  $X \rightarrow Bwd.Fwd = X \rightarrow Fwd$ ;  $X.Fwd \rightarrow Bwd = X \rightarrow Bwd$ ;
- c.  $X.Bwd \rightarrow Fwd = X.Bwd$ ;  $x \rightarrow Fwd.Bwd = X.Bwd$ ;
- d.  $X \rightarrow Bwd \rightarrow Fwd = X \rightarrow Bwd$ ;  $X \rightarrow Fwd \rightarrow Bwd = X \rightarrow Fwd$ ;

isro2018 linked-list data-structures

Answer key

### 35.8.9 Linked List: ISRO-2013-4



The following steps in a linked list

```
p = getnode()  
info(p) = 10  
next(p) = list  
list = p
```

result in which type of operation?

- A. Pop operation in stack
- B. Removal of a node
- C. Inserting a node
- D. Modifying an existing node

isro2013 linked-list

Answer key

## 35.9

### Priority Queue (1)

#### 35.9.1 Priority Queue: ISRO CSE 2023 | Question: 69



In a priority queue, insertion and deletion can be done at

- A. Front
- B. Back
- C. Middle
- D. Any position

isro-cse-2023 priority-queue data-structures

Answer key

## 35.10

### Queue (1)

#### 35.10.1 Queue: ISRO CSE 2014 | Question: 36



Consider a standard Circular Queue implementation (which has the same condition for Queue Full and Queue Empty) whose size is 11 and the elements of the queue are  $q[0], q[1], \dots, q[10]$ .

The front and rear pointers are initialized to point at  $q[2]$ . In which position will the ninth element be added?

- A.  $q[0]$
- B.  $q[1]$
- C.  $q[9]$
- D.  $q[10]$

Answer key 

35.11

Stack (6)

35.11.1 Stack: ISRO CSE 2007 | Question: 16, ISRO CSE 2009 | Question: 30, ISRO CSE 2014 | Question: 43

The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is



A. A

B. B

C. C

D. D

isro2007 stack isro2009 isro2014

Answer key 

35.11.2 Stack: ISRO CSE 2008 | Question: 70

Stack A has the entries a, b, c (with a on top). Stack B is empty. An entry popped out of stack A can be printed immediately or pushed to stack B. An entry popped out of the stack B can be only be printed. In this arrangement, which of the following permutations of a, b, c are not possible?



A. b a c

B. b c a

C. c a b

D. a b c

isro2008 data-structures stack

Answer key 

35.11.3 Stack: ISRO CSE 2017 | Question: 59

The best data structure to check whether an arithmetic expression has balanced parenthesis is a:



A. Queue

B. Stack

C. Tree

D. List

isro2017 data-structures stack easy

Answer key 

35.11.4 Stack: ISRO CSE 2020 | Question: 72

A stack is implemented with an array of ' $A[0...N-1]$ ' and a variable ' $pos$ '. The push and pop operations are defined by the following code.



```
push (x)
    A[pos] <- x
    pos <- pos + 1
end push
pop()
    pos <- pos - 1
    return A[pos]
end pop
```

Which of the following will initialize an empty stack with capacity  $N$  for the above implementation

A.  $pos \leftarrow -1$ B.  $pos \leftarrow 0$ C.  $pos \leftarrow 1$ D.  $pos \leftarrow N - 1$ 

isro-2020 data-structures stack normal

Answer key 

35.11.5 Stack: ISRO2015-11

If the sequence of operations - push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values





- A. 2,2,1,1,2      B. 2,2,1,2,2      C. 2,1,2,2,1      D. 2,1,2,2,2

isro2015   data-structures   stack   easy

Answer key 

### 35.11.6 Stack: ISRO2015-15



The queue data structure is to be realized by using stack. The number of stacks needed would be

- A. It cannot be implemented      B. 2 stacks  
C. 4 stacks      D. 1 stack

isro2015   data-structures   stack

Answer key 

## 35.12

### Tree (3)

#### 35.12.1 Tree: ISRO CSE 2011 | Question: 19



If node A has three siblings and B is parent of A, what is the degree of A?

- A. 0      B. 3      C. 4      D. 5

isro2011   data-structures   tree

Answer key 

#### 35.12.2 Tree: ISRO CSE 2014 | Question: 73



How many different trees are there with four nodes A, B, C and D?

- A. 30      B. 60      C. 90      D. 120

isro2014   data-structures   tree   combinatory

Answer key 

#### 35.12.3 Tree: ISRO CSE 2020 | Question: 71



Of the following, which best approximates the ratio of the number of nonterminal nodes in the total number of nodes in a complete  $K$ -ary tree of depth  $N$ ?

- A.  $1/N$       B.  $N - 1/N$       C.  $1/K$       D.  $K - 1/K$

isro-2020   data-structures   tree   normal

Answer key 

## 35.13

### Tree Traversal (4)

#### 35.13.1 Tree Traversal: ISRO CSE 2011 | Question: 26



The in-order traversal of a tree resulted in FBGADCE. Then the pre-order traversal of that tree would result in

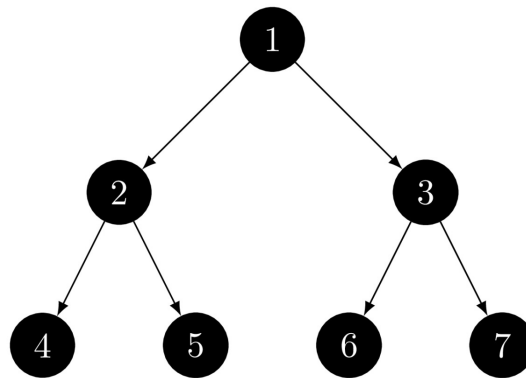
- A. FGBDECA      B. ABFGCDE      C. BFGCDEA      D. AFGBDEC

isro2011   data-structures   tree-traversal

Answer key 

#### 35.13.2 Tree Traversal: ISRO CSE 2017 | Question: 50





If the post order traversal gives  $ab - cd * +$  then the label of the nodes 1,2,3.. will be

- A. +, -, \*, a,b,c,d
- B. a, -,b,+,c,\*,d
- C. a,b,c,d,-,\*,+
- D. -,a,b,+,\*,c,d

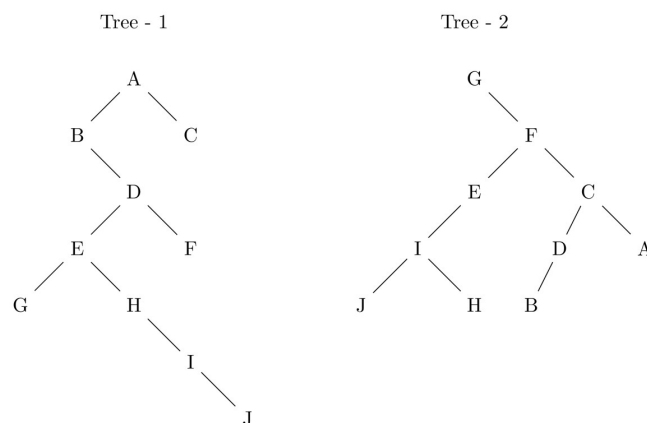
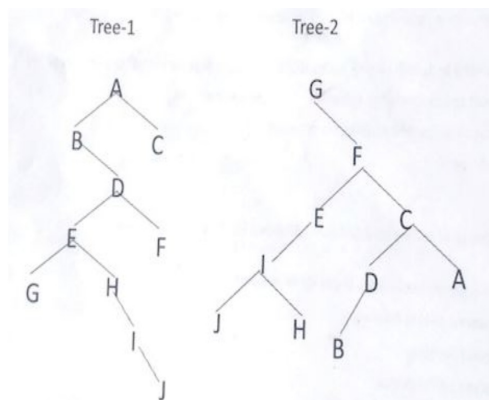
isro2017 data-structures binary-tree tree-traversal

Answer key

### 35.13.3 Tree Traversal: ISRO CSE 2018 | Question: 80



If Tree-1 and Tree-2 are the trees indicated below:



Which traversals of Tree-1 and Tree-2, respectively, will produce the same sequence?

- a. Preorder, postorder
- b. Postorder, inorder
- c. Postorder, preorder
- d. Inorder, preorder

isro2018 data-structures tree-traversal

Answer key

**35.13.4 Tree Traversal: ISRO CSE 2020 | Question: 23**

The post-order traversal of binary tree is ACEDBHIGF. The pre-order traversal is

- A. ABCDEFGHI  
C. FABCDEGHI

- B. FBADCEGHIH  
D. ABDCEFGIHH

isro-2020 data-structures binary-tree tree-traversal normal

Answer key

## Answer Keys

|         |   |         |   |         |     |         |     |         |     |
|---------|---|---------|---|---------|-----|---------|-----|---------|-----|
| 35.0.1  | A | 35.0.2  | C | 35.0.3  | D   | 35.0.4  | B   | 35.0.5  | A   |
| 35.0.6  | D | 35.0.7  | C | 35.1.1  | B   | 35.1.2  | C   | 35.1.3  | D   |
| 35.2.1  | B | 35.2.2  | A | 35.3.1  | C   | 35.3.2  | D   | 35.4.1  | C   |
| 35.4.2  | B | 35.4.3  | B | 35.4.4  | B   | 35.4.5  | C   | 35.4.6  | C   |
| 35.4.7  | C | 35.4.8  | D | 35.5.1  | D   | 35.5.2  | C   | 35.5.3  | D   |
| 35.5.4  | B | 35.5.5  | C | 35.6.1  | Q-Q | 35.6.2  | B   | 35.6.3  | Q-Q |
| 35.7.1  | A | 35.7.2  | A | 35.7.3  | B   | 35.7.4  | A   | 35.7.5  | B   |
| 35.8.1  | B | 35.8.2  | A | 35.8.3  | C   | 35.8.4  | C   | 35.8.5  | B   |
| 35.8.6  | X | 35.8.7  | A | 35.8.8  | A   | 35.8.9  | C   | 35.9.1  | Q-Q |
| 35.10.1 | A | 35.11.1 | D | 35.11.2 | C   | 35.11.3 | B   | 35.11.4 | D   |
| 35.11.5 | A | 35.11.6 | B | 35.12.1 | X   | 35.12.2 | Q-Q | 35.12.3 | C   |
| 35.13.1 | B | 35.13.2 | A | 35.13.3 | X   | 35.13.4 | B   |         |     |

## 36.0.1 ISRO 2008-ECE exit() function



The library function `exit ( )` causes as exit from

- (a) the loop in which it occurs
- (b) the block in which it occurs
- (c) the functions in which it occurs
- (d) the program in which it occurs

isro-ece programming

Answer key

## 36.0.2 ISRO 2008- ECE GOTO statement



The "**go to** statement" causes control to go to

- (a) An operator
- (b) A label
- (c) A variable
- (d) A function

isro-ece

Answer key

## 36.0.3 ISRO-DEC2017-14



Type-IV *JDBC* driver is a driver

- A. which is written in *C++*
- B. which requires an intermediate layer
- C. which communicates through Java sockets
- D. which translates *JDBC* function calls into *API* not native to *DBMS*

isrodec2017

Answer key

## 36.0.4 ISRO-DEC2017-19



```
//Consider the program
void function(int n)
{
    int i,j,count=0;
    for(i=n/2;i<=n;i++)
        for(j=1;j<=n;j=j*2)
            count++;
}
```

The complexity of the program is

- A.  $O(\log n)$
- B.  $O(n^2)$
- C.  $O(n^2 \log n)$
- D.  $O(n \log n)$

Answer key **36.0.5 ISRO-DEC2017-46**

In a compact one-dimensional array representation for lower triangular matrix (all elements above diagonal are zero) of size  $n \times n$ , non zero elements of each row are stored one after another, starting from first row, the index of  $(i, j)^{th}$  element in this niw representation is

- A.  $i+j$                       B.  $j + \frac{i(i-1)}{2}$                       C.  $i+j-1$                       D.  $i + \frac{j(j-1)}{2}$

Answer key **36.0.6 ISRO-DEC2017-65**

Which one of the following are essential features of object-oriented language?

- A. Abstraction and encapsulation  
 B. Strictly-typed  
 C. Type-safe property coupled with sub-type rule  
 D. Polymorphism in the presence of inheritance

- A. A and B only                      B. A, D and B only  
 C. A and D only                      D. A, C and D only

Answer key **36.0.7 ISRO-DEC2017-74**

Consider the function

```
int func(int num)
{
    int count=0;
    while(num)
    {
        count++;
        num>>=1;
    }
    return(count);
}
```

For  $func(435)$  the value returned is

- A. 9                      B. 8                      C. 0                      D. 10

Answer key **36.0.8 ISRO-DEC2017-73**

Consider the function

```
int fun(x: integer)
{
    If x>100 then fun=x-10;
    else
        fun=fun(fun(x+11));
}
```



Answer key **36.2.3 Array: ISRO CSE 2020 | Question: 18**

Consider a 2-dimensional array  $x$  with 10 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row-major format. If the first element  $x[0][0]$  occupies the memory location with address 1000 and each element occupies only one memory location, which all locations (in decimal) will be holding a value of 10?

- a. 1018,1019                      b. 1022,1041                      c. 1013,1014                      d. 1000,1399

isro-2020 programming array normal

Answer key **36.2.4 Array: ISRO CSE 2020 | Question: 61**

What is the output of the code given below?

```
# include<stdio.h>
int main()
{
    char name[]="satellites";
    int len;
    int size;
    len= strlen(name);
    size = sizeof(name);
    printf("%d",len*size);
    return 0;
}
```

- A. 100                      B. 110                      C. 40                      D. 44

isro-2020 programming array normal

Answer key **36.3****Array of Pointers (1)****36.3.1 Array of Pointers: ISRO CSE 2023 | Question: 68**

Pointer  $p$  to an array of integers (of size 2 Bytes) is initialised to 200. What is the value of  $p + 3$ ?

- A. 206                      B. 203                      C. 212                      D. 204

isro-cse-2023 array-of-pointers easy

Answer key **36.4****Functions (2)****36.4.1 Functions: ISRO CSE 2020 | Question: 78**

In the following procedure

```
Integer procedure P(X,Y);
Integer X,Y;
value x;
begin
    K=5;
    L=8;
    P=x+y;
end
```

$X$  is called by value and  $Y$  is called by name. If the procedure were invoked by the following program fragment

```
K=0;
L=0;
Z=P(K, L);
```

then the value of  $Z$  will be set equal to

- A. 5                      B. 8                      C. 13                      D. 0

isro-2020   programming   functions   normal

Answer key 

### 36.4.2 Functions: ISRO2015-70



The following program

```
main()
{
    inc(); inc(); inc();
}
inc()
{
    static int x;
    printf("%d", ++x);
}
```

- A. prints 012  
B. prints 123  
C. prints 3 consecutive, but unpredictable numbers  
D. prints 111

isro2015   programming-in-c   functions

Answer key 

## 36.5

### Identify Function (1)

#### 36.5.1 Identify Function: ISRO-2013-63



Consider the following  $C$  code.

```
#include <stdio.h>
#include <math.h>
void main ()
{
    double pi = 3.1415926535;
    int a = 1;
    int i;
    for (i=0; i < 3; i++)
        if (a = cos(pi * i/2))
            printf("%d", 1);
        else printf("%d", 0);
}
```

What would the program print?

- A. 000                      B. 010                      C. 101                      D. 111

isro2013   programming-in-c   identify-function   output

Answer key 

## 36.6

### Loop (7)





### 36.6.1 Loop: ISRO CSE 2011 | Question: 63



What is the output of the following C code?

```
#include <stdio.h>

int main()
{
    int index;
    for(index=1; index<=5; index++)
    {
        printf("%d", index);
        if (index==3)
            continue;
    }
}
```

- A. 1245                      B. 12345                      C. 12245                      D. 12354

isro2011   programming-in-c   loop

Answer key

### 36.6.2 Loop: ISRO CSE 2014 | Question: 38



How many lines of output does the following C code produce?

```
#include<stdio.h>
float i=2.0;
float j=1.0;
float sum = 0.0;
main()
{
    while (i/j > 0.001)
    {
        j+=j;
        sum=sum+(i/j);
        printf("%f\n", sum);
    }
}
```

- A. 8                      B. 9                      C. 10                      D. 11

isro2014   programming-in-c   loop   output

Answer key

### 36.6.3 Loop: ISRO CSE 2017 | Question: 62



What will be the output of the following C code?

```
#include <stdio.h>
main()
{
    int i;
    for(i=0;i<5;i++)
    {
        int i=10;
        printf("%d" , i);
        i++;
    }
    return 0;
}
```

- A. 10 11 12 13 14                      B. 10 10 10 10 10  
C. 0 1 2 3 4                      D. Compilation error

isro2017   programming-in-c   loop   output

Answer key

#### 36.6.4 Loop: ISRO CSE 2017 | Question: 63



What does the following program do when the input is unsigned 16 bit integer?

```
#include<stdio.h>
main() {
    unsigned int num;
    int i;
    scanf("%u", &num);
    for(i=0; i<16; i++) {
        printf("%d", (num<<i&1<<15)?1:0);
    }
}
```

- A. It prints all even bits from num  
B. It prints all odd bits from num  
C. It prints binary equivalent of num  
D. None of above

isro2017 programming-in-c loop

Answer key

#### 36.6.5 Loop: ISRO2015-71



Consider the following program fragment

```
i=6720; j=4;
while (i%j)==0
{
    i=i/j;
    j=j+1;
}
```

On termination j will have the value

- A. 4  
B. 8  
C. 9  
D. 6720

isro2015 programming-in-c loop

Answer key

#### 36.6.6 Loop: ISRO2015-73



The output of the following program is

```
main()
{
    static int x[] = {1,2,3,4,5,6,7,8}
    int i;
    for (i=2; i<6; ++i)
        x[x[i]]=x[i];
    for (i=0; i<8; ++i)
        printf("%d", x[i]);
}
```

- A. 1 2 3 3 5 5 7 8  
B. 1 2 3 4 5 6 7 8  
C. 8 7 6 5 4 3 2 1  
D. 1 2 3 5 4 6 7 8

isro2015 programming-in-c loop output

Answer key

#### 36.6.7 Loop: ISRO2015-75



The for loop

```
for (i=0; i<10; ++i)
printf("%d", i&1);
```

prints

- A. 0101010101  
B. 0111111111  
C. 0000000000  
D. 1111111111

Answer key 

## 36.7

## Loop Invariants (1)

## 36.7.1 Loop Invariants: ISRO CSE 2007 | Question: 15



Consider the following pseudo-code

```
x:=1;
i:=1;
while (x <= 1000)
begin
    x:=2^x;
    i:=i+1;
end;
```

What is the value of i at the end of the pseudo-code?

- A. 4                      B. 5                      C. 6                      D. 7

isro2007 loop-invariants

Answer key 

## 36.8

## Macros (2)

## 36.8.1 Macros: ISRO CSE 2014 | Question: 3



What is the output of the following C program?

```
#include<stdio.h>
#define SQR(x) (x*x)

int main()
{
    int a;
    int b=4;
    a=SQR(b+2);
    printf("%d\n",a);
    return 0;
}
```

- A. 14                      B. 36                      C. 18                      D. 20

programming-in-c macros isro2014

Answer key 

## 36.8.2 Macros: ISRO2015-76



Consider the following statements

```
#define hypotenuse (a, b) sqrt (a*a+b*b);
```

The macro call hypotenuse(a+2,b+3);

- A. Finds the hypotenuse of a triangle with sides  $a+2$  and  $b+3$       B. Finds the square root of  $(a+2)^2$  and  $(b+3)^2$   
 C. Is invalid                      D. Find the square root of  $3*a+4*b+5$

programming-in-c macros isro2015

Answer key 

## 36.9

## Non Gatecse (3)

### 36.9.1 Non Gatecse: ISRO CSE 2008 | Question: 80



In C, what is the effect of a negative number in a field width specifier?

- A. the values are displayed right justified
- B. the values are displayed centered
- C. the values are displayed left justified
- D. the values are displayed as negative numbers

isro2008 programming programming-in-c non-gatecse

Answer key

### 36.9.2 Non Gatecse: ISRO-2017



Q.A strictly binary tree with 10 leaves

- A) cannot have more than 19 nodes
- B) has exactly 19 nodes
- C)has exactly 17 nodes
- D) has exactly 20 nodes

isro2017 non-gatecse

Answer key

### 36.9.3 Non Gatecse: ISRO-2017



Consider the code segment

```
int i, j, x, y, m, n;
n=20;
for (i=0; i<n; i++)
{
    for (j=0; j<n; j++)
    {
        if (i%2)
        {
            x+= ( (4*j) +5*i) ;
            y+= (7+4*j) ;
        }
    }
}
m=x+y;
```

Which one of the following is false ?

- A) The code contains loop invariant computation
- B) There is scope of common sub-expression elimination in this code
- C)There is scope of strength reduction in this code
- D)There is scope of dead code elimination in this code

isro2017 non-gatecse

Answer key

## 36.10

## Object Oriented Programming (1)

### 36.10.1 Object Oriented Programming: ISRO 2008-ECE programming



A program having features such as data abstraction, encapsulation and data hiding, polymorphism inheritance is called

- (a) Structured program
- (b) Object oriented program
- (c) Open source program

(d) Real time program

isro-ece programming object-oriented-programming

Answer key 

36.11

Output (5)

36.11.1 Output: ISRO CSE 2014 | Question: 32



What is the output of the following C program?

```
#include<stdio.h>
void main(void){
    int shifty;
    shifty=0570;
    shifty=shifty>>4;
    shifty=shifty<<6;
    printf("The value of shifty is %o \n",shifty);
}
```

- A. The value of shifty is 15c0  
C. The value of shifty is 5700

- B. The value of shifty is 4300  
D. The value of shifty is 2700

programming-in-c isro2014 output

Answer key 

36.11.2 Output: ISRO CSE 2017 | Question: 64



What is the output of the following program?

```
#include<stdio.h>
int tmp=20;
main()
{
    printf("%d", tmp);
    func();
    printf("%d", tmp);
}
func()
{
    static int tmp=10;
    printf("%d", tmp);
}
```

A. 20 10 10

B. 20 10 20

C. 20 20 20

D. 10 10 10

isro2017 programming-in-c functions output

Answer key 

36.11.3 Output: ISRO CSE 2018 | Question: 1



Consider the following program

```
{
    int x=1;
    printf("%d", (*char(char*)&x));
}
```

Assuming required header files are included and if the machine in which this program is executed is little endian, then the output will be

A. 0

B. 99999999

C. 1

D. unpredictable

isro2018 programming output

Answer key 

### 36.11.4 Output: ISRO CSE 2018 | Question: 60



What is the output of the following program?

```
main() {  
    int x=2, y=5;  
    if(x<y) return (x=x+y);  
    else printf("z1");  
    printf("z2");  
}
```

- A. z2  
B. z1z2  
C. Compilation error  
D. None of these

isro2018 programming output

Answer key 

### 36.11.5 Output: ISRO CSE 2018 | Question: 8



A language with string manipulation facilities uses the following operations.

head(*s*)- returns the first character of the string *s*

tail(*s*)- returns all but the first character of the string *s*

concat(*s*<sub>1</sub>, *s*<sub>2</sub>)- concatenates string *s*<sub>1</sub> with *s*<sub>2</sub>.

The output of concat(head(*s*), head(tail(tail(*s*)))), where *s* is *acbc* is:

- a. ab                      b. ba                      c. ac                      d. aa

isro2018 programming output

Answer key 

## 36.12

## Parameter Passing (1)

### 36.12.1 Parameter Passing: GATE CSE 2006 | Question: 56, ISRO2009-58



Consider the following code written in a pass-by-reference language like FORTRAN and these statements about the code.

```
subroutine swap(ix,iy)  
    it = ix  
L1 : ix = iy  
L2 : iy = it  
end  
ia = 3  
ib = 8  
call swap (ia, ib+5)  
print *, ia, ib  
end
```

**S1:** The compiler will generate code to allocate a temporary nameless cell, initialize it to 13, and pass the address of the cell to swap

**S2:** On execution the code will generate a runtime error on line L1

**S3:** On execution the code will generate a runtime error on line L2

**S4:** The program will print 13 and 8

**S5:** The program will print 13 and -2

Exactly the following set of statement(s) is correct:

- A. S1 and S2                      B. S1 and S4                      C. S3                      D. S1 and S5

gatecse-2006 programming parameter-passing normal out-of-syllabus-now isro2009

Answer key 

**36.13.1 Pointers: GATE CSE 2005 | Question: 1, ISRO2017-55**

What does the following C-statement declare?

```
int (*f) (int * );
```

- A. A function that takes an integer pointer as argument and returns an integer
- B. A function that takes an integer as argument and returns an integer pointer
- C. A pointer to a function that takes an integer pointer as argument and returns an integer
- D. A function that takes an integer pointer as argument and returns a function pointer

gatecse-2005 programming programming-in-c pointers easy isro2017

Answer key

**36.13.2 Pointers: ISRO CSE 2014 | Question: 13**

Which of the following is true with respect to Reference?

- A. A reference can never be NULL
- B. A reference needs an explicit dereferencing mechanism
- C. A reference can be reassigned after it is established
- D. A reference and pointer are synonymous

programming isro2014 pointers

Answer key

**36.13.3 Pointers: ISRO CSE 2016 | Question: 20**

What is the output of this C code?

```
#include<stdio.h>
void main()
{
    int k=5;
    int *p=&k;
    int **m=&p;
    printf("%d %d %d", k, *p, **m);
}
```

- A. 5 5 5
- B. 5 5 junk
- C. 5 junk junk
- D. compile time error

programming-in-c pointers isro2016

Answer key

**36.13.4 Pointers: ISRO CSE 2016 | Question: 25**

What will be output of the following program? Assume that you are running this program in little-endian processor.

```
#include<stdio.h>
int main()
{
    short a=320;
    char *ptr;
    ptr=(char *) &a;
    printf("%d", *ptr);
    return 0;
}
```

- A. 1                      B. 320                      C. 64                      D. Compilation error

programming-in-c   pointers   isro2016   little-endian-big-endian

Answer key 

### 36.13.5 Pointers: ISRO CSE 2020 | Question: 62



What is output of the following 'C' code assuming it runs on a byte addressed little endian machine?

```
#include<stdio.h>
int main()
{
    int x;
    char *ptr;
    x=622,100,101;
    printf("%d", (*(char *)&x)*(x%3));
    return 0;
}
```

- A. 622                      B. 311                      C. 22                      D. 110

isro-2020   programming   programming-in-c   normal   pointers

Answer key 

### 36.13.6 Pointers: ISRO CSE 2020 | Question: 63



What is the output in a 32 bit machine with 32 bit compiler?

```
#include<stdio.h>
rer(int **ptr2, int **ptr1)
{
    int *ii;
    ii=*ptr2;
    *ptr2=*ptr1;
    *ptr1=ii;
    **ptr1=**ptr2;
    **ptr2+=**ptr1;
}

void main(){
    int var1=5, var2=10;
    int *ptr1=&var1,*ptr2=&var2;
    rer(&ptr1,&ptr2);
    printf("%d %d",var2,var1);
}
```

- A. 60,70                      B. 50,50                      C. 50,60                      D. 60,50

isro-2020   programming   programming-in-c   normal   pointers

Answer key 

### 36.13.7 Pointers: ISRO2015-72



Consider the following declaration:

```
int a, *b=&a, **c=&b;
```

The following program fragment

```
a=4;        **c=5;
```

- A. does not change the value of a                      B. assigns address of c to a  
C. assigns the value of b to a                      D. assigns 5 to a

pointers   programming-in-c   isro2015

Answer key 



## 36.14.1 Programming In C: ISRO CS 2018



```
#include <stdio.h> int main() {  
  
    int i,j,x;  
  
    scanf("%d",&x);  
  
    i=1;  
  
    j=1;  
  
    while(i<10){  
  
        j=j*i;  
  
        i=i+1;  
  
        if(i==x) break;  
  
    }  
  
    return 0;  
  
}
```

isro2018 programming-in-c

Answer key

## 36.14.2 Programming In C: ISRO CSE 2014 | Question: 47



The following three 'C' language statements is equivalent to which single statement?

```
y=y+1;  
z=x+y;  
x=x+1
```

- A.  $z = x + y + 2;$                       B.  $z = (x++) + (++y);$   
C.  $z = (x++) + (y++);$               D.  $z = (x++) + (++y) + 1;$

isro2014 programming-in-c

Answer key

## 36.14.3 Programming In C: ISRO CSE 2016 | Question: 79



Which one of the following is correct about the statements given below?

- I. All function calls are resolved at compile time in C lang  
II. All function calls are resolved at compile time in C++ lang

- A. Only II is correct                      B. Both I and II are correct  
C. Only I is correct                      D. Both I and II are incorrect

isro2016 programming-in-c

Answer key

## 36.14.4 Programming In C: ISRO CSE 2018 | Question: 18



Consider the following C code segment:

```
#include <stdio.h>
main()
{
    int i, j, x;
    scanf("%d", &x);
    i=1; j=1;
    while (i<10) {
        j =j*i;
        i= i+1;
        if (i==x) break;
    }
}
```

For the program fragment above, which of the following statements about the variables i and j must be true after execution of this program? [ !(exclamation) sign denotes factorial in the answer]

- $(j = (x - 1)!) \wedge (i \geq x)$
- $(j = 9!) \wedge (j = 10)$
- $((j = 10!) \wedge (i = 10)) \vee ((j = (x - 1)!) \wedge (i = x))$
- $(j = 9!) \wedge (i \geq 10) \vee ((j = (x - 1)!) \wedge (i = x))$

isro2018 programming-in-c programming

Answer key 

### 36.14.5 Programming In C: ISRO CSE 2018 | Question: 70



Consider the following C program:

```
main()
{
    float sum= 0.0, j=1.0,i=2.0;
    while(i/j>0.001) {
        j=j+1;
        sum=sum+i/j;
        printf("%f/n", sum);
    }
}
```

- |                            |                                 |
|----------------------------|---------------------------------|
| A. 0 - 9 lines of output   | B. 10 - 19 lines out output     |
| C. 20 - 29 lines of output | D. More than 29 lines of output |

isro2018 programming-in-c programming

Answer key 

### 36.14.6 Programming In C: ISRO2015-68



Consider the following program fragment

```
if (a > b) if (b > c) s1; else s2;
```

s2 will be executed if

- |                              |                           |
|------------------------------|---------------------------|
| A. $a \leq b$                | B. $b > c$                |
| C. $b \geq c$ and $a \leq b$ | D. $a > b$ and $b \leq c$ |

isro2015 programming programming-in-c

Answer key 

### 36.14.7 Programming In C: ISRO2015-74



Which of the following has the compilation error in C?

- |                             |                              |
|-----------------------------|------------------------------|
| A. <code>int n = 17;</code> | B. <code>char c = 99;</code> |
|-----------------------------|------------------------------|

C. float f = (float)99.32;

programming-in-c isro2015

D. #include <stdio.h>

Answer key 

36.15

Recursion (3)

36.15.1 Recursion: GATE CSE 2004 | Question: 31, ISRO2008-40



Consider the following C function:

```
int f(int n)
{
    static int i = 1;
    if(n >= 5) return n;
    n = n+i;
    i++;
    return f(n);
}
```

The value returned by  $f(1)$  is:

A. 5

B. 6

C. 7

D. 8

gatecse-2004 programming programming-in-c recursion easy isro2008

Answer key 

36.15.2 Recursion: ISRO CSE 2018 | Question: 69



Let  $P$  be a procedure that for some inputs calls itself (i.e. is recursive). If  $P$  is guaranteed to terminate, which of the following statement(s) must be true?

- I.  $P$  has a local variable
- II.  $P$  has an execution path where it does not call itself
- III.  $P$  either refers to a global variable or has at least one parameter

a. I only

b. II only

c. III only

d. II and III only

isro2018 programming recursion

Answer key 

36.15.3 Recursion: ISRO CSE 2020 | Question: 59



Consider the following recursive C function that takes two arguments

```
unsigned int rer(unsigned int n, unsigned int r){
    if(n>0) return(n%r + rer(n/r,r));
    else return 0;
}
```

What is the return value of the function  $rer$  when it is called as  $rer(513, 2)$ ?

A. 9

B. 8

C. 5

D. 2

isro-2020 programming normal recursion

Answer key 

36.16

Runtime Environment (3)

36.16.1 Runtime Environment: ISRO CSE 2007 | Question: 57



Which of the following programming language(s) provides garbage collection automatically

A. Lisp

B. C++

C. Fortran

D. C

Answer key **36.16.2 Runtime Environment: ISRO CSE 2014 | Question: 39**

If only one memory location is to be reserved for a class variable, no matter how many objects are instantiated, then the variable should be declared as

- A. extern                      B. static                      C. volatile                      D. const

Answer key **36.16.3 Runtime Environment: ISRO CSE 2017 | Question: 70**

We use malloc and calloc for:

- A. Dynamic memory allocation                      B. Static memory allocation  
C. Both dynamic memory allocation and static memory allocation                      D. None of these

Answer key **36.17 Semantic Analysis (1)****36.17.1 Semantic Analysis: ISRO CSE 2018 | Question: 10**

Which of the following comparisons between static and dynamic type checking is incorrect?

- Dynamic type checking slows down the execution
- Dynamic type checking offers more flexibility to the programmers
- In contrast to Static type checking, dynamic type checking may cause failure in runtime due to type errors
- Unlike static type checking dynamic type checking is done during compilation

Answer key **36.18 Structure (2)****36.18.1 Structure: ISRO CSE 2018 | Question: 2**

Consider the following declaration :

```
struct addr {
    char city[10];
    char street[30];
    int pin;
};
struct {
    char name[30];
    int gender;
    struct addr locate;
} person, *kd = &person;
```

Then  $*(kd \rightarrow name + 2)$  can be used instead of:

- A.  $person.name + 2$                       B.  $kd \rightarrow (name + 2)$   
C.  $((*kd).name + 2)$                       D. either A) or B), not C)

Answer key 

### 36.18.2 Structure: ISRO CSE 2020 | Question: 74



Following declaration of an array of struct, assumes size of byte, short, int and long are 1, 2, 3 and 4 respectively. Alignment rule stipulates that  $n$  – byte field must be located at an address divisible by  $n$ , the fields in the struct are not rearranged, padding is used to ensure alignment. All elements of array should be of same size.

```
Struct complx
    Short s
    Byte b
    Long l
    Int i
End Complx
Complx C[10]
```

Assuming  $C$  is located at an address divisible by 8, what is the total size of  $C$ , in bytes?

- A. 150                      B. 160                      C. 200                      D. 240

isro-2020   programming   normal   structure

Answer key

### 36.19 Undefined Behaviour (1)

#### 36.19.1 Undefined Behaviour: ISRO2015-69



If  $n$  has 3, then the statement  $a[++n]=n++$ ;

- A. assigns 3 to  $a[5]$                       B. assigns 4 to  $a[5]$   
C. assigns 4 to  $a[4]$                       D. what is assigned is compiler dependent

isro2015   programming-in-c   non-gatecse   undefined-behaviour

Answer key

### 36.20 Union (1)

#### 36.20.1 Union: GATE CSE 2000 | Question: 1.17, ISRO2015-79



Consider the following C declaration:

```
struct {
    short x[5];
    union {
        float y;
        long z;
    } u;
} t;
```

Assume that the objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable  $t$ , ignoring alignment consideration, is:

- A. 22 bytes                      B. 14 bytes                      C. 18 bytes                      D. 10 bytes

gatecse-2000   programming   programming-in-c   easy   isro2015   union

Answer key

## Answer Keys

|        |     |        |     |        |   |        |   |        |     |
|--------|-----|--------|-----|--------|---|--------|---|--------|-----|
| 36.0.1 | Q-Q | 36.0.2 | Q-Q | 36.0.3 | C | 36.0.4 | D | 36.0.5 | B   |
| 36.0.6 | C   | 36.0.7 | A   | 36.0.8 | C | 36.0.9 | D | 36.1.1 | C   |
| 36.2.1 | B   | 36.2.2 | C   | 36.2.3 | X | 36.2.4 | B | 36.3.1 | Q-Q |

|         |     |
|---------|-----|
| 36.4.1  | B   |
| 36.6.3  | B   |
| 36.7.1  | B   |
| 36.9.3  | Q-Q |
| 36.11.4 | D   |
| 36.13.3 | A   |
| 36.14.1 | Q-Q |
| 36.14.6 | D   |
| 36.16.1 | A   |
| 36.18.2 | X   |

|         |     |
|---------|-----|
| 36.4.2  | B   |
| 36.6.4  | C   |
| 36.8.1  | A   |
| 36.10.1 | Q-Q |
| 36.11.5 | A   |
| 36.13.4 | C   |
| 36.14.2 | B   |
| 36.14.7 | X   |
| 36.16.2 | B   |
| 36.19.1 | D   |

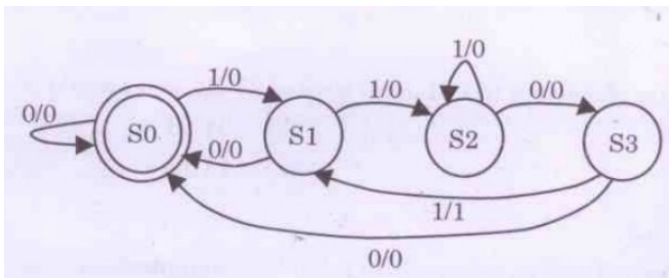
|         |   |
|---------|---|
| 36.5.1  | C |
| 36.6.5  | C |
| 36.8.2  | D |
| 36.11.1 | D |
| 36.12.1 | B |
| 36.13.5 | D |
| 36.14.3 | D |
| 36.15.1 | C |
| 36.16.3 | A |
| 36.20.1 | C |

|         |   |
|---------|---|
| 36.6.1  | B |
| 36.6.6  | A |
| 36.9.1  | C |
| 36.11.2 | B |
| 36.13.1 | C |
| 36.13.6 | D |
| 36.14.4 | D |
| 36.15.2 | D |
| 36.17.1 | D |

|         |     |
|---------|-----|
| 36.6.2  | D   |
| 36.6.7  | A   |
| 36.9.2  | Q-Q |
| 36.11.3 | X   |
| 36.13.2 | A   |
| 36.13.7 | D   |
| 36.14.5 | D   |
| 36.15.3 | D   |
| 36.18.1 | C   |



## 37.0.1 ISRO 2015-EC Finite State Machine



The following Finite State Machine (FSM) is used to detect a particular pattern in input data stream. Whenever the pattern is matched at input, output is set to '1' or else output is cleared to '0'. For which of the following data stream, output goes to '1' twice?

- (a) 0010011010010101
- (b) 0011011010010101
- (c) 0101011000010101
- (d) 1100100101001010

isro2015-ece isro-ece theory-of-computation

Answer key

## 37.0.2 ISRO-DEC2017-24



Let  $L_1$  be regular language,  $L_2$  be a deterministic context free language and  $L_3$  a **recursively enumerable language**, but not recursive. Which one of the following statements is false ?

- A.  $L_3 \cap L_1$  is recursive
- B.  $L_1 \cap L_2 \cap L_3$  is recursively enumerable
- C.  $L_1 \cup L_2$  is context free
- D.  $L_1 \cap L_2$  is context free

isrodec2017

Answer key

## 37.0.3 ISRO-DEC2017-22



Consider the grammar with productions

$$S \rightarrow aSb \mid SS \mid \varepsilon$$

This grammar is

- A. not context-free, not linear
- B. not context-free, linear
- C. context-free, not linear
- D. context-free, linear

isrodec2017

Answer key

## 37.0.4 ISRO-DEC2017-25



Let  $L = \{a^p \mid p \text{ is a prime}\}$ . Then which of the following is true

- A. It is not accepted by a Turing Machine
- B. It is regular but not context-free
- C. It is context-free but not regular
- D. It is neither regular nor context-free, but accepted by a Turing Machine

isrodec2017

Answer key 

### 37.0.5 ISRO-DEC2017-21



For  $\Sigma = \{a, b\}$  the regular expression  $r = (aa)^*(bb)^*b$  denotes

- A. Set of strings with 2  $a$ 's and 2  $b$ 's
- B. Set of strings with 2  $a$ 's 2  $b$ 's followed by  $b$
- C. Set of strings with 2  $a$ 's followed by  $b$ 's which is a multiple of 3
- D. Set of strings with even number of  $a$ 's followed by odd number of  $b$ 's

isrodec2017

Answer key 

### 37.0.6 ISRO-DEC2017-26



Which of the following are context-free?

$$A = \{a^n b^n a^m b^m \mid m, n \geq 0\}$$

$$B = \{a^m b^n a^m b^n \mid m, n \geq 0\}$$

$$C = \{a^m b^n \mid m \neq 2n, m, n \geq 0\}$$

- A. A and B only
- B. A and C only
- C. B and C only
- D. C only

isrodec2017

Answer key 

### 37.0.7 ISRO CSE 2023 | Question: 1



Consider the context-free grammar  $G$  below for arithmetic expressions :

$$E \rightarrow E + E \mid E \times E \mid \text{id}$$

Which of the following statements is TRUE:

- A. The string " id + id  $\times$  id " has no parse tree according to  $G$
- B. The string " id + id  $\times$  id " has only one parse tree according to  $G$
- C. The string " id + id  $\times$  id " has exactly two parse trees according to  $G$
- D. The string " id + id  $\times$  id " has more than two parse trees according to  $G$

isro-cse-2023 theory-of-computation ambiguous-grammar

Answer key 

## 37.1

### Closure Property (2)

#### 37.1.1 Closure Property: ISRO CSE 2017 | Question: 77



If  $L$  and  $P$  are two recursively enumerable languages then they are not closed under

- A. Kleene star  $L^*$  of  $L$
- B. Intersection  $L \cap P$
- C. Union  $L \cup P$
- D. Set difference

isro2017 set-theory theory-of-computation recursive-and-recursively-enumerable-languages closure-property

Answer key 



### 37.1.2 Closure Property: ISRO CSE 2018 | Question: 25



*CFG* (Context Free Grammar) is not closed under:

- A. Union
- B. Complementation
- C. Kleene star
- D. Product

isro2018 closure-property context-free-language theory-of-computation

Answer key

## 37.2

### Context Free Grammar (3)

#### 37.2.1 Context Free Grammar: ISRO CSE 2018 | Question: 27



A *CFG* (Context Free Grammar) is said to be in Chomsky Normal Form (*CNF*), if all the productions are of the form  $A \rightarrow BC$  or  $A \rightarrow a$ . Let  $G$  be a *CFG* in *CNF*. To derive a string of terminals of length  $x$ , the number of products to be used is:

- a.  $2x - 1$
- b.  $2x$
- c.  $2x + 1$
- d.  $2^x$

isro2018 context-free-grammar theory-of-computation

Answer key

#### 37.2.2 Context Free Grammar: ISRO CSE 2020 | Question: 39



The language which is generated by the grammar  $S \rightarrow aSa \mid bSb \mid a \mid b$  over the alphabet of  $\{a, b\}$  is the set of

- A. Strings that begin and end with the same symbol
- B. All odd and even length palindromes
- C. All odd length palindromes
- D. All even length palindromes

isro-2020 theory-of-computation context-free-grammar normal

Answer key

#### 37.2.3 Context Free Grammar: ISRO-DEC2017-23



Identify the language generated by the following grammar

$$S \rightarrow AB$$

$$A \rightarrow aAb \mid \varepsilon$$

$$B \rightarrow bB \mid b$$

- A.  $\{a^m b^n \mid n \geq m, m > 0\}$
- B.  $\{a^m b^n \mid n \geq m, m \geq 0\}$
- C.  $\{a^m b^n \mid n > m, m > 0\}$
- D.  $\{a^m b^m \mid n > m, m \geq 0\}$

isrodec2017 theory-of-computation context-free-grammar

Answer key

## 37.3

### Context Free Language (3)

#### 37.3.1 Context Free Language: GATE CSE 2009 | Question: 12, ISRO2016-37



$$S \rightarrow aSa \mid bSb \mid a \mid b$$

The language generated by the above grammar over the alphabet  $\{a, b\}$  is the set of:

- A. all palindromes  
 C. strings that begin and end with the same symbol
- B. all odd length palindromes  
 D. all even length palindromes

gatecse-2009 theory-of-computation context-free-language easy isro2016

Answer key 

### 37.3.2 Context Free Language: ISRO CSE 2020 | Question: 37



Context free languages are closed under

- A. union, intersection  
 C. intersection, complement
- B. union, kleene closure  
 D. complement, kleene closure

isro-2020 theory-of-computation context-free-language easy

Answer key 

### 37.3.3 Context Free Language: ISRO CSE 2023 | Question: 56



Which of the following statements is FALSE?

- A. The intersection of a regular language and a context-free language is context-free  
 B. The intersection of a regular language and context-free language is regular  
 C. The union of two context-free languages is context-free  
 D. The union of two regular languages is regular

isro-cse-2023 theory-of-computation context-free-language regular-language

Answer key 

## 37.4 Context Sensitive (1)

### 37.4.1 Context Sensitive: ISRO CSE 2017 | Question: 11



Given the following statements

- S1 : Every context-sensitive language  $L$  is recursive
- S2 : There exists a recursive language that is not context-sensitive

Which statements are true?

- A. Only S1 is correct  
 C. Both S1 and S2 are not correct
- B. Only S2 is correct  
 D. Both S1 and S2 are correct

isro2017 theory-of-computation context-sensitive

Answer key 

## 37.5 Finite Automata (5)

### 37.5.1 Finite Automata: ISRO CSE 2014 | Question: 2



The number of states required by a Finite State Machine, to simulate the behavior of a computer with a memory capable of storing 'm' words, each of length 'n' bits is?

- A.  $m \times 2^n$   
 B.  $2^{m+n}$   
 C.  $2^{mn}$   
 D.  $m + n$

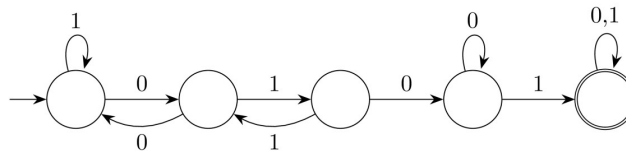
theory-of-computation finite-automata isro2014

Answer key 

### 37.5.2 Finite Automata: ISRO CSE 2014 | Question: 79



Consider the following Deterministic Finite Automaton  $M$ .



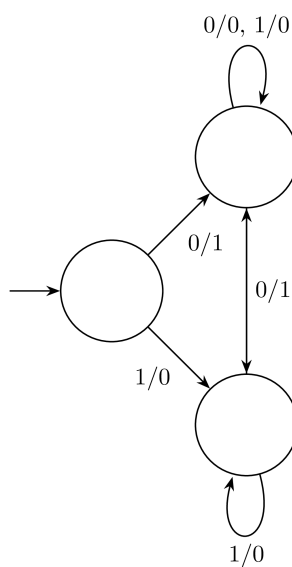
Let  $S$  denote the set of eight length bit strings whose second, third, sixth and seventh bits are 1. The number of strings in  $S$  that are accepted by  $M$  is

- A. 0                      B. 1                      C. 2                      D. 3

isro2014   theory-of-computation   finite-automata

Answer key

### 37.5.3 Finite Automata: ISRO CSE 2018 | Question: 26



The  $FSM$  (Finite State Machine) machine pictured in the figure above

- A. Complements a given bit pattern                      B. Finds  $2^i$ 's complement of a given bit pattern  
C. Increments a given bit pattern by 1                      D. Changes the sign bit

isro2018   finite-automata   theory-of-computation

Answer key

### 37.5.4 Finite Automata: ISRO CSE 2020 | Question: 41



Minimum number of states required in DFA accepting binary strings not ending in "101" is

- A. 3                      B. 4                      C. 5                      D. 6

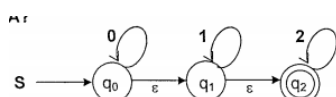
isro-2020   theory-of-computation   finite-automata   normal

Answer key

### 37.5.5 Finite Automata: ISRO-2013-73



What are the final states of the DFA generated from the following NFA?



- A.  $q_0, q_1, q_2$   
 C.  $q_0, [q_1, q_2]$

- B.  $[q_0, q_1], [q_0, q_2], []$   
 D.  $[q_0, q_1], q_2$

isro2013 theory-of-computation finite-automata

Answer key 

## 37.6

## Grammar (3)

### 37.6.1 Grammar: ISRO CSE 2008 | Question: 7



Consider the grammar

- $S \rightarrow ABCc \mid bc$
- $BA \rightarrow AB$
- $Bb \rightarrow bb$
- $Ab \rightarrow ab$
- $Aa \rightarrow aa$

Which of the following sentences can be derived by this grammar?

- A. abc                      B. aab                      C. abcc                      D. abbc

isro2008 theory-of-computation context-free-language grammar

Answer key 

### 37.6.2 Grammar: ISRO CSE 2011 | Question: 68



Which of the following sentences can be generated by

$S \rightarrow aS \mid bA$

$A \rightarrow d \mid cA$

- A. bccdd                      B. abbcca                      C. abcabc                      D. abcd

isro2011 theory-of-computation context-free-language grammar

Answer key 

### 37.6.3 Grammar: ISRO CSE 2023 | Question: 77



Consider the context-free grammar  $G$  below. There  $S$  is the starting non terminal symbol, while  $a$  and  $b$  are terminal symbols.

$S \rightarrow aaSb \mid T$

$T \rightarrow Tb \mid a$

Which of the following statements is true about the language  $L(G)$  generated by  $G$ ?

- A.  $aabbaabb$  belongs to  $L(G)$  but  $aabb$  does not  
 B.  $aaaaabbb$  belongs to  $L(G)$  but  $aaaabb$  does not  
 C.  $aaaabb$  belongs to  $L(G)$  but  $aabbaabb$  does not  
 D.  $aaabb$  belongs to  $L(G)$  but  $aaaaabbb$  does not

isro-cse-2023 theory-of-computation grammar

Answer key 

## 37.7

## Identify Class Language (2)

### 37.7.1 Identify Class Language: ISRO CSE 2016 | Question: 38



What is the highest type number that can be assigned to the following grammar?

$$S \rightarrow Aa, A \rightarrow Ba, B \rightarrow abc$$

- A. Type 0                      B. Type 1                      C. Type 2                      D. Type 3

theory-of-computation   identify-class-language   isro2016

Answer key 

### 37.7.2 Identify Class Language: ISRO CSE 2020 | Question: 40



Which of the following classes of languages can validate an IPv4 address in dotted decimal format? It is to be ensured that the decimal values lie between 0 and 255.

- A. RE and higher                      B. CFG and higher  
C. CSG and higher                      D. Recursively enumerable language

isro-2020   theory-of-computation   normal   identify-class-language

Answer key 

## 37.8 Minimal State Automata (1)

### 37.8.1 Minimal State Automata: ISRO CSE 2014 | Question: 12



How many states are there in a minimum state deterministic finite automaton accepting the language  $L = \{w \mid w \in \{0,1\}^*, \text{number of 0's is divisible by 2 and number of 1's is divisible by 5, respectively}\}$ ?

- A. 7                      B. 9                      C. 10                      D. 11

theory-of-computation   isro2014   minimal-state-automata

Answer key 

## 37.9 Pushdown Automata (1)

### 37.9.1 Pushdown Automata: GATE CSE 2009 | Question: 16, ISRO2017-12



Which one of the following is FALSE?

- A. There is a unique minimal DFA for every regular language  
B. Every NFA can be converted to an equivalent PDA.  
C. Complement of every context-free language is recursive.  
D. Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

gatecse-2009   theory-of-computation   easy   isro2017   pushdown-automata

Answer key 

## 37.10 Recursive and Recursively Enumerable Languages (2)

### 37.10.1 Recursive and Recursively Enumerable Languages: GATE CSE 2008 | Question: 13, ISRO2016-36



If  $L$  and  $\bar{L}$  are recursively enumerable then  $L$  is

- A. regular                      B. context-free  
C. context-sensitive                      D. recursive

gatecse-2008   theory-of-computation   easy   isro2016   recursive-and-recursively-enumerable-languages

Answer key 

**37.10.2 Recursive and Recursively Enumerable Languages: ISRO CSE 2011 | Question: 79**

A problem whose language is recursion is called?

- A. Unified problem  
C. Recursive problem  
B. Boolean function  
D. Decidable

isro2011 theory-of-computation recursive-and-recursively-enumerable-languages

Answer key

**37.11****Regular Expression (3)****37.11.1 Regular Expression: GATE CSE 1995 | Question: 1.9 , ISRO2017-13**

In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If  $L$  and  $D$  denote the sets of letters and digits respectively, which of the following expressions defines an identifier?

- A.  $(L + D)^+$   
B.  $(L.D)^*$   
C.  $L(L + D)^*$   
D.  $L(L.D)^*$

gate1995 theory-of-computation regular-expression easy isro2017

Answer key

**37.11.2 Regular Expression: ISRO CSE 2016 | Question: 33**

Let  $L = \{w \in (0 + 1)^* \mid w \text{ has even number of 1's}\}$ , i.e.  $L$  is the set of all bit strings with even number of 1's. Which one of the regular expression below represents  $L$ ?

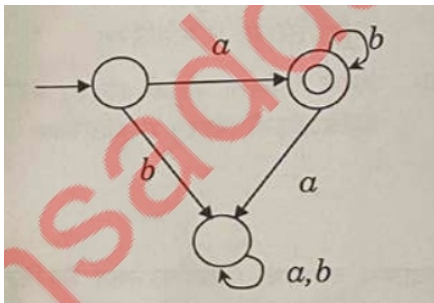
- A.  $(0^*10^*1)^*$   
B.  $0^*(10^*10^*)^*$   
C.  $0^*(10^*1^*)^*0^*$   
D.  $0^*1(10^*1)^*10^*$

theory-of-computation regular-expression isro2016

Answer key

**37.11.3 Regular Expression: ISRO CSE 2023 | Question: 13**

Consider the deterministic finite-state automaton (DFA) below. The alphabet is  $\{a, b\}$ . The state with a small incoming arrow is the initial state, while the double circle state denotes a final state.



Which of the following regular expressions defines the language accepted by the DFA?

- A.  $ab^*$   
B.  $a^*b^*$   
C.  $(ab)^*$   
D.  $a^*b$

isro-cse-2023 theory-of-computation regular-expression

Answer key

**37.12****Regular Language (3)**

### 37.12.1 Regular Language: ISRO CSE 2018 | Question: 24



Choose the correct statement -

- a.  $A = \{a^n b^n \mid n = 1, 2, 3, \dots\}$  is a regular language
- b. The set  $B$ , consisting of all strings made up of only  $a$ 's and  $b$ 's having equal number of  $a$ 's and  $b$ 's defines a regular language
- c.  $L(A^* B) \cap B$  gives the set  $A$
- d. None of the above

isro2018 theory-of-computation regular-language

Answer key

### 37.12.2 Regular Language: ISRO CSE 2020 | Question: 38



Which of the following is true?

- A. Every subset of a regular set is regular
- B. Every finite subset of non-regular set is regular
- C. The union of two non regular set is not regular
- D. Infinite union of finite set is regular

isro-2020 theory-of-computation regular-language easy

Answer key

### 37.12.3 Regular Language: ISRO2015-43



Let  $R_1$  and  $R_2$  be regular sets defined over the alphabet, then

- A.  $R_1 \cap R_2$  is not regular
- B.  $R_1 \cup R_2$  is not regular
- C.  $\Sigma^* - R_1$  is regular
- D.  $R_1^*$  is not regular

isro2015 theory-of-computation regular-language

Answer key

## 37.13

## Turing Machine (2)

### 37.13.1 Turing Machine: ISRO CSE 2014 | Question: 59



Which of the following is FALSE with respect to possible outcomes of executing a Turing Machine over a given input?

- A. it may halt and accept the input
- B. it may halt by changing the input
- C. it may halt and reject the input
- D. it may never halt

isro2014 theory-of-computation turing-machine

Answer key

### 37.13.2 Turing Machine: ISRO CSE 2016 | Question: 32



AN FSM(finite state machine) can be considered to be a turing machine of finite tape length

- A. without rewinding capability and unidirectional tape movement
- B. rewinding capability and unidirectional tape movement
- C. without rewinding capability and bidirectional tape movement
- D. rewinding capability and bidirectional tape movement

theory-of-computation isro2016 turing-machine

## Answer Keys

|         |     |
|---------|-----|
| 37.0.1  | Q-Q |
| 37.0.6  | B   |
| 37.2.2  | C   |
| 37.4.1  | D   |
| 37.5.5  | A   |
| 37.7.2  | A   |
| 37.11.1 | C   |
| 37.12.3 | C   |

|         |     |
|---------|-----|
| 37.0.2  | A   |
| 37.0.7  | Q-Q |
| 37.2.3  | X   |
| 37.5.1  | C   |
| 37.6.1  | X   |
| 37.8.1  | C   |
| 37.11.2 | B   |
| 37.13.1 | B   |

|         |     |
|---------|-----|
| 37.0.3  | C   |
| 37.1.1  | D   |
| 37.3.1  | B   |
| 37.5.2  | C   |
| 37.6.2  | D   |
| 37.9.1  | D   |
| 37.11.3 | Q-Q |
| 37.13.2 | A   |

|         |     |
|---------|-----|
| 37.0.4  | D   |
| 37.1.2  | B   |
| 37.3.2  | B   |
| 37.5.3  | X   |
| 37.6.3  | Q-Q |
| 37.10.1 | D   |
| 37.12.1 | D   |

|         |     |
|---------|-----|
| 37.0.5  | D   |
| 37.2.1  | A   |
| 37.3.3  | Q-Q |
| 37.5.4  | B   |
| 37.7.1  | D   |
| 37.10.2 | D   |
| 37.12.2 | B   |



**38.0.1 What are the wrong answers in ISRO 2016 official key?**

ISRO official key is released on 06 July 2016

What are the wrong answers in ISRO 2016 official key?

Just comment here.

According to ISRO

"CANDIDATES MAY RAISE OBJECTIONS TO THE ANSWER KEYS, IF ANY, WITH VALID JUSTIFICATIONS, WITHIN 5(FIVE) WORKING DAYS, I.E. WITHIN 14.07.2016, BY 06.00 PM. THE OBJECTIONS MAY BE SENT TO [icrb@isro.gov.in](mailto:icrb@isro.gov.in) WITH THE SUBJECT \*OBJECTIONS SCIENTIST/ENGINEER 'SC'-2016\*. ANNEXURES/ENCLOSURES MAY BE IN PDF FORMAT ONLY. CANDIDATES SHOULD MENTION THE BRANCH(ELECTRONICS/MECHANICAL/COMPSCIENCE), AND BOOKLET CODE FOR WHICH OBJECTIONS ARE RAISED.OBJECTIONS RECEIVED WITHOUT VALID JUSTIFICATIONS AND OBJECTIONS RECEIVED AFTER 14.07.2016 WILL NOT BE CONSIDERED."

isro2016

Answer key

## Answer Keys

38.0.1

Q-Q